CALIBRATION
A TECHNICIAN'S GUIDE

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Mike Cable

ISA TECHNICIAN SERIES



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TABLE OF CONTENTS

About the Author ix

Acknowledgements xi

Introduction xiii

Chapter 1 Calibration Principles 1

1.1 What Is Calibration? 1

1.2 What Are the Characteristics of a Calibration? . . . 2

1.3 Why Is Calibration Required? 6

1.4 Who Performs Calibrations? – The Control System Technician 8

1.5 Characteristics of a Control System Technician . 9

1.6 Loop Calibration vs. Individual Instrument Calibration 10

1.7 Bench Calibration vs. Field Calibration 11

1.8 Classification of Instruments 12

Chapter Summary 13

Review Questions 14

Chapter 2 Documentation 17

2.1 Calibration Procedure Content 19

2.2 Calibration Data Sheets (See Examples in Appendix A-4) 21

2.3 P&IDs (See Example in Appendix A-1) 23

2.4 Loop Diagrams (See Examples in Appendix A-2) 25

2.5 Instrument Specification Forms (See Examples in Appendix A-3) 26
2.6 Project Specifications 27
2.7 Manufacturer's Specifications 27
2.8 Calibration Intervals 28
2.9 Safety Considerations 29
2.10 Calibration Status Labels 30
Review Questions 31

Chapter 3 **Temperature Instrument Calibration** 33
3.1 What is Temperature? 33
3.2 Temperature Sensors 34
3.3 Signal Conversion 35
Review Questions 44

Chapter 4 **Pressure Instrument Calibration** 49
4.1 What is Pressure? 49
4.2 Challenges When Calibrating Pressure 50
4.3 Calibrating Pressure Gauges 52
4.4 Calibrating Pressure Transmitters 55
4.5 Calibrating Pressure Switches 55
Review Questions 56

Chapter 5 **Level Instrument Calibration** 61
5.1 Types of Level Instruments 61
5.2 Special Considerations with Level Calibration . 64
5.3 Calibrating a Differential Pressure Level Transmitter 67
5.4 Calibrating a Capacitance Level Instrument . . . 71
5.5 Calibrating an Ultrasonic Level Instrument . . . 72
Review Questions 74

Chapter 6 **Flow Instrument Calibration** 77
6.1 Types of Flow Instruments 77

6.2 Calibration of Flowmeters 82
6.3 D/P Transmitter Calibration 83
6.4 Example: Magnetic Flowmeter Calibration 84
6.5 Flowmeter Calibration Using a Master Meter (Prover) 86
6.6 Gravimetric Method for Flowmeter Calibration (Measurement by Weight) 87
Review Questions 88

Chapter 7 Final Control Devices Calibration 89
7.1 Calibration of an I/P Transducer 95
7.2 Calibration of a Valve Positioner 96
7.3 Calibration of a Control Valve 97
Review Questions 97

Chapter 8 Process Analytical Instrument Calibration 101
8.1 pH Calibration Procedure 105
8.2 Diagnostic Test for pH Electrodes 106
8.3 Measuring Conductivity 107
8.4 Calibration Procedure with Explanations 109
Review Questions 111

Bibliography 113
Appendix A-1. Piping & Instrument Diagrams (P&IDs) 117
Appendix A-2. Loop Diagrams 119
Appendix A-3. Instrument Specifications 123
Appendix A-4. Calibration Procedures 145
Appendix A-5. Test Equipment 227
Appendix A-6. RTD and Thermocouple Tables 229
Appendix A-7. Conversion Tables 245
Appendix B. Answers to Chapter Review Questions 249
Index 271

ABOUT THE AUTHOR

Michael Cable is a Level 3 Certified Control System Technician. He is currently the Validation Manager at Argos Therapeutics in Durham, NC where his responsibilities include managing the Calibration Program. Michael started his career as an Electronics Technician in the Navy Nuclear Power Program, serving as a Reactor Operator and Engineering Watch Supervisor aboard the USS Los Angeles submarine and then at the AIW prototype in Idaho Falls. After leaving the Navy, he started his civilian career at Performance Solutions performing technical services for the pharmaceutical industry. His 11 years there was highlighted by an assignment to Eli Lilly Corporate Process Automation managing Instrument Qualification projects and then starting up a Calibration Services division within Performance Solutions. His practical expertise in instrumentation and controls led him to his current career path in Validation, which has been his main focus for the last 7 years.

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INTRODUCTION

ISA's Certified Control System Technician (CCST) program requirements were developed based on a Job Analysis Report initiated by ISA – The Instrumentation, Systems, and Automation Society and Instrument Technicians Labor-Management Cooperation Fund. Response to the survey used to validate the Job Analysis Report indicated that *calibration* was the most important, most critical, and most frequently performed of all seven domains for each of the three levels of certification. This is not to minimize the importance of the other six domains, because each is important to all technicians. However, it is obvious that calibration is what most of us do every day. The purpose of this book is to serve as a:

- Study guide for the Calibration domain in CCST certification.
- Reference for technicians who perform process instrument calibrations.
- Reference in the classroom for students pursuing studies related to instrumentation.

This text is applicable for control system technicians performing maintenance and calibrations within the process industry. Although most of the principles would apply, this is not meant for metrologists performing calibrations of test standards in a standards laboratory.

Mike Cable

1

CALIBRATION PRINCIPLES

After completing this chapter, you should be able to:

Define key terms relating to calibration and interpret the meaning of each.

Understand traceability requirements and how they are maintained.

Describe characteristics of a good control system technician.

Describe differences between bench calibration and field calibration. List the advantages and disadvantages of each.

Describe the differences between loop calibration and individual instrument calibration. List the advantages and disadvantages of each.

List the advantages and disadvantages of classifying instruments according to process importance—for example, critical, non-critical, reference only, OSHA, EPA, etc.

1.1 WHAT IS CALIBRATION?

There are as many definitions of calibration as there are methods. According to ISA's *The Automation, Systems, and Instrumentation Dictionary*, the word calibration is defined as "a test during which known values of measurand are applied to the transducer and corresponding output readings are recorded under specified conditions." The definition includes the capability to adjust the instrument to zero and to set the desired span. An interpretation of the definition would say that a calibration is a comparison of measuring equipment against a standard instrument of higher accuracy to detect, correlate, adjust, rectify and document the accuracy of the instrument being compared.

Typically, calibration of an instrument is checked at several points throughout the calibration range of the instrument. The *calibration range* is defined as "the region between the limits within which a quantity is measured, received or transmitted, expressed by stating the lower and

upper range values." The limits are defined by the zero and span values. The *zero* value is the lower end of the range. *Span* is defined as the algebraic difference between the upper and lower range values. The calibration range may differ from the *instrument range*, which refers to the capability of the instrument. For example, an electronic pressure transmitter may have a nameplate instrument range of 0–750 pounds per square inch, gauge (psig) and output of 4-to-20 milliamps (mA). However, the engineer has determined the instrument will be calibrated for 0-to-300 psig = 4-to-20 mA. Therefore, the calibration range would be specified as 0-to-300 psig = 4-to-20 mA. In this example, the zero input value is 0 psig and zero output value is 4 mA. The input span is 300 psig and the output span is 16 mA.

Different terms may be used at your facility. Just be careful not to confuse the range the instrument is capable of with the range for which the instrument has been calibrated.

1.2 WHAT ARE THE CHARACTERISTICS OF A CALIBRATION?

Calibration Tolerance: Every calibration should be performed to a specified tolerance. The terms tolerance and accuracy are often used incorrectly. In ISA's *The Automation, Systems, and Instrumentation Dictionary*, the definitions for each are as follows:

Accuracy: The ratio of the error to the full scale output or the ratio of the error to the output, expressed in percent span or percent reading, respectively.

Tolerance: Permissible deviation from a specified value; may be expressed in measurement units, percent of span, or percent of reading.

As you can see from the definitions, there are subtle differences between the terms. It is recommended that the tolerance, specified in measurement units, is used for the calibration requirements performed at your facility. By specifying an actual value, mistakes caused by calculating percentages of span or reading are eliminated. Also, tolerances should be specified in the units measured for the calibration.

For example, you are assigned to perform the calibration of the previously mentioned 0-to-300 psig pressure transmitter with a specified calibration tolerance of ±2 psig. The output tolerance would be:

$$\begin{array}{r} 2 \text{ psig} \\ \div\, 300 \text{ psig} \\ \times\, 16 \text{ mA} \\ \hline 0.1067 \text{ mA} \end{array}$$

The calculated tolerance is rounded down to 0.10 mA, because rounding to 0.11 mA would exceed the calculated tolerance. It is recommended that both ±2 psig and ±0.10 mA tolerances appear on the calibration data sheet if the remote indications and output milliamp signal are recorded.

Note the manufacturer's specified accuracy for this instrument may be 0.25% full scale (FS). Calibration tolerances should not be assigned based on the manufacturer's specification only. Calibration tolerances should be determined from a combination of factors. These factors include:

- Requirements of the process
- Capability of available test equipment
- Consistency with similar instruments at your facility
- Manufacturer's specified tolerance

Example: The process requires ±5°C; available test equipment is capable of ±0.25°C; and manufacturer's stated accuracy is ±0.25°C. The specified calibration tolerance must be between the process requirement and manufacturer's specified tolerance. Additionally the test equipment must be capable of the tolerance needed. A calibration tolerance of ±1°C might be assigned for consistency with similar instruments and to meet the recommended accuracy ratio of 4:1.

Accuracy Ratio: This term was used in the past to describe the relationship between the accuracy of the test standard and the accuracy of the instrument under test. The term is still used by those that do not understand uncertainty calculations (uncertainty is described below). A good rule of thumb is to ensure an accuracy ratio of 4:1 when performing calibrations. This means the instrument or standard used should be four times more accurate than the instrument being checked. Therefore, the test

equipment (such as a field standard) used to calibrate the process instrument should be four times more accurate than the process instrument, the laboratory standard used to calibrate the field standard should be four times more accurate than the field standard, and so on.

With today's technology, an accuracy ratio of 4:1 is becoming more difficult to achieve. Why is a 4:1 ratio recommended? Ensuring a 4:1 ratio will minimize the effect of the accuracy of the standard on the overall calibration accuracy. If a higher level standard is found to be out of tolerance by a factor of two, for example, the calibrations performed using that standard are less likely to be compromised.

Suppose we use our previous example of the test equipment with a tolerance of ±0.25°C and it is found to be 0.5°C out of tolerance during a scheduled calibration. Since we took into consideration an accuracy ratio of 4:1 and assigned a calibration tolerance of ±1°C to the process instrument, it is less likely that our calibration performed using that standard is compromised.

The out-of-tolerance standard still needs to be investigated by reverse traceability of all calibrations performed using the test standard. However, our assurance is high that the process instrument is within tolerance. If we had arbitrarily assigned a calibration tolerance of ±0.25°C to the process instrument, or used test equipment with a calibration tolerance of ±1°C, we would not have the assurance that our process instrument is within calibration tolerance. This leads us to traceability.

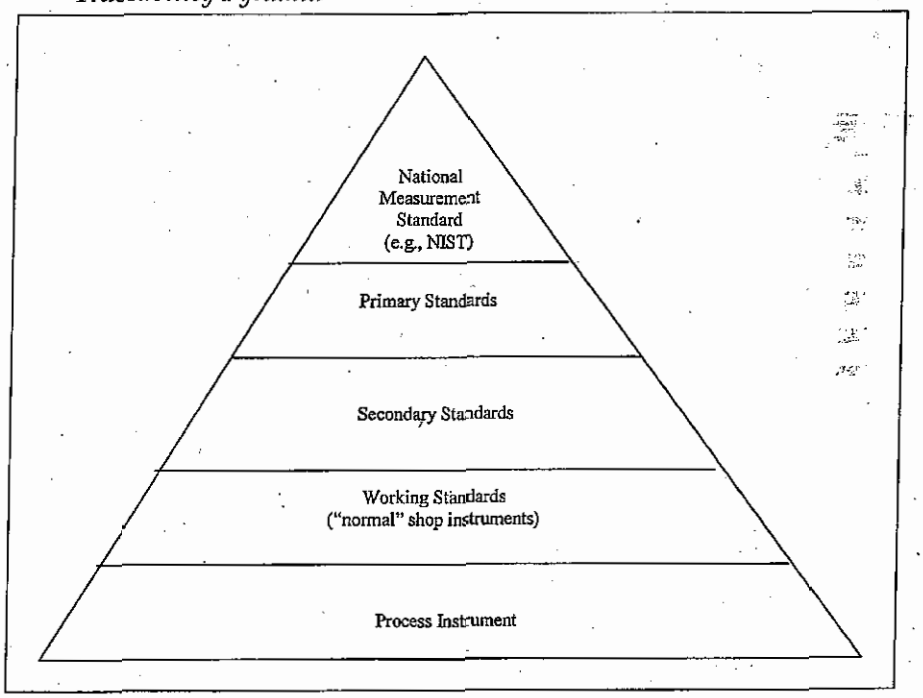
Traceability: All calibrations should be performed traceable to a nationally or internationally recognized standard. For example, in the United States, the National Institute of Standards and Technology (NIST), formerly National Bureau of Standards (NBS), maintains the nationally recognized standards. *Traceability* is defined by ANSI/NCSL Z540-1-1994 (which replaced MIL-STD-45662A) as "the property of a result of a measurement whereby it can be related to appropriate standards, generally national or international standards, through an unbroken chain of comparisons." Note this does not mean a calibration shop needs to have its standards calibrated with a primary standard. It means that the calibrations performed are traceable to NIST through all the standards used to calibrate the standards, no matter how many levels exist between the shop and NIST.

Traceability is accomplished by ensuring the test standards we use are routinely calibrated by "higher level" reference standards. Typically the standards we use from the shop are sent out periodically to a standards lab which has more accurate test equipment. The standards

from the calibration lab are periodically checked for calibration by "higher level" standards, and so on until eventually the standards are tested against Primary Standards maintained by NIST or another internationally recognized standard.

The calibration technician's role in maintaining traceability is to ensure the test standard is within its calibration interval and the unique identifier is recorded on the applicable calibration data sheet when the instrument calibration is performed. Additionally, when test standards are calibrated, the calibration documentation must be reviewed for accuracy and to ensure it was performed using NIST traceable equipment.

FIGURE 1-1.
Traceability Pyramid



Uncertainty: Parameter, associated with the result of a measurement that characterizes the dispersion of the values that could reasonably be attributed to the measurand. Uncertainty analysis is required for calibration labs conforming to ISO 17025 requirements. Uncertainty analysis is performed to evaluate and identify factors associated with the calibration equipment and process instrument that affect the calibration accuracy. Calibration technicians should be aware of basic uncertainty analysis factors, such as environmental effects and how to combine

multiple calibration equipment accuracies to arrive at a single calibration equipment accuracy. Combining multiple calibration equipment or process instrument accuracies is done by calculating the square root of the sum of the squares, illustrated below:

Calibration equipment combined accuracy

$$\sqrt{(\text{calibrator1 error})^2 + (\text{calibrator2 error})^2 + (\text{etc. error})^2}$$

Process instrument combined accuracy

$$\sqrt{(\text{sensor error})^2 + (\text{transmitter error})^2 + (\text{indicator error})^2 + (\text{etc. error})^2}$$

1.3 WHY IS CALIBRATION REQUIRED?

It makes sense that calibration is required for a new instrument. We want to make sure the instrument is providing accurate indication or output signal when it is installed. But why can't we just leave it alone as long as the instrument is operating properly and continues to provide the indication we expect?

Instrument error can occur due to a variety of factors: drift, environment, electrical supply, addition of components to the output loop, process changes, etc. Since a calibration is performed by comparing or applying a known signal to the instrument under test, errors are detected by performing a calibration. An error is the algebraic difference between the indication and the actual value of the measured variable. Typical errors that occur include:

FIGURE 1-2.
Span Error

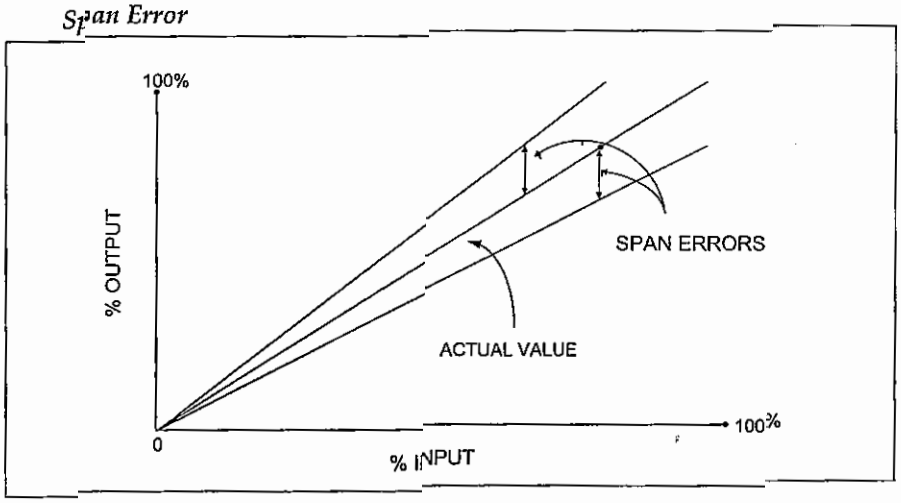


FIGURE 1-3.
Zero Error

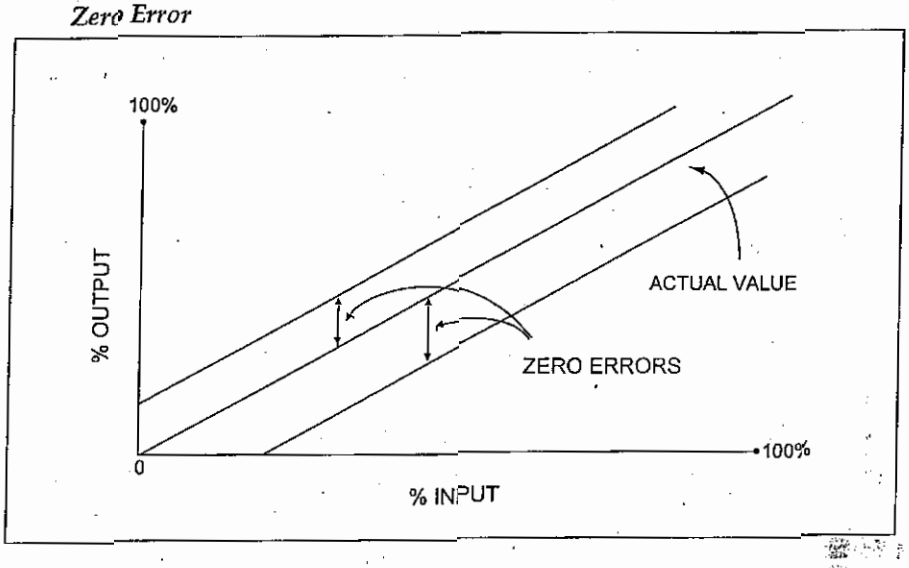


FIGURE 1-4.
Combined Zero and Span Error

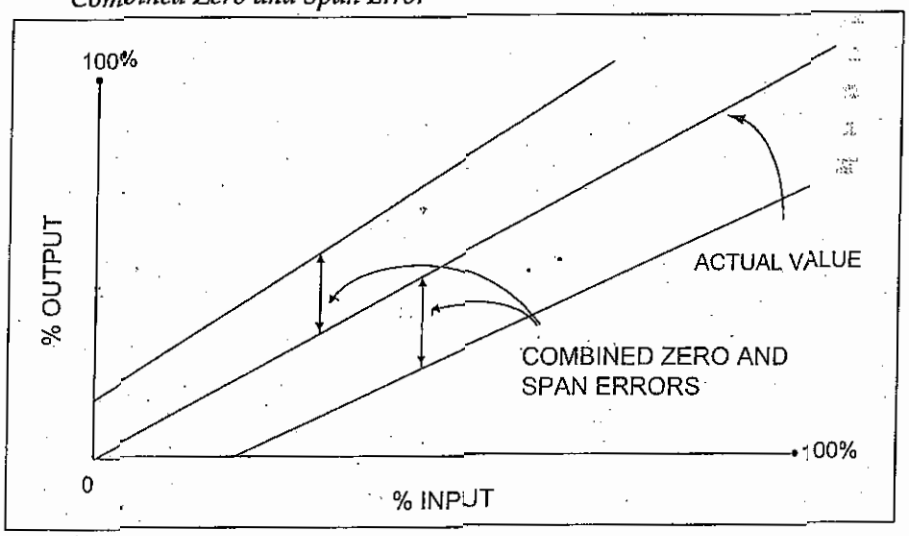
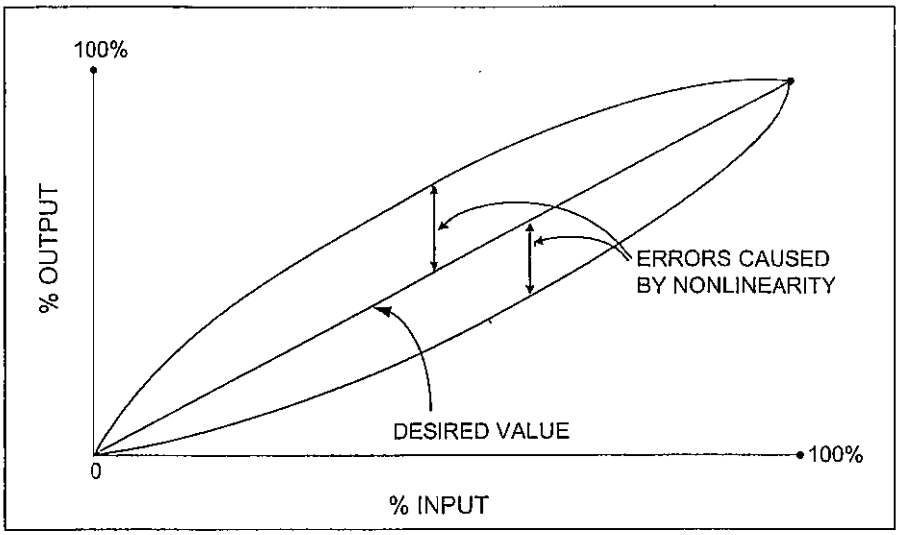


FIGURE 1-5.
Linearization Error



Zero and span errors are corrected by performing a calibration. Most instruments are provided with a means of adjusting the zero and span of the instrument, along with instructions for performing this adjustment. The zero adjustment is used to produce a parallel shift of the input-output curve. The span adjustment is used to change the slope of the input-output curve. Linearization error may be corrected if the instrument has a linearization adjustment. If the magnitude of the nonlinear error is unacceptable and it cannot be adjusted, the instrument must be replaced.

To detect and correct instrument error, periodic calibrations are performed. Even if a periodic calibration reveals the instrument is perfect and no adjustment is required, we would not have known that unless we performed the calibration. And even if adjustments are not required for several consecutive calibrations, we will still perform the calibration check at the next scheduled due date. Periodic calibrations to specified tolerances using approved procedures are an important element of any quality system.

1.4 WHO PERFORMS CALIBRATIONS? – THE CONTROL SYSTEM TECHNICIAN

A control system technician (CST) is a skilled craftsperson who knows pneumatic, mechanical, and electrical instrumentation. He or she understands process control loops and process control systems, including

those that are computer-based. Typically, he or she has received training in such specialized subjects as theory of control, analog and/or digital electronics, microprocessors and/or computers, and the operation and maintenance of particular lines of field instrumentation.

A CST performs calibration, documentation, loop checks, troubleshooting, and repair or replacement of instrumentation. These tasks relate to systems that measure and control level, temperature, pressure, flow, force, power, position, motion, physical properties, chemical composition and other process variables.

1.5 CHARACTERISTICS OF A CONTROL SYSTEM TECHNICIAN

Honesty and Integrity: A CST must possess honesty and integrity above all else. Most technicians work independently much of the time. Calibrations must be performed in accordance with procedures and must be properly documented. Additionally, the calibration department may be understaffed and production schedules may demand unrealistic completion requirements. These factors can have a real impact on proper performance and documentation of calibrations. Remember: Nobody can take away your integrity; only you can give it away.

Attention to Detail: Calibrations should be performed in accordance with detailed instructions. Each different make/model instrument is adjusted differently. Each instrument is installed in a different physical and loop configuration. Because of these and many other differences, attention to detail is very important. The minute a technician is not paying attention to detail, safety and proper performance are jeopardized.

Excellent Documentation Practices: In many facilities, the impression of quality is determined by the content and appearance of documentation. Many technicians complain the paperwork is 90% of the work. In today's world of ISO9000, cGMPs, A2LA, and other quality standards, documentation is essential. If it isn't documented, it wasn't done. Calibration Data Sheets must be neat, complete, signed and, if required, reviewed in a timely manner. When changes occur, all related documentation, such as drawings, manuals, specifications and databases must also be updated.

Understanding of Processes: One thing that sets technicians apart is an understanding of the process, particularly how the instruments monitor and control the process. There is a difference between calibrating an individual component and calibrating an instrument as part of the bigger process control loop. For example, knowing when a controller can be placed in manual without affecting the process and what to do while that controller is in manual, requires an understanding of the process. Additionally, when an operator says there is a problem with his indication, a technician who knows the instrument loop and process will be more capable of identifying the cause of the problem.

Some basic concepts on how calibrations should be performed need to be discussed before we go on. Some of these may be new concepts not used in your facility, but you should be familiar with them. Some of these practices are industry dependent. Although calibrations are generally performed the same, some different practices have developed. These practices are:

- Loop Calibration vs. Individual Instrument Calibration
- Bench Calibration vs. Field Calibration
- Classification of Instruments as Critical, Non-Critical, For Reference Only, etc.

1.6 LOOP CALIBRATION VS. INDIVIDUAL INSTRUMENT CALIBRATION

An *individual instrument calibration* is a calibration performed only on one instrument. The input and output are disconnected. A known source is applied to the input, and the output is measured at various data points throughout the calibration range. The instrument is adjusted, if necessary, and calibration is checked.

DISADVANTAGES OF INDIVIDUAL CALIBRATION	ADVANTAGES OF INDIVIDUAL CALIBRATION
1. Entire loop is not verified within tolerance 2. Mistakes on re-connect 3. Less efficient use of time to do one calibration for each loop instrument as opposed to one calibration for the loop	1. Correct instrument will be adjusted 2. More compatible with multifunction calibrators

A *loop calibration* is performed from the sensor to all loop indications with all the loop components connected. For example, a temperature sensor connected to a temperature transmitter would be inserted in a temperature bath/block. (Note: Either the bath/block would be calibrated or a temperature standard would be used in the bath/block for traceability.) The temperature of the bath/block would be adjusted to each data point required to perform the calibration. All local and remote indications would be recorded. It is also recommended to record the transmitter output. If all indications and transmitter output are within tolerance, the loop is within tolerance. If any loop component is not within tolerance, then a calibration is performed on that instrument. Do not adjust a transmitter to correct a remote indication.

ADVANTAGES OF LOOP CALIBRATION	DISADVANTAGES OF LOOP CALIBRATION
1. Entire loop, including sensor, is verified within tolerance 2. Mistakes on re-connect minimized 3. More efficient use of time to do one calibration for loop as opposed to one calibration for each loop instrument	1. Wrong instrument may be adjusted to bring the loop within calibration 2. Not as compatible with multifunction calibrators used for "paperless" data collection

1.7 BENCH CALIBRATION VS. FIELD CALIBRATION

A *bench calibration* is performed in the shop on the bench with power supplied from an external source, if required. Bench calibrations may be performed upon receipt of new instruments prior to installation. This provides assurance the instrument is received undamaged. This also allows configuration and calibration in a more favorable environment. Some companies perform the periodic calibrations on the bench. In this case the process instrument is removed from service, disconnected and taken to the shop for calibration. In some instances, a spare is installed in its place so the process downtime is minimized. For example, critical flow sensors might be sent out to a certified flow calibration facility. To prevent shutting the process down for several weeks, a replacement flow sensor would be installed.

Field calibrations are performed "in-situ," or in place, as installed. The instrument being calibrated is not removed from the installed location. Field calibrations may be performed after installation to ensure proper

connections and configuration. Periodic calibrations are more likely to be performed in the field. Field calibrations are performed in the environment in which the instrument operates. If the instrument is installed in a harsh environment it is calibrated for that environment. If the instrument is removed for a bench calibration and then returned, some error may be introduced due to the ambient conditions and orientation.

ADVANTAGES OF BENCH CALIBRATION	ADVANTAGES OF FIELD CALIBRATION
1. Removed, cleaned, inspected 2. Better work environment 3. Fixed calibration setup and utilities (electrical, air, vacuum) available	1. May save time 2. May identify and allow trouble-shooting of installation problems 3. Performed in actual ambient environment

1.8 CLASSIFICATION OF INSTRUMENTS

In some industries or even within individual companies it may be advantageous to classify your instruments in a way that indicates the instruments' "importance." There are two schools of thought here. Some say that no instrument is more important than any other instrument. However, in some processes, the undetected error in an instrument may result in product rejections or even product recalls. Additionally, some instruments have calibration requirements specified by outside agencies. For these reasons, it is recommended that each instrument is assigned a classification. ISA-TR91.00.02-2003, *Criticality Classification Guideline for Instrumentation*, is an excellent resource to assist with establishing classification of instrumentation. One example used for classifying instruments is outlined below.

Critical: An instrument which, if not conforming to specification, could potentially compromise product or process quality.

Non-critical: An instrument whose function is not critical to product or process quality, but whose function is more of an operational significance. Example: An instrument that is not classified as critical, but the reading obtained from the instrument is recorded in operating logs.

Reference Only: An instrument whose function is not critical to product quality, not significant to equipment operation, and not used for making

quality decisions. Routine calibration may be less frequent and verification of proper operation will be performed if suspect of error.

OSHA: Calibration of the instrument is mandated by the Occupational Safety and Health Administration.

EPA: Calibration of the instrument is mandated by the EPA. Example: Calibration of the flow totalizer for the wastewater treatment system may be required by EPA.

The above classifications may be helpful in assigning calibration frequencies. For example, you might assign a calibration frequency of six months to a "Critical" pressure transmitter. The same pressure transmitter assigned as "Non-critical" might be calibrated every 12 months.

Another advantage to assigning classifications is to investigate out-of-tolerance calibrations more efficiently. In many industries, out-of-tolerance calibrations are formally reported to the Quality Department. If classification of instruments is not used, all out-of-tolerance calibrations must be investigated for the effect on product. If the instruments are classified, and classifications are approved by the Quality Department, the investigation is performed only for Critical instruments. Of course, the Calibration Department should investigate all out-of-tolerance conditions, but the release of product would not be held up due to an unnecessary investigation.

CHAPTER SUMMARY

In this chapter we covered the What, Why, Who, and How as an introduction to Calibration. We've covered some definitions and concepts that calibration technicians need to be familiar with. It should be emphasized that not all of these concepts are applicable to your facility. Although it would be convenient if we all ran our calibration programs exactly the same way, it just isn't so. Most of what will be presented in this book are examples that do not fit every situation.

REVIEW QUESTIONS

1. Match the term on the left with the definition on the right.

___ Calibration	A. permissible deviation from specified value
___ Instrument Range	B. upper and lower values specified for facility
___ Calibration Range	C. algebraic difference between the upper and lower range value
___ Accuracy	D. adjustment used to produce a parallel shift of the input-output curve
___ Tolerance	E. comparison of instrument to a known value
___ Traceability	F. percent error
___ Zero	G. characterizes the dispersion of the values that could reasonably be attributed to the measurand
___ Span	H. upper and lower values specified by manufacturer
___ Uncertainty	I. measurement related to standards through an unbroken chain of comparisons

2. Which of the following errors is typically not correctable?

   A. Zero
   B. Span
   C. Linearity
   D. Zero, span, and linearity errors are always correctable

3. Why should a calibration technician have:

   A. Honesty and Integrity?

   B. Attention to detail?

   C. Excellent documentation practices?

   D. Understanding of processes?

4. What are the advantages of performing a field calibration? Disadvantages?

5. What are the advantages of performing a bench calibration? Disadvantages?

6. What are the advantages of performing a loop calibration? Disadvantages?

7. What are the advantages of performing an individual instrument calibration? Disadvantages?

8. What are the advantages of classifying instruments by their "importance/criticality" to a process?

9. Arrange the traceability hierarchy below, beginning with lowest level and ending with the highest. ___, ___, ___, ___, ___

   A. Primary Standards
   B. Working Standards
   C. Process Instrument
   D. NIST (or recognized national standard)
   E. Secondary Standards

2

DOCUMENTATION

After completing this chapter, you should be able to:

List the fundamental elements of a calibration procedure.

List the fundamental elements of a calibration data sheet.

Define the relevant calibration information contained in the following resource documents: P&ID, Loop Diagram, Instrument Specification Sheet, Project Specifications, and Manufacturer's Specifications.

List the resources for determination of initial calibration frequency for an instrument.

Identify safety considerations relating to calibration.

Describe the use of calibration status labels and what information is required on each.

Because of quality system requirements throughout industry, documentation has become as important as the actual performance of a calibration. This chapter will summarize documentation that all calibration technicians should be familiar with. Another ISA text, *Instrumentation and Control Systems Documentation*, details the documentation mentioned in this chapter, and more.

The accuracy and reliability of instrumentation in a facility is maintained through the development and implementation of a quality calibration program. In addition to inventorying instruments, determining calibration parameters and intervals, and purchasing appropriate test standards, a calibration program includes written procedures for performing calibrations. The level of detail contained in calibration procedures can vary considerably—from a generic procedure used to calibrate an instrument type to a very specific procedure used to calibrate one particular instrument. Different types of calibration procedures are discussed below and examples of each are included in Appendix A-4.

Technical Manual procedure: Typically, a manufacturer's technical manual is provided for each similar model of instrument. Calibration instructions for the instrument are usually given in the technical manual. These instructions can be adopted or adapted as the calibration procedure for the applicable instruments. Most often, the manufacturer's calibration procedure is used to develop a company calibration procedure approved by management and the quality department. However, the actual technical manual procedure may be used or referenced as the calibration procedure if this practice is approved.

TABLE 2-1.
Using Calibration Procedures from a Technical Manual

ADVANTAGES	DISADVANTAGES
Very little time/resources required to develop procedures	Does not contain all necessary elements of a calibration procedure
Technically accurate and detailed instructions for the specific instrument	Applies to instrument only, not taking into account application/process

Generic procedure: Generic calibration procedures can be developed for each instrument type. For example, one procedure could be developed for Electronic Pressure/Vacuum Transmitters and another procedure for Pneumatic Temperature Controllers. You could even go more generic and develop a procedure for Pressure Instrument Calibration, which would include gauges, transmitters, etc. for pressure, differential pressure, and vacuum. Generic procedures should recommend using the manufacturer's technical manual to perform any necessary adjustments.

TABLE 2-2.
Generic Calibration Procedures

ADVANTAGES	DISADVANTAGES
Limits number of procedures to a manageable level	Inconsistent methods by different technicians using same procedure
Can be a good first step for new facility start-up until more effort can be devoted to procedure development	Inexperienced technicians need more detail

Specific procedure for an instrument or a manufacturer/model: In most facilities, some calibration procedures will have to be very specific to particular instruments. In some cases, specific detailed procedures are required for each instrument. Analytical instruments for such parameters as

conductivity (resistivity), oxygen, and lab instrumentation typically require specific procedures for each type due to the unique differences in the way each is calibrated. Also, if your generic procedure for a particular pressure instrument does not adequately address the proper method, a specific procedure should be developed. Obviously, if safety could be compromised by using an inadequate generic procedure, a new procedure addressing the specific safety issues must be developed.

TABLE 2-3.
Specific Calibration Procedures

ADVANTAGES	DISADVANTAGES
Calibrations are performed the same way by all technicians	Increases resources required to develop, maintain, and track procedures
May take into account the effect on process	
Technically accurate and detailed instructions for the specific instrument	

Some "old school" technicians initially dislike the use of calibration procedures. They say, "We've always done it this way and never had any problems. Now we've got all this paperwork, it's a wonder we can get any actual work done." Well, in today's regulated environment, you'll be out of business if you don't have a documented calibration program in place. Your customers must be assured of a certain quality product based on parameters you said you could maintain. Most of those assurances are provided from accurate process instrumentation. It has been demonstrated that instrument accuracy deteriorates during use. The only way to keep track of accuracy at any given time is to verify, adjust, and document the calibration data.

2.1 CALIBRATION PROCEDURE CONTENT

What information should be included in a calibration procedure? First, the format should follow the format required by your company procedures. Also any governing documents, such as the Calibration Policy, and any applicable procedures subordinate to the Calibration Policy must be followed.

The calibration procedure typically includes most or all of the following sections:

- *Purpose*: Clearly states the reason for the procedure such as: "The purpose of this procedure is to provide standardized instructions for the calibration of temperature instruments."
- *Scope*: Clearly states to what and to whom the procedure applies such as "This procedure applies to the calibration of all analog pressure gauges at the ABC Company calibrated by employee and contract technicians."

  Note: The purpose and scope can be combined into one section of the procedure. The information in these sections is typically obvious to those of us who perform calibrations, but the managers need this so they know what they're approving.
- *Definitions*: Contains brief descriptions of key terms, as applicable, for clarity. Acronyms and abbreviations used in the procedure are noted in this section to document their meaning throughout the text.
- *References/Attachments*: Identifies other documents, including attachments, that are required to be used in conjunction with the procedure, or allow the user to gain further information regarding the procedural content.
- *Test Equipment/Materials Required*: Identifies the test equipment and materials required to perform the procedure. Listing specific test equipment in this section helps to ensure uncertainty requirements are met and/or the desired accuracy ratio is achieved, particularly if the minimum tolerance achievable is specified in the Scope or Title of the procedure. Note that if the specified test equipment is not available, the technician must notify the supervisor prior to performing the calibration.
- *Safety*: Provide information on potential human health hazards and potential hazards to the facility, equipment, or process. All Safety Work Permit requirements and Material Safety Data Sheet (MSDS) references are included in this section.
- *Prerequisites/Initial Conditions* (optional): Provides any conditions that should be met prior to performing the calibration, such as tank drained, controller in manual, or system shutdown. Alternatively,

these conditions can be included in the test procedure and/or notes printed on the calibration data sheet.

- *Test Procedure*: This is the meat of the procedure which outlines the procedure in a clear, concise, step-by-step manner. If any steps of the calibration procedure cannot be performed as specified, the technician must return the instrument to a safe condition and notify the supervisor.
- *Acceptance Criteria*: The pass/fail criteria may be included in the Test Procedure section or as a separate step at completion, evaluating the results obtained against the tolerances specified.
- *Approvals*: The author and approval signatures or approval authority should be included on each calibration procedure.

As you can see there is a lot of flexibility in what information is included in the calibration procedure and where the information is located in the procedure. Much of this depends on the culture at your facility and experience level of the technicians. Personally, I like to develop good generic procedures with the instrument specifics printed on the calibration data sheet used to record calibration data. There are some example procedures in the reference section. Some comply with what has been described above, some do not.

2.2 CALIBRATION DATA SHEETS

(See Examples in Appendix A-4)

You may refer to this as something else, but what we're talking about here is the form where the as-found and as-left calibration data is recorded when a calibration is performed. As of now, most calibration data sheets are still hardcopy, usually printed out from a calibration software package. Some are still totally manual with no pre-printed information, and that's OK, it works. We are seeing increased use of paperless calibrations where the data is either automatically collected by a documenting calibrator or manually entered into a handheld device as an electronic record. In any case, the following information should be included as part of a calibration data sheet:

- *Instrument Tag Number/Instrument Identification Number*: This is a unique identifier or unique combination used as the main tracking number for each instrument. In most cases the tag number is the

P&ID tag number if applicable (P&IDs are discussed later in this chapter). If the instrument is not associated with a P&ID, there should be some consistent tag number system at your facility that utilizes the ISA-5.1-1984-(R1992) standard.

Several facilities use an additional identification number which is sequentially assigned as an additional tracking number. The tag number references the instrument location within a system, and the instrument identification number stays with the instrument. This way the history of any instrument installed in the instrument location (Tag Number) is traceable and the history of any instrument is traceable. Many instruments stay in the same location for the life of the equipment, in which case this is not as important. But, failures occur and instruments need to be replaced. In other instances, spares are installed temporarily to keep a process running when an instrument is removed for calibration. This traceability is important, for example, in a pharmaceutical facility that produces penicillin. Any instrument removed from a penicillin manufacturing area cannot be used in any other manufacturing area. The instrument identification number used to track where an instrument has been is a useful tool in ensuring that instrument is not installed where it shouldn't be. Even if you don't have similar needs, it's good practice to use both types of identification numbers.

- *Nameplate Data*: The *manufacturer*, *model number*, and *serial number* should be listed on the Calibration Data Sheet.
- *Calibration Range* and *Calibration Tolerance*: This defines the upper and lower limit used for calibration. Ideally, this is the input and output range, if applicable. For example, a good format for the calibration range of a temperature transmitter would be 0-to-100°C = 4-to-20 mA with a calibration tolerance of ±1.0°C/±0.16 mA. As we saw in Chapter 1, the calibration range is not always the same as the instrument range or capability of the instrument.
- *Location*: Be as specific as possible about location of the instrument. You don't want your new technician wasting hours searching.
- *Calibration Procedure Number*: This is the calibration procedure that is used to perform the calibration. In some facilities, the entire calibration procedure is printed on the calibration data sheet.

- *Last Calibration Date, Calibration Due Date,* and *Calibration Interval:* This information should be included to ensure the calibration is being performed periodically as required.
- *As-found data* and *As-left data*: Relates the test points specified with the corresponding test standard value. If all as-found data is within tolerance and no adjustments are made, the as-left data would be N/A or the same as the as-found data. Note: Every effort should be made to record as-found data for failed instruments prior to making any adjustments, in order to provide the most data for evaluation.
- *Test Standards*: Record the unique identification of any test standards used to perform the calibration and, if required by procedure, record the calibration due date of the standard(s). It would be best to record this prior to beginning the calibration to ensure each standard is within its calibration periodicity (but, of course you checked this when you obtained the standard from the shop). The most important reason for documenting the test standards is for reverse traceability in case a standard is found to be out of tolerance. If a test standard is found out of tolerance on its next calibration, it is critical that any calibration performed using that standard since its last calibration is known and evaluated to determine a course of action.
- *Comments*: The technician needs someplace to record any comments or observations.
- *Technician Signature* and *Date of Calibration*
- *Supervisor or Reviewer Signature* and *Date*

Example Calibration Data Sheets are included in Appendix A-4. In addition, some example calibration procedures include a data sheet as an attachment.

2.3 P&IDs (See Example in Appendix A-1)

Process/Piping and Instrument Diagrams/Drawings (P&IDs) are drawings that provide a detailed overview of a process system. They include major components, utilities, flowpaths, supporting equipment, and instrumentation. Although P&IDs are commonly used throughout industry, there is not yet a "standard" P&ID. Process Industry Practices

(www.pip.org) has developed a P&ID practice; however PIP is not a developer of standards. Draft-Standard-for-Trial-Use, ISA-DSTU-5.07.01, Piping and Instrumentation Diagram Documentation Criteria, was issued in May 2002. ISA-DSTU-5.07.01 is the result of a cooperative effort between ISA (SP5.7) and Process Industries Practices (PIP) by which ISA will develop an American National Standard covering the requirements, design, and graphic elements of that class of engineering drawings called a P&ID, or Piping & Instrumentation Diagram. ISA-DSTU-5.07.01 is based on existing PIP Practice PIC001.

ISA standards ISA-5.1-1984-(R1992), *Instrumentation Symbols and Identification*, and ISA-5.3-1983, *Graphic Symbols for Distributed Control Shared Display Instrumentation, Logic, and Computer Systems*, are the most generally accepted guides for developing symbolism for instrumentation and control systems.

From the P&ID of a particular system, a technician can determine pertinent information about the instrumentation and controls applicable to performing calibrations and understanding the system operation. The most important of these include:

- Components of an instrument loop
- Functional identification
- Methods of signal transmission (pneumatic, electronic, hydraulic, software link, etc.)
- Controller input(s) and output(s)
- Control valve characteristics (fail position, direct/reverse acting)
- Flow sensor types

P&IDs are also used to organize project documentation. Using P&IDs as the base for all information in a large project provides a single reference point for data provided on other documents. This makes sense since P&IDs are used to define system boundaries. It is very important that P&IDs are controlled and maintained up to date. During start-ups the "master" P&IDs are marked up and highlighted to reflect the state of a system at any time. If the technician discovers any discrepancy at any time, the technician must take the responsibility to ensure the controlled drawing is properly revised.

An example of a P&ID is in Appendix A-1. Let's look at the P&ID and see what we can determine about the temperature loop for this process.

We can see from the P&ID that the temperature loop includes TE-300, TT-300, TIC-300, TY-300, and TV-300. Using ISA-5.1 as a reference, we

know that the TE is a temperature sensor. We do not know if the temperature sensor is an RTD or thermocouple. That would be determined from the specification or physical inspection. The TT is a temperature transmitter that provides the only input to TIC-300. The TIC is a shared display, shared control temperature-indicating controller. The TY is not as obvious if you're not familiar with the use of this symbol. If you think about it though, you see that the TY is between the electronic output of the TIC and the pneumatic input of the valve, TV. Since the "Y" in TY refers to relay, compute, or convert, according to Table 1 of ISA-5.1, the TY must be a conversion device in the temperature loop. Most likely the device is an I/P transducer which converts a 4-to-20 mA signal from the TIC to a 3-to-15 psig signal for the valve. The TW in the loop is a temperature well, or thermowell, that provides a physical boundary between the temperature sensor and the process.

Side note: The thermowell is not a necessary component in the signal conversion of the temperature signal or control. However, because the thermowell consists of a specified thickness of metal it will slow down the response time of the temperature signal. It is also very important to ensure the temperature sensor is the correct length to make contact with the end of the well. Otherwise the air gap will cause an inaccurate reading of the actual process temperature.

2.4 LOOP DIAGRAMS (See Examples in Appendix A-2)

A loop is a combination of interconnected instruments that measures and/or controls a process variable. An instrument loop diagram is a composite representation of instrument loop information containing all associated electrical and piping connections. Instrument loop diagrams are developed in accordance with standard ISA-5.4-1991, *Instrument Loop Diagrams*. Example loop diagrams are included in the Reference Section, Appendix A-2. The minimum content requirements include:

- Identification of the loop and loop components
- Point-to-point interconnections with identifying numbers or colors of electrical and/or pneumatic wires and tubing, including junction boxes, terminals, bulkheads, ports, and grounding connections
- General location of devices such as field, panel, I/O cabinet, control room, etc.

- Energy sources of devices
- Control action or fail-safe conditions

Additional content requirements, format, and examples are included in ISA-5.4-1991.

A formalized loop check should be documented prior to placing any loop in service. This formalized program should include verification of installation against the loop diagram and simulation of signals to verify output responses and indications throughout the range. Why is this important? A significant percentage of instrument loops have some problem, which may result in hidden failures. For example, a temperature transmitter output wired to a programmable logic controller (PLC) analog input provides a temperature display on an operator interface. The transmitter is calibrated from 0-100°C to provide a proportional 4-20 mA output. If the PLC programmer writes the code for this input as 0-150°C for a 4–20 mA input and a loop check is not performed, an inaccurate displayed value will go undetected. Other typical problems found and corrected by performing loop checks include wiring connected to the wrong points, ground loops, and broken wires. Loop checking is another of the CCST domains and is covered in more detail in another ISA Technician Guide.

2.5 INSTRUMENT SPECIFICATION FORMS

(See Examples in Appendix A-3)

Instrument specification forms contain the information necessary to obtain vendor quotes and purchase instrument devices. ISA-TR20 Specification Forms include device specifications for many temperature, pressure, level, flow instruments in Microsoft Word format. The information included on each form is specific to the instrument type but typically includes dimensions, materials of construction, design temperature/pressure, connection sizes, ambient conditions, indicator detail, instrument ranges/tolerances, etc. Examples of instrument specifications are included in the reference section, Appendix A-3. When instruments are received from the vendor, each should be verified against the specification to the maximum extent practicable. This should be documented using a formalized receipt verification process.

For facilities that do not have instrument specification forms from the system design, these forms can be developed using ISA-TR20 or in accordance with company procedure. Some calibration departments

develop their own forms to specify the information needed to perform calibrations. These specifications should be approved by users and the Quality Department to ensure process and quality requirements are considered. An example of this form is included at the end of Appendix A-3.

2.6 PROJECT SPECIFICATIONS

Any new manufacturing facility or large expansion project will have a set of project specifications. They include architectural, plumbing, mechanical, electrical, and instrumentation specifications, etc. For instrumentation, the project specifications are usually general, but include content the technician should be aware of, such as:

- Instrument tagging requirements
- Size and type of pneumatic tubing
- Approved methods of mounting instruments
- Wire labeling conventions
- Intrinsic safety standards

For some facilities, the project specifications may include detailed instrument specification forms and an original master instrument list. Although the calibration technician is not usually installing the devices, any discrepancies observed should be brought to the attention of a supervisor.

2.7 MANUFACTURER'S SPECIFICATIONS

In the manufacturer's technical manual or product literature, there is almost always a specification section. Although most manufacturers provide very good specifications, they can vary from inadequate to overly complex. Understanding what the manufacturer's specifications mean is important to purchasing an instrument acceptable for the process requirements and the technician maintaining the instrument. The most important specification for the calibration technician is the instrument range and accuracy. The range capability of the instrument is usually understandable; accuracy is another matter. For example, accuracy may be specified as 0.25%. What does this mean? Is it 0.25% of the range or the reading? For the calibration of a 0-100 psig pressure transmitter at 20 psig,

this can mean the difference between 0.25 psig and 0.05 psig. In addition, the specified accuracy is usually different at different ranges of the instrument, varying ambient conditions, and may not even be referred to as accuracy. The lesson is, if you're not sure, ask a technical resource from the manufacturer what the specification means.

2.8 CALIBRATION INTERVALS

Probably the #1 question asked at calibration seminars is "How do I determine the *initial* calibration intervals?" The answer to this question is difficult at first for someone new to developing a calibration program. It ends up being pretty simple. Initially we try to use a variety of resources which include:

- Manufacturer recommendation
- National Conference of Standards Laboratories Recommended Practice RP-1
- Past experience
- Intervals of similar existing instruments

In reality, it is a combination of all the above, but mostly past experience. As an example, in my experience, electronic transmitters have a calibration interval of 6 months and analog gauges have an interval of a year. Many manufacturers' specifications contain a 6-month stability specification. This stability specification, in effect, only guarantees the accuracy specification for 6 months. Also, electronic transmitters are typically installed in applications that are "more important" to the process. Even though these instruments are more reliable than analog gauges and fail calibration less often, we check the calibration on a more frequent basis. This means we set our calibration intervals based on how much risk we are willing to take. If we wanted an almost 100% assurance that our instruments were within calibration tolerance, we'd have to check the calibration almost every day. Obviously, that would be impracticable. So we assume some risk that every once in a while a calibration is not going to pass. Of course our managers and quality department don't want to hear that, but it happens and we need to educate other disciplines that it does happen.

Don't be alarmed if you calibrate more or less often than described above. It simply means you're willing to take more or less risk based on the process and quality standards at your facility or you have more history

to base your calibration intervals on. Of course not all instruments fit into the same category. Some instruments, particularly analytical instrumentation, are calibrated more frequently, even to the point that the user performs a calibration check prior to each use. On the other hand, some instruments may have an interval of two years or more.

Calibration intervals may be adjusted over time. Once several calibrations have been performed, the calibration history of the device may be used to adjust the calibration interval. If the as-found calibration data of a particular instrument does not meet the calibration tolerance, the calibration interval may be shortened. If several instruments with a particular manufacturer/model number are always well within the calibration tolerance, the interval is increased.

2.9 SAFETY CONSIDERATIONS

Obviously, performing calibrations safely is very important. One lapse on safety could cost you or your co-worker your lives. Even if it's not a life lost, minor injuries caused by unsafe work practices are preventable. Safety of the product is also of concern when performing calibrations in a manufacturing environment. There are many resources available from ISA on the topic of safety, including Chapter 1 of *Troubleshooting: A Technician's Guide*, by William L. Mostia. Here are a few things we, as calibration technicians, can do to improve safety in our day-to-day work activities.

- Include specific safety considerations in each calibration procedure. For example, if we know there is a tank that does not have a thermowell installed for the resistance temperature detector (RTD), highlight this fact. Better yet, if possible, get a thermowell installed.
- Keep the shop and work areas clean and free of trip hazards.
- Work with a partner or at least make sure someone knows where you are working at all times.
- Some instruments are always installed at difficult places to reach. If it's possible to install some permanent platform, have it done. Otherwise use safety harnesses, ladders, and lifts properly.
- Technicians may be exposed to lethal electrical voltages. Know what the high voltage areas are, de-energize electrical circuits that are not required, and use proper electrical safety practices

(insulated floor mat, rubber electrical safety gloves, roped off area, safety man outside the area with a rope tied around you).

- Ensure electrical power cords are properly insulated. Ensure equipment is properly grounded.

2.10 CALIBRATION STATUS LABELS

Calibration status labels are used to provide a visual indication of the calibration status of an instrument. Many different label styles are in use throughout industry. The main information that must be displayed showing the calibration status are the Instrument Identification (such as Tag Number, Instrument ID number, or serial number), date of calibration, next calibration due date, and the technician who performed the calibration (initials, employee ID, etc.). An example is illustrated below.

ABC Company – Calibration Status
ID: _____
Cal Date: _____
Cal Due: _____
Technician: _____

Other amplifying information may be included on the label or a separate label. Examples include:

- Customized labels with company name and color coded for classification as described in Chapter 1.
- Limited Calibration Label for instruments that are not calibrated throughout the range of the indication. An example of this would be a compound gauge that is not calibrated for the vacuum range because the process does not operate under a vacuum at any time.
- Another example would be an instrument that does not meet tolerance requirements at the extremes, but the process would never operate at these extremes.
- Other labels such as "Do Not Use" or "Calibration Not Required" can be used, if applicable.
- Sometimes tamper-proof seals are used to cover exposed adjustments so that any unauthorized adjustments are detected.

This chapter has summarized the most common documentation required or utilized by the calibration technician. These subjects were not discussed in excruciating detail, because other resources are available. However, it is important to expose technicians to the concepts and provide some examples to help put the big picture together before we start with the actual performance of the calibrations.

REVIEW QUESTIONS

1. List the advantages of each procedure development method below.

   A. Straight from technical manual

   B. Generic procedure for an instrument type

   C. Procedure developed for a specific manufacturer/model or specific instrument in the plant

2. If test equipment with the specified accuracy is not available to perform the calibration, what should you do?

3. If a step of a calibration procedure cannot be followed as specified or a procedure does not exist, what actions would you take?

4. What elements of a calibration data sheet reflect that the calibration is NIST-traceable?

5. For an established facility, what is the most likely resource for determining the calibration frequency of a new instrument?

6. For a new facility, what is the most likely resource for determining initial calibration frequency?

7. What justification can be used to increase the calibration interval (performing calibration less often) of all instruments of the same type?

8. What event(s) can lead to decreasing the calibration interval (performing calibration more often)?

9. Match the resources on the left with content on the right.

___ P&ID	A. Includes general instrument specifications for the facility
___ Instrument Specification Sheet	B. Includes instrument range capability and instrument accuracy
___ Loop diagram	C. Detailed overview of a process system
___ Project specifications	D. Detailed device requirements
___ Manufacturer's Specifications	E. Includes all associated electrical and piping connections

10. What is the purpose of a calibration seal?

11. What is the minimum information required on a calibration status label?

12. When should a Limited Calibration status label be used?

13. What criteria must be met for test equipment to be used for calibration?

3

TEMPERATURE INSTRUMENT CALIBRATION

After completing this chapter, you should be able to:

Describe the different types of temperature sensors, including important advantages and disadvantages of each.

Calibrate the following temperature instrument types (to ISA standards, where applicable) and determine acceptability:

- *Glass and Dial Thermometers*
- *Temperature sensors (RTDs, thermocouples, and thermistors)*
- *Temperature transmitters*
- *Digital temperature indicators and controllers*
- *Temperature switches*

Select proper calibration procedure and calibration data sheet

Select appropriate certified test equipment.

Properly setup/connect test equipment to the Device Under Test (DUT) for calibration.

Properly isolate temperature devices and/or remove from service for field calibration.

Return equipment to service following calibration.

Complete and properly maintain calibration documentation.

3.1 WHAT IS TEMPERATURE?

Anything that moves has kinetic energy, the energy of motion. Atoms and molecules have kinetic energy because they are always moving. The faster a molecule moves, the greater its kinetic energy. Heat is the measure of the total quantity of kinetic energy due to molecular motion in a body of matter. Temperature measures the intensity of heat due to the average kinetic energy of the molecules.

3.2 TEMPERATURE SENSORS

There are several different types of sensors used to measure temperature. Some of the more common sensor types are summarized in this section.

Thermocouples

Thermocouples are very common due to the ruggedness, low cost, response time, and relatively good accuracy, but mostly because of the versatility. Thermocouples can be used over a wide range of temperatures, whereas RTDs are useful only over a certain temperature range. Thermocouples are based on the principle that joining two dissimilar metals will produce a voltage signal proportional to temperature. Although very commonly used, thermocouples are least understood due to the complexity of different metals used, reference junction, and methods of compensation. The Reference Section of the *Omega Temperature Handbook* provides a very good explanation of the details.

There are several types of thermocouples, based on the types of metals that make up the thermocouple. For example, a Type T thermocouple is made from copper and constantan wires, whereas Type K is made from chromel and alumel. Each type is color coded and, per U.S. standards, the red lead is always negative. Due to the many thermocouple types, a technician must verify the thermocouple type used and ensure the correct reference table and/or test equipment setup is used for that thermocouple type. (For examples, see the reference tables in Appendix A-6.)

Resistance Temperature Detectors (RTDs)

RTDs are most commonly made from platinum, due to its stability and linearity throughout the range of use. However, RTDs exhibit a slower response time to temperature changes than thermocouples. All RTDs have a positive temperature coefficient, which means the resistance increases as temperature increases. The most widely accepted type of RTD conforms to DIN 43760. This RTD has a resistance of 100 ohms at 0°C with a temperature coefficient (α) of 0.00385 (spoken as three-eighty-five alpha).

The alpha coefficient is determined from the slope of the line between 0 degrees C and 100 degrees C and also the resistance of the RTD at 0 degrees C. The alpha is expressed in units of ohms per ohm per degree C where the second "ohms" refers to the resistance at 0 degrees C. For the DIN 43760 RTD, the ohms for 0 degrees C is 100 and the ohms for 100

degrees C is 138.50. The change in ohms for 100 degrees is 38.50. The change for 1 degree is 0.385 ohms. The alpha is 0.385 ohms per degree C divided by the resistance at 0 degrees (100 ohms) which equals 0.003850 ohms per ohm per degree C. The resistance output for an RTD is not perfectly linear, and the ohms for a given temperature cannot be determined from the alpha coefficient alone. The complete equation for the RTD output also includes other coefficients that define the non-linearity.

RTDs are connected electrically in a bridge configuration to offset the effects of lead length resistance. RTDs are purchased as 2-wire, 3-wire, or 4-wire RTDs. Again, the *Omega Handbook* provides a good explanation of the details. The most important thing for a technician to understand is the RTDs must be connected correctly during initial installation and upon completion of calibration. Also the technician must be careful to use the correct RTD table as a reference when performing calibrations.

Thermistors

Thermistors are based on a resistance change in a ceramic semiconductor. Thermistors are much more temperature sensitive that thermocouples or RTDs. This allows thermistors to detect very small changes in temperature which would not be detected by RTDs or thermocouples. Most thermistors have a negative temperature coefficient, which means the resistance decreases as the temperature increases. However, thermistors are very non-linear and therefore must be used over a small range to provide a linear response. Thermistors are also susceptible to drift and are very fragile.

Filled Bulb

Filled-bulb sensors are a closed system filled with a liquid that expands as the temperature increases and contracts when temperature decreases, producing a proportional change in temperature indication or a control valve response.

3.3 SIGNAL CONVERSION

Signal conversion is used to convert the voltage of a thermocouple or resistance of an RTD/thermistor to a signal usable by an instrument system. For example a temperature transmitter may convert the RTD resistance to a 4–20 mA signal for a PLC input or a recording device (or both). A separate signal conversion device is not always used since many

devices (controllers, indicators, PLC inputs) can be configured for direct sensor input.

Now that we have some idea of how temperature is sensed, let's look at some examples.

Example Temperature Calibration #1 – Calibration of a Dial Thermometer

A dial thermometer consists of a sensing probe connected to a dial indicator. The temperature sensed causes movement of the dial pointer across a graduated scale. The graduated scale provides an indication over a range of temperatures. The goal of the calibration is to verify that the pointer "points" to the correct value over the range of the instrument. Our example thermometer has a range of 0 to 100°C, and we'll be checking the calibration at 5 points: 0°C, 25°C, 50°C, 75°C, and 100°C.

The test equipment we'll use is a refrigerated temperature bath and a temperature standard. An ice bath could be used to check the 0°C point if a refrigerated bath is not available. An ice bath is prepared by using crushed ice made from distilled water and adding distilled water to make a "slush." It is best to make the ice bath in an insulated container. The ice bath should be adequately mixed just prior to placing the sensor in the ice bath and, if possible, while the sensor is in the ice bath.

In some cases, the temperature bath may have a temperature indication which could be used as the temperature standard. The temperature indication on the temperature bath can only be used as a standard if it has been calibrated itself. It is still not the best practice to use the bath temperature indication as a reference since the location of the sensor is not the same as the device being calibrated. In most cases, a separate temperature standard must be used to obtain the required accuracy ratio (accuracy ratio is discussed in Chapter 1).

To perform the calibration, the thermometer is removed from the system and placed in the temperature bath (or ice bath) at the minimum test point of 0°C. Our temperature standard is also placed in the temperature bath (or ice bath) at the same depth such that the tips of the thermometer and standard are as close together as possible. Once the readings have stabilized, the "as-found" readings are recorded for the temperature standard and the unit under test. No adjustments are made until all "as-found" readings are recorded for all test points. Now we change the temperature bath setpoint to 25°C. (If an ice bath was used at 0°C, move the thermometer and standard to the bath.) Once readings have stabilized, record the "as-found" values for the thermometer and standard on the calibration form. Note that we do not need to record the bath

temperature indication, if provided, since this is not our standard and has no bearing on the calibration. Obtain the "as-found" readings for the thermometer and the standard at 50°C bath setpoint, then 75°C setpoint and finally 100°C setpoint.

Now that the "as-found data has been collected, let's evaluate the data. The table below shows our as-found results.

TEST POINT	AS-FOUND DATA		
	TEST STANDARD READING	UNIT UNDER TEST READING	ERROR
0°C	0.12°C	1°C	+ 0.88
25°C	25.08°C	26°C	+ 0.92
50°C	50.17°C	51.5°C	+ 1.33
75°C	74.99°C	76.5°C	+ 1.51
100°C	100.02°C	102°C	+ 1.98

Let's say our calibration tolerance is 2°C and we find the thermometer is adjustable. Some are not adjustable, in which case the thermometer would need to be replaced if the "as-found" data was not within the acceptable tolerance. Since this thermometer has one adjustment using an external screw, how much should we adjust the thermometer? If the process, monitored by the thermometer, normally operates at a specific setpoint, it is recommended to adjust the thermometer to read correctly at that value. If the process operates over the range of the instrument, adjust the thermometer based on the average error such that all readings are as close as possible. In any case, the thermometer should be adjusted such that when you've completed the adjustments all "as-left" data is within the specified tolerance.

We find the process operates at 50°C. Therefore, adjust the temperature bath setpoint to 50°C. Once readings have stabilized, adjust the thermometer to read the same as the temperature standard. Once the adjustment is complete, record the stabilized thermometer and standard reading at each test point as the "as-left" readings. Verify all values are within tolerance and the desired results were achieved. Below is a table of our results.

TEST POINT	AS-FOUND DATA			AS-LEFT DATA		
	TEST STANDARD READING	UNIT UNDER TEST READING	ERROR	TEST STANDARD READING	UNIT UNDER TEST READING	ERROR
0°C	0.12°C	1°C	+ 0.88	0.10°C	– 0.5°C	– 0.60
25°C	25.08°C	26°C	+ 0.92	25.05°C	24.5°C	– 0.55
50°C	50.17°C	51.5°C	+ 1.33	50.13°C	50°C	– 0.13
75°C	74.99°C	76.5°C	+ 1.51	74.91°C	75°C	+ 0.09
100°C	100.02°C	102°C	+ 1.98	99.97°C	100.5°C	+ 0.53

Note: A Calibration Seal should be affixed to the external adjustment to detect any unauthorized adjustment to the thermometer.

Once you actually perform a calibration as described above, you'll see it can be very time consuming to wait for the temperature bath to stabilize at each of the five points (depending on the type of bath used). It may take all day to perform this one calibration. There are several things you can do to increase productivity.

1. Have several thermometers scheduled for calibration at the same time. It is typically the supervisor's job to properly schedule the work, but you can look to see if there are other thermometers due the same month and obtain authorization to calibrate them.

2. If there are other temperature baths available, set up the baths at different temperature setpoints. It is recommended to use the same temperature standard for all test points, but it does not take as long for the standard to stabilize as it does for the baths.

3. Use your spare time waiting during setpoint changes to perform other work.

Example Temperature Calibration #2 – RTD Calibration Check

Although temperature sensors are usually checked as part of a calibration of the connected device, it may be necessary to check a suspect temperature sensor or to verify the accuracy of a new RTD. The principles of this RTD calibration check would apply to any sensor type. The only exception would be a thermocouple calibration check if the test equipment used to read the thermocouple does not have automatic reference junction temperature compensation. All modern test equipment performs the temperature compensation, but it is important to verify. If you perform a thermocouple calibration check and do not have temperature compensation, you'll need a second temperature standard to measure the reference junction ambient temperature and compensate manually.

For this example, let's say we need to check the RTD at five points, from 50°F to 250°F. Just like the thermometer in example #1, we'll need a temperature bath capable of achieving the desired range of 50°F to 250°F, a temperature standard, and a test instrument capable of reading resistance (such as a multimeter or multi-function calibrator). Will we need a refrigerated bath (or an ice bath) like we needed for the thermometer calibration in example #1? Yes, since the minimum test point of 50°F is below normal ambient temperature, we will need a refrigerated bath.

We'll also need the correct RTD table to determine our expected resistance at 50°F, 100°F, 150°F, 200°F, and 250°F. Typically, we would use the applicable RTD table. However it is not unusual for the manufacturer to provide an RTD table specific to an individual RTD. If the manufacturer provides a specific RTD table, it should be added to the documentation file for this device, used for the calibration check of this RTD, and used for the calibration of any device connected to this RTD. For our example, we will use the standard RTD table for our RTD, which is a 100 ohm, platinum, 385 alpha RTD. (Refer to the correct RTD table in Appendix A-6.)

To perform the calibration check, place the temperature standard and RTD in the temperature bath. Adjust the bath temperature to each of the test points. Record the temperature standard reading and RTD resistance at each test point. Fill in the expected resistance readings for the five-point calibration check from the applicable table in Appendix A-6. If you have difficulty, an explanation follows the table.

TEST POINT (°F)	TEMPERATURE STANDARD READING (°F)	CONVERSION TO °C	EXPECTED RTD RESISTANCE (OHMS)	ACTUAL RTD RESISTANCE (OHMS)	ERROR (OHMS)
50	50.0			103.88	
100	100.0			114.70	
150	150.0			125.40	
200	200.0			136.00	
250	250.0			146.50	

First of all, we base our expected resistance values on the actual temperature standard reading obtained, not the test point. We made this easier by using nice even temperature standard readings, which will not be the case in reality. However, since the RTD tables reference temperature in degrees Celsius, we did add a degree of difficulty specifying the test points in degrees Fahrenheit. For the first data point of 50.0°F we must first convert to °C using the conversion °C = (°F - 32) x 5/9,

which results in an answer of 10.0°C. The remaining conversions are 37.78°C, 65.56°C, 93.33°C, and 121.11°C, respectively.

For the expected resistances, we can obtain the value for 10°C directly from the RTD table. You should have 103.90 ohms for the expected resistance at 50°F (10°C). The remaining test points are not as simple. They require interpolation. For the 37.78°C data point you must first find the resistances for 37°C and 38°C, which are 114.38 ohms and 114.77 ohms, respectively. Next, find the difference between these resistance values, which is 0.39 ohms in this case. To perform this interpolation, understand that we are basically finding the resistance value, which is 78/100ths of the way between 114.38 and 114.77. This is simply done by multiplying 0.78 x 0.39, and adding the result to 114.38. The expected resistance for 37.78°C is 114.68 ohms.

To find the expected resistance for 65.56°C (100.0°F), multiply 0.56 x 0.38 and add the result to 125.17 ohms. The value for the expected resistance at the 100°F is 125.38 ohms. For 93.33°C, multiply 0.33 x 0.38 and add the result to 135.85 ohms for an expected resistance of 135.98 ohms. For 121.11°C, multiply 0.11 x 0.37 and add the result to 146.45 ohms, for an expected resistance of 146.49 ohms.

Complete the table. Assuming our allowable tolerance is 0.1% of reading, did our calibration check pass? At the expected resistance of 103.90, the allowable tolerance is 103.90 x 0.001, or 0.1 ohms. The remaining acceptable tolerances are calculated by multiplying 0.001 by expected resistance.

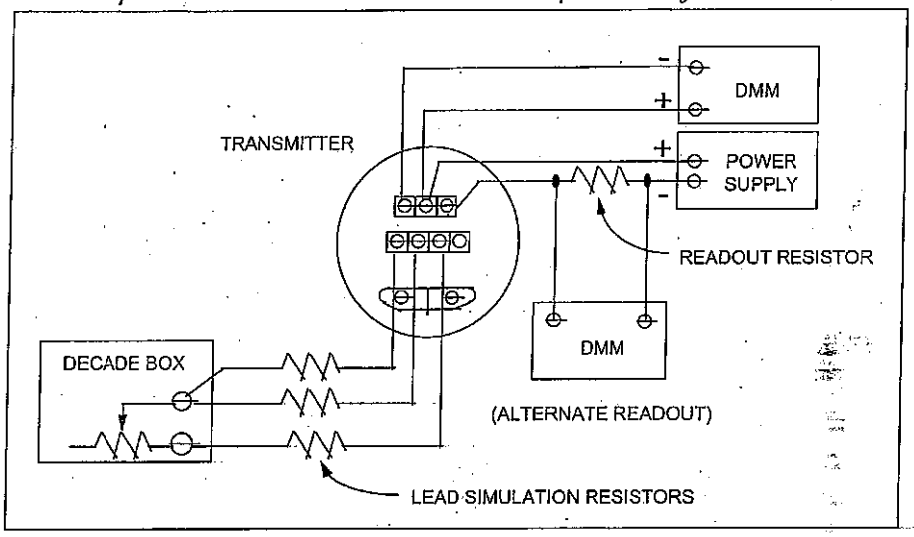
Example Temperature Calibration #3 – Temperature Transmitter

There are two basic methods to calibrate a temperature transmitter in-situ. The first is to calibrate the sensor and transmitter together by placing the sensor in a temperature bath/block and measuring the transmitter output. The second basic method is to disconnect the sensor from the transmitter and use a simulator in place of the sensor. If the sensor is an RTD, a decade box or RTD simulator would be connected to the transmitter input in place of the RTD. If the sensor is a thermocouple, a thermocouple simulator of the correct type would be connected to the transmitter input. Any local or remote indications connected to the transmitter output could be calibrated along with the transmitter as a loop.

It is recommended that, if a loop calibration is performed, the transmitter output should still be measured and recorded. The reason for measuring the transmitter output for a loop calibration is to determine which device, transmitter or indicator, would require adjustment if an

adjustment were necessary. Either method, individual instrument calibration or loop calibration, is acceptable depending on your industry practices and procedures. The advantages and disadvantages of each method are discussed in Chapter 1.

FIGURE 3-1.
Temperature Transmitter Calibration Setup (Courtesy Rosemount, Inc.)



For our example let's perform a calibration of TT-300 using a decade box per the calibration procedure SOP-CAL-08, Method B in Appendix A-4. The specification for TT-300 is included in Appendix A-3. Use the information from procedure SOP-CAL-08, TT-300 specification, and RTD Table in Appendix A-6 for 385-alpha to complete the tables below.

TABLE 3-1.
Preliminary Operating Point Check

TEST POINT	TEMPERATURE STANDARD	ACTUAL TRANSMITTER OUTPUT	CONVERTED TEMPERATURE	ERROR
70°C	70.12°C	11.35 mA	68.90°C	-1.22°C

TEST POINT	SIMULATED TEMPERATURE (°C)	INPUT RESISTANCE (OHMS)	EXPECTED TRANSMITTER OUTPUT (mA)	ACTUAL TRANSMITTER OUTPUT (mA)	ERROR
10%	15.0	105.85	5.60	5.50	-0.10 mA
50%	75.0	128.98	12.00	11.88	-0.12 mA
90%	135.0	157.31	18.40	18.27	-0.13 mA

Given the actual transmitter output above, does the transmitter require calibration? If the manufacturer's specification of 0.2% is used, the transmitter requires calibration. If the calibration tolerance is ±1%, which would equate to 1.5°C and 0.16 mA, the instrument is within tolerance, but should be adjusted since it is close. A good rule of thumb is to adjust if the instrument is > ½ the specified tolerance. You should not get in the practice of always performing adjustments to get the instrument calibration "right on," because even performing adjustments can lead to deteriorated performance.

You may be asking yourself why the procedure requires test points of 10%, 50%, and 90%. Zero percent, 50%, and 100% may be the normal test points that you are used to and that is fine. For loops that include an operator interface, the reading may not go below the minimum or above the maximum. Therefore, if the 0% and 100% test points were used, you would not know if the reading was actually below the 0% reading or above the 100% reading.

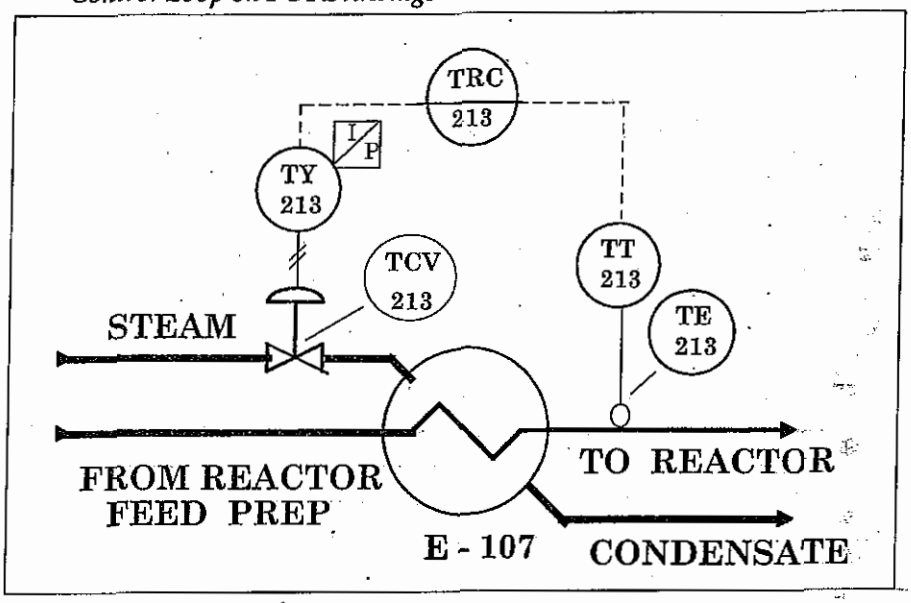
Example Temperature Calibration #4 – Temperature Controller Loop

A temperature controller can be a stand-alone device or a function of the computer control system, such as a programmable logic controller (PLC) or distributed control system (DCS). If the temperature controller is a function of a computer control system, it is typically not calibrated because it is not really a device subject to going out of calibration. A stand-alone controller usually has many capabilities such as multi-function configuration, alarm setpoints, analog and on/off outputs. For our example, we'll assume we have a Moore 352 controller configured for 4-20 mA input with a corresponding display of 0 to 100°C. The controller is also configured for a low alarm of 30°C and a high alarm of 40°C. The 4-20 mA output is supplied to an I/P (current-to-pneumatic) transducer for control of a steam valve.

For our calibration we only need to concern ourselves with the controller itself. The temperature transmitter that supplies the mA input is calibrated separately. The I/P transducer is also calibrated separately. It's a good idea to have the calibration of all loop components scheduled together. You could also perform a calibration for the entire loop as previously discussed, if desired. However, to start with it is very important that the control system technician understands how the controller interfaces with the process. First of all, the system must be in a state which will allow the safe performance of the calibration. The system controlled by this controller should be shut down and the controller

placed in manual operation. If the controller remained in automatic during the calibration, the simulated signals would cause output responses which could be detrimental to equipment and process.

FIGURE 3-2.
Control Loop on P&I Drawings



Once the system is in a safe condition and the controller is placed in manual, we can disconnect the controller input leads and connect a milliamp simulator to the input. To perform the calibration, adjust the milliamp simulator to the desired setpoints and record the controller display. If the display requires calibration, follow the manufacturer's procedures for calibration. For microprocessor based instruments, the calibration is usually not performed by turning a potentiometer, but rather adjusting the input to a pre-determined value and pushing buttons so the microprocessor can make the adjustments.

Once the display is properly calibrated, it's a good idea to verify alarm setpoints and control output action. This is not required and many calibration procedures would not include it. To perform the alarm checks, the milliamp simulator is adjusted as necessary to determine the temperature at which the low and high alarms trip and reset. To check the controller output for proper operation, adjust the milliamp simulator above and below the setpoint safely to verify the proper output action is occurring.

Example Temperature Calibration #5 – Temperature Switch

A temperature switch is a device that senses temperature and changes state at the programmed or adjusted setpoint. A temperature switch usually has at least one set of normally open (NO) and one set of normally closed (NC) contacts, but may have only a NC contact, a NO contact, or more than one set of each. NO and NC refer to the state of that contact with the switch de-energized. The tricky part of calibrating a temperature switch is to know whether the switch should "trip" with increasing or decreasing temperature. Typically, a high temperature switch would "trip" with increasing temperature, and vice versa.

Basically the calibration of the switch is checked by placing the sensor in a bath and measuring across the applicable contact with a multimeter. The bath temperature is increased/decreased to the setpoint. The multimeter will read the change of voltage or resistance when the switch changes state. In other words, the multimeter will read 0 volts with the switch closed and supply voltage with the switch open. If the contact is a "dry" contact (no voltage), the multimeter will read close to 0 ohms with the contact closed and infinite ohms with the switch open. (Hint: You may want to disconnect and electrically insulate the leads to check the switch by itself. This removes any parallel resistances and external circuits from interfering. As an alternative, use the remote indication as confirmation of proper switch operation. Be cautious of circuit and display time delays.)

REVIEW QUESTIONS

Situation for Questions 1 – 10: TT-300 is installed in a process tank, which is currently in production. This is the first calibration after initial startup. Use the references in Appendix-A to answer the following questions.

1. What is the correct calibration range and manufacturer's specified accuracy for TT-300? (Appendix A-3)

2. Select the correct procedure(s) that could be used for this calibration of TT-300? (Appendix A-4)

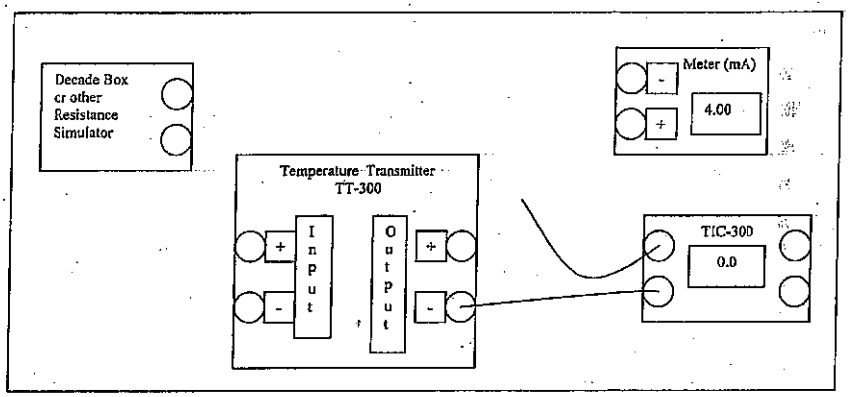
3. What local/remote indications should be recorded during calibration of TT-300? (Appendix A-1)

4. What must be done prior to removing RTD for calibration of TT-300? (initial conditions of standard operating procedure, SOP)

5. What are the correct resistance values to input for 10%, 50%, and 90%? (refer to the correct RTD table in Appendix A-6)

6. What are the expected transmitter output values for the resistance inputs from step 5?

7. Indicate the correct test equipment hookup for calibration of TT-300.



8. Assume a calibration tolerance of ± 0.5°C/0.05 mA. With the results indicated, what instrument(s) in the loop require(s) adjustment?

% INPUT	mA OUTPUT	TIC-300 INDICATION
10%	5.70	15.9
50%	12.10	75.9
90%	18.50	135.9

9. What type of error is indicated by the results of question 8?

   A. Zero error
   B. Span error
   C. Zero and Span error
   D. Linearity error

10. Following adjustment, all "as-left" data is within tolerance. What must be performed to place the instrument loop back in service?

Situation for questions 11 – 15: You will be performing an initial calibration of TT-200 on the bench, prior to installation. Use the references in Appendix A to answer the following questions.

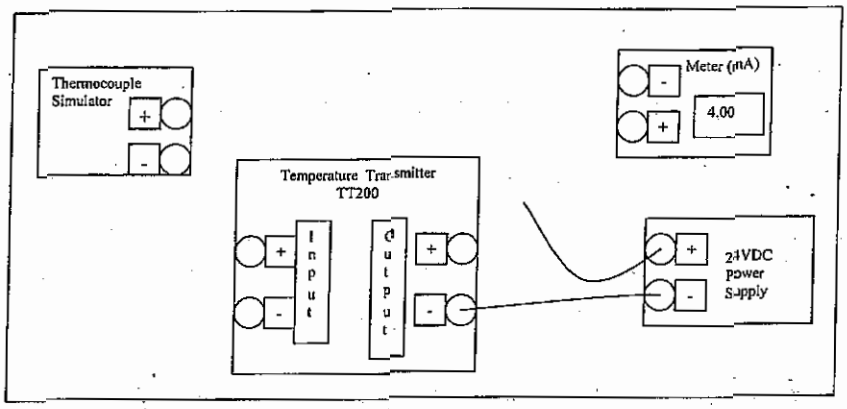
11. What are the correct reference temperatures and corresponding millivolt values to input for a calibration check at the following test points?

% INPUT	TEMPERATURE	MILLIVOLTS
0%		
25%		
50%		
75%		
100%		

12. What are the expected transmitter outputs for the following simulated inputs?

% INPUT	OUTPUT (mA)
0%	
25%	
50%	
75%	
100%	

13. Indicate the correct test equipment hookup for calibration of TT-200.



14. Assume a calibration tolerance of ± 1.0°F/0.08 mA. With the results indicated, what must be done to bring the instrument to within tolerance?

% INPUT	OUTPUT (mA)
0%	4.00
25%	7.98
50%	11.96
75%	15.94
100%	19.92

15. Following adjustment, all "as-left" data is within tolerance. What steps remain to complete the bench calibration?

Situation for questions 16 – 20: The operator suspects that TI-302 is not reading the correct temperature of Reactor 300. You will be performing a calibration of TI-302. Use the references in Appendix A to answer the following questions.

16. What test equipment will be used for calibration of TI-302?

17. What must be performed to safely remove TI-302 from the process?

18. What important consideration must be taken into account for a proper calibration using a temperature block?

19. The thermometer has one external adjustment. With the results indicated, what must be done to bring the instrument to within tolerance?

% INPUT	THERMOMETER INDICATION (°F)
0%	2
25%	34
50%	66
75%	99
100%	>130

20. What would you do if the thermometer was out of tolerance and the thermometer does not have an adjustment available?

For questions 21 – 23 you will be performing a bench calibration of a temperature switch, TSH-205.

21. What is the specified trip point and reset point of TS-205?

    Trip _____ Reset _____

22. What type of sensor does TS-205 use?

23. When performing an initial calibration on the bench, is it important to know what function TS-205 will perform once installed? Why or why not?

4

PRESSURE INSTRUMENT CALIBRATION

After completing this chapter, you should be able to:

Calibrate the following pressure instrument types (to ISA standards, where applicable) and determine acceptability:

- *Gauges*
- *Transmitters*
- *Switches*

Select proper calibration procedure and calibration data sheet.

Select appropriate certified test equipment.

Properly set up/connect test equipment to DUT for calibration.

Properly isolate pressure devices and/or remove from service for field calibration.

Return equipment to service following calibration.

Complete and properly maintain calibration documentation.

4.1 WHAT IS PRESSURE?

Air pressure is the force exerted on you by the weight of tiny particles of air (air molecules). Although air molecules are invisible, they still have weight and take up space. Atmospheric pressure is approximately 14.7 pounds per square inch (psi) at sea level. This means that if we could put one square inch of air from the ground to the upper atmosphere on a scale, it would weigh 14.7 pounds. We do not feel this pressure because it also acts internally and is thus balanced. Since there's a lot of "empty" space between air molecules, air can be compressed to fit in a smaller volume. When it's compressed, air is said to be "under high pressure."

There are two ways to look at pressure: (1) the small-scale action of individual air molecules, or (2) the large-scale action of a large number of

molecules. Starting with the small-scale action, a gas is composed of a large number of molecules that are very small relative to the distance between molecules. The molecules are in constant, random motion and frequently collide with each other and with the walls of any container. The molecules possess the physical properties of mass, momentum, and energy. As the gas molecules collide with the walls of a container, the molecules impart momentum to the walls, producing a force perpendicular to the wall. The sum of the forces of all the molecules striking the wall divided by the area of the wall is defined to be the pressure. The pressure of a gas is then a measure of the linear momentum of the molecules of a gas.

4.2 CHALLENGES WHEN CALIBRATING PRESSURE

The first thing we have to deal with when calibrating a pressure instrument is the *unit of measure* we're dealing with. There seems to be dozens of units. In the U.S. we usually deal with pounds per square inch (psi), but even with that we need to know if it's absolute or gauge pressure (psia or psig). And although we usually deal with psig, we'll routinely deal with inches of water (H_2O or "w.c.") for low pressure applications. Other units of measure we will be exposed to are: bars (1 bar = 1 atmosphere), inches of mercury (Hg), millimeters of mercury (mm of Hg), millimeters of water (mm of H_2O), microns, torr, pascals (Pa), and dynes/cm^2. Some conversion factors for pressure units are included in Appendix A-7.

Many of the modern pressure calibrators allow us to select the desired units of measure, but we still need to understand the relative pressure we're dealing with to ensure the correct test equipment, pressure module, and tubing/fittings rated for the maximum test pressure are used.

When I first started calibrating, I had a pressure gauge that read about 15 psi high at all test points. I went through a tremendous amount of effort to drain the glycerin from the gauge, remove the cover, and attempt to adjust the gauge needle. When I couldn't get the gauge to calibrate properly, I took it to my supervisor to order a new replacement gauge. My supervisor informed me that the gauge read in psia, my "as-found" data was acceptable, and I ruined a perfectly good gauge. The gauge was ruined because the vacuum that had existed in the gauge case was eliminated when the gauge was opened. This vacuum was present to

provide the gauge with a reference to absolute zero pressure. So, if you calibrate a gauge and it reads about 15 psi with no pressure applied, take it from me, the gauge reading is probably in psia, not psig. Know the unit of measure for the device you are calibrating!

Another challenge with pressure calibrations is having the *correct fittings* to connect the pressure source, pressure standard, and the unit under test. Even with a few drawers full of fittings in the shop and access to more adapters in the mechanical shop and spare parts stock, we never seem to have the right fittings. To minimize this problem, enter the fittings needed in the calibration notes that print out with the calibration data sheet or work order. The important thing is to minimize the number of connections in the test setup in order to minimize the potential for leakage.

Leakage is another problem we have to deal with when performing pressure calibrations. This problem is amplified when we use a hand pump as our pressure/vacuum source. Small leaks in the test setup can be compensated for by using a constant pressure source such as a N_2 bottle, a spare plant air connection, or vacuum pump. However, leakage should be eliminated or minimized for all pressure calibrations even when using a constant pressure source. Even the smallest leak is frustrating when using a hand pump, so those connections have to be minimized and tight.

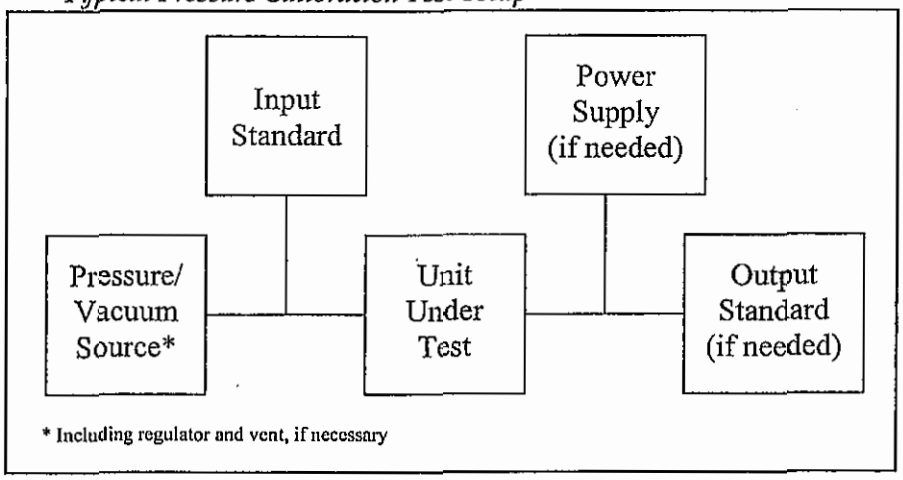
Hysteresis is the measure of the difference in response of a device as the input signal increases from minimum value to a maximum value, and, subsequently decreases from maximum to minimum over the same range. Hysteresis is expressed in percent of full scale (% FS). Instruments with mechanical movements such as pressure gauges and current-to-pneumatic (I/P) transducers develop hysteresis error. Often this is caused by friction and wear of the mechanical components. There is no way to adjust for hysteresis error during calibration. Hysteresis can be evaluated by obtaining calibration data with the input applied in both the increasing and decreasing direction as illustrated below.

APPLIED INPUT (PSIG)	AS-FOUND VALUES (↑)	AS-FOUND VALUES (↓)	HYSTERESIS (%FS)
10	9	10	1%
25	24	25	1%
50	49	51	2%
75	74	76	2%
90	90	91	1%

When evaluating hysteresis, it is important to approach the target value without overshooting the applied input. If the input value exceeds the increasing target value, decrease the applied input to the previous test point and retry. If the input value exceeds the decreasing target value, increase the applied input above the previous test point and retry.

Special *safety considerations* must be considered when calibrating pressure instruments. First of all, ensure the system pressure has been released prior to attempting to remove any pressure devices. Also, you must know what potential hazards exist with the process material. Some process material may be toxic or may cause harm to the environment if exposed. Verify with the supervisor of the area before removing any device for calibration.

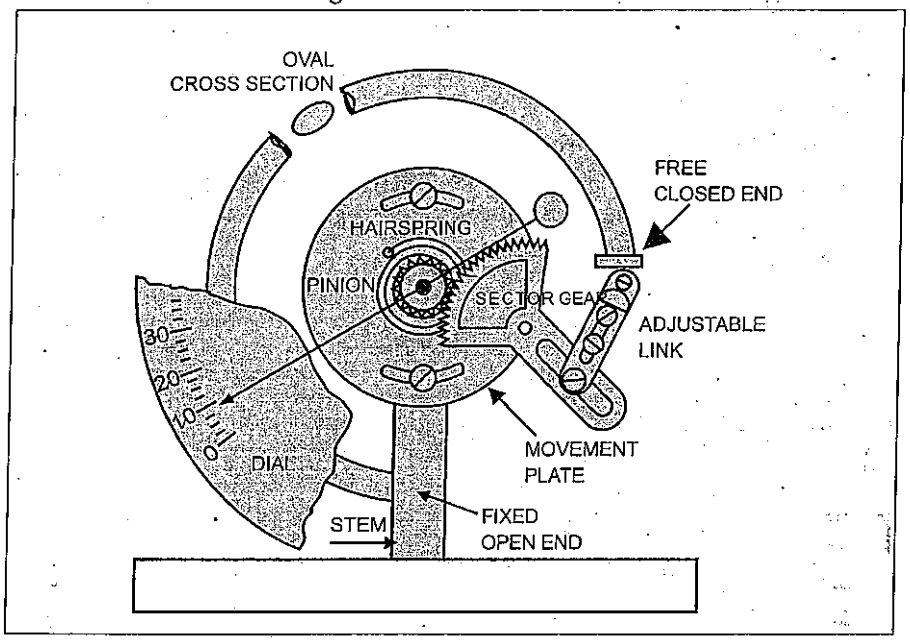
FIGURE 4-1.
Typical Pressure Calibration Test Setup



4.3 CALIBRATING PRESSURE GAUGES

Pressure gauges are often used as local indicators of process pressure. An analog pressure gauge, because of its links, levers, and elastic pressure sensing element, requires periodic calibration checks. Pressure changes applied to the gauge cause the elastic element to expand and contract. The movement of the element is translated into movement of the pointer through links, levers, and gears. The measurement values of the gauge are read directly on the gauge scale from the position of the pointer.

FIGURE 4-2.
Bourdon Pressure Gauge



Calibrating a pressure gauge includes adjustment of these components until the gauge accurately represents the input. The power supply and output standard illustrated above are not used for calibrating a pressure gauge.

The instrument under test determines the calibration standards. First, you need a source of pressure such as a hand pump or regulated pressure source (vacuum hand pump or regulated vacuum source if calibrating a vacuum gauge). A regulator with the smallest range available that exceeds the gauge's range is most appropriate. This allows the gauge's full range to be tested as accurately as possible as well as precise adjustment of the pressure input. Select the proper input standard for the pressure range tested. The pressure standard may be a precision gauge or digital test standard with the correct pressure module, manometer, or dead weight tester.

Use a tee to connect the input test standard to the pressure source and the gauge under test. Be sure the gauge under test is mounted in the same orientation as in the process. Positioning a gauge vertically for calibration and then reinstalling the gauge in its horizontal configuration in the process will introduce error.

Determine the five test points used for the upscale and downscale checks of the gauge under test. As discussed earlier in this chapter, with

any mechanical instrument, it is important to accurately determine whether hysteresis is present in the instrument. This means that you will begin your upscale check from 0% and approach the first test point, 10%, from below. Approach each increasing test point from below and do not overshoot. If you overshoot, reduce pressure to the previous test point and repeat. Similarly, start the downscale check by increasing the input to 100%, then approach 90% from above. Now that the "as-found" data for our 0-200 psig gauge has been collected, let's evaluate the data. The table below shows our "as-found" results.

TEST POINT	AS-FOUND DATA					
	TEST STANDARD READING (PSIG)	UNIT UNDER TEST READING (↑ PSIG)	ERROR (↑) (PSIG)	UNIT UNDER TEST READING (↓) (PSIG)	ERROR (↓) (PSIG)	HYSTERESIS ERROR (↓)
10%	20.0	19	−1	19	−1	0%
25%	50.0	49	−1	50	0	0.5%
50%	100.0	99	−1	100	0	0.5%
75%	150.0	149	−1	151	+1	1%
90%	180.0	179	−1	180	0	0.5%

The test results are then checked against the allowable tolerance. If the results are outside the allowable tolerance, determine the type of errors present (linearity, zero, span, and hysteresis), and adjust per the manufacturer's instructions or use the general instructions, as applicable, that follow.

On most motion balance instruments, adjust linearity first. The movement of the elastic element in a pressure gauge causes a proportional movement in the linkage. On a properly calibrated gauge, the linkage angle will be exactly 90° when the input to the gauge is at 50% of total range. Therefore, apply 50% input and use a template to check the 90° angle.

With linearity adjusted, position the pointer so the gauge reads mid-scale. You may need to remove the pointer and reposition it on the shaft. If removing the pointer is necessary, be sure to use the proper tool. Now lower the input to 10% and adjust the zero so the gauge reading equals the applied input.

Now correct the span error. Increase the input pressure to 90% and adjust the gauge to read the input value. For span adjustments, loosen the screws and rotate the entire adjustment mechanism.

Repeat the zero and span adjustment until the readings at 10% and 90% are accurate. Zero and span interact in the mechanical device, so rechecking them is necessary. When zero and span require no further adjustments, recheck the linearity to make sure it is still properly adjusted. There are no adjustments for hysteresis. After adjusting linearity, zero and span, perform another full scale check and record as-left calibration data.

4.4 CALIBRATING PRESSURE TRANSMITTERS

Calibrating a pressure transmitter is similar to calibrating a pressure gauge, except we must measure the output signal using an appropriate measuring standard. For example, we would use a milliammeter to measure a 4-to-20 mA transmitter output signal. If necessary, we may also have to provide the transmitter output power source, such as 24 volts direct current (VDC). Most modern calibration standards provide the ability to supply the transmitter power so a separate power source is not required.

Adjust the input values to the required test points and record "as-found" values for the milliamp output. If the transmitter includes a local readout, record the display. Typically, it is not necessary to check a transmitter in both the upscale and downscale directions (follow your company procedures). Once the "as-found" data has been recorded, evaluate the error against the acceptable tolerance to determine if adjustment is required. If adjustment is required, perform the zero and span adjustments per the manufacturer's procedures. Sometimes this is adjustment of a zero and span potentiometer, sometimes it's a pushbutton adjustment, while other transmitters require the use of a handheld interface. Because some zero and span adjustments interact, it is important to recheck the zero and span until no further adjustment is required.

Once no further adjustments are required, repeat the five point check and record "as-left" values.

4.5 CALIBRATING PRESSURE SWITCHES

A pressure switch is a device that senses pressure and changes state at the programmed or adjusted setpoint. A single pole double throw

(SPDT) pressure switch has one set of normally open (NO) and one set of normally closed (NC) contacts. A double pole double throw pressure switch is furnished with two SPDT switches. NO and NC refer to the state of that contact with the switch de-energized. The tricky part of calibrating a pressure switch is to know whether the switch should "trip" with increasing or decreasing pressure and which contact is used for the circuit. Typically a high pressure switch would "trip" with increasing pressure, and would use the NO contact. This means at normal pressure the switch would be closed and would open when the high pressure setpoint is reached. Using the NO contact provides a fail-safe condition if the circuit failed (power failure or broken wire). However the requirements of the alarm circuit must be considered.

The calibration setup of a pressure switch is similar to a pressure gauge and pressure transmitter, except a multimeter set to measure voltage or resistance, as applicable, is connected to the applicable set of contacts (such as the common and NO terminals). The applied input pressure is increased to the setpoint. The multimeter will read the change of voltage or resistance when the switch changes state. In other words, the multimeter will read 0 volts with the switch closed and supply voltage with the switch open. If the contact is a "dry" contact (no voltage), the multimeter will read close to 0 ohms with the contact closed and infinite ohms with the switch open. (Hint: You may want to disconnect and electrically insulate the leads to check the switch by itself. This removes any parallel resistances and external circuits from interfering. As an alternative, use the remote indication as confirmation of proper switch operation. Be cautious of circuit and display time delays.)

REVIEW QUESTIONS

Use the information in Chapter 4 and the references in Appendix A to answer the following questions.

1. Arrange the following in order from lowest to highest pressure:

___/ ___/ ___/ ___/ ___/ ___

A. 14.7 psia
B. 14.7 psig
C. 500 millitorr
D. 30 "Hg
E. 500 "H_2O
F. 500 mmHG

2. What is the correct calibration range and manufacturer's specified accuracy for PIT-303?

3. Select the correct procedure to be used for this calibration of PIT-303.

4. What local/remote indications should be recorded during calibration of PIT-303?

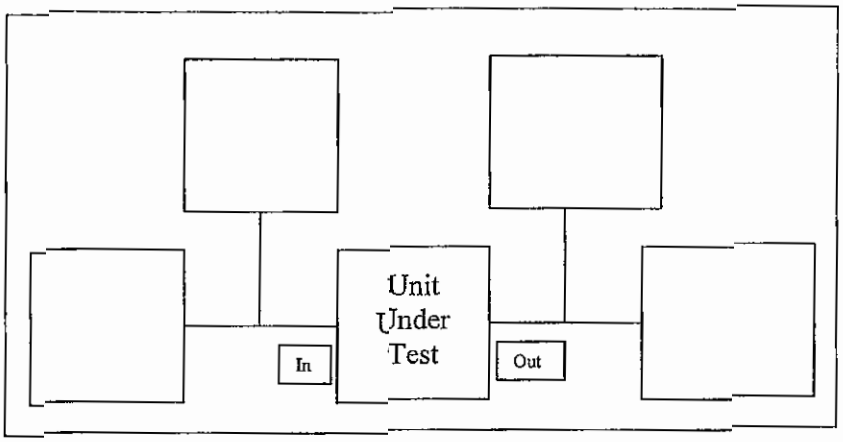
5. What must be done prior to removing pressure sensor for calibration of PIT-303?

6. What pressure values will be applied to perform calibration of PIT-303?

7. What are the expected transmitter output values for the pressures applied in the previous step?

8. Indicate the correct test equipment hookup for calibration of PIT-303 by inserting the correct item from the list below into the appropriate box:

Power supply
Reference standard pressure indicator
Milliammeter
Pressure source



9. What safety considerations must be taken into account when performing a pressure instrument calibration?

10. Assume a calibration tolerance of ± 0.3 psig/0.10 mA. With the results indicated, what instrument(s) in the loop requires adjustment?

% INPUT	mA OUTPUT	PIC-303 INDICATION
10%	5.50	5.0
50%	11.90	25.0
90%	18.30	45.0

11. How would the calibration required in the previous step be performed?

12. Following adjustment, all "as-left" data is within tolerance. What must be performed to place the instrument loop back in service?

13. What is the importance of checking the calibration of a pressure gauge while increasing and decreasing pressure applied?

14. PI-304 is a compound pressure gauge which measures/indicates both vacuum and positive pressure. What test equipment will be required to perform this calibration?

15. What safety considerations must be taken into account when removing a pressure gauge for calibration?

16. PI-304 is out of tolerance on the vacuum portion and within tolerance on the pressure portion. With no available replacements, under what conditions could this gauge be used?

17. What are the specified trip points of PDIS-405? Is the output of the switch connected to the normally open or normally closed contacts?

5

LEVEL INSTRUMENT CALIBRATION

After completing this chapter, you should be able to:

Calibrate the following level instrument types (to ISA standards, where applicable) and determine acceptability:

- *Differential pressure*
- *Capacitance probe*
- *Ultrasonic*

Select the proper calibration procedure and calibration data sheet.

Select appropriate certified test equipment.

Properly set up/connect test equipment to device under test (DUT) for calibration.

Properly isolate level devices and/or remove from service for field calibration.

Return equipment to service following calibration.

Complete and properly maintain calibration documentation.

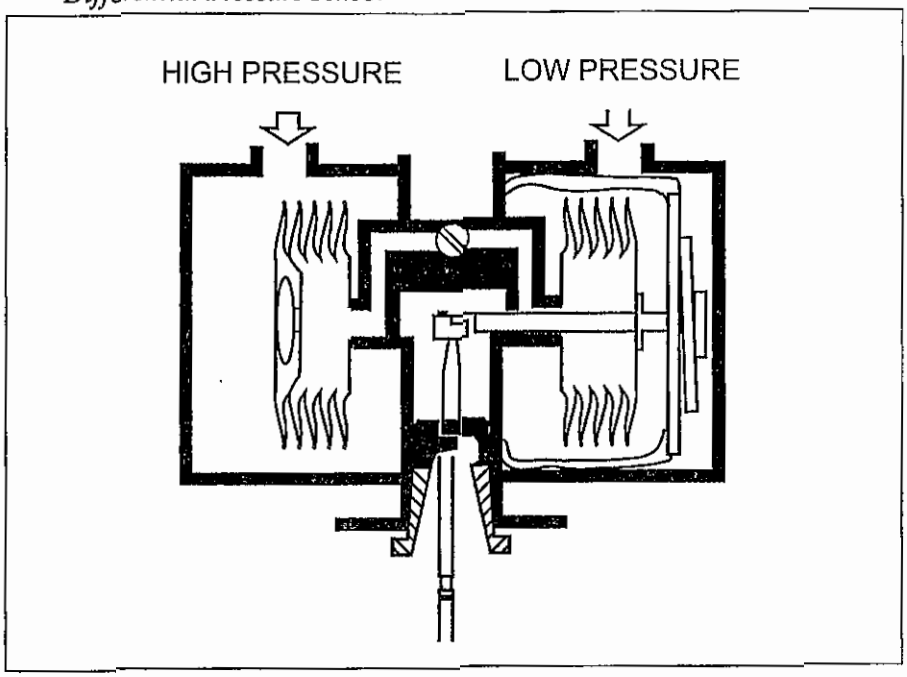
5.1 TYPES OF LEVEL INSTRUMENTS

There are numerous types of level sensors and instrumentation. The sensor type used in a particular application depends on many factors including the process fluid measured, density, vented/pressurized, direct or indirect detection, agitated, continuous or point measurement, accessibility, maintenance requirements, etc. We will briefly overview a few of the major sensor types:

Differential Pressure

Differential pressure sensors have a high pressure side and a low pressure side with a sensing element, such as a bellows, separating the two sides. The high pressure side of the sensor is usually connected to the

FIGURE 5-1.
Differential Pressure Sensor



bottom portion of the vessel and, therefore, senses the pressure exerted by the weight of the process fluid in the vessel plus the tank pressure (if pressurized). The low pressure side is usually connected to the vapor space at the top of a pressurized vessel or is open to atmosphere if the vessel is vented or open to atmosphere. The low pressure side only senses the vessel pressure (or atmospheric pressure) and not the weight of the process fluid. Differential pressure (d/p) measurement is based on the principle that the difference between the two pressures is equal to the height of liquid, multiplied by the specific gravity of the fluid.

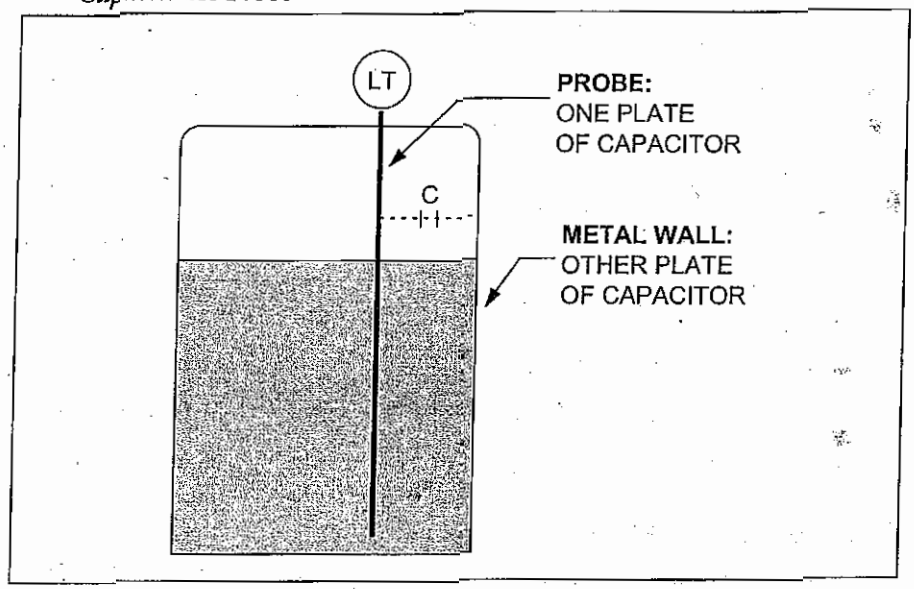
Capacitance/Radio Frequency (RF)

A capacitor consists of two plates electrically isolated from each other by a dielectric (nonconductor). The plates have an area (A) and are separated by a distance (D). Different mediums have a specific dielectric constant (K). Capacitance (C) = KA/D. Therefore, if the plate area and gap distance remain constant, capacitance varies as a function of the dielectric constant of the substance filling the gap.

For capacitance level measurement, a capacitor is formed when a level-sensing electrode is installed in a vessel. The metal rod of the electrode acts as one plate of the capacitor, and the tank wall acts as the

other plate. If the tank wall is not metallic, a reference electrode is used. Referring to Figure 5-2, you can see that, as the liquid level rises, the air or gas around the electrode is displaced by the process material, which has a different dielectric constant. Therefore, there is a corresponding change in capacitance between the probe and the vessel wall. The measured capacitance is proportional to liquid level. The unit of capacitance is the farad. Capacitance level probes typically measure in the pico-farad range (10^{-12} farads).

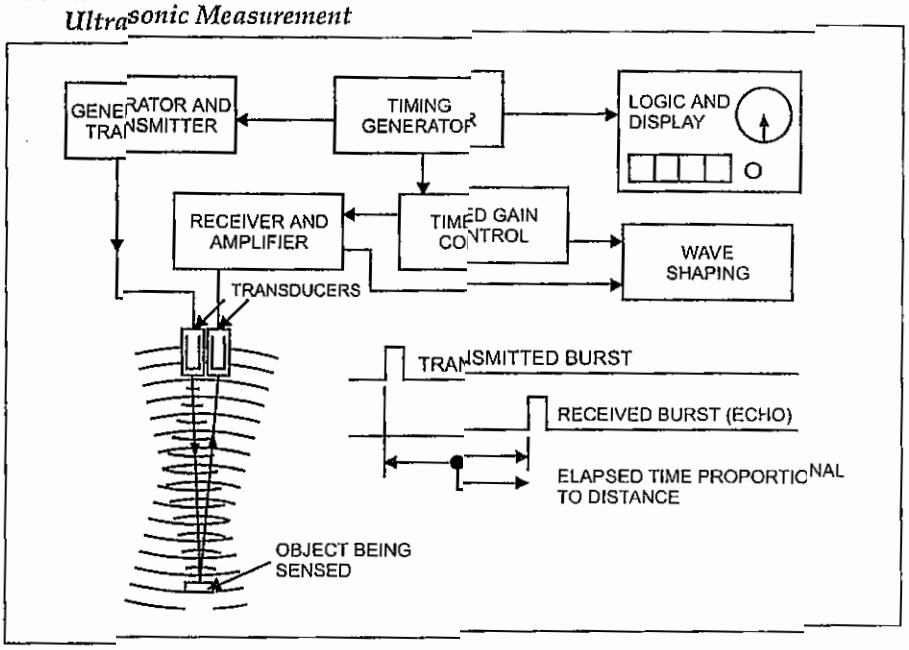
FIGURE 5-2.
Capacitance Probe



Ultrasonic

Level is measured using the ultrasonic method by transmitting a sound pulse from the instrument to the surface of a material. The sound pulse is reflected off the surface, and the echo returns to the instrument where it is detected. A shorter time for the transmitted signal to return means the level is closer to the sensor and, therefore, the level is higher in the tank. The amount of time for the return signal is inversely proportional to the level.

FIGURE 5-3.
Ultrasonic Measurement



5.2 SPECIAL CONSIDERATIONS WITH LEVEL CALIBRATION

Suppressed/Elevated Zero

If the d/p cell is not located at an elevation that corresponds to 0% level in the tank, it must be calibrated to account for the difference in elevation. This calibration adjustment is called zero elevation when the cell is located above the lower tap; it is called zero suppression when the cell is located below the lower tap.

To calculate the amount of elevation or suppression, the distance between the zero reference level and the high pressure port of the d/p cell must be accurately measured. Taking into account that 27.72" H_2O = 1 psig, calibrate the transmitter for the distance measured in direction necessary to result in an accurate output.

Specific Gravity

Specific gravity is ratio of density of a process material to the density of water or air. It is particularly important to consider the specific gravity of the process material when performing calibrations of differential pressure level devices. If specific gravity is assumed to be 1.0, or not

considered at all, significant error will be introduced if the process material is not water or a specific gravity of 1.0. Water has a density of 62.4 lbs./cu. ft.

FIGURE 5-4.
Liquid Head Measurement

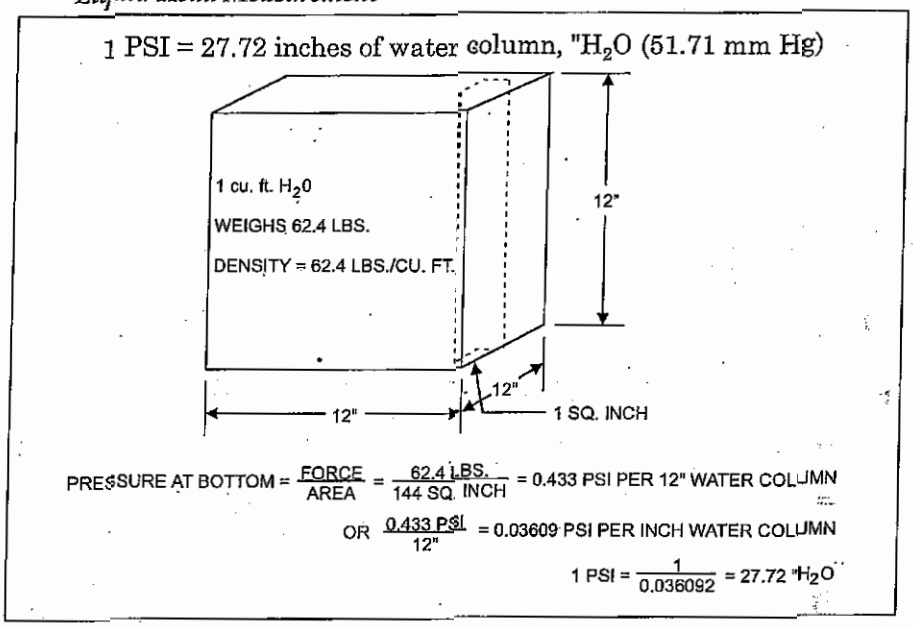
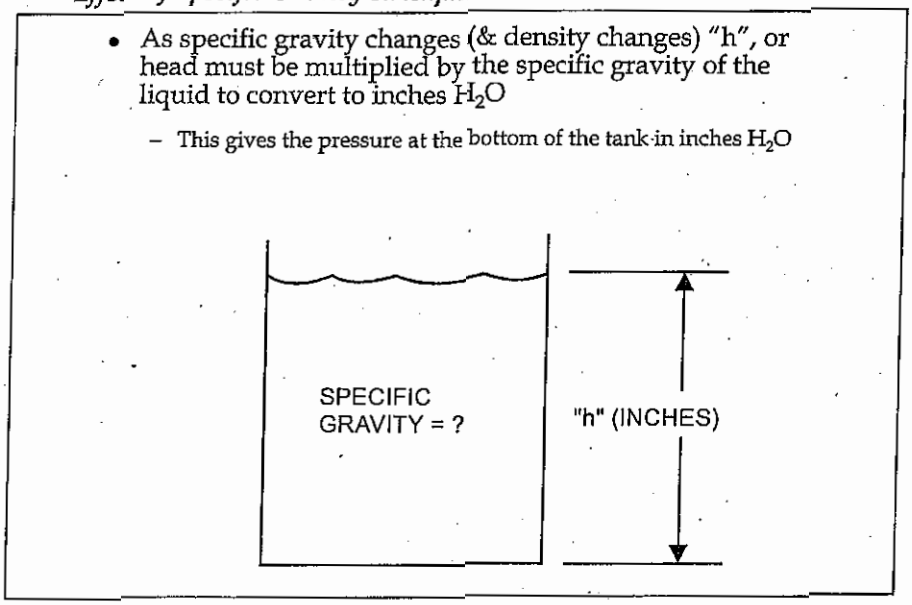


FIGURE 5-5.
Effect of Specific Gravity on Liquid Head Measurement



To determine the specific gravity of a process material, simply divide the density of the process material by the density of water. For example, the density of petroleum is 54.8 lbs/cu. ft. Therefore the specific gravity of petroleum is 54.8 ÷ 62.4, or 0.88. To properly calibrate a d/p transmitter for a sensor monitoring the level of petroleum, the nominal test points for input pressure must be multiplied by 0.88. Specifically, if the transmitter is calibrated for 0 to 100" = 4-to-20 mA output, you would use the following to properly adjust the transmitter for the process fluid specific gravity of 0.88:

NOMINAL (IN. H_2O)	ACTUAL TEST PRESSURE (IN. H_2O) NOM. × 0.88	EXPECTED OUTPUT (MA)
10	8.8	5.60
25	22.0	8.00
50	44.0	12.00
75	66.0	16.00
100	88.0	20.00

Dry and Wet Legs

For differential pressure measurements, the pressure of the vapor space in pressurized vessels is connected to the low pressure side of the d/p cell. When process vapors are non-condensing, non-corrosive, and non-plugging, the leg external to the vessel that is connected to the d/p cell is a dry leg. If any condensate accumulates in this leg, significant error would result in the level indication. Can you see why? If there is condensate in the dry leg, the weight of the condensate adds pressure to the low pressure side of the d/p cell. This would reduce the d/p sensed and result in a level indication or output that is lower than the actual level.

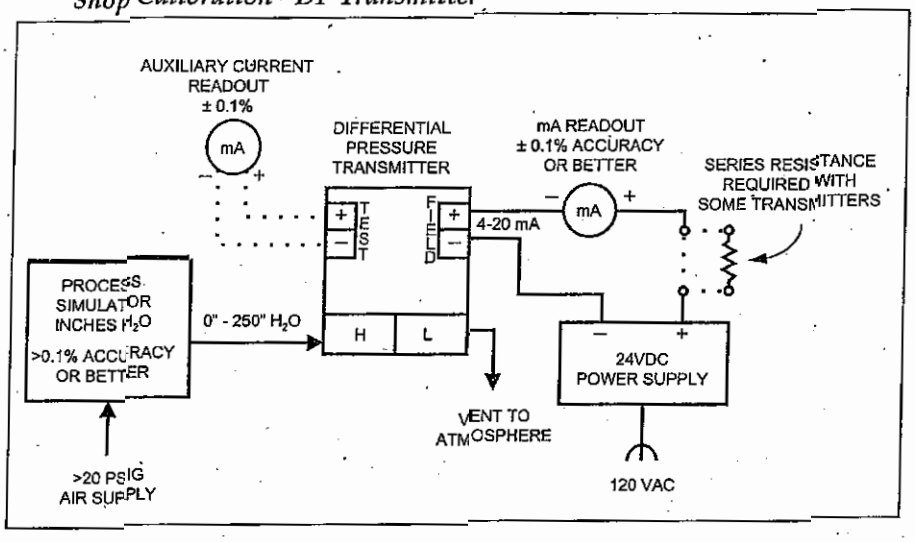
On the other hand, if the process vapors condense, the reference leg can be filled to form a wet leg. The specific gravity and height of the reference column must be accurately determined and accounted for in any calibration of the d/p cell.

Interface

For purposes of level instrumentation, "interface" typically refers to the point at which two non-mixable fluids meet, such as oil and water. Special considerations must be made to accurately measure the level of each fluid and/or the level of the interface.

5.3 CALIBRATING A DIFFERENTIAL PRESSURE LEVEL TRANSMITTER

FIGURE 5-6.
Shop Calibration - DP Transmitter



The illustration above indicates a typical test setup for performing a bench calibration of a differential pressure transmitter. Calibrating a d/p transmitter is similar to calibrating a pressure transmitter discussed in Chapter 4. Measured pressure is applied to the high pressure port of the d/p cell, and the low pressure side is vented to atmosphere. Be sure to remove any caps, cleanliness plugs, or obstructions from the low pressure port. (This is sometimes overlooked, particularly for new instruments.) The calibrator and transmitter must be at the same elevation to prevent inducing head which would introduce error. A milliammeter is connected in series to measure a 4-to-20 mA transmitter output signal. If necessary, we may also have to provide the transmitter output power source, such as 24 volts direct current (VDC). Again, most modern calibration standards provide the ability to supply the transmitter power so a separate power source is not required

The d/p transmitter in Figure 5-6 is calibrated for a 0-250" H_2O input = 4-to-20 mA output. On the bench, with no remote displays to over range, we can perform the five-point check at 0%, 25%, 50%, 75%, and 100%. The calibration data table is illustrated below.

TABLE 5-1.
Calibration Data Table for 0-250 in. H_2O D/P Transmitter

% SPAN	ACTUAL INPUT ("H_2O)	IDEAL OUTPUT (mA)	AS-FOUND OUTPUT (mA)	% DEVIATION (ERROR)	AS-LEFT OUTPUT (mA)	% DEVIATION (ERROR)
0%	0.0	4.00				
25%	62.5	8.00				
50%	125.0	12.00				
75%	187.5	16.00				
100%	250.0	20.00				

Once the test setup is established, adjust the input values to the required test points and record "as-found" values for the milliamp output. Typically, it is not necessary to check a transmitter in both the upscale and downscale directions. Once the "as-found" data has been recorded, evaluate the error against the acceptable tolerance to determine if adjustment is required. The following "as-found" values were obtained. The error is calculated by [(Actual Output–Ideal Output) ÷ (Ideal Output)] x 100%.

TABLE 5-2.
Calibration Data Table for 0-250 in. H_2O = 4-20 mA D/P Transmitter (Note: The % error is calculated based on the reading and not % span in this case)

% SPAN	ACTUAL INPUT ("H_2O)	IDEAL OUTPUT (MA)	AS-FOUND OUTPUT (MA)	% RDG DEVIATION (ERROR)	AS-LEFT OUTPUT (MA)	% RDG DEVIATION (ERROR)
0%	0.0	4.00	4.01	0.25%		
25%	62.5	8.00	8.02	0.25%		
50%	125.0	12.00	12.03	0.25%		
75%	187.5	16.00	16.05	0.3125%		
100%	250.0	20.00	20.08	0.40%		

If adjustment is required, perform the zero and span adjustments per the manufacturer's procedures. Sometimes this is adjustment of a zero and span potentiometer, sometimes it's a pushbutton adjustment, while other transmitters require the use of a handheld interface. Because some zero and span adjustments interact, it is important to recheck the zero and span until no further adjustment is required.

With the "as-found" readings obtained above, what adjustments will likely be required? The zero is adjusted to achieve a 4.00 mA output with the input vented to atmospheric pressure. The span is adjusted to achieve an output of 20.00 mA with 250.0" H_2O applied to the input. The zero and span adjustments are repeated until no further adjustments are required. Once no further adjustments are required, repeat the five-point check and record "as-left" values. The "as-left" values obtained are recorded below.

TABLE 5-3.
Calibration Data Table for 0-250 in. H_2O D/P Transmitter w/ As-Left Data

% SPAN	ACTUAL INPUT ("H_2O)	IDEAL OUTPUT (mA)	AS-FOUND OUTPUT (mA)	% DEVIATION (ERROR)	AS-LEFT OUTPUT (mA)	% DEVIATION (ERROR)
0%	0.0	4.00	4.01	0.25%	4.00	0.0%
25%	62.5	8.00	8.02	0.25%	8.01	0.13%
50%	125.0	12.00	12.03	0.25%	12.00	0.0%
75%	187.5	16.00	16.05	0.31%	16.01	0.06%
100%	250.0	20.00	20.08	0.40%	20.00	0.0%

How would this calibration change if the process material has a specific gravity of 1.1? If we installed the above d/p transmitter to measure the tank level of a material with specific gravity of 1.1, what would the indicated level be at an actual level of 100"? Since the process material is denser than water by a factor of 1.1, it will exert a pressure of 1.1 times water at the same height. Therefore the indicated level would be 110" at an actual level of 100".

For an accurate level indication, the input pressure must be adjusted to compensate for the specific gravity of the process material. This compensation is calculated by multiplying the input pressure in inches of H_2O times the specific gravity. This calculated input pressure is then applied at the corresponding % span using the same ideal output mA values as before. The Calibration Data Table compensated for a specific gravity of 1.1 is illustrated below.

TABLE 5-4.
Calibration Data Table for 0-250 in. D/P Transmitter @ Specific Gravity = 1.1

% SPAN	ACTUAL INPUT ("H_2O)	IDEAL OUTPUT (mA)	AS-FOUND OUTPUT (mA)	% DEVIATION (ERROR)	AS-LEFT OUTPUT (mA)	% DEVIATION (ERROR)
0%	0.0	4.00				
25%	68.75	8.00				
50%	137.5	12.00				
75%	206.25	16.00				
100%	275.0	20.00				

How would this calibration change if it were done in the field? There is no change in the calibration principles. However, some of the considerations that must be made are listed below:

- Before doing anything, the process conditions must be evaluated. This is also true if the instrument is removed from service for a bench calibration. Is the system shutdown with the tank drained? Is the process operating in automatic level control? If so, the system must be placed in manual control and an alternate means of manual level control employed.
- Can the d/p cell be isolated from the system? If not, the system must be shutdown and drained. If so, isolate the d/p cell properly before making test connections. If equipped, the bypass valve is always opened first and closed last to prevent damage to the d/p cell.
- Is the process material hazardous? If so, take appropriate precautions.
- The test pressure input is connected to the high pressure side, and the low pressure side is disconnected, if necessary, and vented to atmosphere.
- Most likely, the mA output will only need to be measured, without the need for providing a power source. The output signal circuit usually provides the power source.
- Record the remote display(s) if required by procedure for a loop calibration. If the remote indication(s) does not go below the 0"

value or above the 100% value, check the instrument at 10% and 90% instead of 0% and 100%. All other test points remain the same.

5.4 CALIBRATING A CAPACITANCE LEVEL INSTRUMENT

The initial calibration of a capacitance level instrument following installation is very vendor-specific. It usually involves adjusting the actual tank level to one point, accurately measuring that level, and entering the actual level in the calibration setup (or pushing a button at the zero level). The tank level is then adjusted to a second point, accurately measuring that level, and entering the actual level in the calibration setup (or pushing a button at the 100% level. The calibration points must be accurately determined and should be as far apart as possible. There may also be linearization values to enter and other factors, depending on the tank shape and probe installation.

To perform a periodic calibration check of the capacitance level, it is best to establish an independent means of determining the level. This could be by:

- Installing a temporary sight glass by connecting some tubing to a tank bottom connection and running up the side of the tank (externally);
- Measuring the actual level using a measuring stick;
- Using a calibrated flowmeter to meter the amount of fluid added to the tank; or
- Weighing out a specified amount and transferring the entire amount to the tank.

You'll have to use your imagination and existing resources to determine a method that works for your application. Once a method has been determined, check the level at a minimum of two different points in addition to the zero level. If the level instrument requires calibration, perform in accordance with the manufacturer's instructions in the instrument technical manual.

Here's what does not work, based on my experience. I've tried to measure the probe capacitance at various levels during initial setup. I thought I could then use these values to simulate a capacitance to the instrument. It just doesn't work out. However, some manufacturers do

provide a calibrator specific to their capacitance level meters. For example, Drexelbrook provides a PC interface to configure and calibrate the electronics. Smart transmitters can be configured and calibrated with smart interface devices such as Rosemount 275. To verify the accurate level measurement, some method indicated above would still be required.

5.5 CALIBRATING AN ULTRASONIC LEVEL INSTRUMENT

There are two methods for calibrating an ultrasonic level transmitter. Ideally the instrument can be removed and the cabling has enough slack to move to an open area with an unobstructed path to a smooth surface (such as a wall or floor). If this is the case, follow Method 1. If this is not the case, follow Method 2.

Method 1:

1. Determine the distance from the face of the ultrasonic transducer to the 100% level. This should already be documented as part of the instrument setup. For this example, we are going to assume the distance is 12".

2. Determine the test point distances. Realize the distance from the test surface is simulating the distance from the top of the liquid level and take into account the distance determined in step 1 above. For example, if the distance from the transducer face to the 100% level is 12 inches and the tank level indication is 0-100 inches, the following test points would be used.

% SPAN	DESIRED LEVEL INDICATION	DISTANCE FROM TEST SURFACE
0%	0"	112"
25%	25"	87"
50%	50"	62"
75%	75"	37"
100%	100"	12"

3. Turn off the transducer.

4. Use appropriate safety precautions; remove the instrument from the top of the tank.
5. Use a tape measure to mark off the distances from the test surface at each desired test point determined in step 2 above.
6. Place the instrument at a distance equal to the first test point from the wall or floor that will be used to bounce the signal back (the test surface simulates the top of the liquid level).
7. Turn the transducer on and record the level reading.
8. Move to the next distance and record the level reading. Repeat until all "as-found" readings have been obtained.
9. If necessary, adjust the instrument in accordance with the manufacturer's instructions in the instrument technical manual.
10. If adjustments were made, repeat steps 6 to 8 above to obtain "as-left" readings.
11. Return the instrument to operational condition.

Method 2:

1. Establish an independent method for determining the actual liquid level in the tank (such as hooking up a temporary sight glass using tubing from a bottom connection and running external to the top of the tank).
2. Adjust the level in the tank to each of the desired setpoints throughout the range of the instrument and record the reading from the ultrasonic instrument at each test point.
3. If necessary, adjust the instrument in accordance with the manufacturer's instructions in the instrument technical manual.
4. If adjustments were made, repeat step 2 above to obtain "as-left" readings.
5. Return the system to operational condition.

REVIEW QUESTIONS

Use the information in Chapter 5 and the references in Appendix A to answer the following questions.

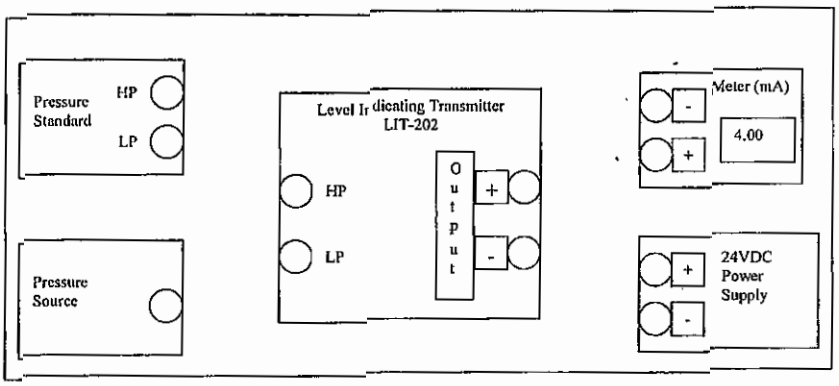
1. What is the correct calibration range and the specified accuracy for LIT-202?

2. Select the correct procedure(s) that could be used for this calibration of LIT-202.

3. The process monitored by LIT-202 is at a specific gravity of 1.0. What are the correct input pressure values for 0%, 25%, 50%, 75%, and 100%? (include units)

4. What are the expected transmitter output values for the pressure inputs of step 3?

5. Indicate the correct test equipment hookup for calibration of LIT-202.



6. If the specific gravity of the process being monitored is 0.8, what would the pressure inputs be in step 3?

7. If a tank located on the roof has a D/P level transmitter installed 100 feet below the zero reference level, what would the applied pressures be in step 3? (s.g. = 1.0)

8. What type of level instrument is LT-305?

9. What is the principle of operation for the level instrument type in step 8?

10. Describe the basic method for initial calibration of LT-305.

11. How is the capacitance probe checked for proper calibration?

12. What two basic methods can be used to perform a calibration of an ultrasonic type level sensor?

6

FLOW INSTRUMENT CALIBRATION

After completing this chapter, you should be able to:

Explain the principles of operation for the following flowmeter types:

- *Differential pressure*
- *Magnetic*
- *Vortex-shedding*
- *Turbine*
- *Coriolis*

Calibrate the following flow instrument types (to ISA standards, where applicable) and determine acceptability:

- *Flow rate meters*
- *Flow totalizers*

Select proper calibration procedure and calibration data sheet.

Select appropriate certified test equipment.

Properly set up and connect test equipment to device under test (DUT) for calibration.

Properly isolate flow devices and/or remove from service for field calibration.

Return equipment to service following calibration.

Complete and properly maintain calibration documentation.

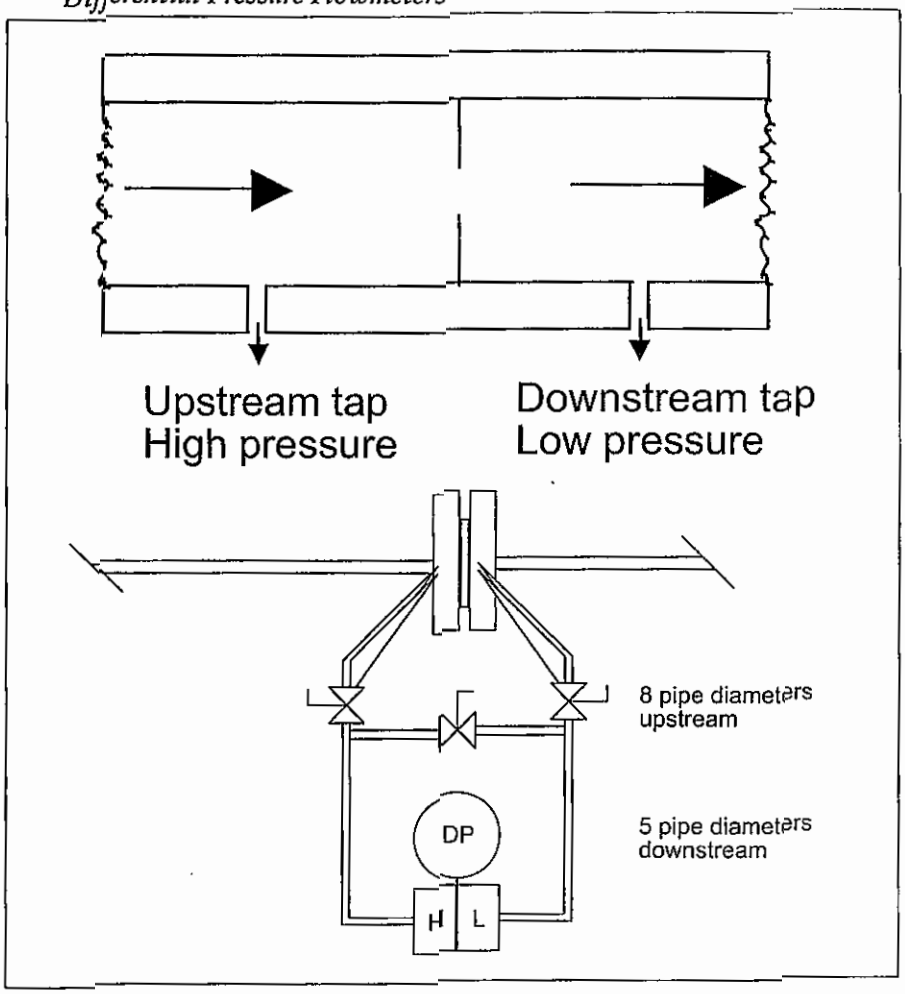
6.1 TYPES OF FLOW INSTRUMENTS

There are numerous sensor technologies for flow instrumentation. The sensor type used in a particular application depends on many factors, including the process fluid measured, pressure, temperature, allowable pressure drop, density, conductivity, viscosity, pipe size and orientation, flow rate and/or flow total required, accuracy requirements, control

system interface, accessibility, maintenance requirements, etc. We will consider briefly a few of the major sensor types:

Differential Pressure

FIGURE 6-1.
Differential Pressure Flowmeters

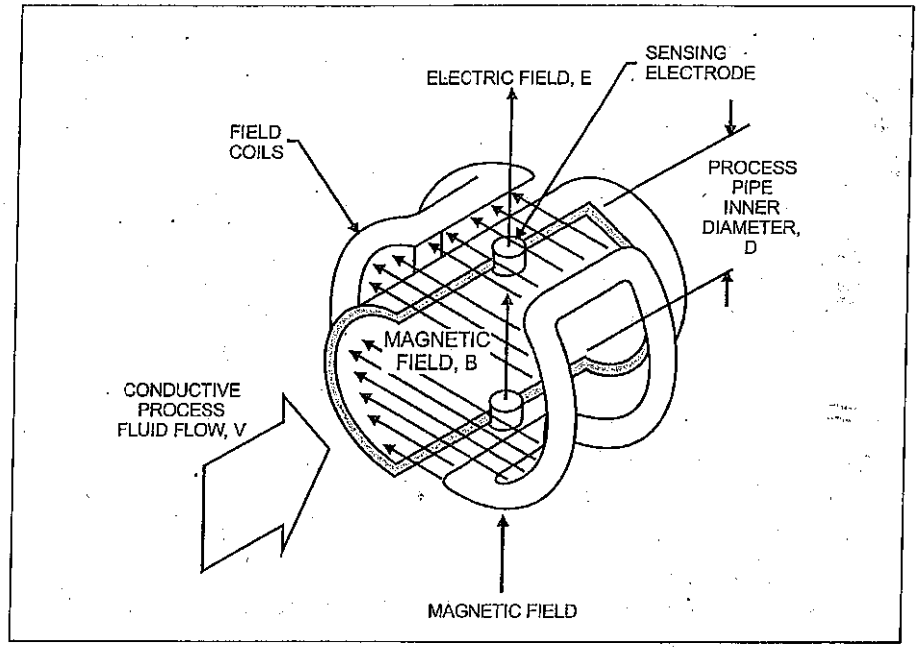


Restriction-type flow instruments are based on the principle that flow rate is proportional to the square root of the differential pressure across the restriction. An orifice plate or venturi is installed inline with the process flow to restrict the liquid flow. This restriction creates a pressure drop that can be converted to flow rate. The upstream pressure is connected to the high pressure port of a d/p (differential pressure) cell. The downstream side is connected to the low pressure port of the d/p cell.

(The d/p cell was described in Chapter 5.) The square root is accounted for in the transmitter signal processing or by the use of square root extractor springs of the d/p cell. However, some older models utilize a separate square root extractor installed in the output signal loop. In this case the installed square root extractor must also be calibrated for the loop to perform within specification.

Magnetic Flowmeter

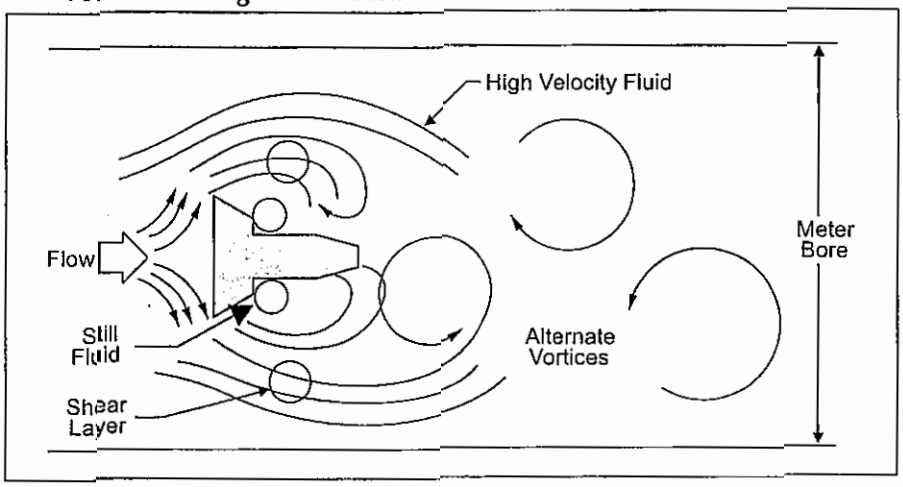
FIGURE 6-2.
Magnetic Flowmeter



The magnetic flowmeter is based on Faraday's Law of electromagnetic induction. When a liquid conductor moves in a pipe having diameter D and travels with an average velocity V through a magnetic field of B intensity, it will induce a voltage (E) according to the relationship E = BVDC, where C is the constant for units conversion. Because the magnetic field, pipe diameter, and conversion constant are fixed values, they can be combined into a calibration factor and the equation reduces to E = KV. Magnetic flowmeters can only be used to measure flow of conductive liquids. As conductive liquid passes through the magnetic field, an electrical potential is induced. The change in potential varies directly with the liquid velocity. This voltage is sensed by electrodes in the walls of the flowmeter.

Vortex-shedding Flowmeter

FIGURE 6-3.
Vortex Shedding Phenomenon

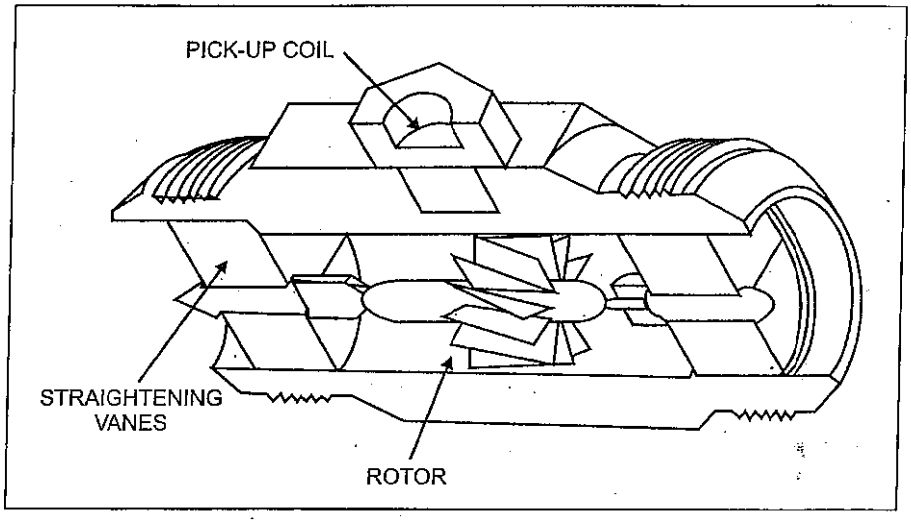


When a flowing medium strikes an obstruction, it separates and moves around the object. At the point of contact with the object, vortex swirls separate from the object on separate sides. This shedding causes a local increase in pressure and decrease in velocity on one side of the object. A local decrease in pressure with corresponding increase in velocity occurs on the other side of the object. After shedding from one side, the process is reversed. The frequency of shedding reversal is proportional to the velocity of the fluid passing the obstruction.

A vortex-shedding flowmeter has three basic parts: a bluff body (or shedder bar), a sensor, and a transmitter. In the process line, flowing liquid strikes the bluff body and vortices are shed alternately on each side of the bluff body. The velocity of the flow determines the frequency at which the vortices are shed. The shedding of the vortices on alternate sides of the bluff body is detected by sensors built into the bluff body. These sensors can be force sensors or temperature sensors (thermistors). The sensor then converts the energy created by the vortices into electrical pulses. These pulses are conditioned and sent to the transmitter, which generates the output signal proportional to the flow rate. The volumetric flow rate (Q) = vortex-shedding frequency (f) × meter coefficient (K).

Turbine Flowmeter

FIGURE 6-4.
Turbine Flowmeter

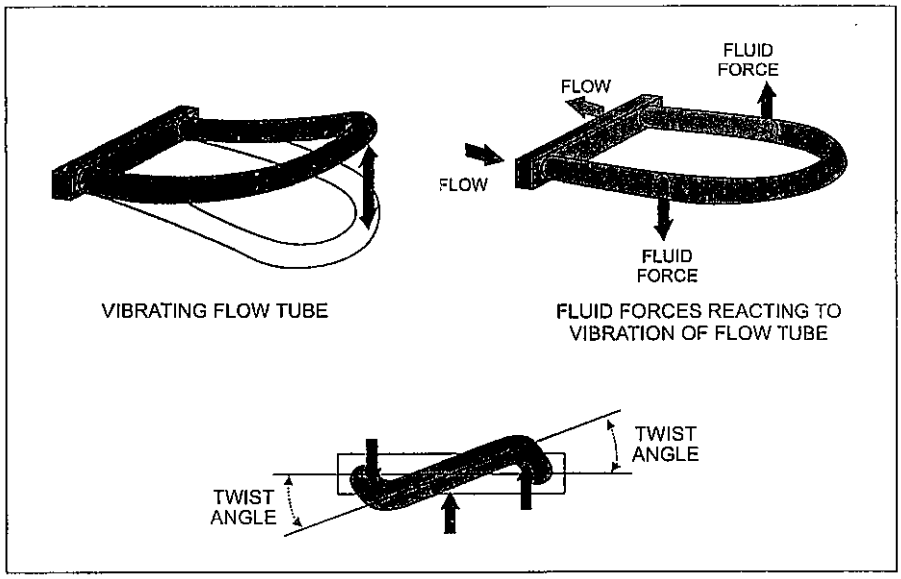


The turbine flowmeter is a mechanical flowmeter that measures flow by means of a spinning turbine or rotor using an arrangement of moving parts. The multi-bladed rotor is mounted at right angles to the flow, suspended in the process fluid stream on a free-running bearing. The speed of rotation is proportional to the volumetric flow rate. Turbine rotation is detected by solid state devices or mechanical sensors. As shown in Figure 6-4, a pickup coil is used to detect the rotary speed for signal processing. As each blade passes the coil, a voltage pulse is generated. Each pulse represents a discrete volume. The number of pulses per unit volume is the meter's K-factor. The output is a continuous sine wave with the frequency proportional to the flow rate.

Coriolis Mass Flowmeter

A mass flowmeter measures flow rate in weight per unit time rather than volume. This measurement compensates for temperature and pressure changes. Fluid moving through a vibrating tube is forced to accelerate as it moves toward the point of peak amplitude of vibration. The fluid decelerates as it moves away from the point of peak amplitude. The acceleration and deceleration cause twisting forces on the flowtube which are proportional to the mass flow.

FIGURE 6-5.
Coriolis Mass Flowmeter



6.2 CALIBRATION OF FLOWMETERS

All meters with moving parts require periodic testing because wear over time will reduce the flowmeter performance. Calibration can be performed either in the lab or in situ (in its original place) using a prover, also called a master meter, or by weighing the flow output. There are several methodologies for flowmeter calibration. Any one of them, and others, may be acceptable depending on the process system configuration, compatibility, availability of test standards, and accuracy requirements.

It is sometimes difficult or impossible to remove a flowmeter from service for calibration. Therefore, field-mounted and inline provers have been developed. Depending on the application and system configuration, other methods can also be developed to check the accuracy of flowmeters. Weighing the flowmeter output collected over a specified time is a common alternative.

The calibration of the signal-processing portion for most flowmeter instruments can be checked by simulating the signal from the flowmeter. These methods do not check the sensor itself. No one generic method works for all flowmeters. Tests must be performed in accordance with the specific manufacturer's instructions. However a few of these methods are discussed below.

GENERAL METHODOLOGY	WHEN TO USE
Calibrate only the electronics (or signal processing) using a test instrument to simulate the sensor.	If it is impossible to perform an in-situ check of the flowmeter sensor using a prover or other methods discussed below and the sensor cannot be removed from the system.
Check the calibration of the flowmeter, sensor and signal processing together, using a prover or some other standard.	If the flowmeter sensor can be checked in situ but test standards are not available for simulating test signal input.
Calibrate the electronics first, and then check the calibration of the flowmeter, including the sensor.	If the required test instruments are available to simulate the flowmeter input signal and the flowmeter sensor can be checked in situ.
Remove the flowmeter and send to a flow calibration lab (internal, manufacturer, or a 3rd party flow calibration lab).	If the system is not compatible with in-situ flowmeter calibration. It may be necessary to install a calibrated spare to keep the process downtime to a minimum.

6.3 D/P TRANSMITTER CALIBRATION

The principles and methods discussed in Chapter 5 for the level d/p transmitter also apply here. The only difference is that you must know how the square root is obtained. Remember that flow rate is proportional to the square root of the differential pressure. As mentioned earlier in this chapter, the square root is usually accounted for in the transmitter signal processing or by the use of a separate square root extractor installed in the output signal loop. Sometimes, but rarely, some models utilize square root extractor springs as part of the d/p cell.

To calibrate the square root extractor in a 4-20 mA transmitter output loop, a milliamp source is connected to the input. A milliammeter is connected in series to the output with a power supply, if necessary. The manufacturer's instructions specify the input values and expected outputs.

Note that this does not check the actual flow sensor. It may be desired or required to check the flow sensor for wear, damage, or obstruction. Later in this chapter we will discuss methods for calibrating the flow sensor in situ.

6.4 EXAMPLE: MAGNETIC FLOWMETER CALIBRATION

You must refer to the manufacturer's instructions for calibration of each specific model of magnetic flowmeter. The manufacturer may also supply specific calibrators for its flowmeters. Here is a typical procedure using a magnetic flowmeter calibrator as an input standard that varies the millivolt signal to simulate the signal from the flowtube electrodes. We'll assume the output is 4 to 20 mA. Therefore, a milliammeter will be used as the output standard.

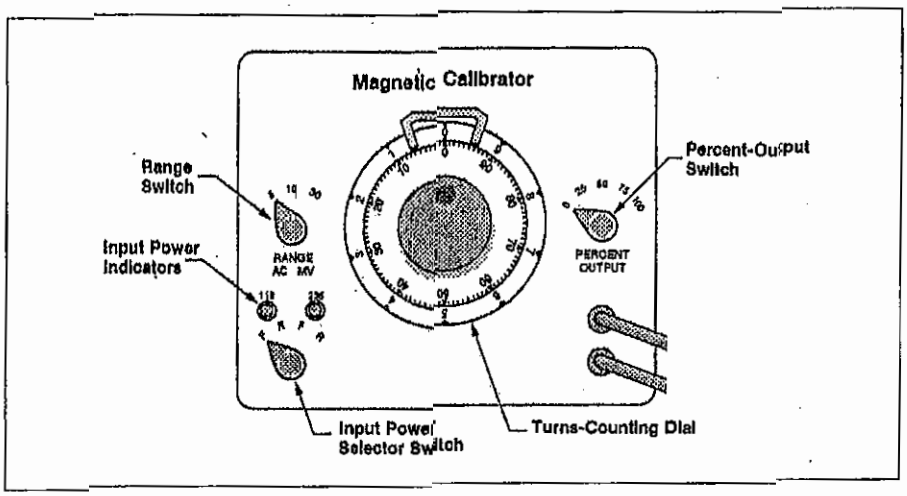
1. Determine the maximum flow rate that will be checked. This is the 100% value.
2. Obtain the mV calibration factor and the phase band factor from the instrument data plate.
3. Calculate the upper range value (URV) as follows:

   URV = (Max Flow Rate) × (mV Calibration Factor)

4. Calculate the maximum input signal value as follows:

   Max Input Signal Value = (URV) × (Phase Band Factor)

5. Disconnect the flowtube from the transmitter and connect the magnetic flowmeter calibrator to the transmitter input.



6. Connect the milliammeter in series with the transmitter output signal.

7. Set the calibrator range knob to select the maximum output signal. For example, with a calculated maximum input signal of 8.32 mV, the range knob would be set to 10 mV.

8. Set the maximum input signal value on the dial. For this example, the dial would be set to 8.32 (outer dial set at 8, inner dial set at 0.32).

9. Use the percent output switch to perform a 5-point check at 0, 25, 50, 75, and 100%. At each position, record the transmitter mA output on the calibration data sheet.

10. At this point you need to evaluate whether to adjust the flowmeter. If you are going to check the flowmeter sensor, it may be prudent to do that prior to making any adjustments. Adjustments may have been made during the last calibration to compensate for slight flow sensor inaccuracy. If you make adjustments prior to checking the flow sensor, you may be "undoing" the correction. If a check of the flowmeter will be performed, use one of the methods described later in this chapter and then proceed with step 11, below.

11. If necessary, calibrate the transmitter by adjusting zero at the 0% input for a 4.00 mA output. Then adjust span at the 100% input for a 20.00 mA output. Since zero and span often interact, recheck and adjust the zero and span until no further adjustment is necessary.

12. If adjustments were made, repeat step 9 above to obtain as-left readings.

Note that this does not check the actual flow sensor. It may be desired or required to check the flow sensor for wear or damage. The next section of this chapter will discuss methods for calibrating the flow sensor in situ.

The calibration of a vortex-shedding or turbine flowmeter would use the same principles as the magnetic flowmeter above. However, the input standard would be a frequency generator. Again, you must use the specific calibration instructions provided by the manufacturer.

Once the calibration of the signal-processing portion of the flowmeter has been checked, a check of the flowmeter in situ is performed. Note that some industries require that all as-found readings be obtained prior to any adjustments. Therefore you may need to evaluate whether you would adjust the electronics prior to checking the actual flowmeter in situ. So far we have only mentioned flow rate meters. Some flowmeters are used as flow totalizers. Some flowmeters are used to obtain both flow rate and flow total. We will have to consider how the flowmeter is used when

performing calibrations. The two methods for in-situ flowmeter calibrations are described below.

6.5 FLOWMETER CALIBRATION USING A MASTER METER (PROVER)

A master meter, or prover, is a calibrated flowmeter used as a calibration standard. The master meter is installed in series with the flowmeter under test, and the readings are compared at various flow rates or flow totals. It is advisable to perform this comparison for at least two different flow rates (or flow totals). Ideally, the process was designed to incorporate a master meter in series with the installed flowmeter. In my experience, this is not usually the case. So you have to get creative about how to install the master meter. It may, for example, involve disconnecting the outlet of the installed flowmeter, adapting the master meter to the output, connecting a discharge hose with a valve at the end of the hose, and directing the hose to a drain or back to the process vessel for proper waste disposal. A discharge valve is important to keep the piping full and maintain some backpressure. Without a discharge valve you may find error in your readings, particularly at low flow conditions.

Once the test setup is made, flow the process material (or water) through the flowmeters and close the discharge valve in order to fill the piping completely. With the piping and flowmeters completely filled and no flow, the flowmeters should read zero flow. If this is a desired test point, record the readings. Sometimes zero flow is difficult to check due to environmental factors, vibration, air pockets, etc. Next, open the discharge valve and establish flow at the desired test point and allow the readings to stabilize. Record the readings and repeat for all desired test points. If adjustment is required, perform it in accordance with manufacturer's instructions and repeat the above procedure to obtain as-left readings.

If the master meter and the installed flowmeter are not capable of reading in the same units, conversion of the results may be required. Some common conversion factors are listed below. Additional conversion factors are included in Appendix A-7.

1 liter = 0.2642 gallons; 1 gallon = 3.785 liters

1 gallon water = 8.35 lbs at 25°C
(measure water temperature used for calibration if significantly different and calculate density for calibration data analysis)

1 lb = 0.45 kg; 1 kg = 2.2 lb

6.6 GRAVIMETRIC METHOD FOR FLOWMETER CALIBRATION (MEASUREMENT BY WEIGHT)

This method involves weighing the amount of fluid that actually flows through the meter into (or out of) a container. For a flow rate meter, the fluid is collected in the container for a specified amount of time, such as one minute. Although it may require as much creativity, the test setup is the same as the prover method, except there is not a master meter in series with the installed flowmeter. The hard part is figuring out how you're going to weigh the amount of fluid. You must use a container that is large enough for the amount of flow to be tested. Likewise, a calibrated weigh scale with adequate capacity must be used. The container will either sit on the weigh scale in close proximity to the flowmeter or the container will need to be transported to the weigh scale. Obviously you don't want to get your container full and then discover that you can't get the container on the scale. So think through the logistics before starting the calibration.

Once the logistics are figured out and the test setup is made, place the empty container on the scale and TARE (zero) the scale reading. It is important to TARE the scale with anything that will be on the scale when weighing the contents of the container. Next, flow the process material (or water) through the flowmeter to a waste container or drain and close the discharge valve in order to fill the piping completely. If zero is a desired test point, read and record the flowmeter reading. Then open the discharge valve and flow to the drain or waste container to establish the desired flow rate. Once the desired flow rate is stabilized, simultaneously start the stopwatch and switch the discharge to the weighing container. Ensure there is no spillage out of the container and verify that the flowmeter reading is maintained at the desired test point. Record the flowmeter reading. Continue discharging the entire output of the flowmeter into the container for the specified time period, such as one minute. It may be desirable to use more time to obtain a measurable amount of fluid (at least ¼ of the container). Once the specified time period has elapsed, simultaneously stop the stopwatch and switch the discharge to the drain or the waste container. Close the discharge valve. Record the stopwatch reading and the weight. Convert the weight to the desired units and divide by the time to obtain the actual flow rate. Compare the flow rate obtained with the flowmeter reading recorded. Empty the weighing container, if necessary, and TARE the scale. Repeat the procedure above for additional test points. If adjustment is required,

perform in accordance with the manufacturer's instructions and repeat the above procedure to obtain as-left readings.

If the flowmeter is used as a flow totalizer, the procedure is essentially the same. Some totalizers can be set to stop the flow when the desired flow total is achieved. Also, instead of using a stopwatch, the totalizer is reset to zero at the same time discharge is switched to the weighing container. If the totalizer cannot be zeroed, record the initial totalizer reading, at the moment the discharge is switched, and the final totalizer reading.

REVIEW QUESTIONS

Use the information from this chapter and the references in Appendix A to answer the following questions:

1. How is a flowmeter calibration performed when accuracy to manufacturer specifications is required and the process system is incompatible with performing an in-situ calibration?

2. How can most flow calibrations be categorized as flow rate or flow totalizer when there are so many different types of flowmeters?

3. What are the two basic calibration methods for checking flowmeters?

4. What methods are used for incorporating the square root of the d/p for venturi-type flow devices?

5. Select the correct procedure to be used for this calibration of FT-301. (Appendix A-4)

6. What local and remote indications must be recorded during calibration of FT-301? (Appendix A-1)

7. What test equipment is required for calibration of FT-301?

7

FINAL CONTROL DEVICES CALIBRATION

After completing this chapter, you should be able to:

Calibrate the following instrument types (to ISA standards, where applicable) and determine acceptability:

- *Current-to-pressure transducers (I/P)*
- *Valve positioners*
- *Control valves*

Select proper calibration procedure and calibration data sheet.

Select appropriate certified test equipment.

Properly set up and connect test equipment to DUT for calibration.

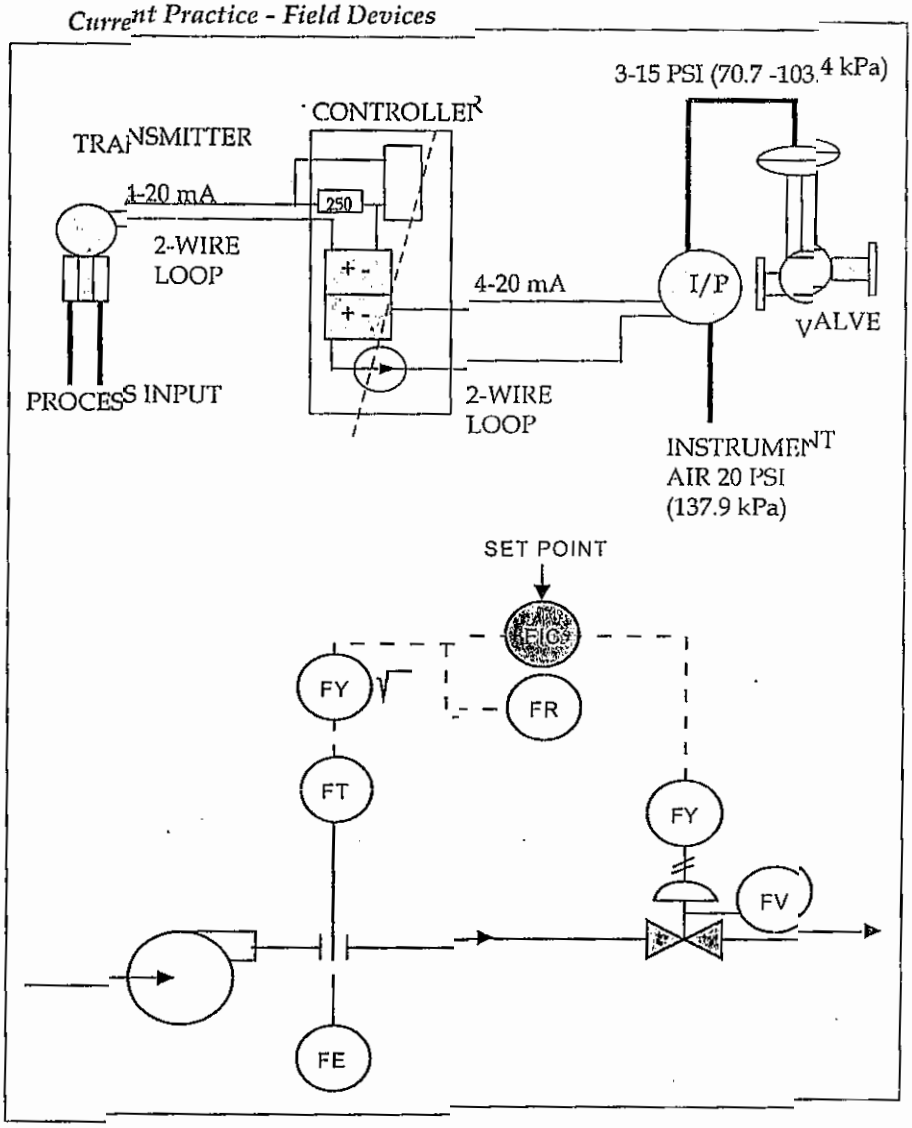
Properly isolate flow devices and/or remove from service for field calibration.

Return equipment to service following calibration.

Complete and properly maintain calibration documentation.

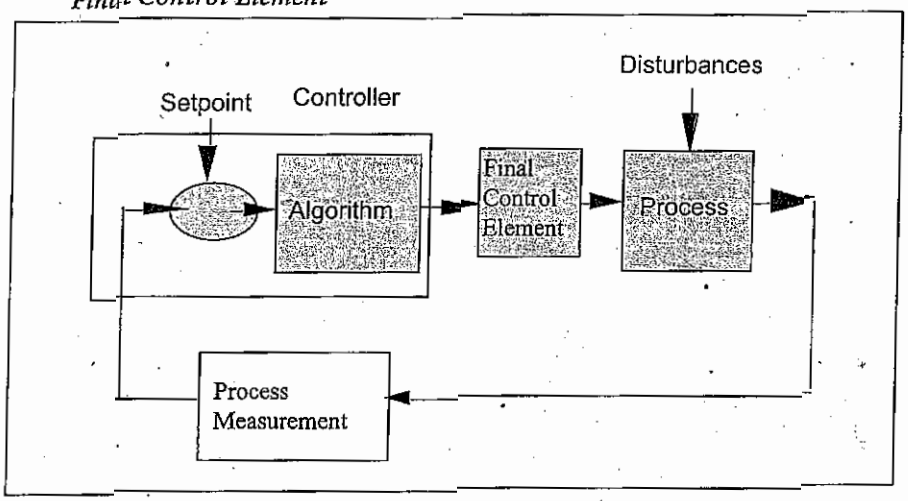
The temperature, pressure, level, and flow instruments covered in the previous chapters all sense a process parameter and produce a signal for indication or controller input. If we want to control a process parameter, the controller output must be converted to a signal that may be used to drive a control valve. The control valve is a final control element. A final control element is any device or element that changes the value of a manipulated variable. Valves and heaters are common examples of final control elements. In this chapter, we will discuss control valves and the devices that process the signal supplied to the control valve.

FIGURE 7-1.
Current Practice - Field Devices



In the illustration above, the controller output sends an electronic signal to the current-to-pressure transducer (I/P), which sends a pneumatic signal to the control valve. The control valve position changes in response to the signal to adjust flow to the setpoint. As the flow changes, it is sensed by the flow transmitter. When the flow sensed is equal to setpoint, the valve position is maintained. Any time there is a disturbance to the system or a change in setpoint, the flow control loop automatically responds to achieve the programmed setpoint. A block diagram of this concept is illustrated below.

FIGURE 7-2.
Final Control Element

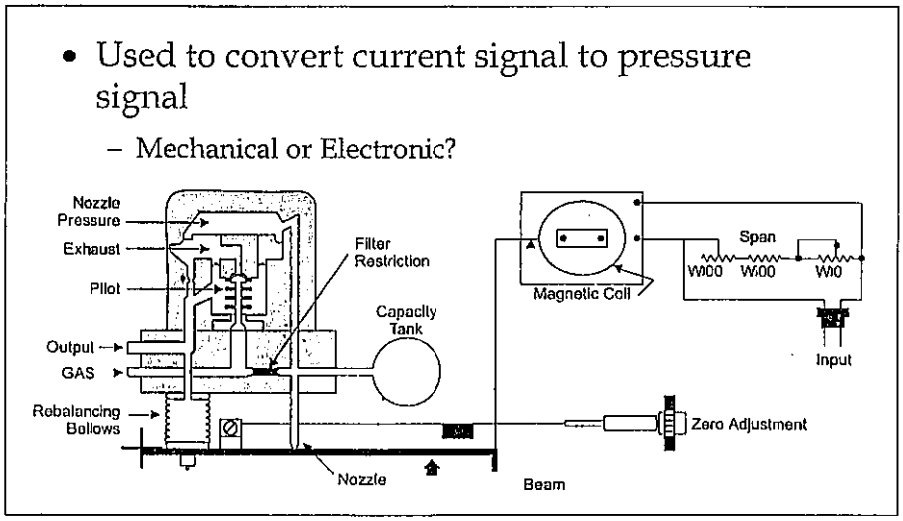


The final control element can be proportional control, as described above, or ON-OFF control. For ON-OFF control, a controller output relay changes the state of the relay contact which completes the circuit for a solenoid valve to energize. The solenoid valve opens to allow air to open (or close) a control valve.

The first component in the final control subsystem is the signal conditioner. The signal conditioner amplifies and, if necessary, converts the signal for compatibility with the actuator. Typical devices used as signal conditioners include current-to-pneumatic transducers, current-to-voltage (I/E) transducers, amplifiers (electronic or pneumatic), relays, digital–to-analog converters, or analog-to-digital converters. The most common signal conditioner in a proportional control loop is an I/P transducer.

A typical I/P transducer is a force balance device in which a coil is suspended in the field of a magnet. Current flowing through the coil generates axial movement of the coil which causes movement of the beam. The beam controls the backpressure against the nozzle by controlling the restriction of air flow through the nozzle. This backpressure acts as a pilot pressure to control the outlet pressure. The zero adjustment causes the beam to move relative to the nozzle. The span adjustment is a potentiometer that limits the current through the coil. The I/P transducer must be supplied with instrument air within the range specified by the manufacturer, usually at least 20 psig. The typical I/P transducer is calibrated for a 4-20 mA input = 3-15 psig output. Most I/P transducers can be configured for direct action (output pressure increases as input

FIGURE 7-3.
I/P Transducer



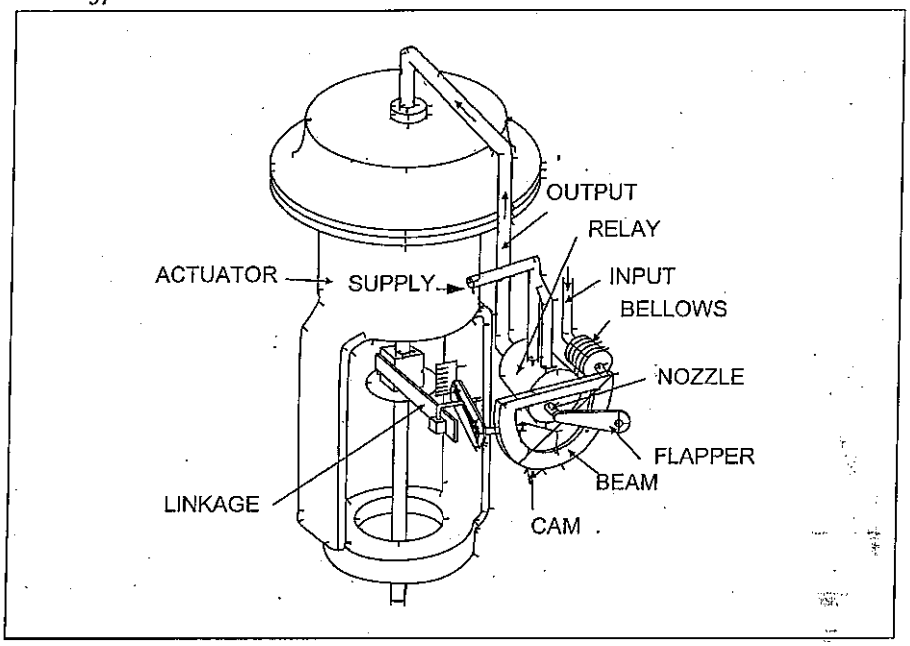
signal increases) or reverse action (output pressure decreases as input signal increases).

The next component in the final control subsystem, if applicable, is the actuator. The actuator receives the conditioned signal and changes it to some form of mechanical energy or motion. Typical devices used as actuators include solenoids, pneumatic valve positioners, AC and DC motors, stepper motors, hydraulic motors, and hydraulic pistons. Many control valves include a pneumatic valve positioner.

A valve positioner is a device used to increase or decrease the air pressure (from the I/P) operating the control valve actuator. Positioners are generally mounted to the control valve actuator and connected mechanically to the valve stem for position indication. A positioner is a type of air relay which acts to overcome hysteresis, packing box friction, and effects of pressure drop across the valve. It assures exact positioning of the valve stem and provides finer control.

There are many types of positioners. The basic principles of operation are similar for all types. The instrument pressure (from an I/P, for example) acts on the input module, which controls the flapper-nozzle system of the relay. Supply pressure is applied to the relay, and the output pressure of the relay is supplied to the control valve actuator. Most positioners can be configured for direct or reverse action. For a direct-acting positioner, increasing the instrument pressure causes the input module to pivot the beam. The beam pivots the flapper and restricts the

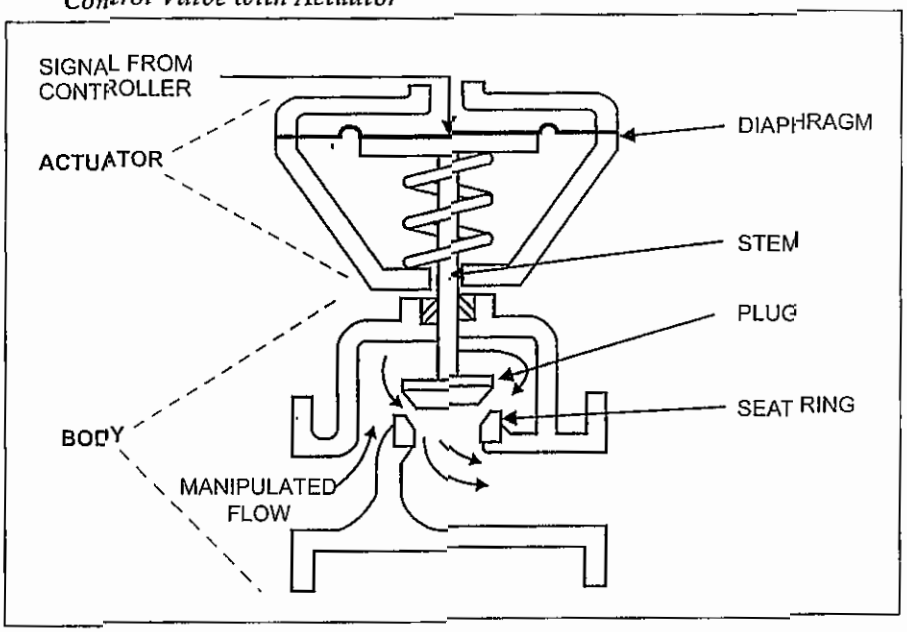
FIGURE 7-4.
Typical Motion-Balance Positioner



nozzle. The nozzle pressure increases and causes the relay assembly to increase output pressure to the actuator. With a direct-acting actuator, the increased pressure moves the actuator stem downward. The positioner is mechanically connected to the stem of the valve. Stem movement is fed back to the beam by means of a feedback lever and range spring, which causes the flapper to pivot slightly away from the nozzle to prevent further increase in relay output pressure. Note that some positioners accept a milliamp input and include an integral I/P transducer.

The last component in the final control subsystem is the final control element. We are only going to discuss control valves. (Other final control elements include servo valves, heaters, conveyors, auger feeds, and hopper gates.) There are many different types, sizes, and applications for control valves. Selecting the correct control valve for a specific application is crucial to proper system performance. Undersizing and oversizing are common problems. Valve selection is not within the scope of this book, but many valuable resources are available to assist with proper selection, for example, a good control valve sales engineer. Let's look at a typical control valve.

FIGURE 7-5.
Control Valve with Actuator

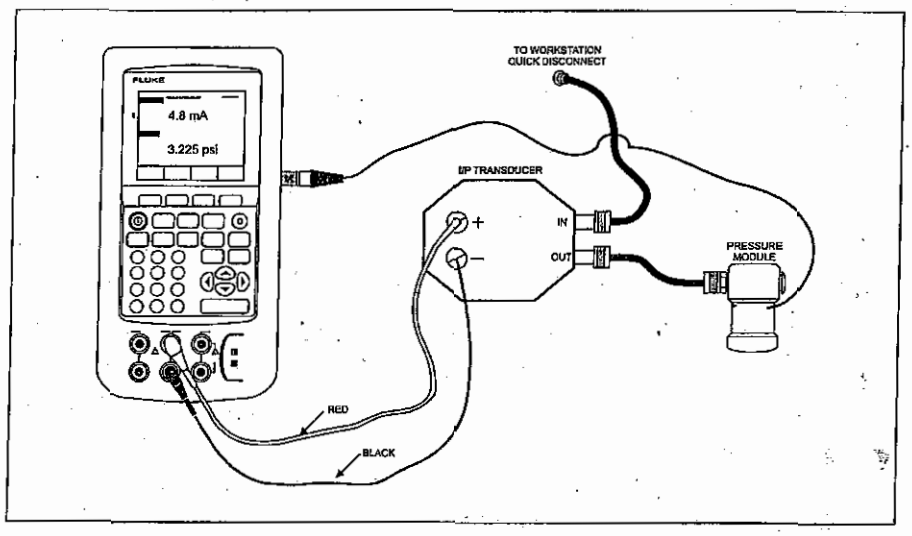


The pneumatic signal from the positioner (or I/P if a positioner is not used) is applied directly to the actuator. For this control valve, the air is supplied above the diaphragm and pushes against spring pressure to close the valve. The valve is fully closed when the plug is fully seated against the seat ring. Notice that as air pressure is decreased, the spring pressure causes the diaphragm, stem, and plug to move upward, opening the valve. This means that a loss of pressure would cause the valve to open. Therefore this is known as a fail-open valve. Different configurations of air inlet, spring location, and valve seat arrangement result in different fail positions and determine whether the valve is direct- or reverse-acting. For example, the same valve pictured above, with the plug below the seat ring (reverse-seated), would open with increased air pressure and would fail closed on loss of air pressure. .

As can be seen from the discussion above, all components in the final control subsystem must be configured correctly for the system to work properly. The fail-safe positions must be correct for the application and the action must produce the desired results. These configurations must be properly documented and utilized during calibration, loop checks, or troubleshooting.

7.1 CALIBRATION OF AN I/P TRANSDUCER

FIGURE 7-6.
I/P Transducer Calibration



The figure above illustrates the setup for a bench calibration of an I/P transducer. The air supply connected to the input must be in accordance with manufacturer's specification (typically 20-100 psig). The pressure standard is connected to the air outlet, and a mA simulator is connected to the current input. It is important for the I/P transducer to be oriented the same way as the installed position in the field. A change in orientation will introduce error in most I/P transducers.

If the calibration is performed in the field, the existing supply air would be used. It is convenient to tee into the air outlet so the control valve position can be checked at the same time. Of course, you need to ensure the system is in a safe condition before you open and close the valve.

Once the setup is established, apply the mA inputs for each desired test point, such as 4.0, 8.0, 12.0, 16.0, and 20.0 mA. Record the corresponding outlet pressure at each test point. For a 4-20 mA input = 3-15 psig output I/P, the corresponding outputs would be 3.0, 6.0, 9.0, 12.0 and 15.0 psig. Some facilities adjust the 0% test point so a slightly higher mA input results in the 0% output. For example, 4.10 mA may result in a 3.0 psig output. This ensures that the valve is closed with a controller output of 4.0 mA.

Once the as-found readings are obtained, evaluate the results against the required specification. If required, perform zero and span adjustments

until no further adjustment is required. Then, repeat all test points to record as-left readings.

Many organizations do not require periodic calibration of I/P transducers, positioners, or control valves. The justification is that the control signal will adjust the output until the required setpoint is achieved based on the process measurement. This is true, but you want to make sure the output loop is performing correctly. The best way to do that is to check the calibration periodically.

7.2 CALIBRATION OF A VALVE POSITIONER

Calibration of the valve positioner can be performed at the same time as the I/P in a loop calibration. Simply tee in the pressure module at the I/P outlet as mentioned above in the I/P calibration. Record the valve position at each test point.

If the valve positioner is calibrated separately, connect an input test pressure regulator or hand pump and monitor the input pressure applied with a pressure standard. If supply air is not provided, connect the required supply air to the positioner. Apply the pressure for the desired test points and record valve position. For example, assume our valve positioner is 3-15 psig input = 0-100% valve position. In this case, apply 3.0, 6.0, 9.0, 12.0, and 15.0 psig. The expected valve positions should be 0, 25, 50, 75, and 100%, respectively. The valve position indicator on the stem is usually indicated in 5% or 10% increments. Therefore a best estimate of the valve position may be all you can obtain. In other cases, a valve position detector provides a remote indication, to a DCS, for example. In this case, ensure both indicators are working properly.

Many organizations do not require calibration of valve positioners for the same reason described above. To be honest, I've calibrated very few positioners myself. However, there is documentation that control valve positioner performance is responsible for significant loss in system efficiency and therefore increased costs. To provide guidance on methods for testing positioners and control valve performance, ISA has developed a standard, ANSI/ISA-75.25.01-2000, *Test Procedure for Control Valve Response Measurement for Step Inputs*. This standard has no application to valve positioner calibration.

7.3 CALIBRATION OF A CONTROL VALVE

As in positioner calibration, a pressure signal is applied to the actuator and the resulting valve position is recorded. This can be performed with the positioner calibration, if applicable. It can also be performed in conjunction with I/P calibration as described above. Just remember to ensure the system is in a safe condition if performing the calibration in the field. Also remember that you should know the correct action, direct or reverse, and fail position before starting.

REVIEW QUESTIONS

Use the information from this chapter and the references in Appendix A as necessary to answer the following questions:

1. From the list below, what is the likely calibration range for TY-300?

   A. 0-20 psig
   B. 3-25 psig = 4-20 mA
   C. 4-20 mA = 3-15 psig
   D. 20 SCFM

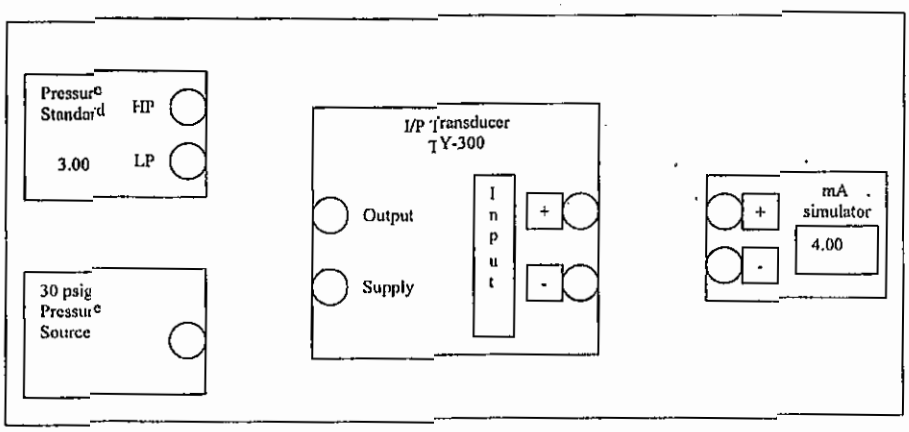
2. Refer to the P&ID in Appendix A-1. If the calibration ranges of FY-301 and PY-301 are the same as TY-300, what if any difference is there between the three I/P transducers?

3. Select the correct procedure to be used for this calibration of TY-300.

4. From your answer to question 1 and the procedure selected in question 3, what are the correct input values to simulate TY-300?

5. What are the expected output values for the inputs applied from question 3?

6. Which is the correct test equipment hookup for calibration of TY-300?



7. With the results indicated, what instrument in the loop requires adjustment?

% INPUT	OUTPUT (PSIG)	VALVE POSITION
0	3.2	2%
50	9.5	54%
100	15.8	100%

8. How would the calibration required in the previous step be performed?

9. If a control valve is being checked during an I/P calibration and the I/P is properly calibrated, what is the most likely cause of an improper valve position (no positioner is installed)?

10. What is the purpose of a valve positioner?

11. Is the action (direct or reverse) verified during a positioner calibration?

12. What is the basic procedure for calibration of a positioner?

13. What is the basic procedure for calibration of a control valve?

8

PROCESS ANALYTICAL INSTRUMENT CALIBRATION

After completing this chapter, you should be able to:

Calibrate the following analytical instrument types (to ISA standards, where applicable) and determine acceptability:

- *pH*
- *Conductivity*
- *Resistivity*

Select the proper calibration procedure and calibration data sheet.

Select appropriate certified test equipment.

Properly set up/connect test equipment to device under test (DUT) for calibration.

Properly maintain pH electrodes and prepare for field calibration.

Return equipment to service following calibration.

Complete and properly maintain calibration documentation.

We can't overlook analytical instruments used to monitor and, in some cases, control processes. There are many specialized analytical instruments used in various processes. These instruments are less common than temperature, pressure, level, flow, and final control instruments—but no less important. Analytical instrumentation is more commonly used in the research or quality control lab.

In the lab environment, these instruments are typically verified against a reference standard by the user. In many facilities, the calibration technician is not responsible for maintaining lab instruments. However, for those control system technicians responsible for lab instrumentation, the same principles discussed for process instruments will apply. We'll discuss pH, conductivity, and resistivity.

Many industrial processes—such as food processing, sewage treatment, water purification, and pharmaceutical production—are sensitive to pH. pH is the measure of the acidity or alkalinity of an aqueous solution. Maintaining the proper pH is essential for living systems. When acids are added to water, they dissociate, either completely or partially, and release hydrogen ions (H^+) into the solution. pH is equal to the negative logarithm of the hydrogen ion concentration, written in equation form as:

$$pH = -\log [H^+]$$

You'll have to study up on logarithms, chemistry, and molarities on your own. To give you an idea of H^+ concentrations and interpreting the associated pH value, the following table is provided. For example, the H^+ concentration of pure water is 1×10^{-7} mole/liter. The log of 1×10^{-7} is -7. Therefore the negative log is 7, and the pH of pure water is 7. A pH of 7 is considered neutral, neither acidic nor basic. Solutions with a pH value less than 7 are acidic; pH values above 7 are basic (sometimes referred to as alkaline). The nominal voltage output at each pH value is described later in the chapter.

HYDROGEN ION CONCENTRATION		pH	NOMINAL VOLTAGE OUTPUT @ 25°C (mV)
EXPONENTIAL	DECIMAL		
ACIDIC			
10^0	1.0	0	414.12
10^{-1}	0.1	1	354.96
10^{-2}	0.01	2	295.8
10^{-3}	0.001	3	236.64
10^{-4}	0.0001	4	177.48
10^{-5}	0.00001	5	118.32
10^{-6}	0.000001	6	59.16
NEUTRAL			
10^{-7}	0.0000001	7	0.0
BASIC			
10^{-8}	0.00000001	8	-59.16
10^{-9}	0.000000001	9	-118.32
10^{-10}	0.0000000001	10	-177.48
10^{-11}	0.00000000001	11	-236.64
10^{-12}	0.000000000001	12	-295.8
10^{-13}	0.0000000000001	13	-354.96
10^{-14}	0.00000000000001	14	-414.12

The instrumentation used to measure pH in a process system consists of a pH meter and electrode. The electrode is actually a combination electrode consisting of a measuring electrode and a reference electrode. The measuring electrode has a thin, fragile glass bulb at its tip that is sensitive to the H+ ions in the surrounding medium. An electrical potential develops between the inner and outer surfaces of the glass bulb. The magnitude of this potential varies, depending on the concentration of hydrogen ions in the solution. The reference electrode provides a stable, constant voltage.

The two electrodes are connected to the pH meter. The meter measures the voltage difference between the reference electrode and the measuring electrode, amplifies the signal, and converts the signal to a pH value.

The difference between the two millivolt signals (Δ mV) is ideally 0 mV at a pH of 7.0. At 25°C the nominal ΔmV signal increases by 59.16 mV for each pH unit measurement below 7.0 pH. For example, ΔmV = 177.48 = 4.0 pH. The ΔmV signal increases by –59.16 mV for each pH unit measurement above 7.0 pH. For example, ΔmV = –177.48 = 10.0 pH.

Since the pH electrode is not a perfect device and performance changes with age, the pH meter contains a calibration circuit. All pH electrodes require calibration from time to time. Many organizations require pH calibration prior to each use, or daily. A two point calibration characterizes an electrode with a specific pH meter. When calibrating, the calibration circuit will adjust the ΔmV output to read 0 mV while the electrode is immersed in a 7.0 pH buffer standardization solution. This is referred to as the "zero point" or "standardize" adjustment.

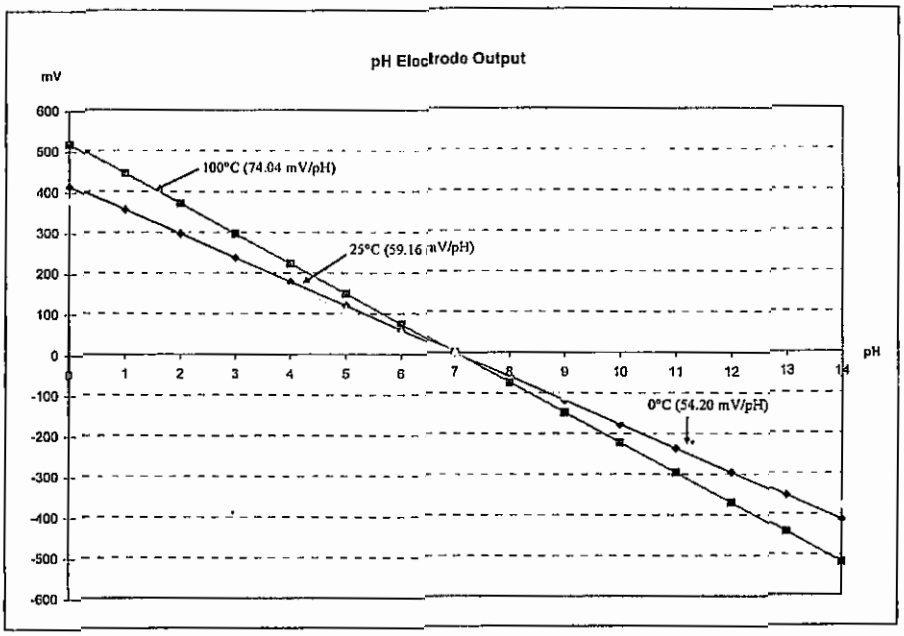
While the electrode is immersed in a 4.0 (or 10.0) pH buffer solution, the slope control adjusts the ΔmV output to read + 177 mV (or –177 mV). This is referred to as the "slope," "span," or "efficiency" adjustment. It is preferable to use a 4.0 pH buffer solution because pH buffer solutions above 7.0 pH are less stable and have a limited life. These high pH buffers will more readily absorb CO_2 from the atmosphere and typically change to a lower pH value when left open.

Temperature is very important when measuring pH. The measuring electrode's response to pH is affected by temperature. Also, the pH of the solution that is being measured may increase or decrease as temperature changes. Temperature compensation is accomplished in one of two ways. Most modern pH meters have an automatic temperature compensating (ATC) probe which measures the sample temperature and compensates the pH reading. This ATC probe may be a separate probe or it may be built into the electrode housing. If the pH meter does not have ATC, the

temperature of the solution is determined with a calibrated temperature indicator or thermometer and the temperature dial on the pH meter is adjusted to this temperature.

The following chart illustrates the linear ideal pH electrode output at 25°C and the effect of temperature on pH electrode output at 0°C and 100°C.

FIGURE 8-1.



pH electrodes must be properly stored when not in use. If not properly stored, the electrode will dry out and this will impair performance or possibly destroy the electrode. The probe should be stored in a 4.0 pH buffer solution or 3.8M KCL solution. There are special storage bottles for long-term secure storage to prevent damage and dehydration. Do not store in deionized water. Prior to returning a pH meter to service after storage, the probe should be thoroughly cleaned with deionized water and calibrated.

8.1 pH CALIBRATION PROCEDURE

Consult the manufacturer's technical manual for the required calibration procedure, especially for microprocessor-based pH meters with auto-calibration features.

1. Rinse the electrode thoroughly with deionized water and gently blot the electrode on a soft tissue.

2. Insert the electrode (and any separate temperature compensation probe) in a 7.00 pH buffer solution. Allow 30 seconds to reach equilibrium. Adjust the pH meter with the standardize or zero control for a pH indication of 7.0.

   Note: If the meter does not have automatic temperature compensation, then place a calibrated temperature indicator in the same buffer solution with the pH electrode. After 30 seconds, adjust the temperature dial on the pH meter to correspond with the temperature indicated. Then adjust the pH meter indication with the standardize/zero control for a pH indication of 7.0.

3. Remove the pH electrode from the buffer solution and repeat step 1.

4. Repeat step 2, except use a 4.01 pH buffer solution and adjust the slope/span control for a pH reading of 4.01.

   Note: As mentioned above, a 10.0 pH buffer solution may be substituted for the 4.01 buffer solution in step 4. However, pH buffer solutions above 7.0 are less stable. Therefore it is preferable to use the 4.01 buffer solution.

5. Repeat steps 1-to-4 to maximize the precision of the calibration.

6. If desired, check the pH meter indication using a 10.0 pH (or 4.01 pH) buffer solution. Typically, there is not an adjustment for this third buffer solution.

   Note: Newer, microprocessor-based pH meters allow the use of any two standard pH buffers that bracket the pH of the samples. Some also allow more than two buffers to determine the slope. Refer to the manufacturer's technical manual for procedures specific to the pH instrument.

8.2 DIAGNOSTIC TEST FOR pH ELECTRODES

The asymmetry potential (AP) and slope (efficiency) can be used as a guideline to judge electrode performance. To perform this test, the pH meter must have a mV readout.

1. Set the pH meter to readout in mV.
2. Connect a shorting plug to the input on the pH meter and adjust the standardize/zero control on the pH meter for a reading of 0.0 mV.
3. Disconnect the shorting plug and connect the pH electrode that will be tested.
4. Rinse the electrode thoroughly with deionized water and gently blot the electrode on a soft tissue
5. Insert the electrode (and any separate temperature compensation probe) in a 7.00 pH buffer solution. Allow 30 seconds to reach equilibrium. Record the mV reading. This is the asymmetry potential (AP).

   Note: If the meter does not have automatic temperature compensation, then place a calibrated temperature indicator in the same buffer solution with the pH electrode. After 30 seconds, adjust the temperature dial on the pH meter to correspond with the temperature indicated. Then record the mV reading to determine the AP.
6. Remove the pH electrode from the buffer solution and repeat step 4.
7. Repeat step 5, except use a 4.01 pH buffer solution and record the mV reading.
8. Determine the mathematical difference between the two mV readings (from steps 5 and 7). This is the electrodes span.
9. Divide the electrode's span by the theoretical span of 176.9 mV and multiply the result by 100.

Example: Reading in a 7.00 pH buffer solution: –5.8 mV
Reading in 4.01 pH buffer solution: +167.5 mV
Asymmetry Potential = –5.8 mV
Span = 167.5 – (–5.8) = 173.3 mV
Slope = (173.3 mV / 176.9 mV) X 100% = 98%

Typically an electrode is replaced when the AP is greater than ± 40mV and/or the slope is < 91%.

8.3 MEASURING CONDUCTIVITY

There are situations where it is useful to measure the combined concentration of all ions in an aqueous solution. The total ionic concentration is determined by measuring the conductivity of the solution. Conductivity is defined as the ability of a material to conduct electrical current. We are used to current flowing through copper wires. Electric current can also flow through ionic solutions. The higher the concentration of ions in a solution, the more conductive the solution.

A simple device to measure conductivity consists of two flat electrodes placed in a sample solution, connected to a battery and an ammeter. The battery generates electric current that can flow only if the sample solution is conductive and completes the current path. Positive ions in the solution are attracted to the negative electrode, and negative ions are attracted to the positive electrode. At the positive electrode, negative ions give up electrons that flow to the positive battery terminal. At the negative electrode, positive ions take electrons. This results in continuing electron flow through the circuit and continuing movement of ions through the solution. The more ions present, the more current detected by the ammeter.

The amount of current flow that is measured by the conductivity instrument will vary depending on the distance between the electrodes (d) and the area of the electrodes (A). For the same ionic concentration, electrodes spaced farther apart or a smaller electrode area would result in less current measured. The relationship of this distance to the surface area is known as the cell constant (K), and is defined as:

$$K = d/A$$

A device with a lower K value is more sensitive than a device with a higher K value. Conductivity probes with different cell constants are necessary for the wide range of applications that require conductivity measurement. A purified water system has fewer ions and therefore requires a more sensitive conductivity probe with a K value of about 0.1. A cell constant of up to 10 would be used for solutions with high conductivities.

Some applications utilize resistivity instruments. Resistivity instruments are the same as conductivity instruments, except resistivity is

simply the inverse of conductivity. Resistivity = 1/conductivity. The units of conductivity are Siemens or mho. In practice, a Siemen is too large, so instruments typically display in microsiemens/micromhos (μS/μmho) or milliSiemens/millimhos (mS/mmho). Resistivity is measured in ohms with typical values in the megohm range. 1 megohm = 1 million ohms. The conductivities and resistivities of common solutions are listed below.

SOLUTION	CONDUCTIVITY	RESISTIVITY
Ultrapure Water	0.055 μS/cm	18.2 MΩ
Distilled Water	1 μS/cm	1 MΩ
Deionized Water	80 μS/cm	12,500 ohm-cm
0.05% NaCl	1000 μS/cm	1000 ohm-cm
Seawater	50,000 μS/cm	20 ohm-cm
30% H_2SO_4	1,000,000 μS/cm	1 ohm-cm

Like pH, temperature has an effect on conductivity. Conductivity probes include a temperature sensor which measures the temperature of the solution and automatically compensates the conductivity reading.

Standard solutions for conductivity calibrations are commercially available. Conductivity instruments can be calibrated by disconnecting the conductivity probe and using a decade box to simulate resistance of the solution. After the instrument calibration is completed and the probe is reconnected, the conductivity system is checked using a standard conductivity solution, if available, or compared to another calibrated conductivity instrument in the same solution. To perform a calibration of a conductivity instrument, you'll need to know the range of the instrument, the cell constant, terminal wiring for the conductivity cell/temperature compensation, and the temperature compensation resistance at 25°C. Also, if the conductivity instrument provides an output signal, the output signal and range must be known. The following example calibration procedure is specific to a Leeds & Northrup 7082 conductivity meter and assumes the following:

- Calibration range: 0 - 10 μS/cm = 4 – 20 mA output.
- Cell Constant = 1 (the actual cell constant for this range would likely be 0.1, but we will assume 1 for ease of explanation).
- Temperature compensation thermistor wired to terminals B and D, conductivity probe wired to terminals A and C.
- Temperature compensation resistance at 25°C is 8550 ohms.

8.4 CALIBRATION PROCEDURE WITH EXPLANATIONS

1. Verify system is shutdown and/or conductivity cell is isolated from the system pressure. Carefully remove the conductivity probe from the system ensuring system pressure is zero.
2. Rinse the probe thoroughly using deionized water.
3. Place the probe in a standard conductivity solution within the range of the instrument or a sample with a calibrated analytical conductivity meter. Record the "as-found" conductivity reading. (This is to record the "as found" condition of the conductivity system, probe and meter, prior to any adjustments. It is difficult to find standard conductivity solutions in this very low conductivity range. Therefore, it may be necessary to use purified water and compare the indication of the instrument under test to another calibrated conductivity instrument.)
4. Disconnect the thermistor from terminals B and D and connect an 8550-ohm resistor or decade box set to 8550 ohms. (This is done to simulate a sample resistance of 25°C during calibration. If the actual temperature compensation sensor was used, the conductivity readings obtained would be uncompensated readings at an unknown temperature.)
5. Disconnect the conductivity probe leads from terminals A and C. Connect a decade box or other calibrated resistance simulator, capable of 100 Kohms to 10 Mohms. (We will use resistance to simulate the conductivity probe. Remember that conductivity = 1/Resistivity.)
6. Connect a milliammeter in series with the 4-20 mA output signal.
7. Set the decade box to 10 Mohms to simulate 0.1 µS/cm. Record the conductivity reading and the mA output. (10 Mohms is used to simulate 0.1 µS/cm because µ (micro) = 10^{-6} and $1 \div 0.1 \times 10^{-6} =$ 10,000,000. Since 1 million is mega (M), 10 Mohms of resistance will simulate 0.1 uS/cm conductivity. Confused? The expected mA output would be 4.16mA. 0.1 of 10 is $1/100^{th}$ and $1/100^{th}$ of 16mA is 0.16 mA. Add this to the zero value of 4 mA and the expected output is 4.16 mA.)
8. Set the decade box to 1 Mohm to simulate 1.0 µS/cm. Record the conductivity reading and mA output. ($1 \div 1.0 \times 10^{-6} = 1{,}000{,}000$. The

expected mA output would be 5.60 mA since 1 µS/cm of 10 µS/cm is $1/10^{th}$ of 4-20 mA.)

9. Set the decade box to 100 Kohm to simulate 10 µS/cm. Record the conductivity reading and mA output. ($1 \div 10.0 \times 10^{-6} = 100{,}000$. The expected mA output would be 20.0 mA.)

10. If adjustment is required, calibrate the indication and/or mA output in accordance with the manufacturer's instruction manual and repeat steps 7-to-9 to obtain "as-left" readings.

11. Disconnect the test equipment and restore the instrument connections.

12. Repeat step 3 and record the "as-left" conductivity reading. (This is to record the "as left" condition of the conductivity system, probe and meter, following any adjustments and to verify instrument is connected properly.)

How would the simulated conductivity readings change if the cell constant (K) is 0.1? A cell constant of 0.1 is 10 times more sensitive than a cell constant of 1. (This means the distance between the electrodes is reduced by a factor of 10, the surface area of the electrodes is increased by a factor of 10, or some combination of both.) The best way to think about this is that if we used the same simulated values as above with a K of 0.1, the conductivity readings would be increased by a factor of 10. Therefore, to perform the calibration properly, we must decrease the simulated resistances by a factor of 10. So, the simulated resistances for the same conductivity instrument with a cell constant of 0.1 would be 1 Mohm, 100 Kohm, and 10 Kohm for conductivities of 0.1 µS/cm, 1 µS/cm, and 10 µS/cm, respectively.

Let's do one more problem. What resistance would we have used to simulate 5 µS/cm with a K =1? Since 5 µS/cm = 5×10^{-6} S/cm and $1 \div 5 \times 10^{-6}$, we would use a resistance of 200,000 ohms or 200 Kohms. That makes sense since 5 µS/cm is ½ of 10 µS/cm and we are using twice the resistance as we used to simulate 10 µS/cm. Got it!

This is all simplified if you are calibrating a resistivity instrument. The procedure is the same, except you do not need to do the conversions from conductivity to resistivity. You do need to account for the cell constant, however.

REVIEW QUESTIONS

Use the reference section, as necessary, to answer the following questions:

1. What are the ideal ΔmV input values for a pH meter at 4, 7, and 10 pH at 25°C?

2. What buffer solution is used to standardize a pH meter?

3. What buffer is preferred for calibration of the slope? Why?

4. What is the purpose of a pH diagnostic test?

5. What are the basic steps of a pH diagnostic test?

6. What is the diagnostic test acceptance criterion that determines whether the electrode should be replaced?

7. What test equipment is used to simulate a conductivity signal to a conductivity meter?

8. How is the resistance value input for a conductivity instrument calibration determined?

9. How is the 25°C temperature simulated during a conductivity or resistivity instrument calibration?

10. Why is it important to know the cell constant for a conductivity calibration?

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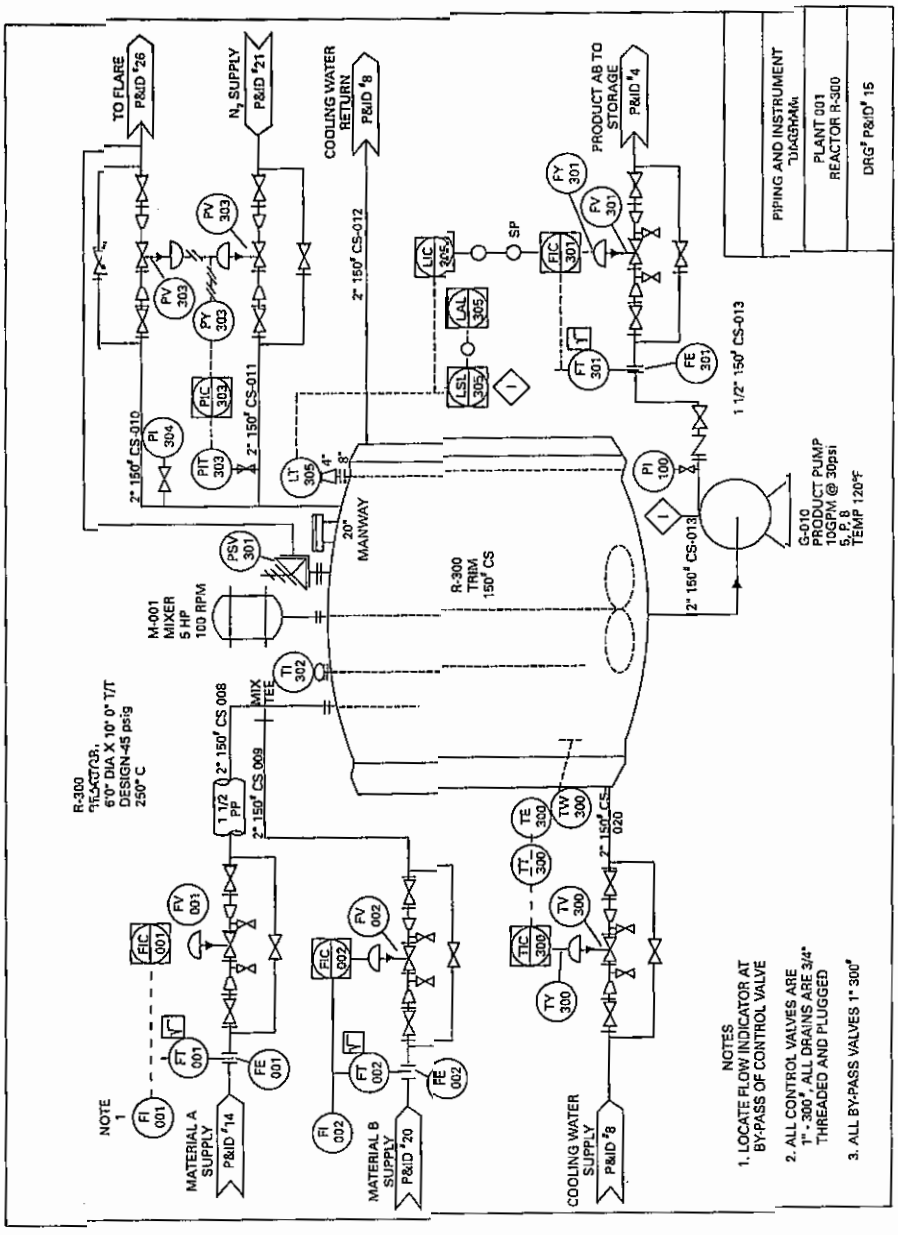
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Appendix A-1

PIPING & INSTRUMENT DIAGRAMS (P&IDs)

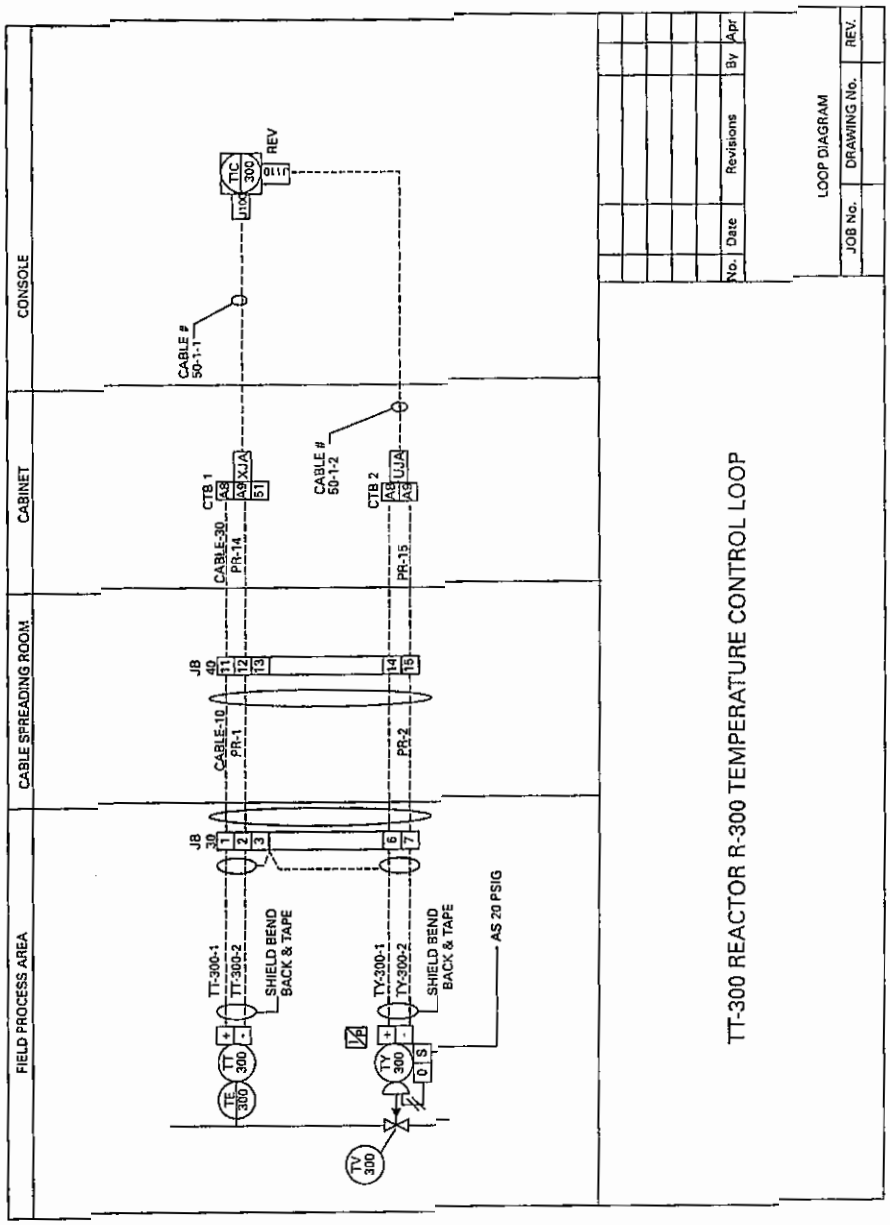
P&ID for the Reactor R-300 System



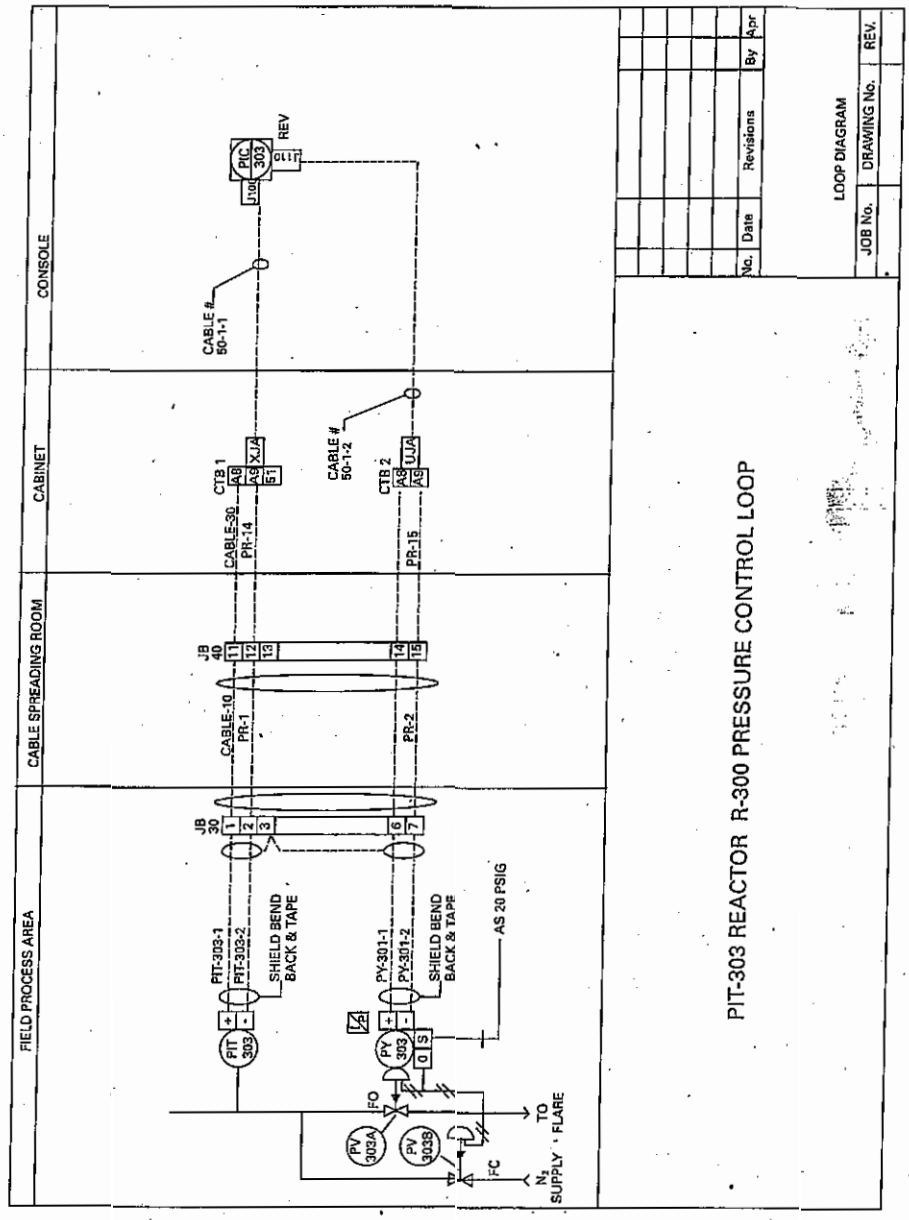
Appendix A-2

LOOP DIAGRAMS

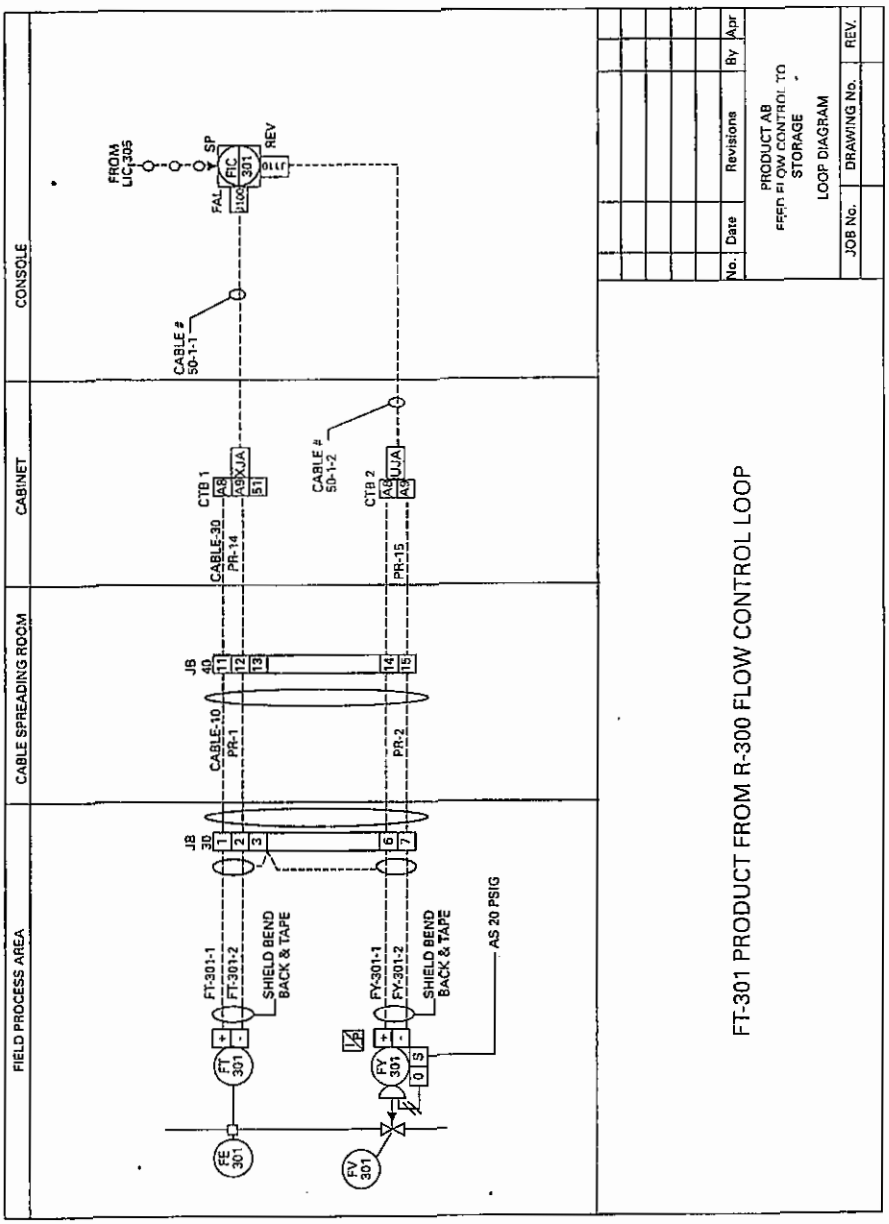
TT-300 Reactor R-300 Temperature Control Loop



PIT-303 Reactor R-300 Pressure Control Loop



FT-301 Product from R-300 Flow Control Loop



Appendix A-3

INSTRUMENT SPECIFICATIONS

The following Instrument Specification Forms are examples only using ISA-TR20. Use the information provided in these specification forms to answer the chapter review questions.

The ABC Company Instrument Specification Data Form at the end of this section is provided as an example for reference only. This form, modified for your use, can be used to document calibration information for each instrument and obtain approval from the user, calibration supervisor, and quality department.

RESPONSIBLE ORGANIZATION	CORIOLIS MASS FLOWMETER w/wo TOTALIZER INDICATOR Device Specification	SPECIFICATION IDENTIFICATIONS
ISA		Document no
		Latest revision / Date
		Issue status

FLOWMETER BODY AND HOUSING	
Body/Housing type	in-line
End conn nominal size	1 1/2 in / Rating
End conn termin type	clamped / Style tri-clamp
Body wetted material	316L SST
Housing material	304 SST
End termination material	316L SST

FLOWTUBE ASSEMBLY	
Flowtube type	parallel curved tubes
Flowtube diameter	1/2 inch
Flowtube material	316L SST

CONNECTION HEAD	
Housing type	manufacturer standard
Enclosure type no/class	NA
Signal termination type	junction box/terminal block
Cert/Approval type	NA
Housing material	epoxy coated aluminum

TRANSMITTER OR COMPUTER	
Housing type	NEMA-type enclosure
Measurement compensation	density
Output signal type	analog current & frequency
Enclosure type no/class	
Span-zero adjustment	hand held interface
Characteristic curve	linear mass flow
Digital communication std	HART protocol
Signal power source	line 115 VAC 60 Hz
Failsafe style	failsafe selectable
Integral indicator style	NA
Signal termination type	terminal block
Cert/Approval type	CSA Hazardous
Mounting location/type	integral with flowmeter
Failure/Diagnostic action	activate alarm relay
Enclosure material	epoxy coated aluminum

TOTALIZER INDICATOR		
Totalizer type		
Enclosure type no/class		
Signal power source		
Contacts arrangement		Quantity
Totalizer reset style		
Integral indicator style		
Cert/Approval type		
Mounting location/type		
Enclosure material		

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	
Flow rate accuracy rating	0.15 % rate		
Density accuracy rating	.005 g/cm3		
Min flow URL	50 kg/min	Max	300 kg/min
Density URL	.01 g/cm3	URL	2 g/cm3
Sns enclosure press rating			
Pressure drop at flow URL	5 psig		
Min ambient working temp		Max	
Contacts ac rating		At max	
Contacts dc rating		At max	
Max sensor to receiver lg			

ACCESSORIES	
Connecting cables length	
Heating kit style	
Calibrator/configurator	

SPECIAL REQUIREMENTS	
Custom tag	ssl permanent attached
Reference specification	
Special preparation	
Compliance standard	
Calibration report	6 pt flow calibration
Software configuration	multivariable configuration
display language	English

PHYSICAL DATA	
Estimated weight	
Face-to-face dimension	
Overall height	
Removal clearance	
Signal conn nominal size	Style
Mfr reference dwg	

CALIBRATIONS AND TEST		INPUT OR TEST			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
FT-301	Mass flow-Analog output	0 kg/min	200 kg/min	direct	4 ma	20 ma
	Mass-Analog output 2					
	Mass-Analog output 3					
	Mass-Freq output 1	0 kg/min	200 kg/min	direct	0 Hz	8000 Hz
	Mass-Freq output 2					
	Measurement-Scale					
	Temp-Digital output					
	Mass-Digital output					
	Density-Digital output					
	Test pressure					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

Form: 20F2*21 Rev 0

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RESPONSIBLE ORGANIZATION	DIFFERENTIAL PRESSURE LEVEL TRANSMITTER - FLANGE MOUNTED Device Specification	SPECIFICATION IDENTIFICATIONS	
ISA		Document no	
		Latest revision	Date
		Issue status	

TRANSMITTER BODY				REMOTE SEAL			
Body/Flange type				Seal type			
Process conn nominal size	1/2 inch	Rating	3000 psi	Process conn nominal size		Rating	
Process conn termn type	fing adapter	Style	NPT (F)	Process conn termn type		Style	
Vent/Drain location	side			Diaphragm extension lg			
Mounting type	body flange			Instrument conn nom size			
Body/Flange material	carbon steel zinc plated			Flushing conn quantity			
Vent/Drain material	316 SST			Capillary/Fitting dia		Length	
Bolting material	carbon steel zinc plated			Diaphragm material			
Flange adapter material	stainless steel			Capillary-armor material			
Gasket/O ring material	teflon			Bolting material			
Mounting kit material	stainless steel			Upper housing material			
				Lower housing/Flange matl			
				Gasket/Oring material			
SENSING ELEMENT				Fill fluid material			
Detector type	piezoresistive sensor						
Min diff pressure span	100 inH2O	Max	400 inH2O	PERFORMANCE CHARACTERISTICS			
Diaphragm/Wetted material	316 SST			Max press at design temp	50 psig	At	250C
Fill fluid material	NA			Min working temp	-40C	Max	85C
				Accuracy rating	0.10% span		
				Diff pressure LRL	0 inH2O	URL	250 inH2O
				Fill fluid sp gr	1.02	At temp	25C
TRANSMITTER				Ambient temperature error			
Output signal type	analog current			Min ambient working temp		Max	
Enclosure type no/class							
Characteristic curve	linear with measurement			ACCESSORIES			
Digital communication std	Hart Protocol			Air set filter style			
Signal power source	loop			Air set gauges			
Transient protection	internal termn block			Heating kit style			
Integral indicator style	analog meter, uniform div			Remote indicator style			
Signal termination type	NA						
Cert/Approval type	NA			SPECIAL REQUIREMENTS			
Span-Zero adjust loc	local			Custom tag	sst permanent attached		
Failure/Diagnostic action	drive output high			Reference specification			
Enclosure material	NA			Special preparation			
				Compliance standard			
				Software configuration			
INTEGRAL SEAL							
Seal type				PHYSICAL DATA			
Process conn nominal size		Rating		Estimated weight			
Process conn termn type		Style		Overall height			
Diaphragm extension lg				Removal clearance			
Diaphragm material				Signal conn nominal size		Style	
Upper housing/Flange matl				Mfr reference dwg			

CALIBRATIONS AND TEST		INPUT OR TEST			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
LIT-202	Diff press-Analog output	0 inH2O	120 inH2O	direct	4 ma	20 ma
	Diff press-Scale					
	Diff press-Digital output					
	Press-Digital output					
	Temp-Digital output					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

Form: 20L2111 Rev 0

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RESPONSIBLE ORGANIZATION	CAPACITANCE OR RF ADMITTANCE LEVEL TRANSMITTER w/wo SWITCHES Device Specification	SPECIFICATION IDENTIFICATIONS
ISA		Document no.
		Latest revision — Date
		Issue status

PROCESS CONNECTION		
Body/Fitting type	sanitary gland	
Process conn nominal size	4 in	Rating
Process conn femma type	clamped	Style tri-clamp
Wetted material	316 SST	
Flange material	carbon steel	
Seal/O ring material	viton	

SENSING ELEMENT	
Configuration type	fully inserted rod
Probe diameter	.37 in
Insertion length	9 ft
Inactive length	0 ft
Integral cable length	25 ft
Probe material	316 SST
Insulation/Sheath matl	polypropylene
Weight material	
ground reference assembly	ground strap

CONNECTION HEAD OR PREAMPLIFIER	
Type	preamplifier
Measurement compensation	varying dielectric
Input signal URL	4000 pf
Output signal type	250 kHz
Enclosure type no/class	NA
Power source	line 115 VAC 60 Hz
Cnrl/Approval type	NA
Mounting location/type	integral with element
Enclosure material	cooled aluminum

TRANSMITTER w/wo SWITCHES	
Housing type	wo enclosure
Measurement compensation	varying dielectric
Output signal type	electric analog & relay
Input signal URL	4000 pf
Enclosure type no/class	4
Characteristic curve	linear with measurement
Signal power source	line 115 VAC 60 Hz
Contact arrangement	DPDT Quantity 2
Failsafe style	selectable low
RFI protection	RFI filter
Integral indicator style	3 1/2 digit LCD / LED
Cnrl/Approval type	NA
Mounting location/type	wall mount
Static discharge prot	
Failure/Diagnostic action	drive input low
Adjustable signal damping	0 - 30 s

TRANSMITTER w/wo SWITCHES Continued	
Switch time delay	off delay
Enclosure material	fiberglass reinforced plastic

PERFORMANCE CHARACTERISTICS			
Max press at design temp	50 psig	At	250C
Min working temperature	-40C	Max	185C
Max discharge to sensor	2A		
Accuracy rating	1% URV		
Min measurement span	30 pf	Max	4000 pf
Min ambient working temp		Max	
Contacts ac rating	6A	At max	250 VAC
Contacts dc rating	1A	At max	30 VDC
Max sensor to receiver ln			

ACCESSORIES	
Connecting cables length	25 ft
Power supply	120 V(ac) / 24 V(dc)
Local indicator style	digital display
Intrinsic safety barrier	NA

SPECIAL REQUIREMENTS	
Custom tag	sst permanent attached
Reference specification	
Compliance standard	
Calibration report	5 point

PHYSICAL DATA		
Estimated weight		
Overall height		
Removal clearance		
Signal conn nominal size		Style
Mfr reference dwg		

CALIBRATIONS AND TEST		INPUT OR SETPOINT			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
LT-305	Level-Analog output	0 in	120 in	direct	4 ma	20 ma
	Level-Scale					
LSL-305 set / reset	Level setpoint 1-Output	20 in	25 in	NC	24 V	0 v
LSH-305 set / reset	Level setpoint 2-Output	100 in	95 in	NC	24 V	0 V
	Level setpoint 3-Output					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

Form: 20L2 101 Rev 0

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RESPONSIBLE ORGANIZATION	DIFFERENTIAL PRESSURE SWITCH w/wo TRANSMITTER Device Specification	SPECIFICATION IDENTIFICATIONS
ISA		Document no
		Latest revision / Date
		Issue status

SWITCH BODY				
Body type	manufacturer standard			
Process conn nominal size	1/4 inch	Rating		
Process conn termn type	threaded	Style	NPT (M)	
Body material	monel			
Seal/O ring material	viton			

SENSING ELEMENT			
Sensor element type	pressure transducer		
Adjustable LRL	0 inH2O	URL	20 inH2O
Diaphragm/Wetted material	316 SST		

SWITCH MECHANISM w/wo TRANSMITTER			
Housing type			
Element Style			
Output signal type	relay & analog		
Enclosure type no/class	4X		
Reset style	manual increasing		
Set point Adjustment type	enclosure internal		
Signal power source			
Measurement type			
Contacts arrangement	SPDT	Quantity	
Failsafe style			
Integral indicator style	LCD display		
Signal termination type	screw terminals		
Cert/Approval type			
Mounting type			
Dead band type	narrow adjustable deadband		
Enclosure material	aluminum		
Exterior coating material			

PERFORMANCE CHARACTERISTICS			
Max press at design temp			A
Min working temperature			Max
Output accuracy rating	1% span		
Repeatability	0.5% span		
Diff press Lower Range-Limit	0 inH2O	URL	20 inH2O
Max overrange limit			
Dead band rating			
Min ambient working temp			Max
Contacts ac rating			At max
Contacts dc rating	1A		At max 30V

ACCESSORIES	
Sealed leads adapter	
Breather/Drain style	

SPECIAL REQUIREMENTS			
Custom tag	sst permanent		
Reference specification			
Special preparation			
Compliance standard			

PHYSICAL DATA			
Estimated weight			
Overall height			
Removal clearance			
Signal conn nominal size		Style	
Mfr reference dwg			

CALIBRATIONS AND TEST		INPUT OR SETPOINT			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
PDIS-405 setpoint / reset	DP setpoint 1-Output	5.0 inH2O	5.5 inH2O	NC	0V	24V
	DP setpoint 2-Output					
	Diff press-Analog output	0 inH2O	20 inH2O	direct	4 ma	20 ma
	Diff pressure-Scale					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

Form: 20P2501 Rev 0

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RESPONSIBLE ORGANIZATION

PRESSURE GAUGE WITH DIAPHRAGM SEAL
Device Specification

SPECIFICATION IDENTIFICATIONS

Document no	
Latest revision	Date
Issue status	

OPERATING PARAMETERS

Project number	Sub project no
Project	
Entity no/no	
Site	
Area	Cell / Unit
Related equipment	
Service	
P&ID/Reference dwg number	
Material name	
Minimum pressure	Full vacuum (FV)
Normal pressure	10 psig
Maximum pressure	45 psig
Normal temperature	37°C
Maximum temperature	130 °C
Material phase	liquid

PROCESS CONNECTION AND CASE

Case type	open front w/ relief
Case style	compound
Gauge size	3 1/2 inch
Process conn nominal size	1 1/2 inch / Style tri-clamp
Process conn location	bottom
Case pressure relief type	pressure relief plug
Ring style	snap
Mounting type	
Case material	304 SST
Ring material	304 SST
Exterior treatment-color	polished stainless steel
Window material	heat treated glass
Stem material	316 SST
Seal/O ring material	buna N
Liquid fill material	NA

PRESSURE ELEMENT AND MOVEMENT

Elastic element type	diaphragm
Movement style	segment shaft & hairspring
Nominal accuracy grade	1A (1.0% span)
Joint type	
Element material	316 SST
Movement material	stainless steel

DIAL AND POINTER

Dial scale type	single scale
Pointer adjustment	adjustable pointer

DIAL AND POINTER Continued

Graduations and color	black marks on white bkgd
Scale range type	compound
Dial material	aluminum

DIAPHRAGM SEAL

Seal type	
Process conn nom size	2 inch / Rating
Process conn termn type	clamped / Style tri-clamp
Diaphragm extension lg	0 inch
Flushing conn quantity	
Capillary diameter	Length
Instrument conn nom size	
Diaphragm/wall material	
Capillary-armor material	
Bolting material	
Upper housing material	
Lower housing material	
Gasket/O ring material	
Fill fluid material	

PERFORMANCE CHARACTERISTICS

Max process design temp	Set
Min working temperature	Max
Ambient temperature error	
Min ambient working temp	Min

ACCESSORIES

Locking device material	NA

SPECIAL REQUIREMENTS

Custom tag	sst permanent attached
Reference specification	
Special preparation	
Compliance standard	
Service design	sanitary gauge

PHYSICAL DATA

Estimated weight	
Maximum thickness	
Max case outside dia	
Mfr reference dwg	

CALIBRATIONS AND TEST

TAG NO/FUNCTIONAL IDENT	PRESSURE OR SCALE	SCALE LRV	SCALE URV
PI-304	Pressure/Scale 1	-15 psig (FV)	50 psig
	Pressure/Scale 2		

COMPONENT IDENTIFICATIONS

COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

Form: 20P2011 Rev 0

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RESPONSIBLE ORGANIZATION	PRESSURE TRANSMITTER	SPECIFICATION IDENTIFICATIONS	
ISA	Device Specification	Document no	
		Latest revision	Date
		Issue status	

TRANSMITTER BODY				PERFORMANCE CHARACTERISTICS			
Body/Flange type	single head			Max press at design temp	50 °C	At	250 °C
Process conn nominal size	1/2 inch	Rating		Min working temperature		Max	
Process conn termn type	threaded	Style	NPT (F)	Accuracy rating	0.1 %span		
Vent/Drain location	side			Pressure LRL		URL	
Mounting type	2 inch pipe flat bracket			Min ambient working temp		Max	
Body/Flange material	316 SST						
Vent/Drain material	316 SST						
Bolting material	carbon steel plated						
Flange adapter material	carbon steel Ni plated						
Gasket/O ring material	viton						
Mounting kit material	carbon steel plated						
SENSING ELEMENT							
Detector type	piezoresistive						
Min pressure span	50 psig	Max	100 psig				
Diaphragm/Wetted material	316L SST						
Fill fluid material	NA						
TRANSMITTER				**ACCESSORIES**			
Output signal type	analog current			Air set filter style			
Enclosure type no/class	4X			Air set gauges			
Characteristic curve	linear			Heating kit style			
Digital communication std	NA			Remote indicator style			
Signal power source	loop			Manifold valve style			
Transient protection	NA						
Integral indicator style	LCD meter 4 digit			**SPECIAL REQUIREMENTS**			
Signal termination type				Custom tag	sst permanent attached		
Cert/Approval type				Reference specification			
Span-Zero adjust lct	local span & zero			Special preparation			
Failure/Diagnostic action	drive ouput high			Compliance standard			
Enclosure material	NA			Software configuration			
				PHYSICAL DATA			
				Estimated weight			
				Overall height			
				Removal clearance			
				Signal conn nominal size		Style	
				Mtr reference dwg			

CALIBRATIONS AND TEST		INPUT			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
PIT-303	Pressure-Analog output	0 psig	50 psig	direct	4 mA	20 mA
	Pressure-Scale					
	Pressure-Digital output					
	Temp-Digital output					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

Form: 20P2201 Rev 0

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1	RESPONSIBLE ORGANIZATION		RTD ASSEMBLY		6	SPECIFICATION IDENTIFICATIONS			
2	ISA		w/wo THERMOWELL		7	Document no			
3			Device Specification		8	Latest revision	0	Date	3-22-03
4					9	Issue status			
5					10				
11	PROTECTIVE SHEATH AND FITTING				60	THERMOWELL OR PROTECTING TUBE			
12	Housing type				61	Construction type	bar stock thermowell		
13	Pad/Collar type				62	Shank style	straight		
14	Fitting conn nominal size		Style		63	Process conn nominal size		Rating	
15	Mounting fitting type				64	Process conn term type		Style	
16	Sheath alignment	straight			65	Internal conn nom size		Style	
17	Sheath outside diameter	0.250 in.	Length	21 in.	66	Bore diameter	0.260 in.		
18	Spring loading	yes			67	Outside dia at support			
19	Sheath material	316L SS			68	Outside dia at tip			
20	Fitting material	316L SS			69	Insertion length "U"	10.5 in		
21					70	Lagging extension lg "T"	0		
22	SENSING ELEMENT				71	Thermowell/Tube material	316L SS		
23	Sensor type				72	Sheath material-thickness			
24	Sensor quantity	2			73				
25	Temperature coefficient	.00385 ohm/ohm/C			74				
26	Tolerance class	IES cl A (0,15 + 0,002t)			75	PERFORMANCE CHARACTERISTICS			
27	Nominal resistance	100 ohm	At temp	0 °C	76	Max press at design temp		At	
28	Configuration-no of wires	4 wire compensating loop			77	Min working temperature		Max	
29	Sensor material	platinum			78	Max fluid velocity limit		At temp	
30	Insulator material	ceramic fiber			79	Temp Lower Range Limit		URL	
31					80	Min insertion length			
32	LEAD WIRE AND HEAD EXTENSION				81	Min ambient working temp		Max	
33	Extension type	NA			82				
34	Ext wire nom size-type				83				
35	Extension/Lead length				84				
36	Nipple-union nom size		Rating		85	ACCESSORIES			
37	Nipple pipe schedule no				86	Moisture seal style			
38	Transition type				87	Bayonet adaptor size			
39	Termination style				88				
40	Connecting wire length				89				
41	Shield - ground wire				90				
42	Nipple material				91				
43	Union/Coupling material				92	SPECIAL REQUIREMENTS			
44	Conduit-armor material				93	Custom tag	sst permanent attached		
45	Extension wire material				94	Reference specification			
46	Ext wire insulation matl				95	Compliance standard			
47					96	Construction code			
48					97	Calculation report	NA		
49	CONNECTION HEAD				98	Calibration report	YES		
50	Housing type	industrial			99				
51	Cover style	screwed & chained			100				
52	Element conn nominal size		Style		101				
53	Signal conn nominal size		Style		102	PHYSICAL DATA			
54	Enclosure type no/class	4			103	Estimated weight			
55	Grounding terminal tcl	grounding screw			104	Overall length			
56	Enclosure material	aluminum cast			105	Removal clearance			
57	Terminal block material	ceramic			106	Mfr reference dwg			
58	Terminal material	nickel plated wires			107				
59					108				

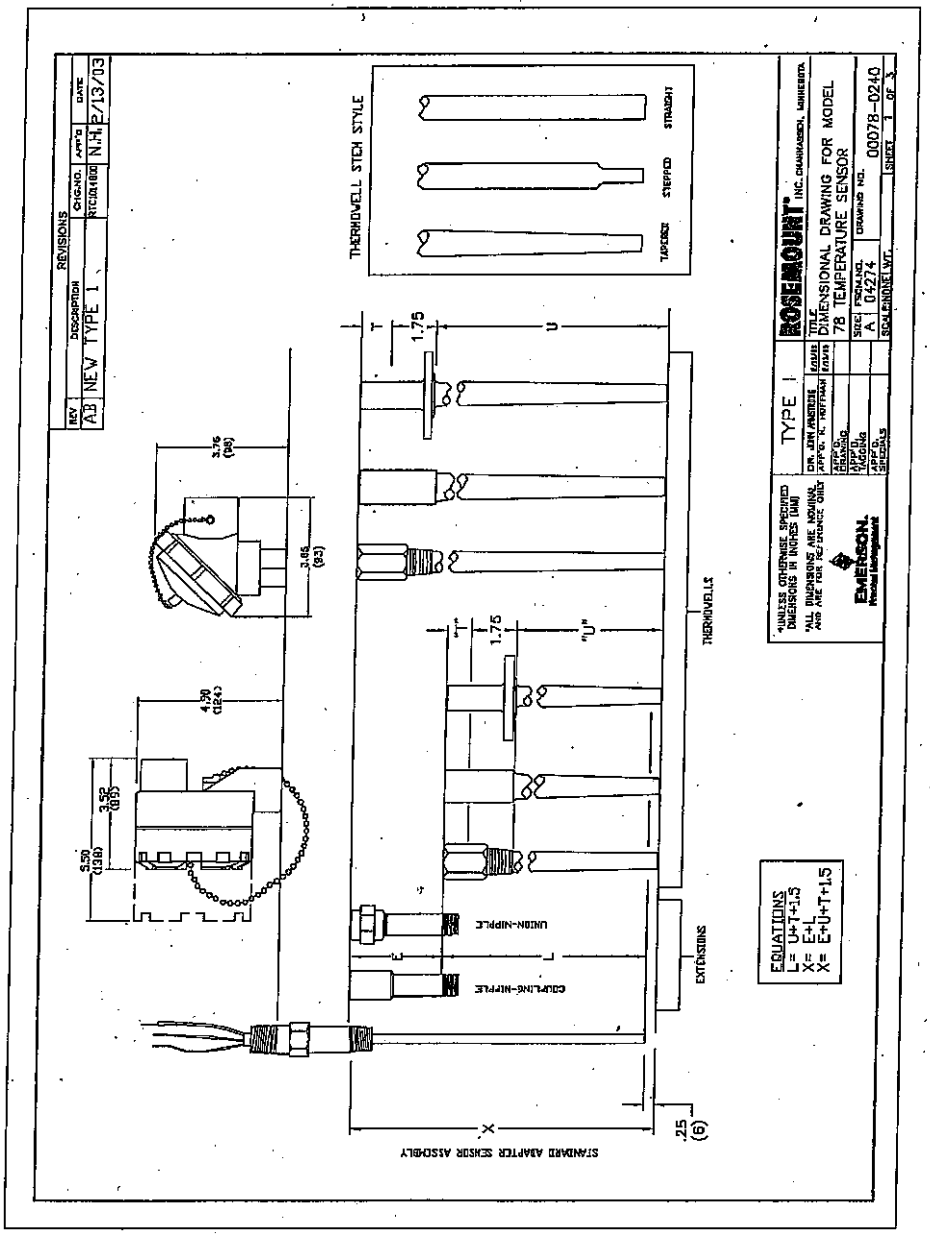
110	CALIBRATIONS AND TEST		INPUT OR TEST		OUTPUT	
111	TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	LRV	URV
112	TW/TE-300	Temp-Output signal	0°C	150°C	100 ohm	157.33 ohm
113		Test pressure				
114						
115						
116						
117						

118	COMPONENT IDENTIFICATIONS		
119	COMPONENT TYPE	MANUFACTURER	MODEL NUMBER
120			
121			
122			
123			
124			
125			

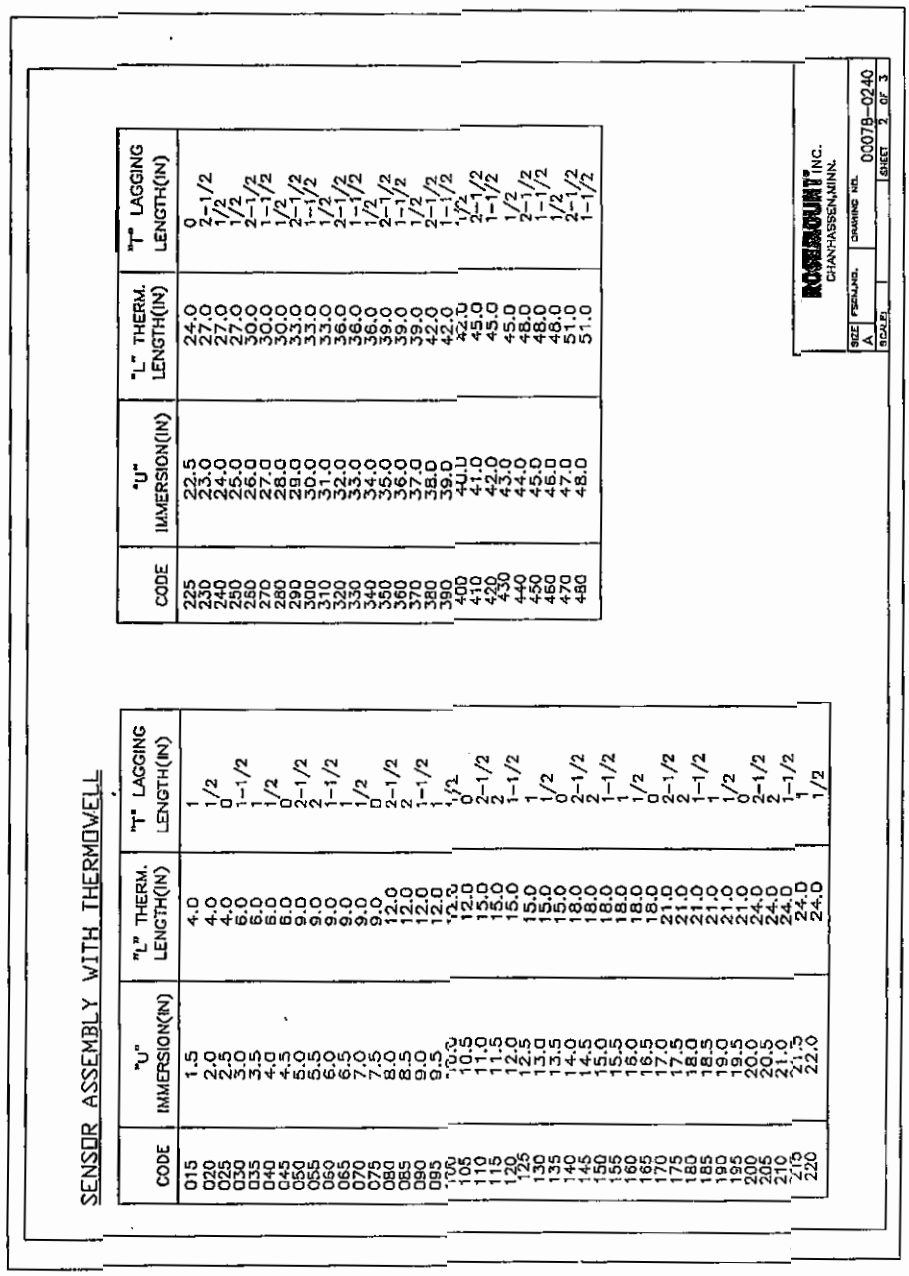
Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

Form: 20T2201 Rev 0

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SENSOR ASSEMBLY WITH THERMOWELL

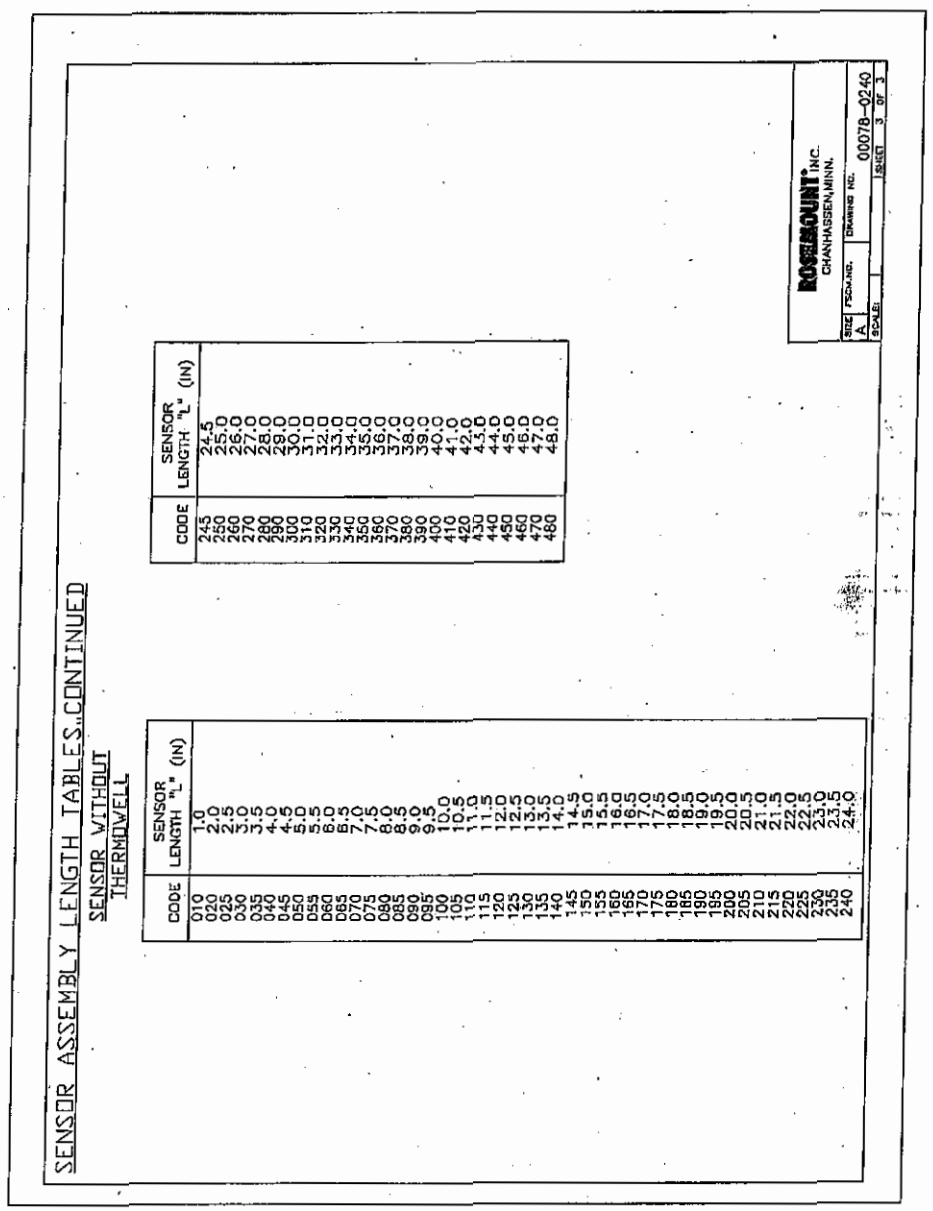
CODE	"U" IMMERSION(IN)	"L" THERM. LENGTH(IN)	"T" LAGGING LENGTH(IN)
015	1.5	4.0	1
020	2.0	4.0	1/2
025	2.5	4.0	0
030	3.0	6.0	1-1/2
035	3.5	6.0	1
040	4.0	6.0	1/2
045	4.5	8.0	0
050	5.0	9.0	2-1/2
055	5.5	9.0	2
060	6.0	9.0	1-1/2
065	6.5	9.0	1
070	7.0	9.0	1/2
075	7.5	9.0	0
080	8.0	12.0	2-1/2
085	8.5	12.0	2
090	9.0	12.0	1-1/2
095	9.5	12.0	1
100	10.0	12.0	1/2
105	10.5	12.0	0
110	11.0	15.0	2-1/2
115	11.5	15.0	2
120	12.0	15.0	1-1/2
125	12.5	15.0	1
130	13.0	15.0	1/2
135	13.5	15.0	0
140	14.0	18.0	2-1/2
145	14.5	18.0	2
150	15.0	18.0	1-1/2
155	15.5	18.0	1
160	16.0	18.0	1/2
165	16.5	18.0	0
170	17.0	21.0	2-1/2
175	17.5	21.0	2
180	18.0	21.0	1-1/2
185	18.5	21.0	1
190	19.0	21.0	1/2
195	19.5	21.0	0
200	20.0	24.0	2-1/2
205	20.5	24.0	2
210	21.0	24.0	1-1/2
215	21.5	24.0	1
220	22.0	24.0	1/2

CODE	"U" IMMERSION(IN)	"L" THERM. LENGTH(IN)	"T" LAGGING LENGTH(IN)
225	22.5	24.0	0
230	23.0	27.0	2-1/2
240	24.0	27.0	1/2
250	25.0	27.0	1/2
260	26.0	30.0	2-1/2
270	27.0	30.0	1-1/2
280	28.0	30.0	1/2
290	29.0	33.0	2-1/2
300	30.0	33.0	1-1/2
310	31.0	33.0	1/2
320	32.0	36.0	2-1/2
330	33.0	36.0	1-1/2
340	34.0	36.0	1/2
350	35.0	39.0	2-1/2
360	36.0	39.0	1-1/2
370	37.0	39.0	1/2
380	38.0	42.0	2-1/2
390	39.0	42.0	1-1/2
400	40.0	42.0	1/2
410	41.0	45.0	2-1/2
420	42.0	45.0	1-1/2
430	43.0	45.0	1/2
440	44.0	48.0	2-1/2
450	45.0	48.0	1-1/2
460	46.0	48.0	1/2
470	47.0	51.0	2-1/2
480	48.0	51.0	1-1/2

ROSEMOUNT INC.
CHANHASSEN,MINN.

SIZE	FSCM NO.	DRAWING NO.	00078-0240
A			
SCALE:		SHEET 2 OF 3	

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SENSOR ASSEMBLY LENGTH TABLES...CONTINUED

SENSOR WITHOUT THERMOWELL

CODE	SENSOR LENGTH "L" (IN)
010	1.0
020	2.0
025	2.5
030	3.0
035	3.5
040	4.0
045	4.5
050	5.0
055	5.5
060	6.0
065	6.5
070	7.0
075	7.5
080	8.0
085	8.5
090	9.0
095	9.5
100	10.0
105	10.5
110	11.0
115	11.5
120	12.0
125	12.5
130	13.0
135	13.5
140	14.0
145	14.5
150	15.0
155	15.5
160	16.0
165	16.5
170	17.0
175	17.5
180	18.0
185	18.5
190	19.0
195	19.5
200	20.0
205	20.5
210	21.0
215	21.5
220	22.0
225	22.5
230	23.0
235	23.5
240	24.0

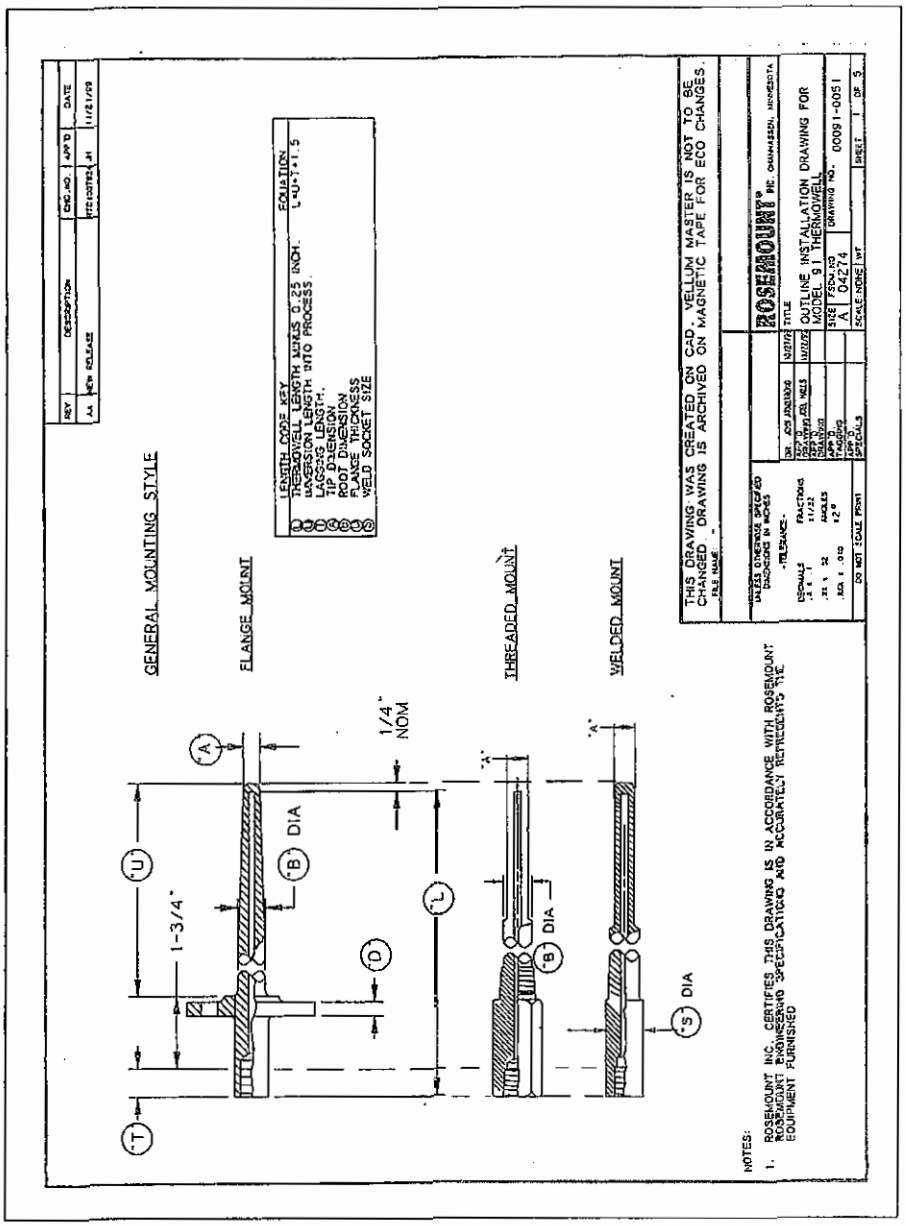
CODE	SENSOR LENGTH "L" (IN)
245	24.5
250	25.0
260	26.0
270	27.0
280	28.0
290	29.0
300	30.0
310	31.0
320	32.0
330	33.0
340	34.0
350	35.0
360	36.0
370	37.0
380	38.0
390	39.0
400	40.0
410	41.0
420	42.0
430	43.0
440	44.0
450	45.0
460	46.0
470	47.0
480	48.0

ROSEMOUNT INC.
CHANHASSEN, MINN.

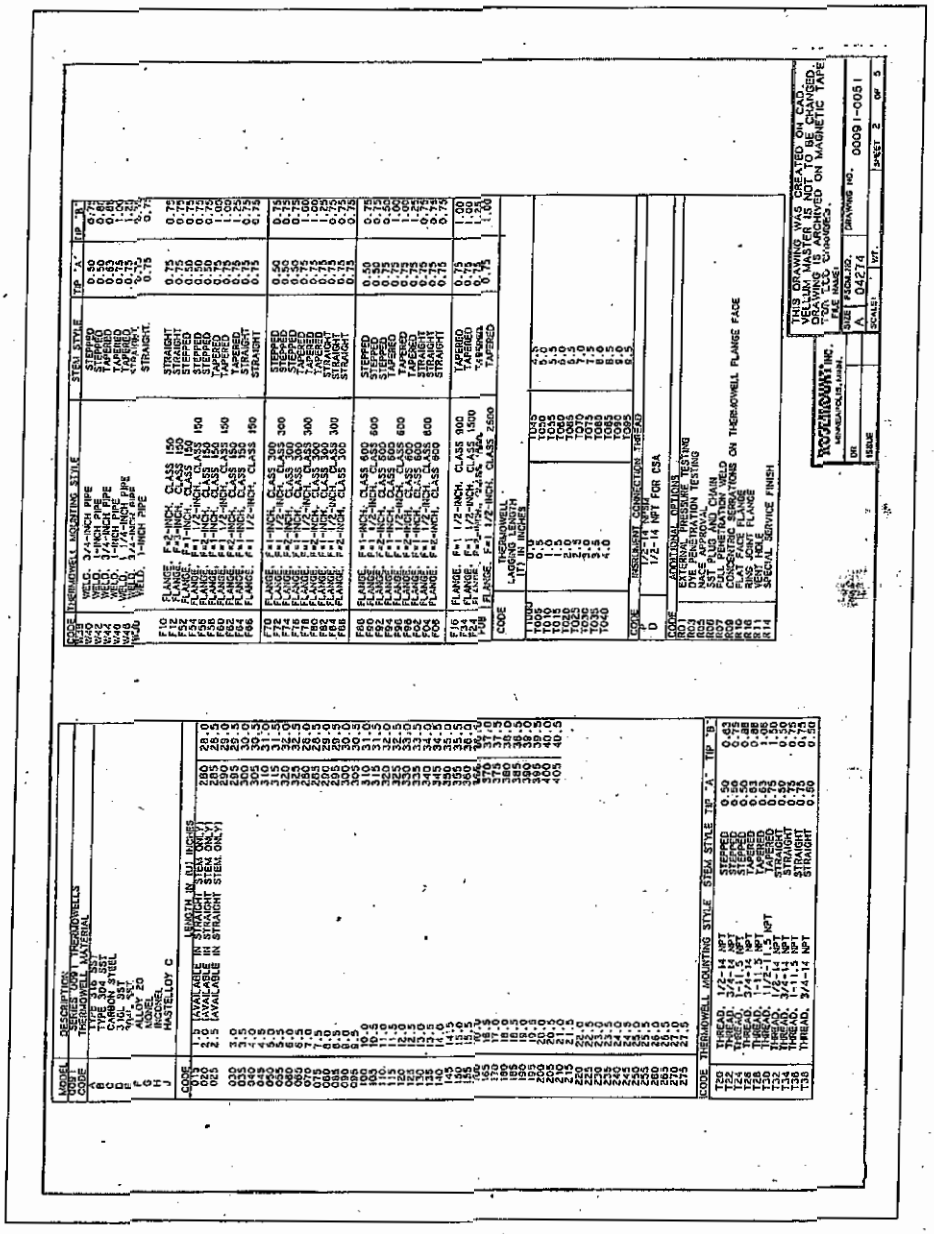
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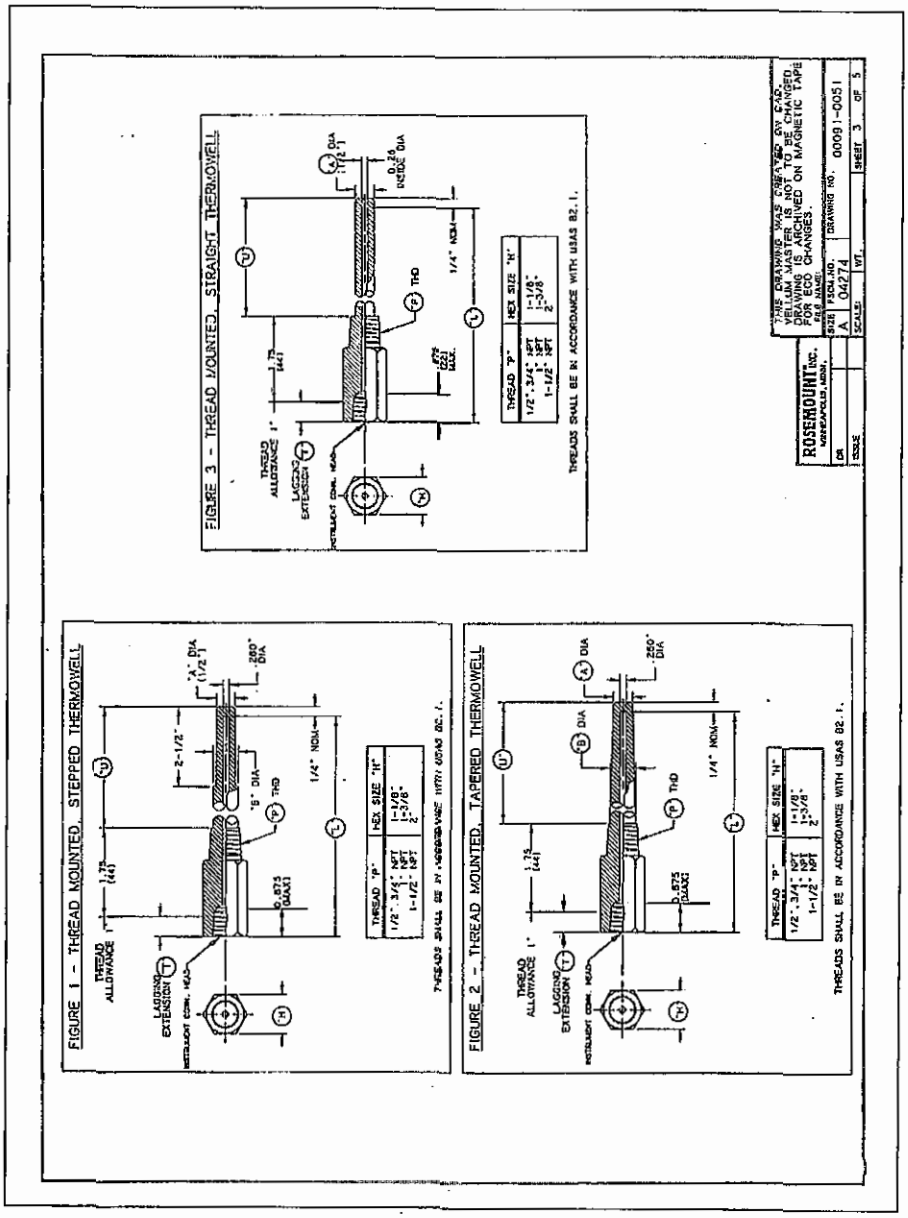
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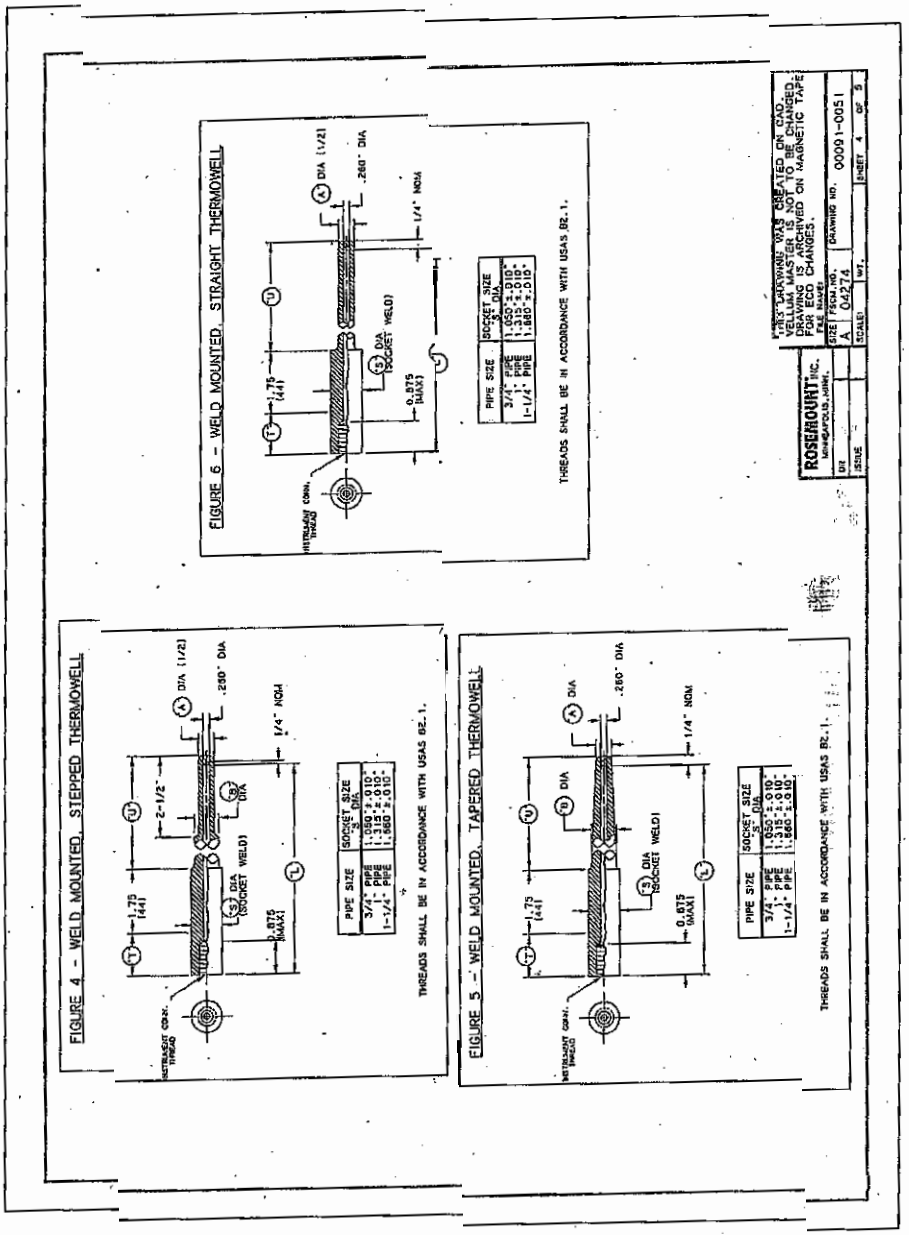
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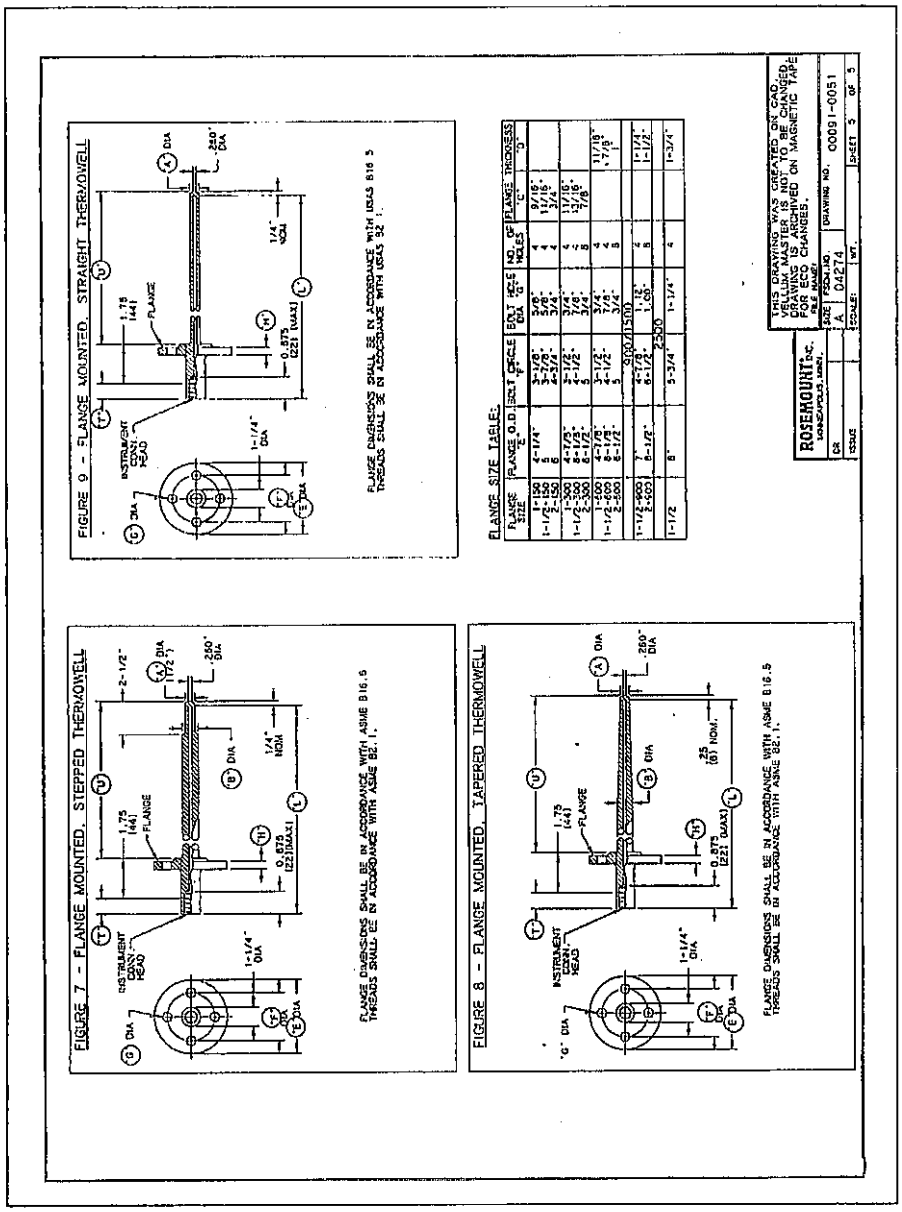
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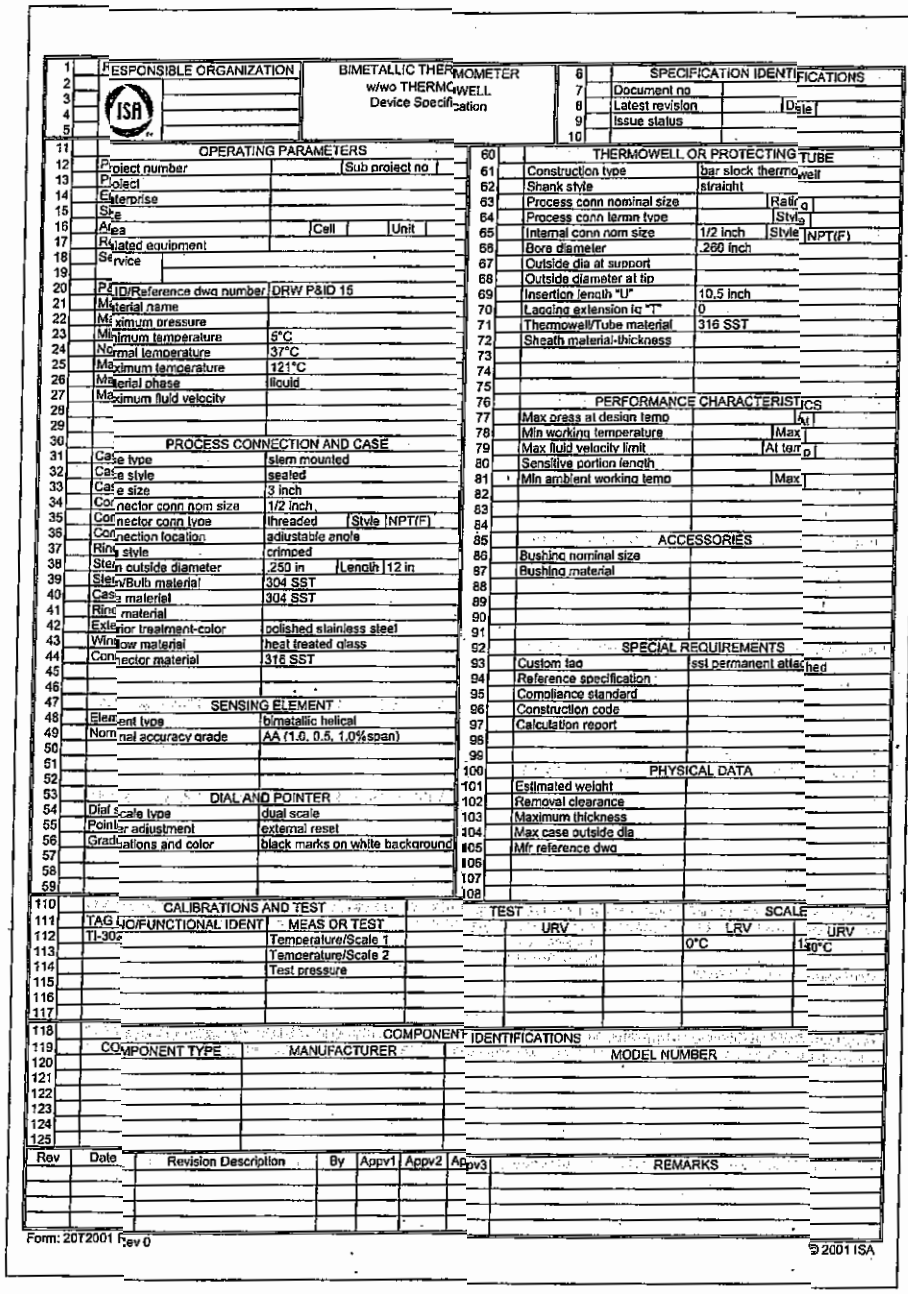
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RESPONSIBLE ORGANIZATION

BIMETALLIC THERMOMETER w/wo THERMOWELL Device Specification

SPECIFICATION IDENTIFICATIONS

Document no	
Latest revision	Date
Issue status	

OPERATING PARAMETERS

Project number	Sub project no
Project	
Enterprise	
Site	
Area	Cell / Unit
Related equipment	
Service	
P&ID/Reference dwg number	DRW P&ID 15
Material name	
Maximum pressure	
Minimum temperature	5°C
Normal temperature	37°C
Maximum temperature	121°C
Material phase	liquid
Maximum fluid velocity	

PROCESS CONNECTION AND CASE

Case type	stem mounted
Case style	sealed
Case size	3 inch
Connector conn nom size	1/2 inch
Connector conn type	threaded / Style NPT(F)
Connection location	adjustable angle
Ring style	crimped
Stem outside diameter	.250 in / Length 12 in
Stem/Bulb material	304 SST
Case material	304 SST
Ring material	
Exterior treatment-color	polished stainless steel
Window material	heat treated glass
Connector material	316 SST

SENSING ELEMENT

Element type	bimetallic helical
Normal accuracy grade	AA (1.0, 0.5, 1.0%span)

DIAL AND POINTER

Dial scale type	dual scale
Pointer adjustment	external reset
Graduations and color	black marks on white background

THERMOWELL OR PROTECTING TUBE

Construction type	bar stock thermowell
Shank style	straight
Process conn nominal size	Rating
Process conn female type	Style
Internal conn nom size	1/2 inch / Style NPT(F)
Bore diameter	.260 inch
Outside dia at support	
Outside diameter at tip	
Insertion length "U"	10.5 inch
Lagging extension lg "T"	0
Thermowell/Tube material	316 SST
Sheath material-thickness	

PERFORMANCE CHARACTERISTICS

Max press at design temp	Max
Min working temperature	At temp
Max fluid velocity limit	
Sensitive portion length	
Min ambient working temp	Max

ACCESSORIES

Bushing nominal size	
Bushing material	

SPECIAL REQUIREMENTS

Custom tag	ssl permanent attached
Reference specification	
Compliance standard	
Construction code	
Calculation report	

PHYSICAL DATA

Estimated weight	
Removal clearance	
Maximum thickness	
Max case outside dia	
Mfr reference dwg	

CALIBRATIONS AND TEST

TAG NO/FUNCTIONAL IDENT	MEAS OR TEST	TEST URV	SCALE LRV	SCALE URV
TI-302	Temperature/Scale 1		0°C	140°C
	Temperature/Scale 2			
	Test pressure			

COMPONENT IDENTIFICATIONS

COMPONENT TYPE	MANUFACTURER	MODEL NUMBER

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3

REMARKS

Form: 20T2001 Rev 0

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RESPONSIBLE ORGANIZATION	FILLED-SYSTEM TEMPERATURE SWITCH w/wo THERMOWELL Device Specification	SPECIFICATION IDENTIFICATIONS		
ISA		Document no		
		Latest revision		Date
		Issue status		

PROCESS CONNECTION AND BULB			
Bulb type	plain bulb		
Stem/Bulb extension style	NA		
Connector conn nom size	1/2 inch		
Connector conn type	threadedNP	Style	
Filling type			
Stem/Bulb diameter	.250 in	Length	6 in
Bulb extension length			
Stem/Bulb material	copper		
Connector material			

CAPILLARY AND SENSING ELEMENT			
Thermal system type	liquid filled		
Compensation style			
Temperature LRL	0°F	URL	75°F
Capillary length	25 feet		
Capillary material	copper		
Coating-armor material			

SWITCH MECHANISM			
Type	dry contact		
Element style			
Enclosure type no/class			
Reset style			
Set point adjustment lct	external		
Signal power source	24 vdc		
Dead band type	narrow adjustable dead band		
Contacts arrangement	SPDT	Quantity	1
Integral indicator style	NA		
Signal termination type	screw terminals		
Cert/Approval type	UL nonclassified		
Mounting type	housing		
Signal conn orientation	right hand		
Enclosure material	aluminum		
Exterior coating material			

THERMOWELL OR PROTECTING TUBE			
Construction type			
Shank style			
Process conn nominal size		Rating	
Process conn termn type		Style	
Internal conn nom size		Style	
Bore diameter			
Outside dia at support			
Outside dia at tip			
Insertion length "U"			
Lagging extension lg "T"			
Thermowell/Tube material			
Sheath material-thickness			

PERFORMANCE CHARACTERISTICS			
Max press at design temp		At	
Min working temperature		Max	
Max fluid velocity limit		At temp	
Repeatability			
Max overrange limit			
Ambient temperature error			
Dead band rating			
Min ambient working temp		Max	
Contacts ac rating		At max	
Contacts dc rating	1 A	At max	30 VDC

ACCESSORIES	
Bushing nominal size	
Bushing material	
Sealed leads adapter	
Breather/Drain style	

SPECIAL REQUIREMENTS	
Custom tag	ssi permanent attached
Reference specification	
Special preparation	
Compliance standard	
Calculation report	

PHYSICAL DATA			
Estimated weight			
Overall height			
Removal clearance			
Signal conn nominal size		Style	
Mfr reference dwg			

CALIBRATIONS AND TEST		INPUT/SETPOINT/TEST			OUTPUT OR SCALE	
TAG NO/FUNCTIONAL IDENT	MEAS OR TEST	LRV	URV	ACTION	LRV	URV
TS-205 setpoint / reset	Temp setpoint 1-Output	32°F	40°F	N.O.	0V	24V
	Temp setpoint 2-Output					
	Temperature-Scale					
	Test pressure					

COMPONENT IDENTIFICATIONS		
COMPONENT TYPE	MANUFACTURER	MODEL NUMBER
filled system temp switch		

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

Form: 20T2111 Rev 0

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1		RESPONSIBLE ORGANIZATION	RTD/THERMOCOUPLE TEMPERATURE TRANSMITTER OR SWITCH Device Specification		6		SPECIFICATION IDENTIFICATIONS	
2		ISA			7		Document no	
3					8		Latest revision	Date
4					9		Issue status	
5					10			

		TRANSMITTER OR SWITCH1			
11					
12		Housing type			
13		Input sensor type	Thermocouple Type T		
14		Output signal type	digital single-variable		
15		Min measurement span	200°F	Max	250°F
16		Temp coef/Tolerance cl	standard		
17		Isolation type			
18		Enclosure type no/class			
19		Adjustment type	front pushbutton		
20		Characteristic curve	linear to measurement		
21		Digital communication std	Hart protocol		
22		Signal power source	loop 12-28 vdc		
23		Measurement type			
24		Configuration-no of wires			
25		Contacts arrangement	NA	Quantity	
26		Failsafe style	failsafe		
27		Transient protection			
28		Integral indicator style	LCD meter 4 digit		
29		Signal termination type	front screw terminals		
30		Cert/Approval type			
31		Mounting type			
32		Failure/Diagnostic action	drive output high		
33		Dead band type			
34		Switch time delay			
35		Temp compensation type	cold junction.internal		
36		Enclosure material			
37		Mounting kit material			
38					
39					
40					
41					
42					
43					
44					
45					
46					
47					
48					
49					
50					
51					

		PERFORMANCE CHARACTERISTICS			
52					
53		Accuracy rating	0.2% span		
54		Measurement LRL	32°F	URL	212°F
55		Min ambient working temp	10°C	Max	50°C
56		Contacts ac rating		At max	
57		Contacts dc rating		At max	
58					
59					
60					
61					
62					
63					
64					
65					
66					
67					
68		ACCESSORIES			
69		Remote indicator style			
70		Indicator enclosure			
71		Air set filter style			
72		Air set gauges			
73					
74					
75		SPECIAL REQUIREMENTS			
76		Custom tag	ssl permanent attached		
77		Reference specification			
78		Compliance standard			
79		Calibration report	5 point		
80		Software configuration	NA		
81					
82					
83					
84		PHYSICAL DATA			
85		Estimated weight			
86		Overall height			
87		Removal clearance			
88		Signal conn nominal size		Style	
89		Mfr reference dwg			
90					
91					
92					

		CALIBRATIONS AND TEST		INPUT OR SETPOINT			OUTPUT OR SCALE	
110								
111		TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
112		TT-200	Temp- Analog output 1	32.0°F	212.0°F	direct	4.00 mA	20.00 mA
113			Temp-Analog output 2					
114			Temp-Scale					
115			Temp diff-Digital output					
116			Temp-Digital output					
117			Terms temp-Digital output					
118								
119			Temp setpoint 1-Output					
120			Temp setpoint 2-Output					
121			Temp setpoint 3-Output					
122			Temp setpoint 4-Output					
123								
124								
125								

		COMPONENT IDENTIFICATIONS		
126				
127		COMPONENT TYPE	MANUFACTURER	MODEL NUMBER
128				
129				
130				
131				
132				
133				

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

Form: 20T2221 Rev 0

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1		RESPONSIBLE ORGANIZATION		RTD/THERMOCOUPLE TEMPERATURE TRANSMITTER OR SWITCH Device Specification		6		SPECIFICATION IDENTIFICATIONS		
2						7		Document no		
3						8		Latest revision		Date
4						9		Issue status		
5						10				
11		TRANSMITTER OR SWITCH				52		PERFORMANCE CHARACTERISTICS		
12		Housing type				53		Accuracy rating	0.2% span	
13		Input sensor type	RTD Pt100			54		Measurement LRL	0°C	URL 150°C
14		Output signal type	analog current			55		Min ambient working temp	10°C	Max 50°C
15		Min measurement span	00°C	Max 200°C		56		Contacts ac rating		At max
16		Temp coef/Tolerance cl	standard			57		Contacts dc rating		At max
17		Isolation type				58				
18		Enclosure type no/class				59				
19		Adjustment type	front multi-turn pot			60				
20		Characteristic curve	linear to measurement			61				
21		Digital communication std	NA			62				
22		Signal power source	loop 12-28 vdc			63				
23		Measurement type				64				
24		Configuration-no of wires	4 wire compensation loop			65				
25		Contacts arrangement	NA	Quantity		66				
26		Failsafe style	failsafe			67				
27		Transient protection				68		ACCESSORIES		
28		Integral indicator style	LCD meter 4 digit			69		Remote indicator style		
29		Signal termination type	front screw terminals			70		Indicator enclosure		
30		Cert/Approval type				71		Air set filter style		
31		Mounting type				72		Air set gauges		
32		Failure/Diagnostic action	drive output high			73				
33		Dead band type				74				
34		Switch time delay				75		SPECIAL REQUIREMENTS		
35		Temp compensation type	NA			76		Custom tag	ast permanent attached	
36		Enclosure material				77		Reference specification		
37		Mounting kit material				78		Compliance standard	IEC 584-1 (ITS-90)	
38						79		Calibration report	5 point	
39						80		Software configuration	NA	
40						81				
41						82				
42						83				
43						84		PHYSICAL DATA		
44						85		Estimated weight		
45						86		Overall height		
46						87		Removal clearance		
47						88		Signal conn nominal size		Style
48						89		Mfr reference dwg		
49						90				
50						91				
51						92				

110		CALIBRATIONS AND TEST		INPUT OR SETPOINT			OUTPUT OR SCALE	
111		TAG NO/FUNCTIONAL IDENT	MEAS/SIGNAL/TEST	LRV	URV	ACTION	LRV	URV
112		TT-300	Temp- Analog output 1	0.0°C	150.0°C	direct	4.00 mA	20.00 mA
113			Temp-Analog output 2					
114			Temp-Scale					
115			Temp diff-Digital output					
116			Temp-Digital output					
117			Temp temp-Digital output					
118								
119			Temp setpoint 1-Output					
120			Temp setpoint 2-Output					
121			Temp setpoint 3-Output					
122			Temp setpoint 4-Output					
123								
124								
125								

126		COMPONENT IDENTIFICATIONS	
127	COMPONENT TYPE	MANUFACTURER	MODEL NUMBER
128			
129			
130			
131			
132			
133			

Rev	Date	Revision Description	By	Appv1	Appv2	Appv3	REMARKS

Form: 20T2221 Rev 0

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ABC Company
Instrument Specification Data Form

Area of Use:	Department:
Room Number:	Owner:
Date Requested:	In Service Date Required:

INSTRUMENT DATA

Manufacturer:	Model Number:
Equipment/System:	Serial Number:
Tag Number:	*Instrument Range:
Description:	Calibration Standard: ☐ YES ☐ NO

* Per manufacturer's specification

PROCESS DATA

Calibration Range:	**Calibration Tolerance:
Use Point (if applicable):	Process Range:

** Calibration tolerance cannot be more stringent than the manufacturer's tolerance. If tighter tolerance is necessary, the instrument must be re-evaluated for the application.

INSTRUMENT CLASSIFICATION (Check one box only)

☐ Critical Instrument – Product impact – A device that, when not performing to specification, could cause product impact. The instrument owner will be required to perform an evaluation on all out of tolerance conditions for Critical devices. An Out of Tolerance Evaluation form will be initiated.

☐ Non-Critical Instrument – Equipment Performance/Safety – A device that is significant to equipment /process operation, or equipment/personal safety, but does not pose an impact to product or process quality. Out od Tolerance Evaluation Forms, are not issued on Non-Critical devices. These instruments are calibrated on a routine basis.

☐ Reference Only Instrument – Trending or Information Only. Not used for quantitative measure, not used for quality decisions. Instruments are calibrated initially and all subsequent calibrations are performed on demand only.

☐ This instrument does not require calibration (if this box is checked attach justification statement)

This area to be completed by the Calibration Department

New Calibration ID Assigned:	Calibration Frequency: ____ months
New Calibration SOP Required?	☐ YES ☐ NO
New Calibration Standard(s) Required?	☐ YES ☐ NO
Is this a Calibration Program Instrument?	☐ YES ☐ NO

Approvals:

Instrument Owner:	Date:
Calibration Supervisor:	Date:
Quality Assurance:	Date:

Appendix A-4

CALIBRATION PROCEDURES

The following calibration standard operating procedures (SOPs) are examples only. Use the information provided in these procedures to answer the chapter review questions.

All calibration procedures in this appendix refer to the manufacturer's technical manual for performing the adjustments. Note that some companies do not allow reference to other documents in calibration procedures per their interpretation of OSHA 1910 and other regulations.

Pressure

SOP NUMBER	TITLE
SOP-CAL-01	Calibration/Calibration Check of Electronic Pressure Transmitter/ Indicator
SOP-CAL-02	Calibration/Calibration Check of Electronic Differential Pressure Transmitter
SOP-CAL-03	Calibration of Differential Pressure Transmitters
SOP-CAL-04	Calibration/Calibration Check of Pressure (Vacuum) Gauges
SOP-CAL-05	Calibration/Calibration Check of a Pressure Switch

Temperature

SOP NUMBER	TITLE
SOP-CAL-06	Calibration/Calibration Check of Temperature Indicator/Recorder
SOP-CAL-07	Calibration/Calibration Check of Dial Thermometers
SOP-CAL-08	Calibration/Calibration Check of RTD Input Electronic Temperature Transmitters
SOP-CAL-09	Calibration Check of Glass Thermometers

Level (see Chapter 5 for additional Level Calibration procedures)

SOP NUMBER	TITLE
SOP-CAL-02	Calibration/Calibration Check of Electronic Differential Pressure Transmitter
SOP-CAL-03	Calibration of Differential Pressure Transmitters
SOP-CAL-10	Calibration/Calibration Check of Capacitance Probe Level Transmitter

Flow (see Chapter 6 for additional Flow Calibration procedures)

SOP NUMBER	TITLE
SOP-CAL-11	Calibration/Calibration Check of Flow Totalizers/Transmitters

Final Control Elements (See Chapter 7 for additional Final Element Calibration Procedures)

SOP NUMBER	TITLE
SOP-CAL-12	Calibration of Current to Pressure Transducers

Analytical Instruments (See Chapter 8 for additional Analytical Instrument Calibration Procedures)

SOP NUMBER	TITLE
SOP-CAL-13	Calibration/Calibration Check of Leeds & Northrup 7082 Conductivity Analyzers

The following example calibration data sheets could be used for a manual system or as a template report for your database system with fields imported for each calibration. This could be used instead of the calibration data sheet attachments to each procedure (in which case you would not set up your system with data sheet attachments to your calibration SOPs).

Example Calibration Data Sheet #1

Instrument ID:	System ID:
Manufacturer:	Location:
Model:	Classification:
Serial Number:	Owner:
Description:	Calibration SOP:
Calibration Range:	Calibration Tolerance:
Calibration Interval:	Calibration Due Date:

Calibration Data

% Input	Standard	As-Found	Deviation	As-Left

Service:
( ) In tolerance
( ) Out of Tolerance
( ) Adj. to within tolerance
( ) Cal by vendor
( ) Repaired
( ) Work Order

Status Change: ( ) N/A
( ) New
( ) Active
( ) Out of service
( ) Tolerance
( ) Classification
( ) Interval

Attachments: ( ) N/A
( ) Data
( ) Chart
( ) Test Report
( ) Certificate
( ) Mfg specifications
( ) Other: _____

Standard(s) Used	ID	Cal Due Date

Calibration Date:	Next Calibration Due Date:

Comments: _____

Performed By: _____ Date: _____

Reviewed By: _____ Date: _____

Example Calibration Data Sheet #2

Tag Number:	Location:
Calibration ID:	Owner/User:
Description:	Calibration Range:
Manufacturer:	Calibration Tolerance:
Model:	Calibration SOP:
Serial Number:	Calibration Interval:
Classification:	Calibration Due Date:

Calibration Data

Target (Std. / DUT) (circle one)	Measured Data (DUT / Standard) (circle one)			
	As-Found	Deviation	As-Left	Deviation

Calibration Date: _____ Next Calibration Due Date: _____

Test Standard(s) Information:

Standard(s) Used	ID	Calibration Due Date

Technician's comments:

Check One:
☐ Found In Tolerance
☐ Found Out of Tolerance
☐ Found Damaged

Check all that apply:
☐ Left in Tolerance
☐ Left Out of Tolerance
☐ Removed from service

Performed By: _____ Date: _____

Reviewed By: _____ Date: _____

ABC Company	Calibration/Calibration Check of Electronic Pressure Transmitter/Indication	SOP-CAL-01 Rev 0
		Effective Date: 3/24/03
	Work Instruction	Page 1 of 4 Attach.: 1

Purpose

To perform a calibration/calibration check of a pressure control loop consisting of sensor, transmitter, and/or digital indication(s) and to operationally check alarms and control outputs as applicable.

Equipment/ materials needed

These items will be needed for performance of this procedure:

- Regulated air or nitrogen pressure source
- Digital pressure gauge accurate to 0.1% of reading
- Multimeter accurate to 0.1% of reading
- Pressure hoses and fittings capable of withstanding applied pressures

Prerequisites

The following prerequisites shall be completed prior to performance of the maintenance item:

- Determine the components of the loop to be checked and record identification (P&ID tag number, bar code number, serial number, or description) in the remarks column.
- Record the required information on the calibration form.

Initial conditions

Perform the following prior to testing:

- Ensure associated system is shut down. If this is not possible, the control functions from the instrument loop under test must be placed in manual or disabled to prevent response to test signal simulations.
- Isolate the pressure sensor, if practical, and bleed off pressure in accordance with the site Lockout/Tagout procedure.

Continued on next page

Procedural steps

Follow the procedural steps shown in the table below:

- Do not perform any adjustments until all "as found" calibration data has been recorded.
- Note any deviation from this procedure in the remarks section of the calibration form.
- Procedural steps marked with a pound sign (#) either require information found on the calibration form used for recording data (Attachment A) or require data to be recorded on the calibration form.
- If the loop being checked does not include a component that this procedure is checking, mark the step N/A on the calibration data form.

Step	Action
1	Make connections between the pressure source, pressure standard, and the transmitter input.
2	Connect the digital multimeter to the current output of the transmitter.
3	Exercise the transmitter from zero to full scale and back to zero.
4#	Adjust the pressure (vacuum) source to 10, 50, and 90% of the calibration range and record the following at each data point: • Digital pressure standard reading • Current output • Remote indication (if applicable)
5#	Verify proper operation of all control outputs and alarms at the proper setpoints. Record all outputs checked and actual values obtained.

Continued on next page

Procedural steps (cont.)

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>6#</td><td>
<table>
<tr><th>If...</th><th>Then...</th></tr>
<tr><td>If transmitter output is not within 1/2 the specified tolerance
manual</td><td>Calibrate the transmitter in - accordance with the applicable manufacturers technical
and repeat step 4.</td></tr>
<tr><td>If remote display is not within 1/2 the specified tolerance
manual</td><td>Calibrate the display in accordance with the manufacturers technical

and repeat step 4.</td></tr>
<tr><td>If any alarm or control output operates improperly or does not actuate within 1/2 the specified tolerance</td><td>Adjust the output and repeat step 5.</td></tr>
</table>
</td></tr>
<tr><td>7</td><td>Disconnect the test equipment from the unit under test and reinstall, if necessary.</td></tr>
<tr><td>8#</td><td>Complete the calibration form and apply the appropriate calibration label.</td></tr>
<tr><td>9</td><td>If all checks are within tolerance, restore the instruments to operational condition and notify appropriate personnel of work performed.</td></tr>
</table>

Continued on next page

Acceptance criteria

Personnel performing this maintenance shall use the following criteria to determine satisfactory completion, subject to review of data by supervision.

- Final readings obtained are within the tolerance specified on the calibration work order form.

If any readings are found to be out of tolerance or if any step cannot be performed:

- Notify the appropriate personnel for resolution.
- Record name of person contacted in comments section of work order form.

Attachments

The following are attachments to this work instruction:

No.	Title	Pages
A	Calibration/Calibration Check of Electronic Pressure Transmitter/Indication Data Sheet	2

References

Manual No.	Manual Title
N/A	Applicable manufacturer's technical manual

Approvals/ author

This work instruction must be approved by:

Manager, ABC Company Date

Quality, ABC Company Date

Author: Mike Cable, Engineering Technician

SOP-CAL-01 Rev 0 Attachment A Page 1 of 2

CALIBRATION/CALIBRATION CHECK OF ELECTRONIC PRESSURE TRANSMITTER/INDICATION DATA SHEET

Step #	Parameter	Required	As-Found Data	As-Left Data
4, 6	Digital Pressure standard at 10%	NA	_____	_____
4, 6	Transmitter output current at 10%	5.60 ± __ mA DC	_____mA	_____mA
4, 6	Remote Indication at 10%	_____ ± _____	_____	_____
4, 6	Digital Pressure standard at 50%	NA	_____	_____
4, 6	Transmitter output current at 50%	12.00 ± __ mA DC	_____mA	_____mA
4, 6	Remote Indication at 50%	_____ ± _____	_____	_____
4, 6	Digital Pressure standard at 90%	NA	_____	_____
4, 6	Transmitter output current at 90%	18.40 ± __ mA DC	_____mA	_____mA
4, 6	Remote Indication at 90%	_____ ± _____	_____	_____
5, 6	Alarm or control output _____	_____ ± _____	_____	_____
5, 6	Alarm or control output _____	_____ ± _____	_____	_____
5, 6	Alarm or control output _____	_____ ± _____	_____	_____

Continued on next page

For Reference Only

Standards Used:

	Description	Serial number or other I.D.	Cal Due Date
1	Digital pressure gauge		
2	Milliammeter		
3			
4			

Remarks:

Performed By: _____ **Date:** _____

Reviewed By: _____ **Date:** _____

ABC Company	Calibration/Calibration Check of Electronic Diffential Pressure Transmitter	SOP-CAL-02 Rev 0
		Effective Date: DRAFT
	Work Instruction	Page 1 of 3 Attach.: 1

Purpose

To perform a calibration check and, if necessary, calibration of a differential pressure transmitter.

Equipment/ materials needed

These items will be needed for performance of this procedure:

- Regulated air or nitrogen pressure source
- Digital pressure gauge accurate to 0.1% of reading
- Multimeter accurate to 0.1% of reading
- Pressure hoses and fittings capable of withstanding applied pressures

Prerequisites

Prior to performing this maintenance item, perform the following activities:

- Record required information on the calibration form.

Procedural steps

Follow the procedural steps shown in the table below:

- Do not perform any adjustments until all "as found" calibration data has been recorded.
- Note any deviation from this procedure in the remarks section of the calibration form.
- Procedural steps marked with a pound sign (#) either require information found on the calibration form used for recording data (Attachment A) or require data to be recorded on the calibration form.

PRESSURE TRANSMITTER CALIBRATION/ CALIBRATION CHECK	
Step	**Action**
1	Make connections between the pressure source, pressure standard, and the transmitter input.
2	Connect the digital multimeter to the current output of the transmitter.

Continued on next page

Procedural steps (cont.)

PRESSURE TRANSMITTER CALIBRATION/ CALIBRATION CHECK (cont.)	
Step	**Action**
3	Exercise the transmitter from zero to full scale and back to zero.
4#	Adjust the pressure (vacuum) source to 0, 25, 50, 75,and 100% of the calibration range and record the following at each data point: • Digital pressure standard reading • Current output
5#	**If...** / **Then...** If transmitter output is not within 1/2 the specified tolerance / Calibrate the transmitter in accordance with the applicable manufacturers technical manual and repeat step 4.
6	Disconnect the test equipment from the unit under test and reinstall, if necessary.
7#	Complete the calibration form and apply the appropriate calibration label.
8	If all checks are within tolerance, restore the instruments to operational condition and notify appropriate personnel of work performed.

Continued on next page

Acceptance criteria

Personnel performing this maintenance shall use the following criteria to determine satisfactory completion, subject to review of data by supervision.

- Final readings obtained are within the tolerance specified on the calibration work order form.

If any readings are found to be out of tolerance or if any step cannot be performed:

- Notify the appropriate personnel for resolution.
- Record name of person contacted in comments section of work order form.

Attachments

The following are attachments to this work instruction:

No.	Title	Pages
A	Calibration/Calibration Check of Differential PressureTransmitter Data Sheet	1

References

Manual No.	Manual Title
N/A	Applicable manufacturer's technical manual

Approvals/ author

This procedure must be approved by:

Manager, ABC Company Date

Quality, ABC Company Date

Writer: Mike Cable, Engineering Technician

SOP-CA-02 Rev 0 Attachment A Page 1 of 1

CALIBRATION/CALIBRATION CHECK OF ELECTRONIC DIFFERENTIAL PRESSURE TRANSMITTER

For: PID Tag # _____

Step #	Parameter	Required	As-Found Data	As-Left Data
4, 5	Output current at 0%	4.00 ± 0.16 mA	_____	_____
4, 5	Output current at 25%	8.00 ± 0.16 mA	_____	_____
4, 5	Output current at 50%	12.00 ± 0.16 mA	_____	_____
4, 5	Output current at 75%	16.00 ± 0.16 mA	_____	_____
4, 5	Output current at 100%	20.00 ± 0.16 mA	_____	_____

Standards Used:

	Description	Serial number or other I.D.	Cal Due Date
1	Digital pressure gauge	_____	_____
2	Milliammeter	_____	_____
3	_____	_____	_____

Remarks:

_____ _____ _____

_____ _____ _____

_____ _____ _____

_____ _____ _____

_____ _____ _____

_____ _____ _____

_____ _____ _____

Performed By: _____ _____ Date: _____

Reviewed By: _____ _____ Date: _____

ABC Company **Standard Operating Procedure**

Document No.: SOP-CAL-03	Revision No.: 00	Effective Date:	Page 1 of 4
Title: **Calibration of Differential Pressure Transmitters**		Supersedes Document No.: None	
Author (Print Name):	Author Signature:		Date:
QA/Compliance Approval (Print Name):	QA/Compliance Approval Signature:		Date:

Table of Contents

1.0 PURPOSE 1
2.0 SCOPE 2
3.0 DEFINITIONS 2
4.0 REFERENCES/ATTACHMENTS 2
5.0 MATERIALS/REAGENTS/EQUIPMENT 2
6.0 SAFETY 2
7.0 PROCEDURE 3
8.0 DOCUMENT REVISION 4

For Reference Only

ABC Company **Standard Operating Procedure**

Document No.: SOP-CAL-03	Revision No.: 00	Effective Date:	Page 2 of 4
Title: Calibration of Differential Pressure Transmitters			

1.0 PURPOSE

1.1 The purpose of this procedure is to provide standardized instruction for the calibration of Differential Pressure Transmitters.

2.0 SCOPE

2.1 This procedure applies to personnel required to calibrate and maintain differential pressure transmitters located at ABC Company.

3.0 DEFINITIONS

3.1 NIST – National Institute of Standards and Technology

3.2 UUT – Unit Under Test

3.3 OOT – Out of Tolerance

4.0 REFERENCES/ATTACHMENTS

4.1 *Calibration System Description SOP*

5.0 MATERIALS/REAGENTS/EQUIPMENT

5.1 NIST Traceable Pressure Calibrating Device

5.2 NIST Traceable Milliammeter

5.3 Pressure Source

5.4 Calibration Labels

6.0 SAFETY

6.1 All procedures will be performed in accordance with and under the constraints of ABC Company safety procedures and applicable Federal, State, and Local safety rules and regulations.

For Reference Only

ABC Company **Standard Operating Procedure**

Document No.: SOP-CAL-03	Revision No.: 00	Effective Date:	Page 3 of 4
Title: **Calibration of Differential Pressure Transmitters**			

7.0 PROCEDURE

7.1 Visually inspect the UUT for damage. Notify the user group and repair or replace as directed.

Note: Always attempt to obtain "As Found" data prior to repair, replacement, or adjustments.

7.2 Warm up/Stabilization Requirement: None.

7.3 UUT Accuracy/Specification: Manufacturer's specification or approved user assigned tolerance.

7.4 "As Found" Data Acquisition

7.4.1 Differential pressure transmitters will be tested at 10%, 50%, and 90% of span unless otherwise specified by instrument owner. Test Standards must maintain a 4:1 accuracy ratio unless otherwise specified on the applicable Instrument Specification Data Form.

7.4.2 Disconnect one of the transmitter output leads and connect the milliammeter in series with the transmitter output.

7.4.3 Connect the pressure source and NIST traceable pressure calibrator to the high pressure port of the UUT.

7.4.4 Apply pressure to the high pressure side of the transmitter. Approach the test point as indicated by the pressure standard. Record the transmitter output current reading and the remote indication, if applicable, on Calibration Data Sheet #2.

Note: Test points must be reached within +/- 5% of targeted test point.

7.4.5 Repeat step 7.4.4 for the remaining test points.

7.4.6 Proceed to section 7.5, "As Found" Data Interpretation

7.5 "As Found" Data Interpretation

7.5.1 Compare the "As Found" data to the tolerance specified on the Calibration Data Sheet.

ABC Company		Standard Operating Procedure	
Document No.: SOP-CAL-03	Revision No.: 00	Effective Date:	Page 4 of 4
Title: Calibration of Differential Pressure Transmitters			

7.5.2 If an "As Found" value is "Out of Tolerance", a Calibration Out of Tolerance Form must be generated per Calibration System Description SOP. Proceed to Adjustments, section 7.6.

7.5.3 To optimize instrument performance or if adjustment is needed, proceed to Adjustments, section 7.6.

7.5.4 If all of the "As Found" data is within the specified tolerance, proceed to section 7.7.

7.6 Adjustments

7.6.1 Adjust and repair transmitter according to manufacturers suggested repair guidelines. If guidelines are unavailable, replace the transmitter.

7.6.2 If the UUT can not be adjusted to specification, remove the device from service

7.6.3 Proceed to section 7.8, Documentation and Labeling.

7.7 No Adjustment Necessary

7.7.1 Return UUT to service.

7.8 Documentation and Labeling

7.8.1 Complete and process the Calibration Data Sheet per the Calibration System Description SOP.

7.8.2 Complete and affix the appropriate calibration labels per the Calibration System Description SOP.

8.0 DOCUMENT REVISION

Step or Section No.	Change(s) Made	Reason for Change(s)
New Issue	N/A	N/A

ABC Company	Calibration/Calibration Check of Pressure (Vacuum) Gauges	SOP-CAL-04 Rev 0
		Effective Date: 3/24/03
	Work Instruction	Page 1 of 3 Attach.: 1

Purpose

To perform a calibration check and, if necessary, calibration of a mechanical pressure, vacuum, or pressure/vacuum compound gauge.

Equipment/ materials needed

These items will be needed for performance of this procedure:

- Regulated air or nitrogen pressure source
- Vacuum source, if applicable
- Digital pressure/vacuum gauge accurate to 0.1% of reading
- Pressure hoses and fittings capable of withstanding applied pressures

Prerequisites

The following prerequisites shall be completed prior to performance of the maintenance item:

- Record the required information on the calibration form.

Initial conditions

Perform the following prior to testing:

- Isolate the gauge and bleed off pressure per the site Lockout/Tagout Procedure.

Procedural steps

Follow the procedural steps shown in the table below:

- Do not perform any adjustments until all "as found" calibration data has been recorded.
- Note any deviation from this procedure in the remarks section of the calibration form.
- Procedural steps marked with a pound sign (#) either require information found on the calibration form used for recording data (Attachment A) or require data to be recorded on the calibration form.

Step	Action
1	• Inspect the gauge for integrity and cleanliness. • Clean as necessary.

Continued on next page

Procedural steps (cont.)

Step	Action
2	Make connections between the pressure source, pressure standard, and the gauge under test.
3	If gauge has been removed, stabilize the gauge in the same orientation as when it is installed.
4	Exercise the gauge from zero to full scale three times.
5#	Record the pressure (vacuum) standard reading and gauge under test reading at 0, 25, 50, 75, and 100% of the calibration range as pressure (vacuum) is increased. Note: If the pressure (vacuum) is increased above the desired reading, decrease pressure (vacuum) to the previous reading taken, and then increase pressure (vacuum) to desired value.
6#	Record the pressure (vacuum) standard reading and gauge under test reading at 75, 50, 25, and 0% of the calibration range as pressure (vacuum) is decreased. Note: If the pressure (vacuum) is decreased below the desired reading, increase pressure (vacuum) to the previous reading taken, then decrease pressure (vacuum) to desired value.
7#	• If the gauge under test indicates values greater than 1/2 the specified tolerance and the gauge is calibratable, adjust the gauge until indications are less than 1/2 the specified tolerance. • If adjustments are made, repeat steps 5 - 6 and record final values.
8	Disconnect the test equipment from the gauge under test and reinstall, if necessary.
9#	Complete the calibration form and apply the applicable calibration label.

Continued on next page

Procedural steps (cont.)

Step	Action
10	If all checks are within tolerance, restore the gauge to operational condition and notify appropriate personnel of work performed.

Acceptance criteria

Personnel performing this maintenance shall use the following criteria to determine satisfactory completion, subject to review of data by supervision.

- Final pressure (vacuum) measurements obtained are within the tolerance specified on the calibration work order form.

If any readings are found to be out of tolerance or if any step cannot be performed:

- Notify the appropriate personnel for resolution.
- Record name of person contacted in comments section of work order form.

Attachments

The following are attachments to this work instruction:

No.	Title	Pages
A	Calibration/Calibration Check of Pressure (Vacuum) Gauges Data Sheet	2

References

Manual No.	Manual Title
N/A	Applicable manufacturer's technical manual

Approvals/ author

This work instruction must be approved by:

Manager, ABC Company _____ Date _____

Quality, ABC Company _____ Date _____

Author: Mike Cable, Engineering Technician

CALIBRATION/CALIBRATION CHECK OF PRESSURE (VACUUM) GAUGE DATA SHEET

Step #	Parameter	Required	As-Found Data	As-Left Data
5,7	0% Increasing pressure (vacuum)	_____ +/- _____	_____	_____
5,7	25% Increasing pressure (vacuum)	_____ +/- _____	_____	_____
5,7	50% Increasing pressure (vacuum)	_____ +/- _____	_____	_____
5,7	75% Increasing pressure (vacuum)	_____ +/- _____	_____	_____
5,7	100% Increasing pressure (vacuum)	_____ +/- _____	_____	_____
6,7	75% Decreasing pressure (vacuum)	_____ +/- _____	_____	_____
6,7	50% Decreasing pressure (vacuum)	_____ +/- _____	_____	_____
6,7	25% Decreasing pressure (vacuum)	_____ +/- _____	_____	_____
6,7	0% Decreasing pressure (vacuum)	_____ +/- _____	_____	_____

Continued on next page

SOP-CAL-04 Rev 0 Attachment A Page 2 of 2

Standards Used:

	Description	Serial number or other I.D.	Cal Due Date
1	Digital pressure/vacuum gauge	_____	_____
2	_____	_____	_____
3	_____	_____	_____

Remarks:

_____
_____
_____
_____
_____
_____

Performed By: _____ **Date:** _____

Reviewed By: _____ **Date:** _____

ABC Company	Calibration/Calibration Check of a Pressure Switch	SOP-CAL-05 Rev 0
		Effective Date: 3/24/03
	Work Instruction	Page 1 of 4 Attach.: 1

Purpose

To perform a calibration check and, if necessary, calibration of a pressure switch.

Equipment/ materials needed

These items will be needed for performance of this procedure:

- Regulated air or nitrogen pressure source
- Digital pressure gauge accurate to 0.1% of reading
- Multimeter capable of reading a change of state in resistance or voltage
- Pressure hoses and fittings capable of withstanding applied pressures

Prerequisites

The following prerequisites shall be completed prior to performance of the maintenance item:

- Record the required information on the calibration form.

Initial conditions

Perform the following prior to testing:

- Isolate the pressure switch and bleed off pressure per the site Lockout/Tagout Procedure.

Procedural steps

Follow the procedural steps shown in the table below:

- Do not perform any adjustments until all "as found" calibration data has been recorded.
- Note any deviation from this procedure in the remarks section of the calibration form.
- Procedural steps marked with a pound sign (#) either require information found on the calibration form used for recording data (Attachment A) or require data to be recorded on the calibration form.

Continued on next page

Procedural steps (cont.)

Step	Action
1	**Caution: Keep power leads separated to prevent circuit shorting and possible equipment damage.** If required, disconnect power leads from pressure switch.
2	**Caution: Confirm sensing line is depressurized.** Make connections between the pressure source, pressure standard, and the pressure switch sensing line.
3	Connect the multimeter leads across the switch contact. • If checking a low pressure switch, go to step 6. • For high pressure switch continue with step 4.
4#	For high pressure switch: • Increase pressure until the switch changes state, as indicated by a change in voltage or resistance reading. • Record the pressure switch trip value as indicated on the digital pressure gauge.
5#	For high pressure switch: • Slowly decrease the pressure until the switch changes state, as indicated by a change in voltage or resistance reading. • Record the pressure switch reset value as indicated on the digital pressure gauge.
6	For low pressure switch: • Increase pressure until the switch resets, as indicated by a change in voltage or resistance reading.
7#	For low pressure switch: • Slowly decrease the pressure until the switch changes state, as indicated by a change in voltage or resistance reading. • Record the pressure switch trip value as indicated on the digital pressure gauge.

Continued on next page

Procedural steps (cont.)

Step	Action
8#	For low pressure switch: • Slowly increase the pressure until the switch changes state, as indicated by a change in voltage or resistance reading. • Record the pressure switch reset value as indicated on the digital pressure gauge.
9#	• If the actual pressure switch trip (and if required, reset) values are greater than 1/2 the specified tolerance, adjust the switch until values are less than 1/2 the specified tolerance. • If adjustments are made, repeat steps 4 and 5 or 6 - 8 as applicable and record final values.
10	Disconnect the test equipment from the switch under test.
11#	Complete the calibration form and affix the applicable calibration label.
12	If all checks are within tolerance, restore the pressure switch to operational condition and notify appropriate personnel of work performed.

Acceptance criteria

Personnel performing this maintenance shall use the following criteria to determine satisfactory completion, subject to review of data by supervision.

- Final trip (and reset) values obtained are within the tolerance specified on the calibration work order form.

If any readings are found to be out of tolerance or if any step cannot be performed:

- Notify the appropriate personnel for resolution.
- Record name of person contacted in comments section of work order form.

Continued on next page

Attachments

The following are attachments to this work instruction:

No.	Title	Pages
A	Calibration/Calibration Check of a Pressure Switch Data Sheet	1

References

Manual No.	Manual Title
N/A	Applicable manufacturer's technical manual

Approvals/ author

This work instruction must be approved by:

Manager, ABC Company Date

Quality, ABC Company Date

Author: Mike Cable, Engineering Technician

SOP-CAL-05 Rev 0 Attachment A Page 1 of 1

CALIBRATION/CALIBRATION CHECK OF A PRESSURE SWITCH DATA SHEET

Step #	Parameter	Required	As-Found Data	As-Left Data
4, 9	High pressure switch trip	_____ +/- _____	_____	_____
5, 9	High pressure switch reset	_____ +/- _____	_____	_____
7, 9	Low pressure switch trip	_____ +/- _____	_____	_____
8, 9	Low pressure switch reset	_____ +/- _____	_____	_____

Standards Used:

	Description	Serial number or other I.D.	Cal Due Date
1	Digital pressure gauge	_____	_____
2	Multimeter	_____	_____
3	_____	_____	_____
4	_____	_____	_____

Remarks

_____
_____
_____
_____
_____
_____

Performed By: _____ **Date:** _____

Reviewed By: _____ **Date:** _____

ABC Company	Calibration/Calibration Check of Temperature Indicator/Recorder	SOP-CAL-06 Rev 0
		Effective Date: 3/24/03
	Work Instruction	Page 1 of 4 Attach.: 1

Purpose

To perform an operational check or calibration check and, if necessary, calibration of a temperature indicator/recorder.

Equipment/ materials needed

These items will be needed for performance of this procedure:

- Digital RTD reference thermometer accurate to +/- 0.5 F (0.2 C)
- Temperature reference bath/block

Procedural steps

Follow the procedural steps shown in the table below:

- Do not perform any adjustments until all "as found" calibration data has been recorded.
- Note any deviation from this procedure in the remarks section of the calibration form.
- Procedural steps marked with a pound sign (#) either require information found on the calibration form used for recording data (Attachment A) or require data to be recorded on the calibration form.

If...	Then perform steps...
Performing only the temperature indicator/ recorder operational check (temperature sensor not accessible)	1 - 3 and 7 - 8.
Performing temperature indicator/recorder calibration check	4 - 8.

Continued on next page

Procedural steps (cont.)

PERFORMING ONLY THE TEMPERATURE INDICATOR/RECORDER OPERATIONAL CHECK	
Step	**Action**
1#	• With temperature of the system at normal operating temperature, measure the system temperature at a point near the system thermometer using the reference RTD thermometer. • Allow reading to stabilize and record this as the required value on the calibration data form.
2#	Record the temperature indication on the calibration data form.
3#	• If the indicator under test indicates values greater than 1/2 the specified tolerance, perform steps 1 and 2 at 10% and 90% of full range. • Adjust the indicator until the correct reading is obtained and repeat steps 1 - 3. • Go to step 7.

PERFORMING TEMPERATURE INDICATOR/RECORDER CALIBRATION CHECK	
Step	**Action**
4	• Verify temperature sensor is in a well and remove from system. • If not in a well, coordinate with the user area and isolate the system to allow removal.
5#	• Place the temperature sensor and digital RTD reference thermometer in the temperature bath/block. • Record both readings at a minimum of two points (such as ice point and process temperature).
6#	• If the indicator under test indicates values greater than 1/2 the specified tolerance, make adjustments at process temperature until indications are less than 1/2 the specified tolerance. • If adjustments are made, repeat step 5 and record final values. • Go to step 7.

Continued on next page

Procedural steps (cont.)

COMPLETION	
Step	**Action**
7#	Complete the calibration form and apply the applicable calibration label.
8	If all checks are within tolerance, restore to operational condition and notify appropriate personnel of work performed.

Acceptance criteria

Personnel performing this maintenance shall use the following criteria to determine satisfactory completion, subject to review of data by supervision.

- Final readings obtained are within the tolerance specified on the calibration work order form.

If any readings are found to be out of tolerance or if any step cannot be performed:

- Notify the appropriate personnel for resolution.
- Record name of person contacted in comments section of work order form.

Attachments

The following are attachments to this work instruction:

No.	Title	Pages
A	Calibration/Calibration Check of Temperature Indicator/Recorder Data Sheet	1

Continued on next page

References

Manual No.	Manual Title
N/A	Applicable manufacturer's technical manual

Approvals/ author

This work instruction must be approved by:

Manager, ABC Company _____ Date

Manager, ABC Company _____ Date

Author: Mike Cable, Engineering Technician

SOP-CAL-06 Rev 0 Attachment A Page 1 of 1

CALIBRATION/CALIBRATION CHECK OF TEMPERATURE INDICATOR/RECORDER DATA SHEET

Step #	Parameter	Required	As-Found Data	As-Left Data
1, 2, 3	Normal Operating Temperature	____ ± ____	____	____
3	10% Range (if required)	____ ± ____ or NA	____	____
3	90% Range (if required)	____ ± ____ or NA	____	____
5, 6	Temperature (ice point)	____ ± ____	____	____
5, 6	Temperature (process)	____ ± ____	____	____
5, 6	Temperature (additional point if desired)	____ ± ____	____	____

Standards Used:

	Description	Serial number or other I.D.	Cal Due Date
1	Reference thermometer	____	____
2	Bath/block	____	____
3	____	____	____

Remarks:

____

____

____

____

Performed By: ____ **Date:** ____

Reviewed By: ____ **Date:** ____

For Reference Only

ABC Company	Calibration/Calibration Check of Dial Thermometers	SOP-CAL-07 Rev 0
		Effective Date: 3/24/03
	Work Instruction	Page 1 of 4 Attach.: 1

Purpose

To perform an operational check or calibration check and, if necessary, calibration of a dial type thermometers.

Equipment/ materials needed

These items will be needed for performance of this procedure:

- Digital RTD reference thermometer accurate to +/- 0.5 F (0.2 C)
- Temperature reference bath/block
- Thermocouple reference (special type T T/C and digital thermocouple reference standard) accurate to +/- 2 F (1 C), if required

Prerequisites

The following prerequisites shall be completed prior to performance of the maintenance item:

- Record the required information on the calibration form.

Procedural steps

Follow the procedural steps shown in the table below:

- Do not perform any adjustments until all "as found" calibration data has been recorded.
- Note any deviation from this procedure in the remarks section of the calibration form.
- Procedural steps marked with a pound sign (#) either require information found on the calibration form used for recording data (Attachment A) or require data to be recorded on the calibration form.

If...	Then perform steps ...
Performing only the thermometer operational check	1 - 3 and 7 - 8.
Performing thermometer calibration check	4 - 8.

Continued on next page

Procedural steps (cont.)

PERFORMING ONLY THE THERMOMETER OPERATIONAL CHECK	
Step	**Action**
1#	With temperature of the system at normal operating temperature, record the thermometer indication on the calibration data form.
2#	• Measure the system temperature at a point near the system thermometer using the reference RTD thermometer or thermocouple. • If the system thermometer is in a well and temperature is constant, remove the dial thermometer and place the reference RTD thermometer in the same well. • Allow reading to stabilize and record on the calibration data form.
3#	• If the thermometer under test indicates values greater than 1/2 the specified tolerance, adjust the thermometer until the correct reading is obtained and perform a complete calibration check in accordance with steps 4 - 6. • Continue with step 7.

PERFORMING THERMOMETER CALIBRATION CHECK	
Step	**Action**
4	• Verify thermometer is in a well and remove from system. • If not in a well, coordinate with the user area and isolate the system to allow removal.
5#	• Place the thermometer and digital RTD reference thermometer in the temperature bath/block. • Record both readings at a minimum of two points (such as ice point and process temperature).

Continued on next page

Procedural steps (cont.)

PERFORMING THERMOMETER CALIBRATION CHECK	
Step	**Action**
6#	• If thermometer under test indicates values greater than 1/2 the specified tolerance and is calibratable, make adjustments at process temperature until indications are less than 1/2 the specified tolerance. • If adjustments are made, repeat step 5 and record final values. • If thermometer is not calibratable but within tolerance, make a note in the remarks section and accept as is. • Continue with step 7.

COMPLETION	
Step	**Action**
7#	Complete the calibration form and apply the applicable calibration label.
8	If all checks are within tolerance, restore the thermometer to operational condition and notify appropriate personnel of work performed.

Acceptance criteria

Personnel performing this maintenance shall use the following criteria to determine satisfactory completion, subject to review of data by supervision.

- Final thermometer indications obtained are within the tolerance specified on the calibration work order form.

If any readings are found to be out of tolerance or if any step cannot be performed:

- Notify the appropriate personnel for resolution.
- Record name of person contacted in comments section of work order form.

Continued on next page

Attachments The following are attachments to this work instruction:

No.	Title	Pages
A	Calibration/Calibration Check of Dial Thermometers Data Sheet	2

References

Manual No.	Manual Title
N/A	Applicable manufacturer's technical manual

Approvals/ author This work instruction must be approved by:

Manager, ABC Company Date

Manager, ABC Company Date

Author: Mike Cable, Engineering Technician

For Reference Only

SOP-CAL-07 Rev 0 Attachment A Page 1 of 2

CALIBRATION/CALIBRATION CHECK OF DIAL THERMOMETER DATA SHEET

Step #	Parameter	Required	As-Found Data	As-Left Data
1, 2, 3	Reference thermometer (normal operating temperature)	NA	_____	_____
1, 2, 3	Dial thermometer (normal operating temperature)	Reference thermometer reading ± _____	_____	_____
5, 6	Reference thermometer (ice point)	NA	_____	_____
5, 6	Dial thermometer (ice point)	Reference thermometer reading ± _____	_____	_____
5, 6	Reference thermometer (process)	NA	_____	_____
5, 6	Dial thermometer (process)	Reference thermometer reading ± _____	_____	_____
5, 6	Reference thermometer (additional point if desired)	NA	_____	_____
5, 6	Dial thermometer (additional point if desired)	Reference thermometer reading ± _____	_____	_____

Continued on next page

SOP-CAL-07 Rev 0 Attachment A Page 2 of 2

Standards Used:

	Description	Serial number or other I.D.	Cal Due Date
1	Reference thermometer		
2	Bath/block		
3	Thermocouple reference (if required)		
4			

Remarks:

Performed By: _____ **Date:** _____

Reviewed By: _____ **Date:** _____

ABC Company	Calibration/Calibration Check of RTD Input Electronic Temperature Controller	SOP-CAL-08 Rev 0
		Effective Date: 3/24/03
	Work Instruction	Page 1 of 7 Attach.: 1

Purpose

To perform a calibration/calibration check of an RTD input electronic temperature transmitter and remote indication(s), in addition to operationally checking other control outputs and alarms.

Equipment/ materials needed

These items will be needed for performance of this procedure:

- RTD simulator (or decade box) accurate to +/- 0.15 ohms (method B only)
- Digital RTD reference thermometer accurate to +/- 0.1 C
- Multimeter accurate to 0.1% of reading
- Temperature reference bath/block

Prerequisites

The following prerequisites shall be completed prior to performance of the maintenance item:

- Determine the method to be used from the work order:
  - Loop check (Method A) or
  - Electronic check (Method B).
- Record the required setpoint on the Calibration form.
- Determine the components of the loop to be checked and record identification (P&ID tag number, bar code number, serial number, or description) in the remarks column.
- Confirm the correct RTD alpha is being used.
- Determine the required transmitter output current (required for performing step 4) using the following equation:

  {(Required Reference Temperature - Instrument Zero Temperature) / Instrument Temperature Range } * 16 mA + 4 mA

  Example: Required reference thermometer temperature is 60 degrees C. Temperature transmitter range is 40 - 90 degrees C.

  { (60 deg - 40 deg) / 50 deg } * 16 mA + 4 mA = 10.40 mA

Continued on next page

Initial conditions

Ensure temperature and humidity conditions in the area being recorded are stable.

If...	Then...
The temperature control loop performs control function which could produce undesirable operating results while simulating test signals (for example, temperature signal provides input for control steam to heat exchanger)	• place controller in manual OR • disable controller output OR • shutdown the system being controlled (coordinate with user area).
RTD is in a well	Remove the RTD from the well
RTD is not in a well	Shutdown and drain the system

Procedural steps

Follow the procedural steps shown in the table below:

- Do not perform any adjustments until all "as found" calibration data has been recorded.
- Note any deviation from this procedure in the remarks section of the calibration form.
- Procedural steps marked with a pound sign (#) either require information found on the calibration form used for recording data (Attachment A) or require data to be recorded on the calibration form.
- If the loop being checked does not include a component that this procedure is checking, mark the step N/A on the calibration data form.

For	perform ...	steps ...
Loop check (method A)	Preliminary Operating Point Check,	1 - 4,
	Loop Check,	5 - 7,
	Final Operating Point Check, and	13, and
	Completion	14 - 15
Electronic check (method B)	Preliminary Operating Point Check,	1 - 4,
	Electronic Check,	8 - 12,
	Final Operating Point Check, and	13, and
	Completion	14 - 15

Continued on next page

For Reference Only

Procedural steps (cont.)

PRELIMINARY OPERATING POINT CHECK (METHOD A AND B)	
Step	**Action**
1	Place RTD and reference thermometer in temperature reference bath/block at or within the process setpoint.
2	Disconnect an output lead from the transmitter and connect multimeter in series to measure output current.
3	Convert the transmitter output current to an equivalent temperature using the following equation: $\frac{(\text{Output current} - 4\text{mA})(\text{Inst temp range})}{16\text{ mA}} + 0\%\text{ value} = X\text{ deg}$ For example: output current for 20 - 100 C transmitter = 13. 62 mA. Therefore [(13.62mA - 4mA)(80 C) / 16mA] + 20 C = 68.1 C.
4#	• Record the reference thermometer indication, transmitter output value (current output and converted temperature), and remote display(s). • Verify current output and remote indication(s) are within the specified tolerance. • Do not make any adjustments.

LOOP CHECK (METHOD A)	
Step	**Action**
5#	Adjust the bath/block temperature to 10, 50, and 90% of the transmitter span and allow temperature to stabilize for 5 minutes at each data point. Record the following at each data point: • Bath/block Temperature • Transmitter output current • Remote display(s), if applicable

Continued on next page

Procedural steps (cont.)

LOOP CHECK (METHOD A) (cont.)	
Step	**Action**
6#	Verify proper actuation of all control outputs and alarms at the proper setpoints. Record all outputs checked and actual values obtained.
7#	

If...	Then...
If transmitter output is not within 1/2 the specified tolerance	Adjust bath/block temperature to 0% and 100% of transmitter span. Adjust zero and span at each point, respectively, until no further adjustments are required. Repeat step 5.
If remote display is not within 1/2 the specified tolerance	Calibrate the display in accordance with the manufacturers technical manual and repeat step 5.
If any control output operates improperly or does not actuate within 1/2 the specified tolerance	Adjust the output and repeat step 6.

ELECTRONIC CHECK (METHOD B)	
Step	**Action**
8	Disconnect the RTD from transmitter input and connect the RTD simulator (or decade box) to transmitter input.

Continued on next page

Procedural steps (cont.)

ELECTRONIC CHECK (METHOD B) (cont.)	
Step	**Action**
9#	Set the RTD simulator to 10, 50, 90% of the transmitter span. Record the following at each data point: • Simulated Temperature • Transmitter output current • Remote display(s), if applicable
10#	Verify proper actuation of all control outputs and alarms at the proper setpoints. Record all outputs checked and actual values obtained.
11#	

If...	Then...
If transmitter output is not within 1/2 the specified tolerance	Calibrate the transmitter in accordance with the applicable manufacturers technical manual and repeat step 9.
If remote display is not within 1/2 the specified tolerance	Calibrate the display in accordance with the manufacturers technical manual and repeat step 9.
If any control output operates improperly or does not actuate within 1/2 the specified tolerance	Adjust the output and repeat step 10.

12	Disconnect the RTD simulator (or decade box) from transmitter input and reconnect the RTD to transmitter input.

Continued on next page

Procedural steps (cont.)

FINAL OPERATING POINT CHECK (METHOD A AND B)	
Step	**Action**
13#	• Repeat steps 1, 3, and 4. • If any readings are not within 1/2 the specified tolerance, confirm the RTD has been properly reconnected. Otherwise, determine and correct cause.

COMPLETION	
Step	**Action**
14#	Complete the calibration form and affix the applicable calibration label.
15	If all checks are within tolerance, restore the instruments to operational condition and notify appropriate personnel of work performed.

Acceptance criteria

Personnel performing this maintenance shall use the following criteria to determine satisfactory completion, subject to review of data by supervision.

- Final readings obtained are within the tolerance specified on the calibration work order form.

If any readings are found to be out of tolerance or if any step cannot be performed:

- Notify the appropriate personnel for resolution.
- Record name of person contacted in comments section of work order form.

Continued on next page

Attachments

The following are attachments to this work instruction:

No.	Title	Pages
A	Calibration/Calibration Check of RTD Input Electronic Temperature Controller Data Sheet	2

References

Manual No.	Manual Title
N/A	Applicable manufacturer's technical manual

Approvals/ author

This procedure must be approved by:

Manager, ABC Company Date

Quality, ABC Company Date

Author: Mike Cable, (Engineering Technician)

SOP-CAL-08 Rev 0 Attachment A Page 1 of 2

CALIBRATION/CALIBRATION CHECK OF RTD INPUT ELECTRONIC TEMPERATURE CONTROLLER DATA SHEET

For: PID Tag # _____

Steps performed for method B are in parenthesis.

Step #	Parameter	Required	As-Found Data	As-Left Data
4, 13	Reference Themometer	_____ +/- _____	_____	_____
4, 13	XMTR Output Current	_____ +/- _____ (corresponding to reference thermometer)	_____ mA	_____ mA
4, 13	Converted Temperature	Reference thermometer reading +/- _____	_____	_____
4, 13	Remote Display	Reference thermometer reading +/- _____	_____	_____
5, 7 (9, 11)	10% Temperature	_____	_____	_____
5, 7 (9, 11)	10% Output Current	5.60 +/- _____ mA	_____ mA	_____ mA
5, 7 (9, 11)	10% Remote Display	_____ +/- _____	_____	_____
5, 7 (9, 11)	50% Temperature	_____	_____	_____
5, 7 (9, 11)	50% Output Current	12.00 +/- _____ mA	_____ mA	_____ mA
5, 7 (9, 11)	50% Remote Display	_____	_____	_____
5, 7 (9, 11)	90% Temperature	_____ +/- _____	_____	_____
5, 7 (9, 11)	90% Output Current	18.40 +/- _____ mA	_____ mA	_____ mA
5, 7 (9, 11)	90% Remote Display	_____ +/- _____	_____	_____

Continued on next page

SOP-CAL-08 Rev 0 Attachment A Page 2 of 2

CALIBRATION/CALIBRATION CHECK OF RTD INPUT ELECTRONIC TEMPERATURE CONTROLLER DATA SHEET (cont.)

Step #	Parameter	Required	As-Found Data	As-Found Data
6, 7	Alarm or control output	_____ +/- _____	_____	_____
(10, 11)	_____			
6, 7	Alarm or control output	_____ +/- _____	_____	_____
(10, 11)	_____			
6, 7	Alarm or control output	_____ +/- _____	_____	_____
(10, 11)	_____			

Standards Used:

	Description	Serial number or other I.D.	Cal Due Date
1	RTD simulator	_____	_____
2	Reference thermometer	_____	_____
3	Milliammeter	_____	_____
4	Bath/Block	_____	_____
5	_____	_____	_____

Remarks:

_____
_____
_____
_____
_____

Performed By: _____ **Date:** _____

Reviewed By: _____ **Date:** _____

For Reference Only

ABC Company **Standard Operating Procedure**

Document No.: SOP-CAL-09	Revision No.: 00	Effective Date:	Page 1 of 4
Title: **Calibration of Glass Thermometers**		Supersedes Document No.: None	
Author (Print Name):	Author Signature:		Date:
QA/Compliance Approval (Print Name):	QA/Compliance Approval Signature:		Date:

Table of Contents

1.0 PURPOSE 2
2.0 SCOPE 2
3.0 DEFINITIONS 2
4.0 REFERENCES/ATTACHMENTS 2
5.0 MATERIALS/REAGENTS/EQUIPMENT 2
6.0 SAFETY 3
7.0 PROCEDURE 3
8.0 DOCUMENT REVISION 4

For Reference Only

ABC Company **Standard Operating Procedure**

Document No.: SOP-CAL-09	Revision No.: 00	Effective Date:	Page 2 of 4
Title: **Calibration of Digital and Glass Thermometers**			

1.0 PURPOSE

1.1 The purpose of this procedure is to provide standardized instruction for the calibration of digital thermometers and glass thermometers.

2.0 SCOPE

2.1 This procedure applies to personnel required to calibrate and maintain glass thermometers and digital thermometers located at ABC Company.

3.0 DEFINITIONS

3.1 NIST – National Institute of Standards and Technology

3.2 UUT – Unit Under Test

3.3 OOT – Out of Tolerance

4.0 REFERENCES/ATTACHMENTS

4.1 *Calibration System Description SOP*

5.0 MATERIALS/REAGENTS/EQUIPMENT

5.1 NIST Traceable Temperature Measuring Device

5.2 Temperature Source (bath)

5.3 Ice

5.4 Calibration Labels

For Reference Only

ABC Company		Standard Operating Procedure	
Document No.: SOP-CAL-09	Revision No.: 00	Effective Date:	Page 3 of 4
Title: Calibration of Digital and Glass Thermometers			

6.0 SAFETY

6.1 All procedures will be performed in accordance with and under the constraints of ABC Company safety procedures and applicable Federal, State, and Local safety rules and regulations.

7.0 PROCEDURE

7.1 Visually inspect the UUT for damage. If damage is found, notify the user group and repair or replace as directed.

Note: Always attempt to obtain "As Found" data prior to repair, replacement, or adjustment.

7.2 Warm up/Stabilization Requirement: None.

7.3 UUT Accuracy/Specification: Manufacturer's specification or approved user-assigned tolerance.

7.4 "As Found" Data Acquisition

Note: Glass thermometers will be tested at 3 points across the operating range of the UUT in the lower, middle and upper third of the range, unless otherwise specified by instrument owner. 0°C will be used as a test point in the appropriate third of the thermometer.

7.4.1 Place a NIST traceable temperature indicator in appropriate bath. Run the bath to the test point as indicated on the test standard. Both the standard and the UUT must be placed at the same depth in the bath. If the UUT is a partial immersion thermometer insert the thermometer in the bath at a depth equal to the immersion line.

7.4.2 After test point is achieved, record the "as found" readings from the test standard and UUT on the Calibration Data Sheet.

7.4.3 Repeat steps 7.4.1 and 7.4.2 for the remaining test points.

7.4.4 Proceed to section 7.5, "As Found" Data Interpretation.

7.5 "As Found" Data Interpretation

7.5.1 Compare the "As Found" data to the tolerance specified on the Calibration Data Sheet.

7.5.2 If an "As Found" value is "Out of Tolerance", a Calibration Out of Tolerance Form must be generated per Calibration System Description SOP. Proceed to section 7.6, Adjustments.

7.5.3 If all of the "As Found" data is within the specified tolerance, proceed to section 7.7.

7.6 Adjustments

7.6.1 Adjust and repair gauge according to manufacturers suggested repair guidelines. If guidelines are unavailable, replace thermometer.

7.6.2 If the UUT can not be adjusted to specification, remove the device from service.

7.6.3 Proceed to section 7.8, Documentation and Labeling.

7.7 No Adjustment Necessary

7.7.1 Return UUT to service.

7.8 Documentation and Labeling

7.8.1 Complete and process the calibration data sheet per the Calibration System Description SOP.

7.8.2 Complete and affix the appropriate calibration labels per the Calibration System Description SOP.

8.0 DOCUMENT REVISION

Step or Section No.	Change(s) Made	Reason for Change(s)
New Issue	N/A	N/A

ABC Company	Calibration/Calibration Check of a Capacitance Probe Level Transmitter	SOP-CAL-10 Rev 0
		Effective Date: 5/24/03
	Work Instruction	Page 1 of 5 Attach.: 1

Purpose

To perform a calibration check and, if necessary, calibration of a capacitance probe level transmitter.

Equipment/ materials needed

These items will be needed for performance of this procedure:

- Multimeter accurate to 0.1% of reading
- Capacitance meter (if desired to check probe capacitance)
- Frequency counter (if desired to check probe frequency)
- Capacitance Decade Box (for Method B using capacitance)
- Function Generator (for Method B using frequency)

Prerequisites

The following prerequisites shall be completed prior to performance of the maintenance item:

- Determine the method to be used from the work order:
  - * Actual Tank Volume (Method A) or
  - * Capacitance/Frequency Simulation (Method B).
- Record the required information on the calibration form.

Procedural steps

Follow the procedural steps shown in the table below:

- Do not perform any adjustments until all "as found" calibration data has been recorded.
- Note any deviation from this procedure in the remarks section of the calibration form.
- Procedural steps marked with a pound sign (#) either require information found on the calibration form used for recording data (Attachment A) or require data to be recorded on the calibration form.

Continued on next page

For Reference Only

Procedural steps (cont.)

If...	Then perform steps ...
Using actual tank volume (method A)	1 - 7 and 15 - 16
Capacitance/frequency simulation (method B)	8 - 14 and 15 - 16

USING ACTUAL TANK VOLUME (METHOD A)

Step	Action
1	Drain the associated tank and fill to 10% capacity.
2#	Record the current output from the transmitter and remote indication. If desired, also record the probe capacitance and/or frequency for future reference.
3#	Fill to 50% total volume. Record transmitter current output and remote indication. If desired, also record the probe capacitance and/or frequency for future references.
4#	Fill to 90% total volume. Record transmitter current output and remote indication. If desired, also record the probe capacitance and/or frequency for future references.
5#	Verify proper operation of all alarm outputs and control functions. Record results.
6#	**IF...** The instrument under test indicates values greater than 1/2 the specified tolerance. **THEN...** Then calibrate in accordance with the manufacturers technical manual until indications are less than 1/2 the specified tolerance and repeat steps 1 - 4.

Continued on next page

Procedural steps (cont.)

USING ACTUAL TANK VOLUME (METHOD A) (cont.)	
Step	**Action**
7	If necessary, drain tank and refill with proper fluid.

CAPACITANCE/FREQUENCY SIMULATION (METHOD B)	
Step	**Action**
8	Disconnect the probe from the transmitter input and install a capacitance decade box or function generator to the transmitter input.
9#	Simulate a capacitance or frequency signal equivalent to 10% level for the probe that was disconnected. Record the current output from the transmitter and remote indication
10#	Simulate a capacitance or frequency signal equivalent to 50% level for the probe that was disconnected. Record the current output from the transmitter and remote indication.
11#	Simulate a capacitance or frequency signal equivalent to 90% level for the probe that was disconnected. Record the current output from the transmitter and remote indication.
12#	Verify proper operation of all alarm outputs and control functions. Record results.
13#	

IF...	THEN...
The instrument under test indicates values greater than 1/2 the specified tolerance.	Then calibrate in accordance with the manufacturers technical manual until indications are less than 1/2 the specified tolerance and repeat steps 9 - 11.

Continued on next page

Procedural steps (cont.)

CAPACITANCE/FREQUENCY SIMULATION (METHOD B)	
Step	**Action**
14	Disconnect capacitance decade box or function generator and restore level transmitter to normal operation.

COMPLETION	
Step	**Action**
15#	Complete the calibration form and affix the applicable calibration label.
16	If all checks are within tolerance, restore the instruments to operational condition and notify appropriate personnel of work performed.

Acceptance criteria

Personnel performing this maintenance shall use the following criteria to determine satisfactory completion, subject to review of data by supervision.

- Final readings obtained are within the tolerance specified on the calibration work order form.

If any readings are found to be out of tolerance or if any step cannot be performed:

- Notify the appropriate personnel for resolution.
- Record name of person contacted in comments section of work order form.

Attachments

The following are attachments to this work instruction:

No.	Title	Pages
A	Calibration/Calibration Check of a Capacitance Probe Level Transmitter Data Sheet	2

Continued on next page

References

Manual No.	Manual Title
N/A	Applicable manufacturer's technical manual

Approvals/ author

This work instruction must be approved by:

Manager, ABC Company Date

Quality, ABC Company Date

Author: Mike Cable, Engineering Technician

For Reference Only

CALIBRATION/CALIBRATION CHECK OF CAPACITANCE PROBE LEVEL TRANSMITTER DATA SHEET

For: PID Tag # _____
Steps performed for method B are in parenthesis.

Step #	Parameter	Required	As-Found Data	As-Left Data
2, 6 (9, 13)	10% Output Current	5.60 +/- _____ mA	_____ mA	_____ mA
2, 6 (9, 13)	10% Remote Display	_____ +/- _____	_____	_____
2, 6 (9, 13)	10% Frequency / Capacitance	_____	_____	_____
3, 6 (10, 13)	50% Output Current	12.00 +/- _____ mA	_____ mA	_____ mA
3, 6 (10, 13)	50% Remote Display	_____ +/- _____	_____	_____
3, 6 (10, 13)	50% Frequency / Capacitance	_____	_____	_____
4, 6 (11, 13)	90% Output Current	18.40 +/- _____ mA	_____ mA	_____ mA
4, 6 (11, 13)	90% Remote Display	_____ +/- _____	_____	_____
4, 6 (11, 13)	90% Frequency	_____	_____	_____

Step #	Parameter	Required	As-Found Data	As-Found Data
5 (12)	Alarm or control output _____	_____ +/- _____	_____	_____
5 (12)	Alarm or control output _____	_____ +/- _____	_____	_____
5 (12)	Alarm or control output _____	_____ +/- _____	_____	_____

Continued on next page

CALIBRATION/CALIBRATION CHECK OF RTD INPUT ELECTRONIC TEMPERATURE CONTROLLER DATA SHEET (cont.)

Standards Used:

	Description	Serial number or other I.D.	Cal Due Date
1	Milliammeter		
2	Capacitance Meter		
3	Frequency Counter		
4	Capacitance Decade Box		
5	Function Generator		

Remarks:

_____

_____

_____

_____

_____

_____

Performed By: _____ **Date:** _____

Reviewed By: _____ **Date:** _____

ABC Company	Calibration/Calibration Check of Flow Totalizers/Transmitters	SOP-CAL-11 Rev 0
		Effective Date: 3/24/03
	Work Instruction	Page 1 of 3 Attach.: 1

Purpose

To perform a calibration/ calibration check of flow totalizer/transmitter.

Equipment/ materials needed

These items will be needed for performance of this procedure:

- Digital scale accurate to 0.1% of reading
- Rosemount 268 interface (for Micromotion, if adjustment is required)
- Container adequate for amount to be dispensed

Prerequisites

The following prerequisites shall be completed prior to performance of the maintenance item:

- Determine a practical method of directing totalizer output into a container.
- Confirm fluid is in the sensor and all air is out of the lines.
- Record the required information on the calibration form.

Procedural steps

Follow the procedural steps shown in the table below:

- Do not perform any adjustments until all "as found" calibration data has been recorded.
- Note any deviation from this procedure in the remarks section of the calibration form.
- Procedural steps marked with a pound sign (#) either require information found on the calibration form used for recording data (Attachment A) or require data to be recorded on the calibration form.
- If the loop being checked does not include a component that this procedure is checking, mark the step N/A on the calibration data form.

Step	Action
1	• Place the container on the scale and zero (tare) the scale. • If a large amount of fluid will be weighed, it may be desirable to place the container on a cart. In this case, ensure the cart is on the scale when tared.
2	• If possible, reset the totalizer indication to zero. • If not possible, note the initial indication.

Continued on next page

Procedural steps (cont.)

Step	Action
3	• Direct all output from the flow totalizer into the container. • Dispense as much fluid as is practical.
4#	Weigh the amount dispensed and record the following: • Weight of amount dispensed • Indication of amount dispensed on totalizer • Remote indication, if applicable
5#	• Using the following conversions, determine the total volume dispensed based on the weight obtained. • Record this as the step 4 required value for totalizer indication and remote indication, if applicable. 1 gallon = 8.35 lb 1 lb = 0.45 kg; 1 kg = 2.2 lb 1 liter = 0.2642 gallons; 1 gallon = 3.785 liters
6	If values obtained are not within the specified tolerance: • Perform a calibration in accordance with the applicable manufacturers technical manual. • Repeat steps 1 - 5.
7	Cómplete the calibration form and apply the appropriate calibration label.
8	If all checks are within tolerance, restore the flow totalizer to operational condition and notify appropriate personnel of work performed.

Continued on next page

Acceptance criteria

Personnel performing this maintenance shall use the following criteria to determine satisfactory completion, subject to review of data by supervision.

- Final volumes agree within the tolerance specified on the calibration work order form.

If any readings are found to be out of tolerance or if any step cannot be performed:

- Notify the appropriate personnel for resolution.
- Record name of person contacted in comments section of work order form.

Attachments

The following are attachments to this work instruction:

No.	Title	Pages
A	Calibration/Calibration Check of Flow Totalizers/ Transmitters Data Sheet	1

References

Manual No.	Manual Title
N/A	Applicable manufacturer's technical manual

Approvals/ Author

This work instruction must be approved by:

Manager, ABC Company _____ Date _____

Quality, ABC Company _____ Date _____

Author: Mike Cable, Engineering Technician

SOP-CAL-11 Rev 0 Attachment A Page 1 of 1

CALIBRATION/CALIBRATION CHECK OF FLOW TOTALIZERS/TRANSMITTERS DATA SHEET

For: PID Tag # _____

Step#	Parameter	Required	As-Found Data	As-Left Data
4, 6	Weight of amount dispensed	NA	_____	_____
4, 6	Totalizer Indication	_____+/-_____	_____	_____
4, 6	Remote Indication (if applicable)	_____+/-_____	_____	_____

Standards Used:

	Description	Serial number or other I.D.	Cal Due Date
1	Digital Scale	_____	_____
2	Rosemount 268 Interface (if applicable)	_____	_____
3	_____	_____	_____
4	_____	_____	_____

Remarks:

_____

_____

_____

_____

Performed By: _____ **Date:** _____

Reviewed By: _____ **Date:** _____

For Reference Only

ABC Company **Standard Operating Procedure**

Document No.: SOP-CAL-12	Revision No.: 00	Effective Date:	Page 1 of 4
Title: **Calibration of Current to Pressure Transducers**		Supersedes Document No.: None	
Author (Print Name):	Author Signature:		Date:
QA/Compliance Approval (Print Name):	QA/Compliance Approval Signature:		Date:

Table of Contents

1.0 PURPOSE 1
2.0 SCOPE 2
3.0 DEFINITIONS 2
4.0 REFERENCES/ATTACHMENTS 2
5.0 MATERIALS/REAGENTS/EQUIPMENT 2
6.0 SAFETY 2
7.0 PROCEDURE 3
8.0 DOCUMENT REVISION 4

For Reference Only

ABC Company		Standard Operating Procedure	
Document No.: SOP-CAL-12	Revision No.: 00	Effective Date:	Page 2 of 4
Title: Calibration of Current to Pressure Transducers			

1.0 PURPOSE

1.1 The purpose of this procedure is to provide standardized instruction for the calibration of Current to Pressure Transducers.

2.0 SCOPE

2.1 This procedure applies to personnel required to calibrate and maintain current to pressure transducers located at ABC Company.

3.0 DEFINITIONS

3.1 NIST – National Institute of Standards and Technology

3.2 UUT – Unit Under Test

3.3 OOT – Out of Tolerance

4.0 REFERENCES/ATTACHMENTS

4.1 *Calibration System Description SOP*

5.0 MATERIALS/REAGENTS/EQUIPMENT

5.1 NIST Traceable Pressure Measuring Device

5.2 NIST Traceable Milliamp simulator

5.3 Pressure Source (if necessary)

5.4 Calibration Labels

6.0 SAFETY

6.1 All procedures will be performed in accordance with and under the constraints of ABC Company safety procedures and applicable Federal, State, and Local safety rules and regulations.

For Reference Only

ABC Company **Standard Operating Procedure**

Document No.: SOP-CAL-12	Revision No.: 00	Effective Date:	Page 3 of 4
Title: **Calibration of Current to Pressure Transducers**			

7.0 PROCEDURE

7.1 Visually inspect the UUT for damage. Notify the user group and repair or replace as directed.

Note: Always attempt to obtain "As Found" data prior to repair, replacement, or adjustments.

7.2 Warm up/Stabilization Requirement: None.

7.3 UUT Accuracy/Specification: Manufacturer's specification or approved user assigned tolerance.

7.4 "As Found" Data Acquisition

7.4.1 Apply the appropriate air supply pressure to the UUT. Measure and document the supply pressure in the comment section of the calibration Data Sheet.

7.4.2 Connect the milliamp simulator to the input of the UUT.

7.4.3 Connect the pressure measuring device to the output of the UUT.

7.4.4 Set the milliamp simulator for 4 mA (or the 0% signal) to the input and record the output pressure on the Calibration Data Sheet.

7.4.5 Repeat step 7.4.4 for input values of 12.0 mA and 20.0 mA.

7.4.6 Proceed to section 7.5, "As Found" Data Interpretation

7.5 "As Found" Data Interpretation

7.5.1 Compare the "As Found" data to the tolerance specified on the Calibration Data Sheet.

7.5.2 If an "As Found" value is "Out of Tolerance", a Calibration Out of Tolerance Form must be generated per Calibration System Description SOP. Proceed to Adjustments, section 7.6.

7.5.3 To optimize instrument performance or if adjustment is needed, proceed to Adjustments, section 7.6.

ABC Company		Standard Operating Procedure	
Document No.: SOP-CAL-12	Revision No.: 00	Effective Date:	Page 4 of 4
Title: Calibration of Current to Pressure Transducers			

7.5.4 If all of the "As Found" data is within the specified tolerance, proceed to section 7.7.

7.6 Adjustments

7.6.1 Adjust and repair transducer according to manufacturers suggested repair guidelines. If guidelines are unavailable, replace the transducer.

7.6.2 If the UUT can not be adjusted to specification, remove the device from service

7.6.3 Proceed to section 7.8, Documentation and Labeling.

7.7 No Adjustment Necessary

7.7.1 Return UUT to service.

7.8 Documentation and Labeling

7.8.1 Complete and process the Calibration Data Sheet per the Calibration System Description SOP.

7.8.2 Complete and affix the appropriate calibration labels per the Calibration System Description SOP.

8.0 DOCUMENT REVISION

Step or Section No.	Change(s) Made	Reason for Change(s)
New Issue	N/A	N/A

For Reference Only

ABC Company	Calibration/Calibration Check of Leads & Northrup 7082 Conductivity Analyzers	SOP-CAL-13 Rev 0
		Effective Date: Draft
	Work Instruction	Page 1 of 5 Attach.: 1

Purpose

To perform a calibration check and, if necessary, calibration of Leeds & Northrup 7082 conductivity analyzers.

Equipment/ materials needed

These items will be needed for performance of this procedure:

- Deionized water
- 8550 ohm precision resistor, Leeds & Northrup P/N 233300 or equivalent
- Decade resistance box, 100 Kohm to 10 Mohm, accurate to +/- 0.1%
- Appropriate beakers for solution and conductivity probe length
- Hydrogen peroxide or other approved sanitizing solution

Prerequisites

The following prerequisites shall be completed prior to performance of the maintenance item:

- Prepare a beaker of deionized water in a clean air environment and let stand for one hour.
- Record the required information on the calibration form.

Initial conditions

Perform the following prior to testing:

- Isolate the conductivity cell and bleed off system pressure per the site Lockout/Tagout Procedure.

Procedural steps

Follow the procedural steps shown in the table below:

- Do not perform any adjustments until all "as found" calibration data has been recorded.
- Note any deviation from this procedure in the remarks section of the calibration form.
- Procedural steps marked with a pound sign (#) either require information found on the calibration form used for recording data (Attachment A) or require data to be recorded on the calibration form.
- If the loop being checked does not include a component that this procedure is checking, mark the step N/A on the calibration data form.

Continued on next page

Procedural steps (cont.)

Step	Action
1	Remove the conductivity cell from the system.
2	Rinse the probe thoroughly using deionized water.
3#	Place the cell in the beaker of prepared deionized water. • Record the conductivity indication. • If available, place a digital conductivity standard in the solution to verify the required indication.
4	Disconnect the temperature compensation leads from terminals B and D of the conductivity cell and connect the 8550 ohm resistor.
5#	• Record the ALARM1 and ALARM2 setpoints on the calibration form.
6#	• At the analyzer, press CELL CONST. • Record the cell constant on the calibration form.
7#	• At the analyzer, press CELL CONST then press CELL CONST and the up arrow simultaneously. • Record the cell calibration factor on the calibration form.
8	Note: This step will remove all stored alarm setpoints and cell constant will default. Use the up and down arrows to change the cell calibration factor to 1.000 and ENTER.
9#	• Disconnect the conductivity cell from terminals A and C and connect the resistance decade box. • Apply values of 10 Mohm, 1 Mohm and 100 Kohm. • Record conductivity indications at each value.

Continued on next page

Procedural steps (cont.)

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>10#</td><td>
<table>
<tr><th>If...</th><th>Then...</th></tr>
<tr><td>If the indicator under test indicates values greater than 1/2 the specified tolerance</td><td>Calibrate the indicator in accordance with the manufacturer's technical manual until indications are less than 1/2 the specified tolerance.</td></tr>
<tr><td>If adjustments are made</td><td>Repeat step 9 and record final values.</td></tr>
<tr><td>If adjustments are made</td><td>Repeat steps 2 and 3. Record final values.</td></tr>
</table>
</td></tr>
<tr><td>11</td><td>• Thoroughly rinse the probe with deionized water and sanitize with hydrogen peroxide or other solution approved by Quality Assurance.
• Reinstall the conductivity probe.</td></tr>
<tr><td>12</td><td>Disconnect the resistance decade box and reconnect the conductivity cell to terminals A and C.</td></tr>
<tr><td>13</td><td>Disconnect the 8550 ohm resistor and connect the temperature compensation to terminals B and D.</td></tr>
<tr><td>14</td><td>At the analyzer, press CELL CONST then press CELL CONST and the up arrow simultaneously.</td></tr>
<tr><td>15#</td><td>• Use the up and down arrows to change the cell calibration factor to the value recorded on step 7.
• Press ENTER.</td></tr>
<tr><td>16#</td><td>• Restore the ALARM1 and ALARM2 to the values recorded on step 5 of the calibration form.</td></tr>
</table>

Continued on next page

Procedural steps (cont.)

Step	Action
17#	• At the analyzer, press CELL CONST. • Use the up and down arrows to change the cell constant to the value recorded on step 6.
18#	Verify proper operation of any alarms and/or control functions and record results.

Step	Action
19#	Complete the calibration form and affix the applicable calibration label.
20	If all checks are within tolerance, restore the conductivity indicator to operational condition and notify appropriate personnel of work performed.

Acceptance criteria

Personnel performing this maintenance shall use the following criteria to determine satisfactory completion, subject to review of data by supervision.

- Final conductivity indications obtained are within the tolerance specified on the calibration work order form.

If any readings are found to be out of tolerance or if any step cannot be performed:

- Notify the appropriate personnel for resolution.
- Record name of person contacted in comments section of work order form.

Attachments

The following are attachments to this work instruction:

No.	Title	Pages
A	Calibration/Calibration Check of Leads & Northrup 7082 Conductivity Analyzers Data Sheet	2

Continued on next page

References

Manual No.	Manual Title
N/A	Applicable manufacturer's technical manual

Approvals/ author

This work instruction must be approved by:

Manager, ABC Company Date

Quality, ABC Company Date

Author: Mike Cable, Engineering Technician

For Reference Only

SOP-CAL-13 Rev 0 Attachment A Page 1 of 2

CALIBRATION/CALIBRATION CHECK OF LEADS & NORTHRUP 7082 CONDUCTIVITY ANALYZERS DATA SHEET

For: PID Tag # _____

Step #	Parameter	Required	As-Found Data	As-Left Data	Tol Code
3, 10	Conductivity in DI water	____ ± 0.1 uMho/cm	_____ uMho/cm	_____ uMho/cm	____
5, 16	Alarm 1 Setpoint	NA	_____	_____	NA
5, 16	Alarm 2 Setpoint	NA	_____	_____	NA
6, 17	Cell Constant	NA	_____	_____	NA
7, 15	Cell Calibration Factor	NA	_____	_____	NA
9, 10	Conductivity at 10 Mohm	0.1 ± 0.1 uMho/cm	_____ uMho/cm	_____ uMho/cm	____
9, 10	Conductivity at 1 Mohm	1.0 ± 0.1 uMho/cm	_____ uMho/cm	_____ uMho/cm	____
9, 10	Conductivity at 100 Kohm	10.0 ± 0.1 uMho/cm	_____ uMho/cm	_____ uMho/cm	____
18	Alarm/Control Functions	Satisfactory	Sat / Unsat (circle one)	NA / Sat (circle one)	____

Standards Used:

	Description	Serial number or other I.D.	Cal Due Date
1	Decade Resistance Box	_____	_____
2	Digital Conductivity Standard	_____	_____
3	_____	_____	_____

Continued on next page

CALIBRATION/CALIBRATION CHECK OF LEADS & NORTHRUP 7082 CONDUCTIVITY ANALYZERS DATA SHEET (cont.)

Worse Case Tolerance Code: ____

Remarks:

_____

_____

_____

_____

Performed By: _____ **Date:** _____

Reviewed By: _____ **Date:** _____

Appendix A-5

TEST EQUIPMENT

Note: This list is for example purposes only and is not recommending a particular manufacturer/model number.

EQUIPMENT TYPE	EXAMPLE	SIMPLIFIED SPECIFICATIONS
Digital multimeter	Fluke 87, or equivalent	DCV ± (.05% +1 digit) DC Current ± (.2% +2 digit) ACV ± (.7% +1 digit) AC Current ± (1.0% +2 digit) Resistance ± (.2% +1 digit)
Analog multimeter	Simpson 560	DCV ± 2% FS DC Current ± 2% FS ACV ± 3% FS Resistance ± 2° of arc
Multi-function calibrator	Fluke 743B with applicable pressure modules	Multiple
Decade box	General Radio 1432A	± 0.05% reading
Thermocouple calibrator	Altek 322	± (0.008% rdg + 0.006 mv)
Temperature bath	Hart 7320	Range: –20C to 150C Stability/Uniformity: ± .005C
Temperature block	Hart 9103	–25C to 140C ± 0.25C
Precision RTD	Hart 1521	± 0.025C
HART communicator	Rosemount 275	NA
Floor scale	Mettler-Toledo	0.1% reading + 1 digit
Hand pressure/vacuum pump	Heise TP-1	N/A
Dead weight tester	Ashcroft 1305D	± 0.1% of reading
12″ analog test gauge	Heise CC	± 0.1% FS
4½ - 6 digit calibrated meter	Heise 910B	± 0.035% span

Appendix A-6

RTD AND THERMOCOUPLE TABLES

RTD vs. Resistance-Table for Alpha = .00385, ITS-90 (DIN 43760)

RTD vs. Resistance Table for Alpha = .00392, ITS-90

Type J Thermocouple Reference Tables

Type T Thermocouple Reference Tables

Type K Thermocouple Reference Tables

RTD Temperature vs. Resistance Table

For European Curve, Alpha = .00385, ITS-90

1° Celsius Increments

°C	Ohms	Diff.	°C	Ohms	Diff.	°C	Ohms	Diff.
-200	18.52		-140	43.96	0.42	-80	68.33	0.41
199	18.96	0.44	139	44.39	0.41	79	68.73	0.40
198	19.49	0.43	138	44.71	0.42	78	69.13	0.40
197	19.92	0.43	137	45.12	0.41	77	69.53	0.40
196	20.35	0.43	136	45.53	0.41	76	69.93	0.40
195	20.78	0.43	135	45.95	0.42	75	70.33	0.40
194	21.21	0.43	134	46.35	0.40	74	70.73	0.40
193	21.64	0.43	133	46.76	0.41	73	71.13	0.40
192	21.07	0.43	132	47.16	0.42	72	71.53	0.40
191	22.50	0.43	131	47.59	0.41	71	71.93	0.40
190	22.93	0.43	130	48.00	0.41	70	72.33	0.40
189	23.36	0.43	129	48.41	0.41	69	72.73	0.40
188	23.79	0.43	128	48.82	0.41	68	73.13	0.40
187	24.12	0.43	127	49.23	0.41	67	73.53	0.40
186	24.55	0.43	126	49.64	0.41	66	73.93	0.40
185	24.97	0.42	125	50.06	0.42	65	74.33	0.40
184	25.39	0.42	124	50.47	0.41	64	74.73	0.40
183	25.82	0.43	123	50.88	0.41	63	75.13	0.40
182	26.25	0.43	122	51.29	0.41	62	75.53	0.40
181	26.67	0.42	121	51.70	0.41	61	75.93	0.40
180	27.10	0.43	120	52.11	0.41	60	76.33	0.40
179	27.52	0.42	119	52.52	0.41	59	76.73	0.40
178	27.95	0.43	118	52.92	0.40	58	77.13	0.40
177	28.37	0.42	117	53.33	0.41	57	77.52	0.39
176	28.80	0.43	116	53.74	0.41	56	77.92	0.40
175	29.22	0.42	115	54.15	0.41	55	78.32	0.40
174	29.65	0.43	114	54.56	0.41	54	78.72	0.40
173	30.07	0.42	113	54.97	0.41	53	79.11	0.39
172	30.49	0.42	112	55.38	0.41	52	79.51	0.40
171	30.92	0.43	111	55.78	0.40	51	79.91	0.40
170	31.34	0.42	110	56.19	0.41	50	80.31	0.40
169	31.76	0.42	109	56.60	0.41	49	80.70	0.39
168	32.18	0.42	108	57.00	0.40	48	81.10	0.40
167	32.61	0.43	107	57.41	0.41	47	81.50	0.40
166	33.03	0.42	106	57.82	0.41	46	81.89	0.39
165	33.45	0.42	105	58.22	0.40	45	82.29	0.40
164	33.86	0.41	104	58.63	0.41	44	82.69	0.40
163	34.28	0.42	103	59.04	0.41	43	83.08	0.39
162	34.70	0.42	102	59.44	0.40	42	83.48	0.40
161	35.12	0.42	101	59.85	0.41	41	83.68	0.40
160	35.54	0.42	100	60.26	0.41	40	84.27	0.39
159	35.96	0.42	99	60.67	0.41	39	84.67	0.40
158	36.38	0.42	98	61.07	0.40	38	85.06	0.39
157	36.80	0.42	97	61.48	0.41	37	85.46	0.40
156	37.22	0.42	96	61.87	0.41	36	85.85	0.39
155	37.63	0.41	95	62.28	0.42	35	86.25	0.40
154	38.05	0.42	94	62.69	0.40	34	86.64	0.39
153	38.47	0.42	93	63.10	0.41	33	87.04	0.40
152	38.89	0.42	92	63.50	0.40	32	87.43	0.39
151	39.31	0.42	91	63.91	0.41	31	87.83	0.40
150	39.72	0.41	90	64.30	0.39	30	88.22	0.39
149	40.14	0.42	89	64.70	0.40	29	88.62	0.40
148	40.56	0.42	88	65.11	0.41	28	89.01	0.39
147	40.97	0.41	87	65.51	0.40	27	89.40	0.39
146	41.39	0.42	86	65.91	0.40	26	89.80	0.40
145	41.80	0.41	85	66.31	0.40	25	90.19	0.39
144	42.22	0.42	84	66.72	0.41	24	90.59	0.40
143	42.64	0.42	83	67.12	0.40	23	90.98	0.39
142	43.05	0.41	82	67.52	0.40	22	91.37	0.39
141	43.46	0.41	81	67.92	0.40	21	91.77	0.40
-20	92.16	0.39	±0	100.00	0.39	+60	123.24	0.38
19	92.55	0.39	+1	100.39	0.39	61	123.62	0.38
18	92.95	0.40	2	100.78	0.39	62	124.01	0.39
17	93.34	0.39	3	101.17	0.39	63	124.39	0.39
16	93.73	0.39	4	101.56	0.39	64	124.77	0.38
15	94.12	0.39	5	101.95	0.39	65	125.17	0.40
14	94.52	0.40	6	102.34	0.39	66	125.55	0.38
13	94.91	0.39	7	102.73	0.39	67	125.93	0.38
12	95.30	0.39	8	103.12	0.39	68	126.32	0.39
11	95.69	0.39	9	103.51	0.39	69	126.70	0.38
10	96.09	0.40	10	103.90	0.39	70	127.08	0.38
9	96.48	0.39	11	104.29	0.39	71	127.46	0.38
8	96.87	0.39	12	104.68	0.39	72	127.85	0.39
7	97.26	0.39	13	105.07	0.39	73	128.23	0.38
6	97.65	0.39	14	105.46	0.39	74	128.61	0.38
5	98.04	0.39	15	105.85	0.39	75	128.99	0.38
4	98.44	0.40	16	106.24	0.39	76	129.38	0.39
3	98.83	0.39	17	106.63	0.39	77	129.76	0.38
2	99.22	0.39	18	107.02	0.39	78	130.14	0.38
1	99.61	0.39	19	107.40	0.38	79	130.52	0.38
			20	107.79	0.39	80	130.90	0.38
			21	108.18	0.39	81	131.28	0.38
			22	108.57	0.39	82	131.67	0.39
			23	108.96	0.39	83	132.05	0.38
			24	109.35	0.39	84	132.43	0.38
			25	109.73	0.38	85	132.81	0.38
			26	110.12	0.39	86	133.19	0.38
			27	110.51	0.39	87	133.57	0.38
			28	110.90	0.39	88	133.95	0.38
			29	111.28	0.38	89	134.33	0.38
			30	111.67	0.39	90	134.71	0.38
			31	112.06	0.39	91	135.09	0.38
			32	112.45	0.39	92	135.47	0.38
			33	112.83	0.38	93	135.85	0.38
			34	113.22	0.39	94	136.23	0.38
			35	113.61	0.39	95	136.61	0.38
			36	113.99	0.38	96	136.99	0.38
			37	114.38	0.39	97	137.37	0.38
			38	114.77	0.39	98	137.75	0.38
			39	115.15	0.38	99	138.13	0.38
			40	115.54	0.39	100	138.51	0.38
			41	115.93	0.39	101	138.89	0.38
			42	116.31	0.38	102	139.27	0.38
			43	116.70	0.39	103	139.65	0.38
			44	117.09	0.39	104	140.03	0.38
			45	117.47	0.39	105	140.39	0.36
			46	117.85	0.38	106	140.77	0.38
			47	118.24	0.39	107	141.15	0.38
			48	118.62	0.38	108	141.53	0.38
			49	119.01	0.39	109	141.91	0.38
			50	119.40	0.39	110	142.29	0.38
			51	119.78	0.38	111	142.66	0.37
			52	120.16	0.38	112	143.04	0.38
			53	120.55	0.39	113	143.42	0.38
			54	120.93	0.38	114	143.80	0.38
			55	121.32	0.39	115	144.18	0.38
			56	121.70	0.38	116	144.56	0.38
			57	122.09	0.39	117	144.94	0.38
			58	122.47	0.38	118	145.32	0.38
			59	122.86	0.39	119	145.69	0.37

(DIN 43 760)

Note: At 100°C, resistance is 138.50 ohms.

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www.omega.com

RTD Temperature vs. Resistance Table

For European Curve, Alpha = .00385, ITS-90 — **1° Celsius Increments**

°C	Ohms	Diff.	°C	Ohms	Diff.	°C	Ohms	Diff.	°C	Ohms	Diff.	°C	Ohms	Diff.	°C	Ohms	Diff.
+120	146.07	0.38	+180	168.48	0.37	+240	190.47	0.36	+300	212.05	0.36	+360	233.21	0.35	+420	253.96	0.34
121	146.45	0.38	181	168.85	0.37	241	190.83	0.36	301	212.40	0.35	361	233.56	0.35	421	254.30	0.34
122	146.82	0.37	182	169.22	0.37	242	191.20	0.37	302	212.76	0.36	362	233.91	0.35	422	254.65	0.35
123	147.20	0.38	183	169.59	0.37	243	191.56	0.36	303	213.12	0.36	363	234.26	0.35	423	254.99	0.34
124	147.58	0.38	184	169.96	0.37	244	191.92	0.36	304	213.47	0.35	364	234.60	0.34	424	255.33	0.34
125	147.95	0.37	185	170.33	0.37	245	192.28	0.36	305	213.83	0.36	365	234.95	0.35	425	255.67	0.34
126	148.33	0.38	186	170.69	0.36	246	192.66	0.38	306	214.19	0.36	366	235.30	0.35	426	256.01	0.34
127	148.71	0.38	187	171.05	0.37	247	193.02	0.36	307	214.55	0.36	367	235.65	0.35	427	256.35	0.34
128	149.08	0.37	188	171.43	0.37	248	193.38	0.36	308	214.90	0.35	368	236.00	0.35	428	256.70	0.35
129	149.46	0.38	189	171.80	0.37	249	193.74	0.36	309	215.26	0.36	369	236.35	0.35	429	257.04	0.34
130	149.83	0.37	190	172.17	0.37	250	194.10	0.36	310	215.61	0.35	370	236.70	0.35	430	257.38	0.34
131	150.21	0.38	191	172.54	0.37	251	194.47	0.37	311	215.97	0.36	371	237.05	0.35	431	257.72	0.34
132	150.58	0.37	192	172.91	0.37	252	194.83	0.36	312	216.32	0.35	372	237.40	0.35	432	258.06	0.34
133	150.96	0.38	193	173.27	0.36	253	195.19	0.36	313	216.68	0.36	373	237.75	0.35	433	258.40	0.34
134	151.34	0.38	194	173.64	0.37	254	195.55	0.36	314	217.03	0.35	374	238.09	0.34	434	258.74	0.34
135	151.71	0.37	195	174.01	0.37	255	195.90	0.35	315	217.39	0.36	375	238.44	0.35	435	259.08	0.34
136	152.09	0.38	196	174.39	0.38	256	196.26	0.36	316	217.73	0.34	376	238.79	0.35	436	259.42	0.34
137	152.46	0.37	197	174.75	0.36	257	196.62	0.36	317	218.08	0.35	377	239.14	0.35	437	259.76	0.34
138	152.84	0.38	198	175.12	0.37	258	196.98	0.36	318	218.44	0.36	378	239.48	0.34	438	260.10	0.34
139	153.21	0.37	199	175.49	0.37	259	197.35	0.37	319	218.79	0.35	379	239.83	0.35	439	260.44	0.34
140	153.58	0.37	200	175.86	0.37	260	197.71	0.36	320	219.15	0.36	380	240.18	0.35	440	260.78	0.34
141	153.95	0.37	201	176.23	0.37	261	198.07	0.36	321	219.50	0.35	381	240.52	0.34	441	261.12	0.34
142	154.32	0.37	202	176.59	0.36	262	198.43	0.36	322	219.85	0.35	382	240.87	0.35	442	261.46	0.34
143	154.71	0.39	203	176.96	0.37	263	198.79	0.36	323	220.21	0.36	383	241.22	0.35	443	261.80	0.34
144	155.08	0.37	204	177.33	0.37	264	199.15	0.36	324	220.56	0.35	384	241.56	0.34	444	262.14	0.34
145	155.46	0.38	205	177.70	0.37	265	199.51	0.36	325	220.91	0.35	385	241.91	0.35	445	262.48	0.34
146	155.83	0.37	206	178.06	0.36	266	199.87	0.36	326	221.27	0.36	386	242.25	0.34	446	262.83	0.35
147	156.21	0.38	207	178.43	0.37	267	200.23	0.36	327	221.62	0.35	387	242.60	0.35	447	263.17	0.34
148	156.58	0.37	208	178.80	0.37	268	200.59	0.36	328	221.97	0.35	388	242.95	0.35	448	263.50	0.33
149	156.96	0.38	209	179.16	0.36	269	200.95	0.36	329	222.32	0.35	389	243.29	0.34	449	263.84	0.34
150	157.33	0.37	210	179.53	0.37	270	201.31	0.36	330	222.68	0.36	390	243.64	0.35	450	264.18	0.34
151	157.71	0.38	211	179.90	0.37	271	201.67	0.36	331	223.03	0.35	391	243.98	0.34	451	264.52	0.34
152	158.08	0.37	212	180.26	0.36	272	202.03	0.36	332	223.38	0.35	392	244.33	0.35	452	264.86	0.34
153	158.45	0.37	213	180.63	0.37	273	202.38	0.35	333	223.73	0.35	393	244.67	0.34	453	265.20	0.34
154	158.83	0.38	214	180.99	0.36	274	202.74	0.36	334	224.09	0.36	394	245.02	0.35	454	265.54	0.34
155	159.20	0.37	215	181.36	0.37	275	203.10	0.36	335	224.45	0.36	395	245.36	0.34	455	265.87	0.33
156	159.56	0.36	216	181.73	0.37	276	203.46	0.36	336	224.80	0.35	396	245.71	0.35	456	266.21	0.34
157	159.94	0.38	217	182.09	0.36	277	203.82	0.36	337	225.15	0.35	397	246.05	0.34	457	266.55	0.34
158	160.31	0.37	218	182.46	0.37	278	204.18	0.36	338	225.50	0.35	398	246.40	0.35	458	266.89	0.34
159	160.68	0.37	219	182.82	0.36	279	204.54	0.36	339	225.85	0.35	399	246.74	0.34	459	267.22	0.33
160	161.05	0.37	220	183.19	0.37	280	204.90	0.36	340	226.21	0.36	400	247.09	0.35	460	267.56	0.34
161	161.43	0.38	221	183.55	0.36	281	205.25	0.35	341	226.56	0.35	401	247.43	0.34	461	267.90	0.34
162	161.80	0.37	222	183.92	0.37	282	205.61	0.36	342	226.91	0.35	402	247.78	0.35	462	268.24	0.34
163	162.17	0.37	223	184.28	0.36	283	205.97	0.36	343	227.26	0.35	403	248.12	0.34	463	268.57	0.33
164	162.54	0.37	224	184.65	0.37	284	206.33	0.36	344	227.61	0.35	404	248.46	0.34	464	268.91	0.34
165	162.91	0.37	225	185.01	0.36	285	206.70	0.37	345	227.96	0.35	405	248.81	0.35	465	269.25	0.34
166	163.28	0.37	226	185.38	0.37	286	207.05	0.35	346	228.31	0.35	406	249.15	0.34	466	269.58	0.33
167	163.66	0.38	227	185.74	0.36	287	207.41	0.36	347	228.66	0.35	407	249.50	0.35	467	269.92	0.34
168	164.03	0.37	228	186.11	0.37	288	207.77	0.36	348	229.01	0.35	408	249.84	0.34	468	270.26	0.34
169	164.40	0.37	229	186.47	0.36	289	208.13	0.36	349	229.36	0.35	409	250.18	0.34	469	270.59	0.33
170	164.77	0.37	230	186.84	0.37	290	208.48	0.35	350	229.72	0.34	410	250.53	0.35	470	270.93	0.34
171	165.14	0.37	231	187.20	0.36	291	208.84	0.36	351	230.07	0.35	411	250.89	0.34	471	271.27	0.34
172	165.51	0.37	232	187.56	0.36	292	209.20	0.36	352	230.42	0.35	412	251.21	0.34	472	271.60	0.33
173	165.88	0.37	233	187.93	0.37	293	209.55	0.35	353	230.77	0.35	413	251.55	0.34	473	271.94	0.34
174	166.25	0.37	234	188.29	0.36	294	209.91	0.36	354	231.12	0.35	414	251.90	0.35	474	272.27	0.33
175	166.62	0.37	235	188.65	0.36	295	210.27	0.36	355	231.47	0.35	415	252.24	0.34	475	272.61	0.34
176	167.00	0.38	236	189.02	0.37	296	210.62	0.35	356	231.81	0.36	416	252.59	0.35	476	272.95	0.34
177	167.37	0.37	237	189.38	0.36	297	210.98	0.36	357	232.16	0.35	417	252.94	0.35	477	273.28	0.33
178	167.74	0.37	238	189.74	0.36	298	211.34	0.36	358	232.51	0.35	418	253.28	0.34	478	273.62	0.34
179	168.11	0.37	239	190.11	0.37	299	211.69	0.35	359	232.86	0.35	419	253.62	0.34	479	273.95	0.33

(DIN 43 760)

Note: At 100°C, resistance is 138.50 ohms.

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RTD Temperature vs. Resistance Table

For European Curve, Alpha = .00385, ITS-90

1° Celsius Increments

°C	Ohms	Diff.	°C	Ohms	Diff.	°C	Ohms	Diff.	°C	Ohms	Diff.	°C	Ohms	Diff.	°C	Ohms	Diff.
+400	274.20	0.34	+542	294.87	0.33	+604	315.00	0.32	+666	334.68	0.32	+728	353.91	0.30	+790	372.71	0.30
401	274.62	0.33	543	295.20	0.33	605	315.32	0.32	667	334.99	0.31	729	354.22	0.31	791	373.01	0.30
402	274.96	0.34	544	295.53	0.33	606	315.64	0.32	668	335.31	0.32	730	354.53	0.31	792	373.31	0.30
403	275.29	0.33	545	295.85	0.32	607	315.96	0.32	669	335.62	0.31	731	354.83	0.30	793	373.61	0.30
404	275.63	0.34	546	296.18	0.33	608	316.28	0.32	670	335.93	0.31	732	355.14	0.31	794	373.91	0.30
405	275.96	0.33	547	296.51	0.33	609	316.60	0.32	671	336.25	0.32	733	355.44	0.30	795	374.21	0.30
406	276.31	0.34	548	296.84	0.33	610	316.92	0.32	672	336.56	0.31	734	355.75	0.31	796	374.51	0.30
407	276.64	0.33	549	297.16	0.32	611	317.24	0.32	673	336.87	0.31	735	356.06	0.31	797	374.80	0.29
408	276.97	0.33	550	297.49	0.33	612	317.56	0.32	674	337.18	0.31	736	356.37	0.31	798	375.10	0.30
409	277.31	0.34	551	297.82	0.33	613	317.88	0.32	675	337.50	0.32	737	356.68	0.31	799	375.40	0.30
410	277.64	0.33	552	298.14	0.32	614	318.20	0.32	676	337.81	0.31	738	356.98	0.30	800	375.70	0.30
411	277.98	0.34	553	298.47	0.33	615	318.52	0.32	677	338.12	0.31	739	357.29	0.31	801	376.00	0.30
412	278.31	0.33	554	298.80	0.33	616	318.85	0.33	678	338.43	0.31	740	357.59	0.30	802	376.29	0.29
413	278.64	0.33	555	299.12	0.32	617	319.17	0.32	679	338.75	0.32	741	357.90	0.31	803	376.59	0.30
414	278.98	0.34	556	299.45	0.33	618	319.49	0.32	680	339.06	0.31	742	358.20	0.30	804	376.89	0.30
415	279.31	0.33	557	299.78	0.33	619	319.81	0.32	681	339.37	0.31	743	358.51	0.31	805	377.19	0.30
416	279.64	0.33	558	300.10	0.32	620	320.12	0.31	682	339.68	0.31	744	358.81	0.30	806	377.49	0.30
417	279.98	0.34	559	300.43	0.33	621	320.44	0.32	683	339.99	0.31	745	359.12	0.31	807	377.79	0.30
418	280.31	0.33	560	300.75	0.32	622	320.76	0.32	684	340.30	0.31	746	359.42	0.30	808	378.09	0.30
419	280.64	0.33	561	301.08	0.33	623	321.08	0.32	685	340.62	0.32	747	359.72	0.30	809	378.39	0.30
420	280.98	0.34	562	301.41	0.33	624	321.40	0.32	686	340.94	0.32	748	360.03	0.31	810	378.68	0.29
421	281.31	0.33	563	301.73	0.32	625	321.72	0.32	687	341.25	0.31	749	360.33	0.30	811	378.98	0.30
422	281.64	0.33	564	302.06	0.33	626	322.03	0.31	688	341.55	0.30	750	360.64	0.31	812	379.28	0.30
423	281.97	0.33	565	302.38	0.32	627	322.34	0.31	689	341.87	0.32	751	360.94	0.30	813	379.57	0.29
424	282.31	0.34	566	302.71	0.33	628	322.66	0.32	690	342.18	0.31	752	361.24	0.30	814	379.87	0.30
425	282.64	0.33	567	303.03	0.32	629	322.98	0.32	691	342.49	0.31	753	361.55	0.31	815	380.17	0.30
426	282.97	0.33	568	303.36	0.33	630	323.30	0.32	692	342.80	0.31	754	361.85	0.30	816	380.46	0.29
427	283.30	0.33	569	303.68	0.32	631	323.61	0.31	693	343.11	0.31	755	362.15	0.30	817	380.76	0.30
428	283.63	0.33	570	304.01	0.33	632	323.93	0.32	694	343.42	0.31	756	362.46	0.31	818	381.05	0.29
429	283.97	0.34	571	304.33	0.32	633	324.25	0.32	695	343.73	0.31	757	362.76	0.30	819	381.35	0.30
430	284.30	0.33	572	304.66	0.33	634	324.57	0.32	696	344.04	0.31	758	363.06	0.30	820	381.65	0.30
431	284.63	0.33	573	304.98	0.32	635	324.88	0.31	697	344.35	0.31	759	363.36	0.30	821	381.94	0.29
432	284.96	0.33	574	305.30	0.32	636	325.21	0.33	698	344.66	0.31	760	363.67	0.31	822	382.24	0.30
433	285.29	0.33	575	305.63	0.33	637	325.53	0.32	699	344.97	0.31	761	363.97	0.30	823	382.53	0.29
434	285.62	0.33	576	305.95	0.32	638	325.85	0.32	700	345.28	0.31	762	364.27	0.30	824	382.83	0.30
435	285.95	0.33	577	306.28	0.33	639	326.16	0.31	701	345.59	0.31	763	364.57	0.30	825	383.12	0.29
436	286.30	0.35	578	306.60	0.32	640	326.48	0.32	702	345.90	0.31	764	364.88	0.31	826	383.42	0.30
437	286.63	0.33	579	306.92	0.32	641	326.79	0.31	703	346.21	0.31	765	365.18	0.30	827	383.71	0.29
438	286.96	0.33	580	307.25	0.33	642	327.11	0.32	704	346.52	0.31	766	365.49	0.31	828	384.01	0.30
439	287.29	0.33	581	307.57	0.32	643	327.43	0.32	705	346.83	0.31	767	365.79	0.30	829	384.30	0.29
440	287.62	0.33	582	307.89	0.32	644	327.74	0.31	706	347.15	0.32	768	366.09	0.30	830	384.60	0.30
441	287.95	0.33	583	308.22	0.33	645	328.06	0.32	707	347.46	0.31	769	366.40	0.31	831	384.89	0.29
442	288.28	0.33	584	308.54	0.32	646	328.38	0.32	708	347.76	0.30	770	366.70	0.30	832	385.18	0.29
443	288.61	0.33	585	308.86	0.32	647	328.69	0.31	709	348.07	0.31	771	367.00	0.30	833	385.48	0.30
444	288.94	0.33	586	309.18	0.33	648	329.01	0.32	710	348.38	0.31	772	367.30	0.30	834	385.77	0.29
445	289.27	0.33	587	309.51	0.32	649	329.32	0.31	711	348.69	0.31	773	367.60	0.30	835	386.07	0.30
446	289.60	0.33	588	309.83	0.32	650	329.64	0.32	712	349.00	0.31	774	367.90	0.30	836	386.37	0.30
447	289.93	0.33	589	310.15	0.32	651	329.95	0.31	713	349.31	0.31	775	368.20	0.30	837	386.66	0.29
448	290.26	0.33	590	310.48	0.33	652	330.27	0.32	714	349.61	0.30	776	368.50	0.30	838	386.96	0.30
449	290.59	0.33	591	310.80	0.32	653	330.58	0.31	715	349.92	0.31	777	368.81	0.31	839	387.25	0.29
450	290.92	0.33	592	311.12	0.32	654	330.90	0.32	716	350.23	0.31	778	369.11	0.30	840	387.55	0.30
451	291.25	0.33	593	311.45	0.33	655	331.21	0.31	717	350.54	0.31	779	369.41	0.30	841	387.84	0.29
452	291.58	0.33	594	311.78	0.33	656	331.53	0.32	718	350.85	0.31	780	369.71	0.30	842	388.13	0.29
453	291.90	0.32	595	312.10	0.32	657	331.84	0.31	719	351.15	0.30	781	370.01	0.30	843	388.42	0.29
454	292.23	0.33	596	312.43	0.33	658	332.16	0.32	720	351.46	0.31	782	370.31	0.30	844	388.72	0.30
455	292.56	0.33	597	312.75	0.32	659	332.47	0.31	721	351.77	0.31	783	370.61	0.30	845	389.01	0.29
456	292.90	0.34	598	313.07	0.32	660	332.79	0.32	722	352.07	0.30	784	370.91	0.30	846	389.31	0.30
457	293.23	0.33	599	313.39	0.32	661	333.10	0.31	723	352.38	0.31	785	371.21	0.30	847	389.61	0.30
458	293.56	0.33	600	313.71	0.32	662	333.41	0.31	724	352.69	0.31	786	371.52	0.31	848	389.90	0.29
459	293.89	0.33	601	314.04	0.33	663	333.73	0.32	725	352.99	0.30	787	371.82	0.30	849	390.19	0.29
460	294.21	0.32	602	314.36	0.32	664	334.04	0.31	726	353.30	0.31	788	372.12	0.30	850	390.48	0.29
461	294.54	0.33	603	314.68	0.32	665	334.36	0.32	727	353.61	0.31	789	372.41	0.29			

Note: At 100°C, resistance is 138.50 ohms.

(DIN 43 760)

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RTD Temperature vs. Resistance Table

For American Curve, Alpha = .00392

1° Celsius Increments

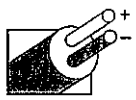
°C	Ohms	°C	Ohms	°C	Ohms	°C	Ohms	°C	Ohms	°C	Ohms	°C	Ohms	°C	Ohms	°C	Ohms
-100	59.57	-38	84.80	24	109.51	86	133.75	148	157.53	210	180.86	272	203.74	334	226.17	396	248.16
-99	59.8	-37	85.20	25	109.90	87	134.14	149	157.91	211	181.23	273	204.11	335	226.53	397	248.51
-98	60.2	-36	85.60	26	110.30	88	134.52	150	158.29	212	181.61	274	204.47	336	226.89	398	248.86
-97	60.6	-35	86.01	27	110.69	89	134.91	151	158.67	213	181.98	275	204.84	337	227.25	399	249.21
-96	61.0	-34	86.41	28	111.09	90	135.30	152	159.05	214	182.35	276	205.20	338	227.61	400	249.56
-95	61.4	-33	86.81	29	111.48	91	135.68	153	159.43	215	182.72	277	205.57	339	227.96	401	249.91
-94	62.0	-32	87.21	30	111.88	92	136.07	154	159.81	216	183.09	278	205.93	340	228.32	402	250.26
-93	62.4	-31	87.61	31	112.27	93	136.46	155	160.19	217	183.47	279	206.30	341	228.68	403	250.61
-92	62.8	-30	88.01	32	112.66	94	136.84	156	160.57	218	183.84	280	206.66	342	229.04	404	250.96
-91	63.2	-29	88.42	33	113.06	95	137.23	157	160.95	219	184.21	281	207.02	343	229.39	405	251.31
-90	63.6	-28	88.82	34	113.45	96	137.62	158	161.33	220	184.58	282	207.39	344	229.75	406	251.66
-89	64.0	-27	89.22	35	113.84	97	138.00	159	161.70	221	184.95	283	207.75	345	230.11	407	252.01
-88	64.5	-26	89.62	36	114.24	98	138.39	160	162.08	222	185.32	284	208.12	346	230.46	408	252.36
-87	64.9	-25	90.02	37	114.63	99	138.77	161	162.46	223	185.70	285	208.48	347	230.82	409	252.71
-86	65.3	-24	90.42	38	115.02	100	139.16	162	162.84	224	186.07	286	208.85	348	231.18	410	253.06
-85	65.7	-23	90.82	39	115.42	101	139.55	163	163.22	225	186.44	287	209.21	349	231.53	411	253.41
-84	66.1	-22	91.22	40	115.81	102	139.93	164	163.60	226	186.81	288	209.57	350	231.89	412	253.76
-83	66.5	-21	91.62	41	116.20	103	140.32	165	163.97	227	187.18	289	209.94	351	232.25	413	254.11
-82	66.9	-20	92.02	42	116.59	104	140.70	166	164.35	228	187.55	290	210.30	352	232.60	414	254.46
-81	67.3	-19	92.42	43	116.99	105	141.09	167	164.73	229	187.92	291	210.66	353	232.96	415	254.80
-80	67.7	-18	92.82	44	117.38	106	141.47	168	165.11	230	188.29	292	211.03	354	233.31	416	255.15
-79	68.1	-17	93.22	45	117.77	107	141.86	169	165.48	231	188.66	293	211.39	355	233.67	417	255.50
-78	68.5	-16	93.62	46	118.16	108	142.24	170	165.86	232	189.03	294	211.75	356	234.03	418	255.85
-77	69.0	-15	94.02	47	118.56	109	142.63	171	166.24	233	189.40	295	212.11	357	234.38	419	256.20
-76	69.4	-14	94.42	48	118.95	110	143.01	172	166.62	234	189.77	296	212.48	358	234.74	420	256.55
-75	69.81	-13	94.82	49	119.34	111	143.39	173	166.99	235	190.14	297	212.84	359	235.09	421	256.89
-74	70.2	-12	95.22	50	119.73	112	143.78	174	167.37	236	190.51	298	213.20	360	235.45	422	257.24
-73	70.6	-11	95.62	51	120.12	113	144.16	175	167.75	237	190.88	299	213.56	361	235.80	423	257.59
-72	71.04	-10	96.02	52	120.51	114	144.55	176	168.12	238	191.25	300	213.93	362	236.16	424	257.94
-71	71.44	-9	96.42	53	120.91	115	144.93	177	168.50	239	191.62	301	214.29	363	236.51	425	258.29
-70	71.85	-8	96.81	54	121.30	116	145.31	178	168.88	240	191.99	302	214.65	364	236.87	426	258.63
-69	72.26	-7	97.21	55	121.69	117	145.70	179	169.25	241	192.35	303	215.01	365	237.22	427	258.98
-68	72.66	-6	97.61	56	122.08	118	146.08	180	169.63	242	192.73	304	215.37	366	237.58	428	259.33
-67	73.07	-5	98.01	57	122.47	119	146.47	181	170.00	243	193.09	305	215.74	367	237.93	429	259.67
-66	73.48	-4	98.41	58	122.86	120	146.85	182	170.38	244	193.46	306	216.10	368	238.28	430	260.02
-65	73.88	-3	98.81	59	123.25	121	147.23	183	170.76	245	193.83	307	216.46	369	238.64	431	260.37
-64	74.29	-2	99.20	60	123.64	122	147.61	184	171.13	246	194.20	308	216.82	370	238.99	432	260.72
-63	74.70	-1	99.60	61	124.03	123	148.00	185	171.51	247	194.57	309	217.18	371	239.35	433	261.06
-62	75.10	0	100.00	62	124.42	124	148.38	186	171.88	248	194.94	310	217.54	372	239.70	434	261.41
-61	75.51	1	100.40	63	124.81	125	148.75	187	172.26	249	195.31	311	217.90	373	240.05	435	261.75
-60	75.91	2	100.80	64	125.20	126	149.15	188	172.63	250	195.67	312	218.26	374	240.41	436	262.10
-59	76.32	3	101.19	65	125.59	127	149.53	189	173.01	251	196.04	313	218.63	375	240.76	437	262.45
-58	76.72	4	101.59	66	125.98	128	149.91	190	173.38	252	196.41	314	218.99	376	241.11	438	262.79
-57	77.13	5	101.99	67	126.37	129	150.29	191	173.76	253	196.78	315	219.35	377	241.47	439	263.14
-56	77.53	6	102.38	68	126.76	130	150.67	192	174.13	254	197.14	316	219.71	378	241.82	440	263.49
-55	77.94	7	102.78	69	127.15	131	151.06	193	174.51	255	197.51	317	220.07	379	242.17	441	263.83
-54	78.34	8	103.18	70	127.54	132	151.44	194	174.88	256	197.88	318	220.43	380	242.53	442	264.18
-53	78.75	9	103.57	71	127.93	133	151.82	195	175.26	257	198.25	319	220.79	381	242.88	443	264.52
-52	79.15	10	103.97	72	128.32	134	152.20	196	175.63	258	198.61	320	221.15	382	243.23	444	264.87
-51	79.56	11	104.37	73	128.71	135	152.58	197	176.01	259	198.98	321	221.51	383	243.58	445	265.21
-50	79.96	12	104.76	74	129.09	136	152.96	198	176.38	260	199.35	322	221.87	384	243.94	446	265.56
-49	80.36	13	105.16	75	129.48	137	153.35	199	176.75	261	199.71	323	222.23	385	244.29	447	265.90
-48	80.77	14	105.56	76	129.87	138	153.73	200	177.13	262	200.08	324	222.59	386	244.64	448	266.25
-47	81.17	15	105.95	77	130.26	139	154.11	201	177.50	263	200.45	325	222.94	387	244.99	449	266.59
-46	81.58	16	106.35	78	130.65	140	154.49	202	177.88	264	200.81	326	223.30	388	245.35	450	266.94
-45	81.98	17	106.74	79	131.04	141	154.87	203	178.25	265	201.18	327	223.66	389	245.70	451	267.28
-44	82.38	18	107.14	80	131.42	142	155.25	204	178.62	266	201.55	328	224.02	390	246.05	452	267.63
-43	82.79	19	107.53	81	131.81	143	155.63	205	179.00	267	201.91	329	224.38	391	246.40	453	267.97
-42	83.19	20	107.93	82	132.20	144	156.01	206	179.37	268	202.28	330	224.74	392	246.75	454	268.31
-41	83.59	21	108.32	83	132.59	145	156.39	207	179.74	269	202.64	331	225.10	393	247.10	455	268.66
-40	83.99	22	108.72	84	132.98	146	156.77	208	180.12	270	203.01	332	225.46	394	247.46	456	269.00
-39	84.40	23	109.11	85	133.36	147	157.15	209	180.49	271	203.38	333	225.81	395	247.81	457	269.35

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Revised Thermocouple Reference Tables

TYPE J

Reference Tables
N.I.S.T. Monograph 175
Revised to ITS-90



Thermocouple Grade

Iron vs. Copper-Nickel



Extension Grade

MAXIMUM TEMPERATURE RANGE

Thermocouple Grade
32 to 1382°F
0 to 750°C

Extension Grade
32 to 392°F
0 to 200°C

LIMITS OF ERROR
(whichever is greater)
Standard: 2.2°C or 0.75%
Special: 1.1°C or 0.4%

COMMENTS, BARE WIRE ENVIRONMENT:
Reducing, Vacuum, Inert; Limited Use in Oxidizing at High Temperatures; Not Recommended for Low Temperatures

TEMPERATURE IN DEGREES °C
REFERENCE JUNCTION AT 0°C

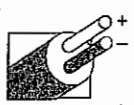
Thermoelectric Voltage in Millivolts

°C	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	°C
-200	-8.095	-8.078	-8.057	-8.037	-8.017	-7.996	-7.976	-7.953	-7.934	-7.912	-7.890	-200
-190	-7.890	-7.868	-7.846	-7.824	-7.801	-7.778	-7.755	-7.731	-7.707	-7.683	-7.659	-190
-180	-7.659	-7.634	-7.610	-7.585	-7.559	-7.534	-7.508	-7.483	-7.458	-7.432	-7.403	-180
-170	-7.403	-7.376	-7.349	-7.321	-7.293	-7.265	-7.237	-7.209	-7.181	-7.152	-7.123	-170
-160	-7.123	-7.094	-7.064	-7.035	-7.005	-6.975	-6.944	-6.914	-6.883	-6.852	-6.821	-160
-150	-6.821	-6.790	-6.758	-6.727	-6.695	-6.663	-6.631	-6.598	-6.566	-6.533	-6.500	-150
-140	-6.500	-6.467	-6.433	-6.400	-6.366	-6.332	-6.298	-6.263	-6.229	-6.194	-6.159	-140
-130	-6.159	-6.124	-6.089	-6.054	-6.018	-5.982	-5.946	-5.910	-5.874	-5.838	-5.801	-130
-120	-5.801	-5.764	-5.727	-5.690	-5.653	-5.616	-5.578	-5.541	-5.503	-5.465	-5.426	-120
-110	-5.426	-5.388	-5.350	-5.311	-5.272	-5.233	-5.194	-5.155	-5.116	-5.076	-5.037	-110
-100	-5.037	-4.997	-4.957	-4.917	-4.877	-4.836	-4.796	-4.755	-4.714	-4.674	-4.633	-100
-90	-4.633	-4.591	-4.550	-4.509	-4.467	-4.425	-4.384	-4.342	-4.300	-4.257	-4.215	-90
-80	-4.215	-4.173	-4.130	-4.088	-4.045	-4.002	-3.959	-3.916	-3.872	-3.829	-3.786	-80
-70	-3.786	-3.742	-3.698	-3.654	-3.610	-3.566	-3.522	-3.478	-3.434	-3.389	-3.344	-70
-60	-3.344	-3.300	-3.255	-3.210	-3.165	-3.120	-3.075	-3.029	-2.984	-2.938	-2.893	-60
-50	-2.893	-2.847	-2.801	-2.755	-2.709	-2.663	-2.617	-2.571	-2.524	-2.478	-2.431	-50
-40	-2.431	-2.385	-2.338	-2.291	-2.244	-2.197	-2.150	-2.103	-2.056	-2.008	-1.961	-40
-30	-1.961	-1.913	-1.865	-1.818	-1.770	-1.722	-1.674	-1.626	-1.578	-1.530	-1.482	-30
-20	-1.482	-1.433	-1.385	-1.336	-1.288	-1.239	-1.190	-1.142	-1.093	-1.044	-0.995	-20
-10	-0.995	-0.946	-0.896	-0.847	-0.798	-0.749	-0.699	-0.650	-0.600	-0.550	-0.501	-10
0	-0.501	-0.451	-0.401	-0.351	-0.301	-0.251	-0.201	-0.151	-0.101	-0.050	0.000	0
0	0.000	0.050	0.101	0.151	0.202	0.253	0.303	0.354	0.405	0.456	0.507	0
10	0.507	0.558	0.609	0.660	0.711	0.762	0.814	0.865	0.916	0.968	1.019	10
20	1.019	1.071	1.122	1.174	1.226	1.277	1.329	1.381	1.433	1.485	1.537	20
30	1.537	1.589	1.641	1.693	1.745	1.797	1.849	1.902	1.954	2.006	2.059	30
40	2.059	2.111	2.164	2.216	2.269	2.322	2.374	2.427	2.480	2.532	2.585	40
50	2.585	2.638	2.691	2.744	2.797	2.850	2.903	2.956	3.009	3.062	3.116	50
60	3.116	3.169	3.222	3.275	3.329	3.382	3.436	3.489	3.543	3.595	3.650	60
70	3.650	3.703	3.757	3.810	3.864	3.918	3.971	4.025	4.079	4.133	4.187	70
80	4.187	4.240	4.294	4.348	4.402	4.456	4.510	4.564	4.618	4.672	4.726	80
90	4.726	4.781	4.835	4.889	4.943	4.997	5.052	5.106	5.160	5.215	5.269	90
100	5.269	5.323	5.378	5.432	5.487	5.541	5.595	5.650	5.705	5.759	5.814	100
110	5.814	5.868	5.923	5.977	6.032	6.087	6.141	6.196	6.251	6.306	6.360	110
120	6.360	6.415	6.470	6.525	6.579	6.634	6.689	6.744	6.799	6.854	6.909	120
130	6.909	6.964	7.019	7.074	7.129	7.184	7.239	7.294	7.349	7.404	7.459	130
140	7.459	7.514	7.569	7.624	7.679	7.734	7.789	7.844	7.900	7.955	8.010	140
150	8.010	8.065	8.120	8.175	8.231	8.286	8.341	8.396	8.452	8.507	8.562	150
160	8.562	8.618	8.673	8.728	8.783	8.839	8.894	8.949	9.005	9.060	9.115	160
170	9.115	9.171	9.226	9.282	9.337	9.392	9.448	9.503	9.559	9.614	9.669	170
180	9.669	9.725	9.780	9.836	9.891	9.947	10.002	10.057	10.113	10.168	10.224	180
190	10.224	10.279	10.335	10.390	10.446	10.501	10.557	10.612	10.668	10.723	10.779	190
200	10.779	10.834	10.890	10.945	11.001	11.056	11.112	11.167	11.223	11.278	11.334	200
210	11.334	11.389	11.445	11.501	11.556	11.612	11.667	11.723	11.778	11.834	11.889	210
220	11.889	11.945	12.000	12.056	12.111	12.167	12.222	12.278	12.334	12.389	12.445	220
230	12.445	12.500	12.556	12.611	12.667	12.722	12.778	12.833	12.889	12.944	13.000	230
240	13.000	13.056	13.111	13.167	13.222	13.278	13.333	13.389	13.444	13.500	13.555	240
250	13.555	13.611	13.666	13.722	13.777	13.833	13.888	13.944	13.999	14.055	14.110	250
260	14.110	14.166	14.221	14.277	14.332	14.388	14.443	14.499	14.554	14.609	14.665	260
270	14.665	14.720	14.776	14.831	14.887	14.942	14.998	15.053	15.109	15.164	15.219	270
280	15.219	15.275	15.330	15.386	15.441	15.496	15.552	15.607	15.663	15.718	15.773	280
290	15.773	15.829	15.884	15.940	15.995	16.050	16.106	16.161	16.216	16.272	16.327	290
300	16.327	16.383	16.438	16.493	16.549	16.604	16.659	16.715	16.770	16.825	16.881	300
310	16.881	16.936	16.991	17.046	17.102	17.157	17.212	17.268	17.323	17.378	17.434	310
320	17.434	17.489	17.544	17.599	17.655	17.710	17.765	17.820	17.876	17.931	17.986	320
330	17.986	18.041	18.097	18.152	18.207	18.262	18.318	18.373	18.428	18.483	18.538	330
340	18.538	18.594	18.649	18.704	18.759	18.814	18.870	18.925	18.980	19.035	19.090	340
350	19.090	19.146	19.201	19.256	19.311	19.366	19.422	19.477	19.532	19.587	19.642	350
360	19.642	19.697	19.753	19.808	19.863	19.918	19.973	20.028	20.083	20.139	20.194	360
370	20.194	20.249	20.304	20.359	20.414	20.469	20.525	20.580	20.635	20.690	20.745	370
380	20.745	20.800	20.855	20.911	20.966	21.021	21.076	21.131	21.186	21.241	21.297	380
390	21.297	21.352	21.407	21.462	21.517	21.572	21.627	21.683	21.738	21.793	21.848	390
400	21.848	21.903	21.958	22.014	22.069	22.124	22.179	22.234	22.289	22.345	22.400	400
410	22.400	22.455	22.510	22.565	22.620	22.676	22.731	22.786	22.841	22.896	22.952	410
420	22.952	23.007	23.062	23.117	23.172	23.228	23.283	23.338	23.393	23.449	23.504	420
430	23.504	23.559	23.614	23.670	23.725	23.780	23.835	23.891	23.946	24.001	24.057	430
440	24.057	24.112	24.167	24.223	24.278	24.333	24.389	24.444	24.499	24.555	24.610	440
450	24.610	24.665	24.721	24.776	24.832	24.887	24.943	24.998	25.053	25.109	25.164	450
460	25.164	25.220	25.275	25.331	25.386	25.442	25.497	25.553	25.608	25.664	25.720	460
470	25.720	25.775	25.831	25.886	25.942	25.998	26.053	26.109	26.165	26.220	26.276	470
480	26.276	26.332	26.387	26.443	26.499	26.555	26.610	26.666	26.722	26.778	26.834	480
490	26.834	26.889	26.945	27.001	27.057	27.113	27.169	27.225	27.281	27.337	27.393	490
°C	0	1	2	3	4	5	6	7	8	9	10	°C

°C	0	1	2	3	4	5	6	7	8	9	10	°C
500	27.393	27.449	27.505	27.561	27.617	27.673	27.729	27.785	27.841	27.897	27.953	500
510	27.953	28.010	28.066	28.122	28.178	28.234	28.291	28.347	28.403	28.460	28.516	510
520	28.516	28.572	28.629	28.685	28.741	28.798	28.854	28.911	28.967	29.024	29.080	520
530	29.080	29.137	29.193	29.250	29.307	29.363	29.420	29.477	29.533	29.590	29.647	530
540	29.647	29.704	29.761	29.818	29.874	29.931	29.988	30.045	30.102	30.159	30.216	540
550	30.216	30.273	30.330	30.387	30.444	30.502	30.559	30.616	30.673	30.730	30.788	550
560	30.788	30.845	30.902	30.960	31.017	31.074	31.132	31.189	31.247	31.304	31.362	560
570	31.362	31.419	31.477	31.535	31.592	31.650	31.708	31.766	31.823	31.881	31.939	570
580	31.939	31.997	32.055	32.113	32.171	32.229	32.287	32.345	32.403	32.461	32.519	580
590	32.519	32.577	32.636	32.694	32.752	32.810	32.869	32.927	32.985	33.044	33.102	590
600	33.102	33.161	33.219	33.278	33.337	33.395	33.454	33.513	33.571	33.630	33.689	600
610	33.689	33.747	33.806	33.865	33.924	33.983	34.043	34.102	34.161	34.220	34.279	610
620	34.279	34.338	34.397	34.457	34.516	34.575	34.635	34.694	34.754	34.813	34.873	620
630	34.873	34.932	34.992	35.051	35.111	35.171	35.230	35.290	35.350	35.410	35.470	630
640	35.470	35.530	35.590	35.650	35.710	35.770	35.830	35.890	35.950	36.010	36.071	640
650	36.071	36.131	36.191	36.252	36.312	36.373	36.433	36.494	36.554	36.615	36.675	650
660	36.675	36.736	36.797	36.858	36.918	36.979	37.040	37.101	37.162	37.223	37.284	660
670	37.284	37.345	37.406	37.467	37.528	37.590	37.651	37.712	37.773	37.835	37.896	670
680	37.896	37.958	38.019	38.081	38.142	38.204	38.265	38.327	38.389	38.450	38.512	680
690	38.512	38.574	38.636	38.698	38.760	38.822	38.884	38.946	39.008	39.070	39.132	690
700	39.132	39.194	39.256	39.318	39.381	39.443	39.505	39.568	39.630	39.693	39.755	700
710	39.755	39.818	39.880	39.943	40.005	40.068	40.131	40.193	40.256	40.319	40.382	710
720	40.382	40.445	40.508	40.570	40.633	40.696	40.759	40.822	40.886	40.949	41.012	720
730	41.012	41.075	41.138	41.201	41.265	41.328	41.391	41.455	41.518	41.581	41.645	730
740	41.645	41.708	41.772	41.835	41.899	41.962	42.026	42.090	42.153	42.217	42.281	740
750	42.281	42.344	42.408	42.472	42.536	42.599	42.663	42.727	42.791	42.855	42.919	750
760	42.919	42.983	43.047	43.111	43.175	43.239	43.303	43.367	43.431	43.495	43.559	760
770	43.559	43.624	43.688	43.752	43.817	43.881	43.945	44.010	44.074	44.139	44.203	770
780	44.203	44.267	44.332	44.396	44.461	44.525	44.590	44.655	44.719	44.784	44.848	780
790	44.848	44.913	44.977	45.042	45.107	45.171	45.236	45.301	45.365	45.430	45.494	790
800	45.494	45.559	45.624	45.688	45.753	45.818	45.882	45.947	46.011	46.076	46.141	800
810	46.141	46.205	46.270	46.334	46.399	46.464	46.528	46.593	46.657	46.722	46.786	810
820	46.786	46.851	46.915	46.980	47.044	47.109	47.173	47.238	47.302	47.367	47.431	820
830	47.431	47.495	47.560	47.624	47.688	47.753	47.817	47.881	47.946	48.010	48.074	830
840	48.074	48.138	48.202	48.267	48.331	48.395	48.459	48.523	48.587	48.651	48.715	840
850	48.715	48.779	48.843	48.907	48.971	49.034	49.098	49.162	49.226	49.290	49.353	850
860	49.353	49.417	49.481	49.544	49.608	49.672	49.735	49.799	49.862	49.926	49.989	860
870	49.989	50.052	50.116	50.179	50.243	50.306	50.369	50.432	50.495	50.559	50.622	870
880	50.622	50.685	50.748	50.811	50.874	50.937	51.000	51.063	51.126	51.189	51.251	880
890	51.251	51.314	51.377	51.439	51.502	51.565	51.627	51.690	51.752	51.815	51.877	890
900	51.877	51.940	52.002	52.064	52.127	52.189	52.251	52.314	52.376	52.438	52.500	900
910	52.500	52.562	52.624	52.686	52.748	52.810	52.872	52.934	52.996	53.057	53.119	910
920	53.119	53.181	53.243	53.304	53.366	53.427	53.489	53.550	53.612	53.673	53.735	920
930	53.735	53.796	53.857	53.918	53.979	54.040	54.101	54.162	54.223	54.284	54.345	930
940	54.345	54.406	54.467	54.528	54.589	54.650	54.711	54.773	54.834	54.895	54.956	940
950	54.956	55.016	55.077	55.138	55.198	55.259	55.319	55.380	55.440	55.501	55.561	950
960	55.561	55.622	55.682	55.742	55.803	55.863	55.923	55.983	56.043	56.104	56.164	960
970	56.164	56.224	56.284	56.344	56.404	56.464	56.524	56.584	56.643	56.703	56.763	970
980	56.763	56.823	56.883	56.942	57.002	57.062	57.121	57.181	57.240	57.300	57.360	980
990	57.360	57.419	57.479	57.538	57.597	57.657	57.716	57.775	57.834	57.893	57.953	990
1000	57.953	58.012	58.072	58.131	58.190	58.249	58.309	58.368	58.427	58.486	58.545	1000
1010	58.545	58.604	58.663	58.722	58.781	58.840	58.899	58.957	59.016	59.075	59.134	1010
1020	59.134	59.193	59.252	59.310	59.369	59.428	59.487	59.545	59.604	59.663	59.721	1020
1030	59.721	59.780	59.838	59.897	59.956	60.014	60.073	60.131	60.190	60.248	60.307	1030
1040	60.307	60.365	60.423	60.482	60.540	60.599	60.657	60.715	60.774	60.832	60.890	1040
1050	60.890	60.949	61.007	61.065	61.123	61.182	61.240	61.298	61.356	61.415	61.473	1050
1060	61.473	61.531	61.589	61.647	61.705	61.763	61.822	61.880	61.938	61.996	62.054	1060
1070	62.054	62.112	62.170	62.228	62.286	62.344	62.402	62.460	62.518	62.576	62.634	1070
1080	62.634	62.692	62.750	62.808	62.866	62.924	62.982	63.040	63.098	63.156	63.214	1080
1090	63.214	63.271	63.329	63.387	63.445	63.503	63.561	63.619	63.677	63.734	63.792	1090
1100	63.792	63.850	63.908	63.966	64.024	64.081	64.139	64.197	64.255	64.313	64.370	1100
1110	64.370	64.428	64.486	64.544	64.602	64.659	64.717	64.775	64.833	64.890	64.948	1110
1120	64.948	65.006	65.064	65.121	65.179	65.237	65.295	65.352	65.410	65.468	65.525	1120
1130	65.525	65.583	65.641	65.699	65.756	65.814	65.872	65.929	65.987	66.045	66.102	1130
1140	66.102	66.160	66.218	66.275	66.333	66.391	66.448	66.506	66.564	66.621	66.679	1140
1150	66.679	66.737	66.794	66.852	66.910	66.967	67.025	67.082	67.140	67.198	67.255	1150
1160	67.255	67.313	67.370	67.428	67.486	67.543	67.601	67.658	67.716	67.773	67.831	1160
1170	67.831	67.888	67.946	68.003	68.061	68.119	68.176	68.234	68.291	68.348	68.406	1170
1180	68.406	68.463	68.521	68.578	68.636	68.693	68.751	68.808	68.865	68.923	68.980	1180
1190	68.980	69.037	69.095	69.152	69.209	69.267	69.324	69.381	69.439	69.496	69.553	1190
°C	0	1	2	3	4	5	6	7	8	9	10	°C

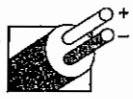
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MAXIMUM TEMPERATURE RANGE
Thermocouple Grade
32 to 1382°F
0 to 750°C
Extension Grade
32 to 392°F
0 to 200°C
LIMITS OF ERROR
(whichever is greater)
Standard: 2.2°C or 0.75%
Special: 1.1°C or 0.4%
COMMENTS, BARE WIRE ENVIRONMENT:
Reducing, Vacuum, Inert; Limited Use in Oxidizing at High Temperatures; Not Recommended for Low Temperatures
TEMPERATURE IN DEGREES °F
REFERENCE JUNCTION AT 32°F



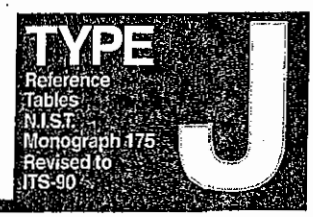
Thermocouple Grade

Iron
vs.
Copper-Nickel



Extension Grade

Revised Thermocouple Reference Tables



Thermoelectric Voltage in Millivolts

°F	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	°F
-340					-8.095	-8.085	-8.074	-8.063	-8.052	-8.041	-8.030	-340
-330	-8.030	-8.019	-8.008	-7.996	-7.985	-7.973	-7.962	-7.950	-7.938	-7.927	-7.915	-330
-320	-7.915	-7.903	-7.890	-7.878	-7.866	-7.854	-7.841	-7.829	-7.816	-7.804	-7.791	-320
-310	-7.791	-7.778	-7.765	-7.752	-7.739	-7.726	-7.713	-7.699	-7.686	-7.672	-7.659	-310
-300	-7.659	-7.645	-7.632	-7.618	-7.604	-7.590	-7.576	-7.562	-7.548	-7.534	-7.519	-300
-290	-7.519	-7.505	-7.491	-7.476	-7.462	-7.447	-7.432	-7.417	-7.403	-7.388	-7.373	-290
-280	-7.373	-7.357	-7.342	-7.327	-7.312	-7.296	-7.281	-7.265	-7.250	-7.234	-7.219	-280
-270	-7.219	-7.203	-7.187	-7.171	-7.155	-7.139	-7.123	-7.107	-7.090	-7.074	-7.058	-270
-260	-7.058	-7.041	-7.025	-7.008	-6.991	-6.975	-6.958	-6.941	-6.924	-6.907	-6.890	-260
-250	-6.890	-6.873	-6.856	-6.839	-6.821	-6.804	-6.787	-6.769	-6.752	-6.734	-6.715	-250
-240	-6.715	-6.699	-6.681	-6.663	-6.645	-6.627	-6.609	-6.591	-6.573	-6.555	-6.536	-240
-230	-6.536	-6.518	-6.500	-6.481	-6.463	-6.444	-6.426	-6.407	-6.388	-6.370	-6.351	-230
-220	-6.351	-6.332	-6.313	-6.294	-6.275	-6.256	-6.236	-6.217	-6.198	-6.179	-6.159	-220
-210	-6.159	-6.140	-6.120	-6.101	-6.081	-6.061	-6.042	-6.022	-6.002	-5.982	-5.962	-210
-200	-5.962	-5.942	-5.922	-5.902	-5.882	-5.862	-5.842	-5.821	-5.801	-5.781	-5.760	-200
-190	-5.760	-5.740	-5.719	-5.699	-5.678	-5.657	-5.637	-5.616	-5.595	-5.574	-5.553	-190
-180	-5.553	-5.532	-5.511	-5.490	-5.469	-5.448	-5.426	-5.405	-5.384	-5.363	-5.341	-180
-170	-5.341	-5.320	-5.298	-5.277	-5.255	-5.233	-5.212	-5.190	-5.168	-5.146	-5.125	-170
-160	-5.125	-5.103	-5.081	-5.059	-5.037	-5.015	-4.992	-4.970	-4.948	-4.926	-4.903	-160
-150	-4.903	-4.881	-4.859	-4.836	-4.814	-4.791	-4.769	-4.746	-4.724	-4.701	-4.678	-150
-140	-4.678	-4.655	-4.633	-4.610	-4.587	-4.564	-4.541	-4.518	-4.495	-4.472	-4.449	-140
-130	-4.449	-4.425	-4.402	-4.379	-4.356	-4.332	-4.309	-4.286	-4.262	-4.239	-4.215	-130
-120	-4.215	-4.192	-4.168	-4.144	-4.121	-4.097	-4.073	-4.050	-4.026	-4.002	-3.978	-120
-110	-3.978	-3.954	-3.930	-3.906	-3.882	-3.858	-3.834	-3.810	-3.786	-3.761	-3.737	-110
-100	-3.737	-3.713	-3.689	-3.664	-3.640	-3.615	-3.591	-3.566	-3.542	-3.517	-3.493	-100
-90	-3.493	-3.468	-3.443	-3.419	-3.394	-3.369	-3.344	-3.320	-3.295	-3.270	-3.245	-90
-80	-3.245	-3.220	-3.195	-3.170	-3.145	-3.120	-3.095	-3.070	-3.044	-3.019	-2.994	-80
-70	-2.994	-2.969	-2.943	-2.918	-2.893	-2.867	-2.842	-2.817	-2.791	-2.766	-2.740	-70
-60	-2.740	-2.714	-2.689	-2.663	-2.638	-2.612	-2.586	-2.560	-2.535	-2.509	-2.483	-60
-50	-2.483	-2.457	-2.431	-2.405	-2.379	-2.353	-2.327	-2.301	-2.275	-2.249	-2.223	-50
-40	-2.223	-2.197	-2.171	-2.145	-2.118	-2.092	-2.066	-2.040	-2.013	-1.987	-1.961	-40
-30	-1.961	-1.934	-1.908	-1.881	-1.855	-1.828	-1.802	-1.775	-1.749	-1.722	-1.695	-30
-20	-1.695	-1.669	-1.642	-1.615	-1.589	-1.562	-1.535	-1.508	-1.482	-1.455	-1.428	-20
-10	-1.428	-1.401	-1.374	-1.347	-1.320	-1.293	-1.266	-1.239	-1.212	-1.185	-1.158	-10
0	-1.158	-1.131	-1.104	-1.076	-1.049	-1.022	-0.995	-0.967	-0.940	-0.913	-0.886	0
0	-0.886	-0.858	-0.831	-0.803	-0.776	-0.748	-0.721	-0.694	-0.666	-0.639	-0.611	0
10	-0.611	-0.583	-0.556	-0.528	-0.501	-0.473	-0.445	-0.418	-0.390	-0.362	-0.334	10
20	-0.334	-0.307	-0.279	-0.251	-0.223	-0.195	-0.168	-0.140	-0.112	-0.084	-0.056	20
30	-0.056	-0.028	0.000	0.028	0.056	0.084	0.112	0.140	0.168	0.196	0.225	30
40	0.225	0.253	0.281	0.309	0.337	0.365	0.394	0.422	0.450	0.478	0.507	40
50	0.507	0.535	0.563	0.592	0.620	0.649	0.677	0.705	0.734	0.762	0.791	50
60	0.791	0.819	0.848	0.876	0.905	0.933	0.962	0.991	1.019	1.048	1.076	60
70	1.076	1.105	1.134	1.162	1.191	1.220	1.249	1.277	1.306	1.335	1.364	70
80	1.364	1.392	1.421	1.450	1.479	1.508	1.537	1.565	1.594	1.623	1.652	80
90	1.652	1.681	1.710	1.739	1.768	1.797	1.826	1.855	1.884	1.913	1.942	90
100	1.942	1.972	2.001	2.030	2.059	2.088	2.117	2.146	2.175	2.205	2.234	100
110	2.234	2.263	2.292	2.322	2.351	2.380	2.409	2.439	2.468	2.497	2.527	110
120	2.527	2.556	2.585	2.615	2.644	2.673	2.703	2.732	2.762	2.791	2.821	120
130	2.821	2.850	2.880	2.909	2.938	2.968	2.997	3.027	3.057	3.086	3.116	130
140	3.116	3.145	3.175	3.204	3.234	3.264	3.293	3.323	3.353	3.382	3.412	140
150	3.412	3.442	3.471	3.501	3.531	3.560	3.590	3.620	3.650	3.679	3.709	150
160	3.709	3.738	3.768	3.798	3.828	3.858	3.888	3.918	3.948	3.977	4.007	160
170	4.007	4.037	4.067	4.097	4.127	4.157	4.187	4.217	4.246	4.276	4.306	170
180	4.306	4.336	4.366	4.396	4.426	4.456	4.486	4.516	4.546	4.576	4.606	180
190	4.606	4.636	4.666	4.696	4.726	4.757	4.787	4.817	4.847	4.877	4.907	190
200	4.907	4.937	4.967	4.997	5.028	5.058	5.088	5.118	5.148	5.178	5.209	200
210	5.209	5.239	5.269	5.299	5.329	5.360	5.390	5.420	5.450	5.480	5.511	210
220	5.511	5.541	5.571	5.602	5.632	5.662	5.692	5.723	5.753	5.783	5.814	220
230	5.814	5.844	5.874	5.905	5.935	5.965	5.996	6.026	6.056	6.087	6.117	230
240	6.117	6.147	6.178	6.208	6.239	6.269	6.299	6.330	6.360	6.391	6.421	240
250	6.421	6.452	6.482	6.512	6.543	6.573	6.604	6.634	6.665	6.695	6.726	250
260	6.726	6.756	6.787	6.817	6.848	6.878	6.909	6.939	6.970	7.000	7.031	260
270	7.031	7.061	7.092	7.122	7.153	7.184	7.214	7.245	7.275	7.306	7.336	270
280	7.336	7.367	7.398	7.428	7.459	7.489	7.520	7.550	7.581	7.612	7.642	280
290	7.642	7.673	7.704	7.734	7.765	7.795	7.826	7.857	7.887	7.918	7.949	290

°F	0	1	2	3	4	5	6	7	8	9	10	°F
300	7.949	7.979	8.010	8.041	8.071	8.102	8.133	8.163	8.194	8.225	8.255	300
310	8.255	8.286	8.317	8.347	8.378	8.408	8.439	8.470	8.501	8.532	8.562	310
320	8.562	8.593	8.624	8.654	8.685	8.716	8.747	8.777	8.808	8.839	8.869	320
330	8.869	8.900	8.931	8.962	8.992	9.023	9.054	9.085	9.115	9.146	9.177	330
340	9.177	9.208	9.238	9.269	9.300	9.331	9.362	9.392	9.423	9.454	9.485	340
350	9.485	9.515	9.546	9.577	9.608	9.639	9.668	9.700	9.731	9.762	9.793	350
360	9.793	9.823	9.854	9.885	9.916	9.947	9.977	10.008	10.039	10.070	10.101	360
370	10.101	10.131	10.162	10.193	10.224	10.255	10.285	10.316	10.347	10.378	10.409	370
380	10.409	10.440	10.470	10.501	10.532	10.563	10.594	10.625	10.655	10.686	10.717	380
390	10.717	10.748	10.779	10.810	10.840	10.871	10.902	10.933	10.964	10.995	11.025	390
400	11.026	11.056	11.087	11.118	11.149	11.180	11.211	11.241	11.272	11.303	11.334	400
410	11.334	11.365	11.396	11.426	11.457	11.488	11.518	11.550	11.581	11.612	11.642	410
420	11.642	11.673	11.704	11.735	11.765	11.797	11.828	11.859	11.889	11.920	11.951	420
430	11.951	11.982	12.013	12.044	12.074	12.105	12.136	12.167	12.198	12.229	12.260	430
440	12.260	12.290	12.321	12.352	12.383	12.414	12.445	12.476	12.506	12.537	12.568	440
450	12.568	12.599	12.630	12.661	12.691	12.722	12.753	12.784	12.815	12.846	12.877	450
460	12.877	12.907	12.938	12.969	13.000	13.031	13.062	13.093	13.123	13.154	13.185	460
470	13.185	13.216	13.247	13.278	13.308	13.339	13.370	13.401	13.432	13.463	13.494	470
480	13.494	13.524	13.555	13.586	13.617	13.648	13.679	13.709	13.740	13.771	13.802	480
490	13.802	13.833	13.864	13.894	13.925	13.956	13.987	14.018	14.049	14.079	14.110	490
500	14.110	14.141	14.172	14.203	14.233	14.264	14.295	14.326	14.357	14.388	14.418	500
510	14.418	14.449	14.480	14.511	14.542	14.573	14.603	14.634	14.665	14.695	14.727	510
520	14.727	14.757	14.788	14.819	14.850	14.881	14.911	14.942	14.973	15.004	15.035	520
530	15.035	15.065	15.096	15.127	15.158	15.189	15.219	15.250	15.281	15.312	15.343	530
540	15.343	15.373	15.404	15.435	15.466	15.496	15.527	15.558	15.589	15.620	15.650	540
550	15.650	15.681	15.712	15.743	15.773	15.804	15.835	15.866	15.897	15.927	15.958	550
560	15.958	15.989	16.020	16.050	16.081	16.112	16.143	16.173	16.204	16.235	16.266	560
570	16.266	16.296	16.327	16.358	16.389	16.419	16.450	16.481	16.512	16.542	16.573	570
580	16.573	16.604	16.635	16.666	16.696	16.727	16.758	16.788	16.819	16.850	16.881	580
590	16.881	16.911	16.942	16.973	17.003	17.034	17.065	17.096	17.126	17.157	17.188	590
600	17.188	17.219	17.249	17.280	17.311	17.341	17.372	17.403	17.434	17.464	17.495	600
610	17.495	17.526	17.556	17.587	17.618	17.649	17.679	17.710	17.741	17.771	17.802	610
620	17.802	17.833	17.863	17.894	17.925	17.955	17.985	18.017	18.048	18.078	18.109	620
630	18.109	18.140	18.170	18.201	18.232	18.262	18.293	18.324	18.354	18.385	18.416	630
640	18.416	18.446	18.477	18.508	18.538	18.569	18.600	18.630	18.661	18.692	18.722	640
650	18.722	18.753	18.784	18.814	18.845	18.876	18.906	18.937	18.968	18.998	19.029	650
660	19.029	19.060	19.090	19.121	19.152	19.182	19.213	19.244	19.274	19.305	19.335	660
670	19.335	19.366	19.397	19.428	19.458	19.489	19.520	19.550	19.581	19.612	19.642	670
680	19.642	19.673	19.704	19.734	19.765	19.795	19.826	19.857	19.887	19.918	19.949	680
690	19.949	19.979	20.010	20.041	20.071	20.102	20.132	20.163	20.194	20.224	20.255	690
700	20.255	20.286	20.316	20.347	20.378	20.408	20.439	20.469	20.500	20.531	20.561	700
710	20.561	20.592	20.623	20.653	20.684	20.715	20.745	20.776	20.806	20.837	20.868	710
720	20.868	20.898	20.929	20.960	20.990	21.021	21.052	21.082	21.113	21.143	21.174	720
730	21.174	21.205	21.235	21.266	21.297	21.327	21.358	21.389	21.419	21.450	21.480	730
740	21.480	21.511	21.542	21.572	21.603	21.634	21.664	21.695	21.726	21.756	21.787	740
750	21.787	21.817	21.848	21.879	21.909	21.940	21.971	22.001	22.032	22.063	22.093	750
760	22.093	22.124	22.154	22.185	22.216	22.246	22.277	22.308	22.338	22.369	22.400	760
770	22.400	22.430	22.461	22.492	22.522	22.553	22.584	22.614	22.645	22.675	22.706	770
780	22.706	22.737	22.768	22.798	22.829	22.860	22.890	22.921	22.952	22.982	23.013	780
790	23.013	23.044	23.074	23.105	23.136	23.166	23.197	22.228	23.258	23.289	23.320	790
800	23.320	23.350	23.381	23.412	23.443	23.473	23.504	23.535	23.565	23.596	23.627	800
810	23.627	23.657	23.688	23.719	23.749	23.780	23.811	23.842	23.872	23.903	23.934	810
820	23.934	23.964	23.995	24.025	24.057	24.087	24.118	24.149	24.180	24.210	24.241	820
830	24.241	24.272	24.303	24.333	24.364	24.395	24.426	24.455	24.487	24.518	24.549	830
840	24.549	24.579	24.610	24.641	24.672	24.702	24.733	24.764	24.795	24.825	24.856	840
850	24.856	24.887	24.918	24.949	24.979	25.010	25.041	25.072	25.103	25.134	25.164	850
860	25.164	25.195	25.226	25.257	25.288	25.318	25.349	25.380	25.411	25.442	25.473	860
870	25.473	25.504	25.534	25.565	25.596	25.627	25.658	25.689	25.720	25.750	25.781	870
880	25.781	25.812	25.843	25.874	25.905	25.936	25.967	25.998	26.028	26.059	26.090	880
890	26.090	26.121	26.152	26.183	26.214	26.245	26.276	26.307	26.338	26.369	26.400	890
900	26.400	26.431	26.462	26.493	26.524	26.555	26.586	26.617	26.648	26.679	26.710	900
910	26.710	26.741	26.772	26.803	26.834	26.865	26.896	26.927	26.958	26.989	27.020	910
920	27.020	27.051	27.082	27.113	27.144	27.175	27.205	27.237	27.268	27.299	27.330	920
930	27.330	27.362	27.393	27.424	27.455	27.486	27.517	27.548	27.579	27.610	27.642	930
940	27.642	27.673	27.704	27.735	27.766	27.797	27.828	27.860	27.891	27.922	27.953	940

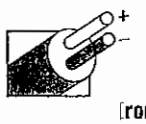
°F	0	1	2	3	4	5	6	7	8	9	10	°F

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Revised Thermocouple Reference Tables

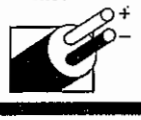
TYPE J

Reference Tables
N.I.S.T. Monograph 175
Revised to ITS-90



Thermocouple Grade

Iron vs. Copper-Nickel



Extension Grade

MAXIMUM TEMPERATURE RANGE

Thermocouple Grade
32 to 1382°F
0 to 750°C

Extension Grade
32 to 392°F
0 to 200°C

LIMITS OF ERROR
(whichever is greater)
Standard: 2.2°C or 0.75%
Special: 1.1°C or 0.4%

COMMENTS, BARE WIRE ENVIRONMENT:
Reducing, Vacuum, Inert; Limited Use in Oxidizing at High Temperatures; Not Recommended for Low Temperatures

TEMPERATURE IN DEGREES °F
REFERENCE JUNCTION AT 32°F

Thermoelectric Voltage in Millivolts

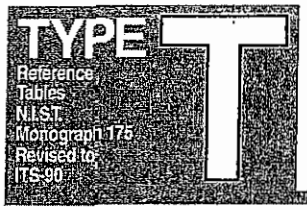
°F	0	1	2	3	4	5	6	7	8	9	10	°F
900	27.953	27.985	28.016	28.047	28.078	28.109	28.141	28.172	28.203	28.234	28.266	900
910	28.266	28.297	28.328	28.359	28.391	28.422	28.453	28.485	28.516	28.547	28.579	910
920	28.579	28.610	28.641	28.672	28.704	28.735	28.767	28.798	28.829	28.861	28.892	920
930	28.892	28.923	28.955	28.986	29.018	29.049	29.080	29.112	29.143	29.175	29.206	930
940	29.206	29.238	29.269	29.301	29.332	29.363	29.395	29.426	29.458	29.489	29.521	940
950	29.521	29.552	29.584	29.616	29.647	29.679	29.710	29.742	29.773	29.805	29.836	950
960	29.836	29.868	29.900	29.931	29.963	29.995	30.026	30.058	30.089	30.121	30.153	960
970	30.153	30.184	30.216	30.248	30.279	30.311	30.343	30.375	30.406	30.438	30.470	970
980	30.470	30.502	30.533	30.565	30.597	30.628	30.660	30.692	30.724	30.756	30.788	980
990	30.788	30.819	30.851	30.883	30.915	30.947	30.979	31.011	31.043	31.074	31.106	990
1000	31.106	31.138	31.170	31.202	31.234	31.266	31.298	31.330	31.362	31.394	31.426	1000
1010	31.426	31.458	31.490	31.522	31.554	31.586	31.618	31.650	31.682	31.714	31.746	1010
1020	31.746	31.778	31.811	31.843	31.875	31.907	31.939	31.971	32.003	32.035	32.068	1020
1030	32.068	32.100	32.132	32.164	32.196	32.229	32.261	32.293	32.325	32.358	32.390	1030
1040	32.390	32.422	32.455	32.487	32.519	32.551	32.584	32.616	32.648	32.681	32.713	1040
1050	32.713	32.746	32.778	32.810	32.843	32.875	32.908	32.940	32.973	33.005	33.037	1050
1060	33.037	33.070	33.102	33.135	33.167	33.200	33.232	33.265	33.298	33.330	33.363	1060
1070	33.363	33.395	33.428	33.460	33.493	33.526	33.558	33.591	33.624	33.656	33.689	1070
1080	33.689	33.722	33.754	33.787	33.820	33.853	33.885	33.918	33.951	33.984	34.016	1080
1090	34.016	34.049	34.082	34.115	34.148	34.180	34.213	34.246	34.279	34.312	34.345	1090
1100	34.345	34.378	34.411	34.444	34.476	34.509	34.542	34.575	34.608	34.641	34.674	1100
1110	34.674	34.707	34.740	34.773	34.806	34.840	34.873	34.906	34.939	34.972	35.005	1110
1120	35.005	35.038	35.071	35.104	35.138	35.171	35.204	35.237	35.270	35.304	35.337	1120
1130	35.337	35.370	35.403	35.437	35.470	35.503	35.536	35.570	35.603	35.636	35.670	1130
1140	35.670	35.703	35.736	35.770	35.803	35.837	35.870	35.903	35.937	35.970	36.004	1140
1150	36.004	36.037	36.071	36.104	36.138	36.171	36.205	36.238	36.272	36.305	36.339	1150
1160	36.339	36.373	36.406	36.440	36.473	36.507	36.541	36.574	36.608	36.642	36.675	1160
1170	36.675	36.709	36.743	36.777	36.810	36.844	36.878	36.912	36.945	36.979	37.013	1170
1180	37.013	37.047	37.081	37.114	37.148	37.182	37.216	37.250	37.284	37.318	37.352	1180
1190	37.352	37.386	37.420	37.454	37.488	37.522	37.556	37.590	37.624	37.658	37.692	1190
1200	37.692	37.726	37.760	37.794	37.828	37.862	37.896	37.930	37.964	37.998	38.033	1200
1210	38.033	38.067	38.101	38.135	38.169	38.204	38.238	38.272	38.306	38.341	38.375	1210
1220	38.375	38.409	38.444	38.478	38.512	38.546	38.581	38.615	38.650	38.684	38.718	1220
1230	38.718	38.753	38.787	38.822	38.856	38.890	38.925	38.959	38.994	39.028	39.063	1230
1240	39.063	39.097	39.132	39.166	39.201	39.235	39.270	39.305	39.339	39.374	39.408	1240
1250	39.408	39.443	39.478	39.512	39.547	39.582	39.616	39.651	39.686	39.720	39.755	1250
1260	39.755	39.790	39.825	39.859	39.894	39.929	39.964	39.998	40.033	40.068	40.103	1260
1270	40.103	40.138	40.173	40.207	40.242	40.277	40.312	40.347	40.382	40.417	40.452	1270
1280	40.452	40.487	40.522	40.556	40.591	40.626	40.661	40.696	40.731	40.766	40.801	1280
1290	40.801	40.836	40.872	40.907	40.942	40.977	41.012	41.047	41.082	41.117	41.152	1290
1300	41.152	41.187	41.222	41.258	41.293	41.328	41.363	41.398	41.433	41.469	41.504	1300
1310	41.504	41.539	41.574	41.610	41.645	41.680	41.716	41.751	41.786	41.821	41.856	1310
1320	41.856	41.892	41.927	41.962	41.998	42.033	42.068	42.104	42.139	42.174	42.210	1320
1330	42.210	42.245	42.281	42.316	42.351	42.387	42.422	42.458	42.493	42.529	42.564	1330
1340	42.564	42.599	42.635	42.670	42.706	42.741	42.777	42.812	42.848	42.883	42.919	1340
1350	42.919	42.954	42.990	43.025	43.061	43.096	43.132	43.167	43.203	43.239	43.274	1350
1360	43.274	43.310	43.346	43.381	43.417	43.452	43.488	43.524	43.559	43.595	43.631	1360
1370	43.631	43.667	43.702	43.738	43.774	43.809	43.845	43.881	43.917	43.953	43.988	1370
1380	43.988	44.024	44.060	44.096	44.131	44.167	44.203	44.239	44.275	44.310	44.346	1380
1390	44.346	44.382	44.418	44.454	44.490	44.525	44.561	44.597	44.633	44.669	44.705	1390
1400	44.705	44.741	44.777	44.812	44.848	44.884	44.920	44.956	44.992	45.028	45.064	1400
1410	45.064	45.099	45.135	45.171	45.207	45.243	45.279	45.315	45.351	45.387	45.423	1410
1420	45.423	45.458	45.494	45.530	45.566	45.602	45.638	45.674	45.710	45.746	45.782	1420
1430	45.782	45.818	45.853	45.889	45.925	45.961	45.997	46.033	46.069	46.105	46.141	1430
1440	46.141	46.177	46.212	46.248	46.284	46.320	46.356	46.392	46.428	46.464	46.500	1440
1450	46.500	46.536	46.571	46.607	46.643	46.679	46.715	46.751	46.786	46.822	46.858	1450
1460	46.858	46.894	46.930	46.966	47.001	47.037	47.073	47.109	47.145	47.181	47.216	1460
1470	47.216	47.252	47.288	47.324	47.360	47.395	47.431	47.467	47.503	47.538	47.574	1470
1480	47.574	47.610	47.646	47.681	47.717	47.753	47.788	47.824	47.860	47.896	47.931	1480
1490	47.931	47.967	48.003	48.038	48.074	48.110	48.145	48.181	48.217	48.252	48.288	1490
1500	48.288	48.324	48.359	48.395	48.430	48.466	48.502	48.537	48.573	48.608	48.644	1500
1510	48.644	48.679	48.715	48.750	48.786	48.822	48.857	48.893	48.928	48.964	48.999	1510
1520	48.999	49.034	49.070	49.105	49.141	49.176	49.212	49.247	49.283	49.318	49.353	1520
1530	49.353	49.389	49.424	49.460	49.495	49.530	49.566	49.601	49.636	49.672	49.707	1530
1540	49.707	49.742	49.778	49.813	49.848	49.883	49.919	49.954	49.989	50.024	50.059	1540

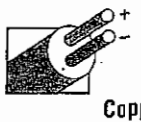
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1500	50.060	50.095	50.130	50.165	50.200	50.235	50.271	50.306	50.341	50.376	50.411	1500
1510	50.411	50.446	50.481	50.517	50.552	50.587	50.622	50.657	50.692	50.727	50.762	1510
1520	50.762	50.797	50.832	50.867	50.902	50.937	50.972	51.007	51.042	51.077	51.112	1520
1530	51.112	51.147	51.181	51.216	51.251	51.286	51.321	51.356	51.391	51.425	51.460	1530
1540	51.460	51.495	51.530	51.565	51.599	51.634	51.669	51.704	51.739	51.773	51.808	1540
1550	51.808	51.843	51.877	51.912	51.947	51.981	52.016	52.051	52.085	52.120	52.154	1550
1560	52.154	52.189	52.224	52.258	52.293	52.327	52.362	52.396	52.431	52.465	52.500	1560
1570	52.500	52.534	52.569	52.603	52.638	52.672	52.707	52.741	52.775	52.810	52.844	1570
1580	52.844	52.879	52.913	52.947	52.982	53.016	53.050	53.085	53.119	53.153	53.188	1580
1590	53.188	53.222	53.256	53.290	53.325	53.359	53.393	53.427	53.462	53.496	53.530	1590
1600	53.530	53.564	53.598	53.632	53.667	53.701	53.735	53.769	53.803	53.837	53.871	1600
1610	53.871	53.905	53.939	53.973	54.007	54.041	54.075	54.109	54.143	54.177	54.211	1610
1620	54.211	54.245	54.279	54.313	54.347	54.381	54.415	54.449	54.483	54.517	54.550	1620
1630	54.550	54.584	54.618	54.652	54.686	54.719	54.753	54.787	54.821	54.855	54.888	1630
1640	54.888	54.922	54.956	54.990	55.023	55.057	55.091	55.124	55.158	55.192	55.225	1640
1650	55.225	55.259	55.293	55.326	55.360	55.393	55.427	55.461	55.494	55.528	55.561	1650
1660	55.561	55.595	55.628	55.662	55.695	55.729	55.762	55.796	55.830	55.863	55.898	1660
1670	55.896	55.930	55.963	55.997	56.030	56.063	56.097	56.130	56.164	56.197	56.230	1670
1680	56.230	56.264	56.297	56.330	56.364	56.397	56.430	56.464	56.497	56.530	56.564	1680
1690	56.564	56.597	56.630	56.663	56.697	56.730	56.763	56.796	56.829	56.863	56.896	1690
1700	56.896	56.929	56.962	56.995	57.028	57.062	57.095	57.128	57.161	57.194	57.227	1700
1710	57.227	57.260	57.293	57.326	57.360	57.393	57.426	57.459	57.492	57.525	57.555	1710
1720	57.558	57.591	57.624	57.657	57.690	57.723	57.756	57.789	57.822	57.855	57.888	1720
1730	57.888	57.920	57.953	57.986	58.019	58.052	58.085	58.118	58.151	58.184	58.217	1730
1740	58.217	58.249	58.282	58.315	58.348	58.381	58.414	58.445	58.478	58.512	58.545	1740
1750	58.545	58.578	58.610	58.643	58.676	58.709	58.741	58.774	58.807	58.840	58.872	1750
1760	58.872	58.905	58.938	58.971	59.003	59.036	59.069	59.101	59.134	59.167	59.199	1760
1770	59.199	59.232	59.265	59.297	59.330	59.363	59.395	59.428	59.460	59.493	59.526	1770
1780	59.526	59.558	59.591	59.623	59.656	59.689	59.721	59.754	59.786	59.819	59.851	1780
1790	59.851	59.884	59.916	59.949	59.982	60.014	60.047	60.079	60.112	60.144	60.177	1790
1800	60.177	60.209	60.242	60.274	60.307	60.339	60.371	60.404	60.436	60.468	60.501	1800
1810	60.501	60.534	60.566	60.599	60.631	60.663	60.696	60.728	60.761	60.793	60.826	1810
1820	60.826	60.858	60.890	60.923	60.955	60.987	61.020	61.052	61.085	61.117	61.149	1820
1830	61.149	61.182	61.214	61.246	61.279	61.311	61.343	61.376	61.408	61.440	61.473	1830
1840	61.473	61.505	61.537	61.570	61.602	61.634	61.667	61.699	61.731	61.763	61.796	1840
1850	61.796	61.828	61.860	61.893	61.925	61.957	61.989	62.022	62.054	62.086	62.118	1850
1860	62.118	62.151	62.183	62.215	62.247	62.280	62.312	62.344	62.376	62.409	62.441	1860
1870	62.441	62.473	62.505	62.537	62.570	62.602	62.634	62.666	62.699	62.731	62.763	1870
1880	62.763	62.795	62.827	62.860	62.892	62.924	62.956	62.988	63.020	63.053	63.085	1880
1890	63.085	63.117	63.149	63.181	63.214	63.246	63.278	63.310	63.342	63.374	63.406	1890
1900	63.406	63.439	63.471	63.503	63.535	63.567	63.599	63.632	63.664	63.696	63.728	1900
1910	63.728	63.760	63.792	63.824	63.856	63.889	63.921	63.953	63.985	64.017	64.049	1910
1920	64.049	64.081	64.113	64.146	64.178	64.210	64.242	64.274	64.306	64.338	64.370	1920
1930	64.370	64.402	64.435	64.467	64.499	64.531	64.563	64.595	64.627	64.659	64.691	1930
1940	64.691	64.723	64.756	64.788	64.820	64.852	64.884	64.916	64.948	64.980	65.012	1940
1950	65.012	65.044	65.076	65.109	65.141	65.173	65.205	65.237	65.269	65.301	65.333	1950
1960	65.333	65.365	65.397	65.429	65.461	65.493	65.525	65.557	65.590	65.622	65.654	1960
1970	65.654	65.686	65.718	65.750	65.782	65.814	65.846	65.878	65.910	65.942	65.974	1970
1980	65.974	66.006	66.038	66.070	66.102	66.134	66.166	66.199	66.231	66.263	66.295	1980
1990	66.295	66.327	66.359	66.391	66.423	66.455	66.487	66.519	66.551	66.583	66.615	1990
2000	66.615	66.647	66.679	66.711	66.743	66.775	66.807	66.839	66.871	66.903	66.935	2000
2010	66.935	66.967	66.999	67.031	67.063	67.095	67.127	67.159	67.191	67.223	67.255	2010
2020	67.255	67.287	67.319	67.351	67.383	67.415	67.447	67.479	67.511	67.543	67.575	2020
2030	67.575	67.607	67.639	67.671	67.703	67.735	67.767	67.799	67.831	67.863	67.895	2030
2040	67.895	67.927	67.959	67.991	68.023	68.055	68.087	68.119	68.150	68.182	68.214	2040
2050	68.214	68.246	68.278	68.310	68.342	68.374	68.406	68.438	68.470	68.502	68.534	2050
2060	68.534	68.566	68.597	68.629	68.661	68.693	68.725	68.757	68.789	68.821	68.853	2060
2070	68.853	68.884	68.916	68.948	68.980	69.012	69.044	69.076	69.108	69.139	69.171	2070
2080	69.171	69.203	69.235	69.267	69.299	69.330	69.362	69.394	69.426	69.458	69.490	2080
2090	69.490	69.521	69.553									2090

°F	0	1	2	3	4	5	6	7	8	9	10	°F

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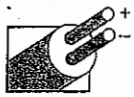
Revised Thermocouple Reference Tables





Thermocouple Grade

Copper vs. Copper-Nickel



Extension Grade

MAXIMUM TEMPERATURE RANGE
Thermocouple Grade
– 328 to 662°F
– 200 to 350°C
Extension Grade
– 76 to 212°F
– 60 to 100°C
LIMITS OF ERROR
(whichever is greater)
Standard: 1.0°C or 0.75% Above 0°C
1.0°C or 1.5% Below 0°C
Special: 0.5°C or 0.4%
COMMENTS, BARE WIRE ENVIRONMENT:
Mild Oxidizing, Reducing Vacuum or Inert; Good Where Moisture is Present; Low Temperature and Cryogenic Applications
TEMPERATURE IN DEGREES °C
REFERENCE JUNCTION AT 0°C

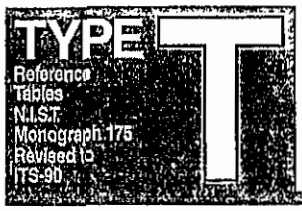
Thermoelectric Voltage in Millivolts

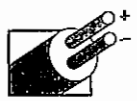
°C	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	°C
-260	-6.256	-6.256	-6.255	-6.253	-6.251	-6.248	-6.245	-6.242	-6.239	-6.236	-6.232	-260
-250	-6.232	-6.230	-6.228	-6.223	-6.219	-6.214	-6.209	-6.204	-6.198	-6.193	-6.187	-250
-240	-6.180	-6.174	-6.167	-6.160	-6.153	-6.146	-6.139	-6.130	-6.122	-6.114	-6.105	-240
-230	-6.095	-6.086	-6.077	-6.068	-6.058	-6.049	-6.039	-6.028	-6.017	-6.007	-5.995	-230
-220	-6.007	-5.986	-5.985	-5.973	-5.962	-5.950	-5.938	-5.926	-5.914	-5.901	-5.888	-220
-210	-5.888	-5.876	-5.863	-5.850	-5.836	-5.823	-5.809	-5.795	-5.782	-5.767	-5.753	-210
-200	-5.753	-5.739	-5.724	-5.710	-5.695	-5.680	-5.665	-5.650	-5.634	-5.619	-5.603	-200
-190	-5.603	-5.587	-5.571	-5.555	-5.539	-5.523	-5.506	-5.489	-5.473	-5.456	-5.439	-190
-180	-5.439	-5.421	-5.404	-5.387	-5.369	-5.351	-5.334	-5.316	-5.297	-5.279	-5.261	-180
-170	-5.261	-5.242	-5.224	-5.205	-5.185	-5.167	-5.148	-5.129	-5.109	-5.089	-5.070	-170
-160	-5.070	-5.050	-5.030	-5.010	-4.990	-4.969	-4.949	-4.928	-4.907	-4.886	-4.865	-160
-150	-4.865	-4.844	-4.823	-4.802	-4.780	-4.759	-4.737	-4.715	-4.693	-4.671	-4.648	-150
-140	-4.648	-4.626	-4.604	-4.581	-4.559	-4.535	-4.512	-4.489	-4.466	-4.443	-4.419	-140
-130	-4.419	-4.395	-4.372	-4.348	-4.324	-4.300	-4.275	-4.251	-4.226	-4.202	-4.177	-130
-120	-4.177	-4.152	-4.127	-4.102	-4.077	-4.052	-4.026	-4.000	-3.975	-3.949	-3.923	-120
-110	-3.923	-3.897	-3.871	-3.844	-3.818	-3.791	-3.765	-3.738	-3.711	-3.684	-3.657	-110
-100	-3.657	-3.629	-3.602	-3.574	-3.547	-3.519	-3.491	-3.463	-3.435	-3.407	-3.379	-100
-90	-3.379	-3.350	-3.322	-3.293	-3.264	-3.235	-3.206	-3.177	-3.148	-3.118	-3.089	-90
-80	-3.089	-3.059	-3.030	-3.000	-2.970	-2.940	-2.910	-2.879	-2.848	-2.818	-2.788	-80
-70	-2.788	-2.757	-2.726	-2.695	-2.664	-2.633	-2.602	-2.571	-2.539	-2.507	-2.476	-70
-60	-2.476	-2.444	-2.412	-2.380	-2.348	-2.316	-2.283	-2.251	-2.218	-2.186	-2.153	-60
-50	-2.153	-2.120	-2.087	-2.054	-2.021	-1.987	-1.954	-1.920	-1.887	-1.853	-1.819	-50
-40	-1.819	-1.785	-1.751	-1.717	-1.683	-1.648	-1.614	-1.579	-1.545	-1.510	-1.475	-40
-30	-1.475	-1.440	-1.405	-1.370	-1.335	-1.299	-1.264	-1.228	-1.192	-1.157	-1.121	-30
-20	-1.121	-1.085	-1.049	-1.013	-0.976	-0.940	-0.904	-0.867	-0.830	-0.794	-0.757	-20
-10	-0.757	-0.720	-0.683	-0.646	-0.608	-0.571	-0.534	-0.496	-0.459	-0.421	-0.383	-10
0	-0.383	-0.345	-0.307	-0.269	-0.231	-0.193	-0.154	-0.116	-0.077	-0.039	0.000	0
0	0.000	0.039	0.078	0.117	0.156	0.195	0.234	0.273	0.312	0.352	0.391	0
10	0.391	0.430	0.470	0.510	0.549	0.589	0.629	0.669	0.709	0.749	0.790	10
20	0.790	0.830	0.870	0.911	0.951	0.992	1.033	1.074	1.114	1.155	1.196	20
30	1.196	1.238	1.279	1.320	1.362	1.403	1.445	1.486	1.528	1.570	1.612	30
40	1.612	1.654	1.696	1.738	1.780	1.823	1.865	1.908	1.950	1.993	2.036	40
°C	0	1	2	3	4	5	6	7	8	9	10	°C

°C	0	1	2	3	4	5	6	7	8	9	10	°C
50	2.036	2.079	2.122	2.165	2.208	2.251	2.294	2.338	2.381	2.425	2.468	50
60	2.468	2.512	2.555	2.600	2.643	2.687	2.732	2.776	2.820	2.864	2.909	60
70	2.909	2.953	2.998	3.043	3.087	3.132	3.177	3.222	3.267	3.312	3.358	70
80	3.358	3.403	3.448	3.494	3.539	3.585	3.631	3.677	3.722	3.768	3.814	80
90	3.814	3.860	3.907	3.953	3.999	4.046	4.092	4.139	4.185	4.232	4.279	90
100	4.279	4.325	4.372	4.419	4.466	4.513	4.561	4.608	4.655	4.702	4.750	100
110	4.750	4.798	4.845	4.893	4.941	4.989	5.036	5.084	5.132	5.180	5.228	110
120	5.228	5.277	5.325	5.373	5.422	5.470	5.519	5.567	5.616	5.665	5.714	120
130	5.714	5.763	5.812	5.861	5.910	5.959	6.008	6.057	6.107	6.156	6.206	130
140	6.206	6.255	6.305	6.355	6.404	6.454	6.504	6.554	6.604	6.654	6.704	140
150	6.704	6.754	6.805	6.855	6.905	6.956	7.006	7.057	7.107	7.158	7.209	150
160	7.209	7.259	7.310	7.361	7.412	7.463	7.515	7.566	7.617	7.668	7.720	160
170	7.720	7.771	7.823	7.874	7.926	7.977	8.029	8.081	8.133	8.185	8.237	170
180	8.237	8.289	8.341	8.393	8.445	8.497	8.550	8.602	8.654	8.707	8.759	180
190	8.759	8.812	8.865	8.917	8.970	9.023	9.076	9.129	9.182	9.235	9.288	190
200	9.288	9.341	9.395	9.448	9.501	9.555	9.608	9.662	9.715	9.769	9.822	200
210	9.822	9.876	9.930	9.984	10.038	10.092	10.146	10.200	10.254	10.308	10.362	210
220	10.362	10.417	10.471	10.525	10.580	10.634	10.688	10.743	10.798	10.853	10.907	220
230	10.907	10.962	11.017	11.072	11.127	11.182	11.237	11.292	11.347	11.403	11.458	230
240	11.458	11.513	11.569	11.624	11.680	11.735	11.791	11.846	11.902	11.958	12.013	240
250	12.013	12.069	12.125	12.181	12.237	12.293	12.349	12.405	12.461	12.516	12.574	250
260	12.574	12.630	12.687	12.743	12.799	12.856	12.912	12.969	13.026	13.082	13.139	260
270	13.139	13.196	13.253	13.310	13.366	13.423	13.480	13.537	13.595	13.652	13.709	270
280	13.709	13.766	13.823	13.881	13.938	13.995	14.053	14.110	14.168	14.226	14.283	280
290	14.283	14.341	14.398	14.456	14.514	14.572	14.630	14.688	14.746	14.804	14.862	290
300	14.862	14.920	14.978	15.036	15.095	15.153	15.211	15.270	15.328	15.386	15.445	300
310	15.445	15.503	15.562	15.621	15.679	15.738	15.797	15.856	15.914	15.973	16.032	310
320	16.032	16.091	16.150	16.209	16.268	16.327	16.387	16.446	16.505	16.564	16.624	320
330	16.624	16.683	16.742	16.802	16.861	16.921	16.980	17.040	17.100	17.159	17.219	330
340	17.219	17.279	17.339	17.399	17.458	17.518	17.578	17.638	17.698	17.758	17.819	340
350	17.819	17.879	17.939	17.999	18.060	18.120	18.180	18.241	18.301	18.362	18.422	350
360	18.422	18.483	18.543	18.604	18.665	18.725	18.786	18.847	18.908	18.969	19.030	360
370	19.030	19.091	19.152	19.213	19.274	19.335	19.396	19.457	19.518	19.579	19.641	370
380	19.641	19.702	19.763	19.825	19.886	19.947	20.009	20.070	20.132	20.193	20.255	380
390	20.255	20.317	20.378	20.440	20.502	20.563	20.625	20.687	20.748	20.810	20.872	390
°C	0	1	2	3	4	5	6	7	8	9	10	°C

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Revised Thermocouple Reference Tables





Thermocouple Grade

Copper vs. Copper-Nickel



Extension Grade

MAXIMUM TEMPERATURE RANGE
Thermocouple Grade
– 328 to 662°F
– 200 to 350°C
Extension Grade
– 76 to 212°F
– 60 to 100°C

LIMITS OF ERROR
(whichever is greater)
Standard: 1.0°C or 0.75% Above 0°C
1.0°C or 1.5% Below 0°C
Special: 0.5°C or 0.4%

COMMENTS, BARE WIRE ENVIRONMENT:
Mild Oxidizing, Reducing Vacuum or Inert; Good Where Moisture Is Present; Low Temperature and Cryogenic Applications

TEMPERATURE IN DEGREES °F REFERENCE JUNCTION AT 32°F

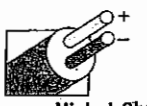
Thermoelectric Voltage in Millivolts

°F	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	°F
-450							-6.258	-6.257	-6.255	-6.253	-6.251	-450
-440	-6.254	-6.253	-6.252	-6.251	-6.250	-6.248	-6.247	-6.245	-6.243	-6.242	-6.240	-440
-430	-6.243	-6.238	-6.236	-6.234	-6.232	-6.230	-6.227	-6.225	-6.222	-6.220	-6.217	-430
-420	-6.217	-6.215	-6.212	-6.210	-6.208	-6.205	-6.202	-6.197	-6.194	-6.191	-6.187	-420
-410	-6.187	-6.184	-6.180	-6.177	-6.173	-6.170	-6.166	-6.162	-6.158	-6.154	-6.150	-410
-400	-6.150	-6.145	-6.141	-6.137	-6.133	-6.128	-6.124	-6.119	-6.115	-6.110	-6.105	-400
-390	-6.105	-6.100	-6.095	-6.090	-6.085	-6.080	-6.075	-6.069	-6.064	-6.059	-6.053	-390
-380	-6.053	-6.047	-6.042	-6.036	-6.030	-6.025	-6.019	-6.013	-6.007	-6.001	-5.994	-380
-370	-5.994	-5.988	-5.982	-5.976	-5.969	-5.963	-5.956	-5.950	-5.943	-5.937	-5.930	-370
-360	-5.930	-5.923	-5.916	-5.909	-5.902	-5.896	-5.888	-5.881	-5.874	-5.867	-5.860	-360
-350	-5.860	-5.853	-5.845	-5.838	-5.830	-5.823	-5.815	-5.808	-5.800	-5.792	-5.785	-350
-340	-5.785	-5.777	-5.769	-5.761	-5.753	-5.745	-5.737	-5.729	-5.721	-5.713	-5.705	-340
-330	-5.705	-5.697	-5.689	-5.680	-5.672	-5.663	-5.655	-5.646	-5.638	-5.629	-5.620	-330
-320	-5.620	-5.612	-5.603	-5.594	-5.585	-5.577	-5.568	-5.559	-5.550	-5.541	-5.532	-320
-310	-5.532	-5.523	-5.513	-5.504	-5.495	-5.486	-5.476	-5.467	-5.458	-5.448	-5.439	-310
-300	-5.439	-5.429	-5.420	-5.410	-5.400	-5.391	-5.381	-5.371	-5.361	-5.351	-5.341	-300
-290	-5.341	-5.331	-5.322	-5.312	-5.301	-5.291	-5.281	-5.271	-5.261	-5.250	-5.240	-290
-280	-5.240	-5.230	-5.219	-5.209	-5.198	-5.188	-5.177	-5.167	-5.156	-5.145	-5.135	-280
-270	-5.135	-5.124	-5.113	-5.102	-5.091	-5.081	-5.070	-5.059	-5.048	-5.036	-5.025	-270
-260	-5.025	-5.014	-5.003	-4.992	-4.980	-4.969	-4.958	-4.946	-4.935	-4.923	-4.912	-260
-250	-4.912	-4.900	-4.889	-4.877	-4.865	-4.854	-4.842	-4.830	-4.818	-4.806	-4.794	-250
-240	-4.794	-4.783	-4.771	-4.759	-4.748	-4.734	-4.722	-4.710	-4.698	-4.685	-4.673	-240
-230	-4.673	-4.661	-4.648	-4.636	-4.624	-4.611	-4.599	-4.586	-4.573	-4.561	-4.548	-230
-220	-4.548	-4.535	-4.523	-4.510	-4.497	-4.484	-4.471	-4.458	-4.445	-4.432	-4.419	-220
-210	-4.419	-4.406	-4.393	-4.380	-4.368	-4.353	-4.340	-4.326	-4.313	-4.300	-4.286	-210
-200	-4.286	-4.273	-4.259	-4.246	-4.232	-4.218	-4.205	-4.191	-4.177	-4.163	-4.149	-200
-190	-4.149	-4.136	-4.122	-4.108	-4.094	-4.080	-4.066	-4.052	-4.037	-4.023	-4.009	-190
-180	-4.009	-3.995	-3.981	-3.966	-3.952	-3.937	-3.923	-3.908	-3.894	-3.879	-3.865	-180
-170	-3.865	-3.850	-3.836	-3.821	-3.806	-3.791	-3.777	-3.762	-3.747	-3.733	-3.717	-170
-160	-3.717	-3.702	-3.687	-3.672	-3.657	-3.642	-3.626	-3.611	-3.596	-3.581	-3.565	-160
-150	-3.565	-3.550	-3.535	-3.519	-3.504	-3.488	-3.472	-3.457	-3.441	-3.426	-3.410	-150
-140	-3.410	-3.394	-3.379	-3.363	-3.347	-3.331	-3.315	-3.299	-3.283	-3.267	-3.251	-140
-130	-3.251	-3.235	-3.219	-3.203	-3.187	-3.171	-3.154	-3.138	-3.122	-3.105	-3.089	-130
-120	-3.089	-3.072	-3.056	-3.040	-3.023	-3.006	-2.990	-2.973	-2.956	-2.940	-2.923	-120
-110	-2.923	-2.906	-2.889	-2.873	-2.856	-2.839	-2.822	-2.805	-2.788	-2.771	-2.754	-110
-100	-2.754	-2.737	-2.719	-2.702	-2.685	-2.668	-2.651	-2.633	-2.616	-2.599	-2.581	-100
-90	-2.581	-2.564	-2.546	-2.529	-2.511	-2.493	-2.476	-2.458	-2.440	-2.422	-2.405	-90
-80	-2.405	-2.387	-2.369	-2.351	-2.334	-2.316	-2.298	-2.280	-2.262	-2.244	-2.225	-80
-70	-2.225	-2.207	-2.189	-2.171	-2.153	-2.134	-2.116	-2.098	-2.079	-2.061	-2.043	-70
-60	-2.043	-2.024	-2.006	-1.987	-1.969	-1.950	-1.931	-1.913	-1.894	-1.875	-1.857	-60
-50	-1.857	-1.838	-1.819	-1.800	-1.781	-1.762	-1.743	-1.724	-1.705	-1.686	-1.667	-50
-40	-1.667	-1.648	-1.629	-1.610	-1.591	-1.572	-1.552	-1.533	-1.514	-1.494	-1.475	-40
-30	-1.475	-1.456	-1.436	-1.417	-1.397	-1.378	-1.358	-1.339	-1.319	-1.299	-1.279	-30
-20	-1.279	-1.260	-1.240	-1.220	-1.200	-1.181	-1.161	-1.141	-1.121	-1.101	-1.081	-20
-10	-1.081	-1.061	-1.041	-1.021	-1.001	-0.980	-0.960	-0.940	-0.920	-0.900	-0.879	-10
0	-0.879	-0.859	-0.839	-0.818	-0.798	-0.777	-0.757	-0.736	-0.716	-0.695	-0.675	0
0	-0.675	-0.654	-0.633	-0.613	-0.592	-0.571	-0.550	-0.530	-0.509	-0.488	-0.467	0
10	-0.467	-0.446	-0.425	-0.404	-0.383	-0.362	-0.341	-0.320	-0.299	-0.278	-0.256	10
20	-0.256	-0.235	-0.214	-0.193	-0.171	-0.150	-0.129	-0.107	-0.086	-0.064	-0.043	20
30	-0.043	-0.022	0.000	0.022	0.043	0.065	0.086	0.108	0.130	0.151	0.173	30
40	0.173	0.195	0.216	0.238	0.260	0.282	0.303	0.325	0.347	0.369	0.391	40
50	0.391	0.413	0.435	0.457	0.479	0.501	0.523	0.545	0.567	0.589	0.611	50
60	0.611	0.634	0.656	0.678	0.700	0.723	0.745	0.767	0.790	0.812	0.834	60
70	0.834	0.857	0.879	0.902	0.924	0.947	0.969	0.992	1.015	1.037	1.060	70
80	1.060	1.083	1.105	1.128	1.151	1.174	1.196	1.219	1.242	1.265	1.288	80
90	1.288	1.311	1.334	1.357	1.380	1.403	1.426	1.449	1.472	1.495	1.519	90
100	1.519	1.542	1.565	1.588	1.612	1.635	1.659	1.682	1.705	1.729	1.752	100
110	1.752	1.776	1.799	1.823	1.846	1.870	1.893	1.917	1.941	1.964	1.988	110
120	1.988	2.012	2.036	2.060	2.083	2.107	2.131	2.155	2.179	2.203	2.227	120
130	2.227	2.251	2.275	2.299	2.323	2.347	2.371	2.395	2.420	2.444	2.468	130
140	2.468	2.492	2.517	2.541	2.565	2.590	2.614	2.639	2.663	2.687	2.712	140
°F	0	1	2	3	4	5	6	7	8	9	10	°F

°F	0	1	2	3	4	5	6	7	8	9	10	°F
150	2.712	2.737	2.761	2.785	2.810	2.835	2.860	2.884	2.909	2.934	2.956	150
160	2.958	2.983	3.008	3.033	3.058	3.082	3.107	3.132	3.157	3.182	3.207	160
170	3.207	3.232	3.257	3.282	3.307	3.333	3.358	3.383	3.408	3.433	3.459	170
180	3.459	3.484	3.509	3.534	3.560	3.585	3.610	3.636	3.662	3.687	3.712	180
190	3.712	3.738	3.763	3.789	3.814	3.840	3.866	3.891	3.917	3.943	3.968	190
200	3.968	3.994	4.020	4.046	4.071	4.097	4.123	4.149	4.175	4.201	4.227	200
210	4.227	4.253	4.279	4.305	4.331	4.357	4.383	4.409	4.435	4.461	4.487	210
220	4.487	4.513	4.540	4.566	4.592	4.618	4.645	4.671	4.697	4.724	4.750	220
230	4.750	4.776	4.803	4.829	4.855	4.882	4.908	4.935	4.962	4.988	5.015	230
240	5.015	5.042	5.068	5.095	5.122	5.148	5.175	5.202	5.228	5.255	5.282	240
250	5.282	5.309	5.336	5.363	5.389	5.416	5.443	5.470	5.497	5.524	5.551	250
260	5.551	5.578	5.605	5.632	5.660	5.687	5.714	5.741	5.768	5.795	5.823	260
270	5.823	5.850	5.877	5.904	5.932	5.959	5.986	6.014	6.041	6.068	6.096	270
280	6.096	6.123	6.151	6.178	6.206	6.233	6.261	6.288	6.316	6.343	6.371	280
290	6.371	6.399	6.426	6.454	6.482	6.510	6.537	6.565	6.593	6.621	6.648	290
300	6.648	6.676	6.704	6.732	6.760	6.788	6.816	6.844	6.872	6.900	6.928	300
310	6.928	6.956	6.984	7.012	7.040	7.068	7.096	7.124	7.152	7.181	7.209	310
320	7.209	7.237	7.265	7.294	7.322	7.350	7.378	7.407	7.434	7.463	7.492	320
330	7.492	7.520	7.549	7.577	7.606	7.634	7.663	7.691	7.720	7.748	7.777	330
340	7.777	7.805	7.834	7.863	7.891	7.920	7.949	7.977	8.006	8.035	8.064	340
350	8.064	8.092	8.121	8.150	8.179	8.208	8.237	8.265	8.294	8.323	8.352	350
360	8.352	8.381	8.410	8.439	8.468	8.497	8.526	8.555	8.584	8.614	8.643	360
370	8.643	8.672	8.701	8.730	8.759	8.789	8.818	8.847	8.876	8.906	8.935	370
380	8.935	8.964	8.994	9.023	9.052	9.082	9.111	9.141	9.170	9.200	9.229	380
390	9.229	9.259	9.288	9.318	9.347	9.377	9.406	9.436	9.465	9.495	9.525	390
400	9.525	9.555	9.584	9.614	9.644	9.673	9.703	9.733	9.762	9.793	9.822	400
410	9.822	9.852	9.882	9.912	9.942	9.972	10.002	10.032	10.062	10.092	10.122	410
420	10.122	10.152	10.182	10.212	10.242	10.272	10.302	10.332	10.362	10.392	10.423	420
430	10.423	10.453	10.483	10.513	10.543	10.574	10.604	10.634	10.664	10.695	10.725	430
440	10.725	10.755	10.786	10.816	10.847	10.877	10.907	10.938	10.968	10.999	11.029	440
450	11.029	11.060	11.090	11.121	11.151	11.182	11.213	11.243	11.274	11.304	11.335	450
460	11.335	11.366	11.396	11.427	11.458	11.489	11.519	11.550	11.581	11.612	11.643	460
470	11.643	11.673	11.704	11.735	11.766	11.797	11.828	11.859	11.890	11.920	11.951	470
480	11.951	11.982	12.013	12.044	12.075	12.106	12.138	12.169	12.200	12.231	12.262	480
490	12.262	12.293	12.324	12.356	12.386	12.418	12.449	12.480	12.511	12.543	12.574	490
500	12.574	12.605	12.636	12.668	12.699	12.730	12.762	12.793	12.824	12.856	12.887	500
510	12.887	12.919	12.950	12.982	13.013	13.045	13.076	13.108	13.139	13.171	13.202	510
520	13.202	13.234	13.265	13.297	13.328	13.360	13.392	13.423	13.455	13.487	13.518	520
530	13.518	13.550	13.582	13.614	13.645	13.677	13.709	13.741	13.772	13.804	13.836	530
540	13.836	13.868	13.900	13.932	13.964	13.995	14.027	14.059	14.091	14.123	14.155	540
550	14.155	14.187	14.219	14.251	14.283	14.315	14.347	14.379	14.411	14.444	14.476	550
560	14.476	14.508	14.540	14.572	14.604	14.636	14.669	14.701	14.733	14.765	14.797	560
570	14.797	14.830	14.862	14.894	14.926	14.959	14.991	15.023	15.055	15.088	15.121	570
580	15.121	15.153	15.185	15.218	15.250	15.283	15.315	15.347	15.380	15.412	15.445	580
590	15.445	15.477	15.510	15.543	15.575	15.608	15.640	15.673	15.705	15.738	15.771	590
600	15.771	15.803	15.836	15.869	15.901	15.934	15.967	15.999	16.032	16.065	16.098	600
610	16.098	16.130	16.163	16.196	16.229	16.262	16.295	16.327	16.360	16.393	16.426	610
620	16.426	16.459	16.492	16.525	16.558	16.591	16.624	16.657	16.690	16.723	16.756	620
630	16.756	16.789	16.822	16.855	16.888	16.921	16.954	16.987	17.020	17.053	17.086	630
640	17.086	17.120	17.153	17.186	17.219	17.252	17.286	17.319	17.352	17.385	17.418	640
650	17.418	17.452	17.485	17.518	17.552	17.585	17.618	17.652	17.685	17.718	17.752	650
660	17.752	17.785	17.818	17.852	17.885	17.919	17.952	17.986	18.019	18.053	18.086	660
670	18.086	18.120	18.153	18.187	18.221	18.254	18.288	18.321	18.355	18.389	18.422	670
680	18.422	18.456	18.490	18.523	18.557	18.591	18.624	18.658	18.692	18.725	18.759	680
690	18.759	18.793	18.827	18.861	18.894	18.928	18.962	18.996	19.030	19.064	19.097	690
700	19.097	19.131	19.165	19.199	19.233	19.267	19.301	19.335	19.369	19.403	19.437	700
710	19.437	19.471	19.505	19.539	19.573	19.607	19.641	19.675	19.709	19.743	19.777	710
720	19.777	19.811	19.845	19.879	19.913	19.947	19.982	20.016	20.050	20.084	20.118	720
730	20.118	20.152	20.187	20.221	20.255	20.289	20.323	20.358	20.392	20.426	20.460	730
740	20.460	20.495	20.529	20.563	20.597	20.632	20.666	20.700	20.735	20.769	20.803	740
750	20.803	20.838	20.872									750
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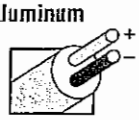
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MAXIMUM TEMPERATURE RANGE
Thermocouple Grade
– 328 to 2282°F
– 200 to 1250°C
Extension Grade
32 to 392°F
0 to 200°C
LIMITS OF ERROR
(whichever is greater)
Standard: 2.2°C or 0.75% Above 0°C
2.2°C or 2.0% Below 0°C
Special: 1.1°C or 0.4%
COMMENTS, BARE WIRE ENVIRONMENT:
Clean Oxidizing and Inert; Limited Use in Vacuum or Reducing; Wide Temperature Range; Most Popular Calibration
TEMPERATURE IN DEGREES °C
REFERENCE JUNCTION AT 0°C



Thermocouple Grade

Nickel-Chromium vs. Nickel-Aluminum



Extension Grade

Revised Thermocouple Reference Tables

TYPE K

Reference Tables
N.I.S.T. Monograph 175
Revised to ITS-90

Thermoelectric Voltage in Millivolts

°C	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	-0	°C
-260	-6.458	-6.457	-6.456	-6.455	-6.453	-6.452	-6.450	-6.448	-6.445	-6.444	-6.441	-260
-250	-6.441	-6.438	-6.435	-6.432	-6.429	-6.425	-6.421	-6.417	-6.413	-6.408	-6.404	-250
-240	-6.404	-6.399	-6.393	-6.388	-6.382	-6.377	-6.370	-6.364	-6.356	-6.351	-6.344	-240
-230	-6.344	-6.337	-6.329	-6.322	-6.314	-6.306	-6.297	-6.289	-6.280	-6.271	-6.262	-230
-220	-6.262	-6.252	-6.243	-6.233	-6.223	-6.213	-6.202	-6.192	-6.181	-6.170	-6.158	-220
-210	-6.158	-6.147	-6.135	-6.123	-6.111	-6.099	-6.087	-6.074	-6.061	-6.048	-6.035	-210
-200	-6.035	-6.021	-6.007	-5.994	-5.980	-5.965	-5.951	-5.936	-5.922	-5.907	-5.891	-200
-190	-5.891	-5.876	-5.861	-5.845	-5.829	-5.813	-5.797	-5.780	-5.763	-5.747	-5.730	-190
-180	-5.730	-5.713	-5.695	-5.678	-5.660	-5.642	-5.624	-5.605	-5.586	-5.569	-5.550	-180
-170	-5.550	-5.531	-5.512	-5.493	-5.474	-5.454	-5.435	-5.415	-5.395	-5.374	-5.354	-170
-160	-5.354	-5.333	-5.313	-5.292	-5.271	-5.250	-5.228	-5.207	-5.185	-5.163	-5.141	-160
-150	-5.141	-5.119	-5.097	-5.074	-5.052	-5.029	-5.006	-4.983	-4.960	-4.936	-4.913	-150
-140	-4.913	-4.889	-4.865	-4.841	-4.817	-4.793	-4.768	-4.744	-4.719	-4.694	-4.669	-140
-130	-4.669	-4.644	-4.618	-4.593	-4.567	-4.542	-4.516	-4.490	-4.463	-4.437	-4.411	-130
-120	-4.411	-4.384	-4.357	-4.330	-4.303	-4.276	-4.249	-4.221	-4.194	-4.165	-4.138	-120
-110	-4.138	-4.110	-4.082	-4.054	-4.025	-3.997	-3.968	-3.939	-3.911	-3.882	-3.852	-110
-100	-3.852	-3.823	-3.794	-3.764	-3.734	-3.705	-3.675	-3.645	-3.614	-3.584	-3.554	-100
-90	-3.554	-3.523	-3.492	-3.462	-3.431	-3.400	-3.368	-3.337	-3.306	-3.274	-3.243	-90
-80	-3.243	-3.211	-3.179	-3.147	-3.115	-3.083	-3.050	-3.018	-2.986	-2.953	-2.920	-80
-70	-2.920	-2.887	-2.854	-2.821	-2.788	-2.755	-2.721	-2.688	-2.654	-2.620	-2.587	-70
-60	-2.587	-2.553	-2.519	-2.485	-2.450	-2.416	-2.382	-2.347	-2.312	-2.278	-2.243	-60
-50	-2.243	-2.208	-2.173	-2.138	-2.103	-2.067	-2.032	-1.996	-1.961	-1.925	-1.889	-50
-40	-1.889	-1.854	-1.818	-1.782	-1.745	-1.709	-1.673	-1.637	-1.600	-1.564	-1.527	-40
-30	-1.527	-1.490	-1.453	-1.417	-1.380	-1.343	-1.306	-1.268	-1.231	-1.194	-1.156	-30
-20	-1.156	-1.119	-1.081	-1.043	-1.006	-0.968	-0.930	-0.892	-0.854	-0.816	-0.778	-20
-10	-0.778	-0.739	-0.701	-0.663	-0.624	-0.586	-0.547	-0.509	-0.470	-0.431	-0.392	-10
0	-0.392	-0.353	-0.314	-0.275	-0.236	-0.197	-0.157	-0.118	-0.079	-0.039	0.000	0
0	0.000	0.039	0.079	0.119	0.158	0.198	0.238	0.277	0.317	0.357	0.397	0
10	0.397	0.437	0.477	0.517	0.557	0.597	0.637	0.677	0.718	0.758	0.798	10
20	0.798	0.838	0.879	0.919	0.960	1.000	1.041	1.081	1.122	1.163	1.203	20
30	1.203	1.244	1.285	1.326	1.366	1.407	1.448	1.489	1.530	1.571	1.612	30
40	1.612	1.653	1.694	1.735	1.776	1.817	1.858	1.899	1.941	1.982	2.023	40
50	2.023	2.064	2.106	2.147	2.188	2.230	2.271	2.312	2.354	2.395	2.436	50
60	2.436	2.478	2.519	2.561	2.602	2.644	2.685	2.727	2.768	2.810	2.851	60
70	2.851	2.893	2.934	2.976	3.017	3.059	3.100	3.142	3.184	3.225	3.267	70
80	3.267	3.308	3.350	3.391	3.433	3.474	3.516	3.557	3.599	3.640	3.682	80
90	3.682	3.723	3.765	3.806	3.848	3.889	3.931	3.972	4.013	4.055	4.096	90
100	4.096	4.138	4.179	4.220	4.262	4.303	4.344	4.385	4.427	4.468	4.509	100
110	4.509	4.550	4.591	4.633	4.674	4.715	4.756	4.797	4.838	4.879	4.920	110
120	4.920	4.961	5.002	5.043	5.084	5.124	5.165	5.206	5.247	5.288	5.328	120
130	5.328	5.369	5.410	5.450	5.491	5.532	5.572	5.613	5.653	5.694	5.735	130
140	5.735	5.775	5.815	5.856	5.896	5.937	5.977	6.017	6.058	6.098	6.138	140
150	6.138	6.179	6.219	6.259	6.299	6.339	6.380	6.420	6.460	6.500	6.540	150
160	6.540	6.580	6.620	6.660	6.701	6.741	6.781	6.821	6.861	6.901	6.941	160
170	6.941	6.981	7.021	7.060	7.100	7.140	7.180	7.220	7.260	7.300	7.340	170
180	7.340	7.380	7.420	7.460	7.500	7.540	7.579	7.619	7.659	7.699	7.739	180
190	7.739	7.779	7.819	7.859	7.899	7.939	7.979	8.019	8.059	8.099	8.138	190
200	8.138	8.178	8.218	8.258	8.298	8.338	8.378	8.418	8.458	8.499	8.539	200
210	8.539	8.579	8.619	8.659	8.699	8.739	8.779	8.819	8.860	8.900	8.940	210
220	8.940	8.980	9.020	9.061	9.101	9.141	9.181	9.222	9.262	9.302	9.343	220
230	9.343	9.383	9.423	9.464	9.504	9.545	9.585	9.626	9.666	9.707	9.747	230
240	9.747	9.788	9.828	9.869	9.909	9.950	9.991	10.031	10.072	10.113	10.153	240
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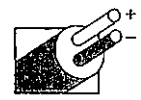
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260	10.561	10.602	10.643	10.684	10.725	10.766	10.807	10.848	10.889	10.930	10.971	260
270	10.971	11.012	11.053	11.094	11.135	11.176	11.217	11.259	11.300	11.341	11.382	270
280	11.382	11.423	11.465	11.506	11.547	11.588	11.630	11.671	11.712	11.753	11.795	280
290	11.795	11.836	11.877	11.919	11.960	12.001	12.043	12.084	12.126	12.167	12.209	290
300	12.209	12.250	12.291	12.333	12.374	12.416	12.457	12.499	12.540	12.582	12.624	300
310	12.624	12.665	12.707	12.748	12.790	12.831	12.873	12.915	12.956	12.998	13.040	310
320	13.040	13.081	13.123	13.165	13.206	13.248	13.290	13.331	13.373	13.415	13.457	320
330	13.457	13.498	13.540	13.582	13.624	13.665	13.707	13.749	13.791	13.833	13.874	330
340	13.874	13.916	13.958	14.000	14.042	14.084	14.125	14.167	14.209	14.251	14.293	340
350	14.293	14.335	14.377	14.419	14.461	14.503	14.545	14.587	14.629	14.671	14.713	350
360	14.713	14.755	14.797	14.839	14.881	14.923	14.965	15.007	15.049	15.091	15.133	360
370	15.133	15.175	15.217	15.259	15.301	15.343	15.385	15.427	15.469	15.511	15.554	370
380	15.554	15.596	15.638	15.680	15.722	15.764	15.806	15.849	15.891	15.933	15.975	380
390	15.975	16.017	16.059	16.102	16.144	16.186	16.228	16.270	16.313	16.355	16.397	390
400	16.397	16.439	16.482	16.524	16.566	16.609	16.651	16.693	16.735	16.778	16.820	400
410	16.820	16.862	16.904	16.947	16.989	17.031	17.074	17.116	17.158	17.201	17.243	410
420	17.243	17.285	17.328	17.370	17.413	17.455	17.497	17.540	17.582	17.624	17.667	420
430	17.667	17.709	17.752	17.794	17.837	17.879	17.921	17.964	18.006	18.048	18.091	430
440	18.091	18.134	18.176	18.218	18.261	18.303	18.345	18.388	18.431	18.473	18.516	440
450	18.516	18.558	18.601	18.643	18.686	18.728	18.771	18.813	18.856	18.898	18.941	450
460	18.941	18.983	19.026	19.068	19.111	19.154	19.196	19.239	19.281	19.324	19.366	460
470	19.366	19.409	19.451	19.494	19.537	19.579	19.622	19.664	19.707	19.750	19.792	470
480	19.792	19.835	19.877	19.920	19.962	20.005	20.048	20.090	20.133	20.175	20.218	480
490	20.218	20.261	20.303	20.346	20.389	20.431	20.474	20.516	20.559	20.602	20.644	490
500	20.644	20.687	20.730	20.772	20.815	20.857	20.900	20.943	20.985	21.028	21.071	500
510	21.071	21.113	21.156	21.199	21.241	21.284	21.326	21.369	21.412	21.454	21.497	510
520	21.497	21.540	21.582	21.625	21.668	21.710	21.753	21.795	21.838	21.881	21.924	520
530	21.924	21.966	22.009	22.052	22.094	22.137	22.179	22.222	22.265	22.307	22.350	530
540	22.350	22.393	22.435	22.478	22.521	22.563	22.606	22.649	22.691	22.734	22.776	540
550	22.776	22.819	22.862	22.904	22.947	22.990	23.032	23.075	23.117	23.160	23.203	550
560	23.203	23.245	23.288	23.331	23.373	23.416	23.458	23.501	23.544	23.586	23.629	560
570	23.629	23.671	23.714	23.757	23.799	23.842	23.884	23.927	23.970	24.012	24.055	570
580	24.055	24.097	24.140	24.182	24.225	24.267	24.310	24.353	24.395	24.438	24.480	580
590	24.480	24.523	24.565	24.608	24.650	24.693	24.735	24.778	24.820	24.863	24.905	590
600	24.905	24.948	24.990	25.033	25.075	25.118	25.160	25.203	25.245	25.288	25.330	600
610	25.330	25.373	25.415	25.458	25.500	25.543	25.585	25.627	25.670	25.712	25.755	610
620	25.755	25.797	25.840	25.882	25.924	25.967	26.009	26.052	26.094	26.136	26.179	620
630	26.179	26.221	26.263	26.306	26.348	26.390	26.433	26.475	26.517	26.560	26.602	630
640	26.602	26.644	26.687	26.729	26.771	26.814	26.856	26.898	26.940	26.983	27.025	640
650	27.025	27.067	27.109	27.152	27.194	27.236	27.278	27.320	27.363	27.405	27.447	650
660	27.447	27.489	27.531	27.574	27.616	27.658	27.700	27.742	27.784	27.826	27.869	660
670	27.869	27.911	27.953	27.995	28.037	28.079	28.121	28.163	28.205	28.247	28.289	670
680	28.289	28.332	28.374	28.416	28.458	28.500	28.542	28.584	28.626	28.668	28.710	680
690	28.710	28.752	28.794	28.835	28.877	28.919	28.961	29.003	29.045	29.087	29.129	690
700	29.129	29.171	29.213	29.255	29.297	29.338	29.380	29.422	29.464	29.506	29.548	700
710	29.548	29.589	29.631	29.673	29.715	29.757	29.798	29.840	29.882	29.924	29.965	710
720	29.965	30.007	30.049	30.090	30.132	30.174	30.216	30.257	30.299	30.341	30.382	720
730	30.382	30.424	30.466	30.507	30.549	30.590	30.632	30.674	30.715	30.757	30.798	730
740	30.798	30.840	30.881	30.923	30.964	31.006	31.047	31.089	31.130	31.172	31.213	740
750	31.213	31.255	31.296	31.338	31.379	31.421	31.462	31.504	31.545	31.586	31.628	750
760	31.628	31.669	31.710	31.752	31.793	31.834	31.876	31.917	31.958	32.000	32.041	760
770	32.041	32.082	32.124	32.165	32.206	32.247	32.289	32.330	32.371	32.412	32.453	770
780	32.453	32.495	32.536	32.577	32.618	32.659	32.700	32.742	32.783	32.824	32.865	780
790	32.865	32.906	32.947	32.988	33.029	33.070	33.111	33.152	33.193	33.234	33.275	790
°C	0	1	2	3	4	5	6	7	8	9	10	°C

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Revised Thermocouple Reference Tables

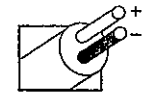
TYPE K

Reference Tables
N.I.S.T. Monograph 175
Revised to ITS-90



Thermocouple Grade

Nickel-Chromium vs. Nickel-Aluminum



Extension Grade

MAXIMUM TEMPERATURE RANGE
Thermocouple Grade
– 328 to 2282°F
– 200 to 1250°C
Extension Grade
32 to 392°F
0 to 200°C

LIMITS OF ERROR
(whichever is greater)
Standard: 2.2°C or 0.75% Above 0°C
2.2°C or 2.0% Below 0°C
Special: 1.1°C or 0.4%

COMMENTS, BARE WIRE ENVIRONMENT:
Clean Oxidizing and Inert; Limited Use in Vacuum or Reducing; Wide Temperature Range; Most Popular Calibration

TEMPERATURE IN DEGREES °C
REFERENCE JUNCTION AT 0°C

Thermoelectric Voltage in Millivolts

°C	0	1	2	3	4	5	6	7	8	9	10	°C
000	00.000	00.039	00.079	00.119	00.158	00.198	00.238	00.277	00.317	00.357	00.397	000
010	00.397	00.437	00.477	00.517	00.557	00.597	00.637	00.677	00.718	00.758	00.798	010
020	00.798	00.838	00.879	00.919	00.960	01.000	01.041	01.081	01.122	01.163	01.203	020
030	01.203	01.244	01.285	01.326	01.366	01.407	01.448	01.489	01.530	01.570	01.611	030
040	01.611	01.652	01.693	01.734	01.776	01.817	01.858	01.899	01.940	01.981	02.022	040
050	02.022	02.064	02.105	02.146	02.188	02.229	02.270	02.312	02.353	02.394	02.436	050
060	02.436	02.477	02.519	02.560	02.601	02.643	02.684	02.726	02.767	02.809	02.850	060
070	02.850	02.892	02.933	02.975	03.016	03.058	03.100	03.141	03.183	03.224	03.266	070
080	03.266	03.307	03.349	03.390	03.432	03.473	03.515	03.556	03.598	03.639	03.681	080
090	03.681	03.722	03.764	03.805	03.847	03.888	03.930	03.971	04.012	04.054	04.095	090
100	04.095	04.137	04.178	04.219	04.261	04.302	04.343	04.384	04.426	04.467	04.508	100
110	04.508	04.549	04.590	04.632	04.673	04.714	04.755	04.796	04.837	04.878	04.919	110
120	04.919	04.960	05.001	05.042	05.083	05.124	05.164	05.205	05.246	05.287	05.327	120
130	05.327	05.368	05.409	05.450	05.490	05.531	05.571	05.612	05.652	05.693	05.733	130
140	05.733	05.774	05.814	05.855	05.895	05.936	05.976	06.016	06.057	06.097	06.137	140
150	06.137	06.177	06.218	06.258	06.298	06.338	06.378	06.419	06.459	06.499	06.539	150
160	06.539	06.579	06.619	06.659	06.699	06.739	06.779	06.819	06.860	06.900	06.940	160
170	06.940	06.980	07.020	07.060	07.100	07.140	07.180	07.220	07.260	07.300	07.340	170
180	07.340	07.380	07.420	07.460	07.500	07.540	07.579	07.619	07.659	07.699	07.739	180
190	07.739	07.779	07.819	07.859	07.899	07.939	07.979	08.019	08.059	08.099	08.138	190
200	08.138	08.178	08.218	08.258	08.298	08.338	08.378	08.418	08.458	08.499	08.539	200
210	08.539	08.579	08.619	08.659	08.699	08.739	08.779	08.819	08.860	08.900	08.940	210
220	08.940	08.980	09.020	09.061	09.101	09.141	09.181	09.222	09.262	09.302	09.343	220
230	09.343	09.383	09.423	09.464	09.504	09.545	09.585	09.626	09.666	09.707	09.747	230
240	09.747	09.788	09.828	09.869	09.909	09.950	09.991	10.031	10.072	10.113	10.153	240
250	10.153	10.194	10.235	10.276	10.316	10.357	10.398	10.439	10.480	10.520	10.561	250
260	10.561	10.602	10.643	10.684	10.725	10.766	10.807	10.848	10.889	10.930	10.971	260
270	10.971	11.012	11.053	11.094	11.135	11.176	11.217	11.259	11.300	11.341	11.382	270
280	11.382	11.423	11.465	11.506	11.547	11.588	11.630	11.671	11.712	11.753	11.795	280
290	11.795	11.836	11.877	11.919	11.960	12.001	12.043	12.084	12.126	12.167	12.209	290
300	12.209	12.250	12.291	12.333	12.374	12.416	12.457	12.499	12.540	12.582	12.624	300
310	12.624	12.665	12.707	12.748	12.790	12.831	12.873	12.915	12.956	12.998	13.040	310
320	13.040	13.081	13.123	13.165	13.206	13.248	13.290	13.331	13.373	13.415	13.457	320
330	13.457	13.498	13.540	13.582	13.624	13.665	13.707	13.749	13.791	13.833	13.874	330
340	13.874	13.916	13.958	14.000	14.042	14.084	14.126	14.167	14.209	14.251	14.293	340
350	14.293	14.335	14.377	14.419	14.461	14.503	14.545	14.587	14.629	14.671	14.713	350
360	14.713	14.755	14.797	14.839	14.881	14.923	14.965	15.007	15.049	15.091	15.133	360
370	15.133	15.175	15.217	15.259	15.301	15.343	15.385	15.427	15.469	15.511	15.554	370
380	15.554	15.596	15.638	15.680	15.722	15.764	15.806	15.849	15.891	15.933	15.975	380
390	15.975	16.017	16.059	16.102	16.144	16.186	16.228	16.270	16.313	16.355	16.397	390
400	16.397	16.439	16.482	16.524	16.566	16.608	16.651	16.693	16.735	16.778	16.820	400
410	16.820	16.862	16.904	16.947	16.989	17.031	17.074	17.116	17.158	17.201	17.243	410
420	17.243	17.285	17.328	17.370	17.413	17.455	17.497	17.540	17.582	17.624	17.667	420
430	17.667	17.709	17.752	17.794	17.837	17.879	17.921	17.964	18.006	18.049	18.091	430
440	18.091	18.134	18.176	18.218	18.261	18.303	18.346	18.388	18.431	18.473	18.516	440
450	18.516	18.558	18.600	18.643	18.685	18.728	18.770	18.813	18.855	18.898	18.940	450
460	18.940	18.983	19.025	19.068	19.110	19.153	19.195	19.238	19.280	19.323	19.365	460
470	19.365	19.408	19.450	19.493	19.535	19.578	19.620	19.663	19.705	19.748	19.790	470
480	19.790	19.833	19.875	19.918	19.960	20.003	20.045	20.088	20.130	20.173	20.215	480
490	20.215	20.258	20.301	20.343	20.386	20.428	20.471	20.513	20.556	20.598	20.641	490
500	20.641	20.683	20.726	20.768	20.811	20.853	20.896	20.938	20.981	21.023	21.066	500
510	21.066	21.108	21.151	21.193	21.236	21.278	21.321	21.363	21.406	21.448	21.491	510
520	21.491	21.533	21.576	21.618	21.661	21.703	21.746	21.788	21.831	21.873	21.916	520
530	21.916	21.958	22.001	22.043	22.086	22.128	22.171	22.213	22.256	22.298	22.341	530
540	22.341	22.383	22.426	22.468	22.511	22.553	22.596	22.638	22.681	22.723	22.766	540
550	22.766	22.808	22.851	22.893	22.936	22.978	23.021	23.063	23.106	23.148	23.191	550
560	23.191	23.233	23.276	23.318	23.361	23.403	23.446	23.488	23.531	23.573	23.616	560
570	23.616	23.658	23.701	23.743	23.786	23.828	23.871	23.913	23.956	23.998	24.041	570
580	24.041	24.083	24.126	24.168	24.211	24.253	24.296	24.338	24.381	24.423	24.466	580
590	24.466	24.508	24.551	24.593	24.636	24.678	24.721	24.763	24.806	24.848	24.891	590
600	24.891	24.933	24.976	25.018	25.061	25.103	25.146	25.188	25.231	25.273	25.315	600
610	25.315	25.358	25.400	25.443	25.485	25.528	25.570	25.613	25.655	25.698	25.740	610
620	25.740	25.783	25.825	25.868	25.910	25.953	25.995	26.038	26.080	26.123	26.165	620
630	26.165	26.208	26.250	26.293	26.335	26.378	26.420	26.463	26.505	26.548	26.590	630
640	26.590	26.633	26.675	26.718	26.760	26.803	26.845	26.888	26.930	26.973	27.015	640
650	27.015	27.058	27.100	27.143	27.185	27.228	27.270	27.313	27.355	27.398	27.440	650
660	27.440	27.483	27.525	27.568	27.610	27.653	27.695	27.738	27.780	27.823	27.865	660
670	27.865	27.907	27.950	27.992	28.035	28.077	28.120	28.162	28.205	28.247	28.289	670
680	28.289	28.332	28.374	28.417	28.459	28.502	28.544	28.587	28.629	28.671	28.714	680
690	28.714	28.756	28.799	28.841	28.883	28.926	28.968	29.011	29.053	29.095	29.138	690
700	29.138	29.180	29.222	29.265	29.307	29.349	29.392	29.434	29.476	29.518	29.561	700
710	29.561	29.603	29.645	29.688	29.730	29.772	29.814	29.857	29.899	29.941	29.983	710
720	29.983	30.026	30.068	30.110	30.152	30.194	30.237	30.279	30.321	30.363	30.405	720
730	30.405	30.448	30.490	30.532	30.574	30.616	30.658	30.701	30.743	30.785	30.827	730
740	30.827	30.869	30.911	30.954	30.996	31.038	31.080	31.122	31.164	31.206	31.248	740
750	31.248	31.290	31.332	31.375	31.417	31.459	31.501	31.543	31.585	31.627	31.669	750
760	31.669	31.711	31.753	31.795	31.837	31.879	31.921	31.963	32.005	32.047	32.089	760
770	32.089	32.131	32.173	32.215	32.257	32.299	32.341	32.383	32.425	32.467	32.509	770
780	32.509	32.551	32.593	32.635	32.677	32.719	32.761	32.803	32.845	32.887	32.929	780
790	32.929	32.971	33.013	33.055	33.097	33.139	33.181	33.223	33.265	33.307	33.349	790
800	33.349	33.391	33.433	33.475	33.517	33.559	33.601	33.643	33.685	33.727	33.769	800
810	33.769	33.811	33.853	33.895	33.937	33.979	34.021	34.063	34.105	34.147	34.189	810
820	34.189	34.230	34.272	34.314	34.356	34.398	34.440	34.482	34.524	34.566	34.608	820
830	34.608	34.650	34.692	34.734	34.776	34.818	34.860	34.902	34.944	34.986	35.028	830
840	35.028	35.070	35.112	35.154	35.196	35.238	35.280	35.322	35.364	35.406	35.448	840
850	35.448	35.490	35.532	35.574	35.616	35.658	35.700	35.742	35.784	35.826	35.868	850
860	35.868	35.910	35.952	35.994	36.036	36.078	36.120	36.162	36.204	36.246	36.288	860
870	36.288	36.330	36.372	36.414	36.456	36.498	36.540	36.582	36.624	36.666	36.708	870
880	36.708	36.750	36.792	36.834	36.876	36.918	36.960	37.002	37.044	37.086	37.128	880
890	37.128	37.170	37.212	37.254	37.296	37.338	37.380	37.422	37.464	37.506	37.548	890
900	37.548	37.590	37.632	37.674	37.716	37.758	37.800	37.842	37.884	37.926	37.968	900
910	37.968	38.010	38.052	38.094	38.136	38.178	38.220	38.262	38.304	38.346	38.388	910
920	38.388	38.430	38.472	38.514	38.556	38.598	38.640	38.682	38.724	38.766	38.808	920
930	38.808	38.850	38.892	38.934	38.976	39.018	39.060	39.102	39.144	39.186	39.228	930
940	39.228	39.270	39.312	39.354	39.396	39.438	39.480	39.522	39.564	39.606	39.648	940
950	39.648	39.690	39.732	39.774	39.816	39.858	39.900	39.942	39.984	40.026	40.068	950
960	40.068	40.110	40.152	40.194	40.236	40.278	40.320	40.362	40.404	40.446	40.488	960
970	40.488	40.530	40.572	40.614	40.656	40.698	40.740	40.782	40.824	40.866	40.908	970
980	40.908	40.950	40.992	41.034	41.076	41.118	41.160	41.202	41.244	41.286	41.328	980
990	41.328	41.370	41.412	41.454	41.496	41.538	41.580	41.622	41.664	41.706	41.748	990
1000	41.748	41.790	41.832	41.874	41.916	41.958	42.000	42.042	42.084	42.126	42.168	1000
1010	42.168	42.210	42.252	42.294	42.336	42.378	42.420	42.462	42.504	42.546	42.588	1010
1020	42.588	42.630	42.672	42.714	42.756	42.798	42.840	42.882	42.924	42.966	43.008	1020
1030	43.008	43.050	43.092	43.134	43.176	43.218	43.260	43.302	43.344	43.386	43.428	1030
1040	43.428	43.470	43.512	43.554	43.596	43.638	43.680	43.722	43.764	43.806	43.848	1040
1050	43.848	43.890	43.932	43.974	44.016	44.058	44.100	44.142	44.184	44.226	44.268	1050
1060	44.268	44.310	44.352	44.394	44.436	44.478	44.520	44.562	44.604	44.646	44.688	1060
1070	44.688	44.730	44.772	44.814	44.856	44.898	44.940	44.982	45.024	45.066	45.108	1070
1080	45.108	45.150	45.192	45.234	45.276	45.318	45.360	45.402	45.444	45.486	45.528	1080
1090	45.528	45.570	45.612	45.654	45.696	45.738	45.780	45.822	45.864	45.906	45.948	1090
1100	45.948	45.990	46.032	46.074	46.116	46.158	46.200	46.242	46.284	46.326	46.368	1100
1110	46.368	46.410	46.452	46.494	46.536	46.578	46.620	46.662	46.704	46.746	46.788	1110
1120	46.788	46.830	46.872	46.914	46.956	46.998	47.040	47.082	47.124	47.166	47.208	1120
1130	47.208	47.250	47.292	47.334	47.376	47.418	47.460	47.502	47.544	47.586	47.628	1130
1140	47.628	47.670	47.712	47.754	47.796	47.838	47.880	47.922	47.964	48.006	48.048	1140
1150	48.048	48.090	48.132	48.174	48.216	48.258	48.300	48.342	48.384	48.426	48.468	1150
1160	48.468	48.510	48.552	48.594	48.636	48.678	48.720	48.762	48.804	48.846	48.888	1160
1170	48.888	48.930	48.972	49.014	49.056	49.098	49.140	49.182	49.224	49.266	49.308	1170
1180	49.308	49.350	49.392	49.434	49.476	49.518	49.560	49.602	49.644	49.686	49.728	1180
1190	49.728	49.770	49.812	49.854	49.896	49.938	49.980	50.022	50.064	50.106	50.148	1190
1200	50.148	50.190	50.232	50.274	50.316	50.358	50.400	50.442	50.484	50.526	50.568	1200
1210	50.568	50.610	50.652	50.694	50.736	50.778	50.820	50.862	50.904	50.946	50.988	1210
1220	50.988	51.030	51.072	51.114	51.156	51.198	51.240	51.282	51.324	51.366	51.408	1220
1230	51.408	51.450	51.492	51.534	51.576	51.618	51.660	51.702	51.744	51.786	51.828	1230
1240	51.828	51.870	51.912	51.954	51.996	52.038	52.080	52.122	52.164	52.206	52.248	1240
1250	52.248	52.290	52.332	52.374	52.416	52.458	52.500	52.542	52.584	52.626	52.668	1250
1260	52.668	52.710	52.752	52.794	52.836	52.878	52.920	52.962	53.004	53.046	53.088	1260
1270	53.088	53.130	53.172	53.214	53.256	53.298	53.340	53.382	53.424	53.466	53.508	1270
1280	53.508	53.550	53.592	53.634	53.676	53.718	53.760	53.802	53.844	53.886	53.928	1280
1290	53.928	53.970	54.012	54.054	54.096	54.138	54.180	54.222	54.264	54.306	54.348	1290
1300	54.348	54.390	54.432	54.474	54.516	54.558	54.600	54.642	54.684	54.726	54.768	1300
1310	54.768	54.810	54.852	54.894	54.936	54.978	55.020	55.062	55.104	55.146	55.188	1310
1320	55.188	55.230	55.272	55.314	55.356	55.398	55.440	55.482	55.524	55.566	55.608	1320
1330	55.608	55.650	55.692	55.734	55.776	55.818	55.860	55.902	55.944	55.986	56.028	1330
1340	56.028	56.070	56.112	56.154	56.196	56.238	56.280	56.322	56.364	56.406	56.448	1340
1350	56.448	56.490	56.532	56.574	56.616	56.658	56.700	56.742	56.784	56.826	56.868	1350
1360	56.868	56.910	56.952	56.994	57.036	57.078	57.120	57.162	57.204	57.246	57.288	1360
1370	57.288	57.330	57.372	57.414	57.456	57.498	57.540	57.582	57.624	57.666	57.708	1370

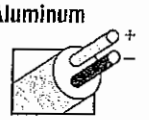
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MAXIMUM TEMPERATURE RANGE
Thermocouple Grade
– 328 to 2282°F
– 200 to 1250°C
Extension Grade
32 to 392°F
0 to 200°C
LIMITS OF ERROR
(whichever is greater)
Standard: 2.2°C or 0.75% Above 0°C
2.2°C or 2.0% Below 0°C
Special: 1.1°C or 0.4%
COMMENTS, BARE WIRE ENVIRONMENT:
Clean Oxidizing and Inert; Limited Use in Vacuum or Reducing; Wide Temperature Range; Most Popular Calibration
TEMPERATURE IN DEGREES °F
REFERENCE JUNCTION AT 32°F



Thermocouple Grade

Nickel-Chromium vs. Nickel-Aluminum



Extension Grade

Revised Thermocouple Reference Tables

TYPE K

Reference Tables
N.I.S.T. Monograph 175
Revised to ITS-90

Thermoelectric Voltage in Millivolts

°F	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	°F
-450							-6.458	-6.457	-6.457	-6.456	-6.456	-450
-440	-6.456	-6.456	-6.455	-6.454	-6.453	-6.452	-6.451	-6.450	-6.449	-6.448	-6.446	-440
-430	-6.446	-6.445	-6.444	-6.443	-6.441	-6.440	-6.438	-6.436	-6.435	-6.433	-6.431	-430
-420	-6.431	-6.429	-6.427	-6.425	-6.423	-6.421	-6.419	-6.416	-6.414	-6.411	-6.408	-420
-410	-6.408	-6.405	-6.404	-6.401	-6.398	-6.395	-6.392	-6.389	-6.386	-6.383	-6.380	-410
-400	-6.380	-6.377	-6.373	-6.370	-6.366	-6.363	-6.359	-6.355	-6.352	-6.348	-6.344	-400
-390	-6.344	-6.340	-6.336	-6.332	-6.328	-6.323	-6.319	-6.315	-6.310	-6.306	-6.301	-390
-380	-6.301	-6.296	-6.292	-6.287	-6.282	-6.277	-6.272	-6.267	-6.262	-6.257	-6.251	-380
-370	-6.251	-6.246	-6.241	-6.235	-6.230	-6.224	-6.218	-6.213	-6.207	-6.201	-6.195	-370
-360	-6.195	-6.189	-6.183	-6.177	-6.171	-6.165	-6.158	-6.152	-6.146	-6.139	-6.133	-360
-350	-6.133	-6.126	-6.119	-6.113	-6.106	-6.099	-6.092	-6.085	-6.078	-6.071	-6.064	-350
-340	-6.064	-6.057	-6.049	-6.042	-6.035	-6.027	-6.020	-6.012	-6.004	-5.997	-5.989	-340
-330	-5.989	-5.981	-5.973	-5.965	-5.957	-5.949	-5.941	-5.933	-5.925	-5.917	-5.908	-330
-320	-5.908	-5.900	-5.891	-5.883	-5.874	-5.866	-5.857	-5.848	-5.840	-5.831	-5.822	-320
-310	-5.822	-5.813	-5.804	-5.795	-5.786	-5.778	-5.767	-5.758	-5.749	-5.739	-5.730	-310
-300	-5.730	-5.721	-5.711	-5.701	-5.691	-5.682	-5.672	-5.662	-5.652	-5.642	-5.632	-300
-290	-5.632	-5.622	-5.612	-5.602	-5.592	-5.581	-5.571	-5.561	-5.550	-5.540	-5.529	-290
-280	-5.529	-5.518	-5.508	-5.497	-5.487	-5.476	-5.465	-5.454	-5.443	-5.432	-5.421	-280
-270	-5.421	-5.410	-5.398	-5.388	-5.377	-5.365	-5.354	-5.343	-5.331	-5.320	-5.308	-270
-260	-5.308	-5.297	-5.285	-5.273	-5.261	-5.250	-5.238	-5.226	-5.214	-5.202	-5.190	-260
-250	-5.190	-5.178	-5.166	-5.153	-5.141	-5.129	-5.117	-5.104	-5.092	-5.078	-5.067	-250
-240	-5.067	-5.054	-5.042	-5.029	-5.016	-5.003	-4.991	-4.978	-4.965	-4.952	-4.939	-240
-230	-4.939	-4.926	-4.913	-4.900	-4.886	-4.873	-4.860	-4.847	-4.833	-4.820	-4.806	-230
-220	-4.806	-4.793	-4.779	-4.766	-4.752	-4.738	-4.724	-4.711	-4.697	-4.683	-4.669	-220
-210	-4.669	-4.655	-4.641	-4.627	-4.613	-4.599	-4.584	-4.570	-4.556	-4.542	-4.527	-210
-200	-4.527	-4.513	-4.498	-4.484	-4.469	-4.455	-4.440	-4.425	-4.411	-4.396	-4.381	-200
-190	-4.381	-4.366	-4.351	-4.336	-4.321	-4.306	-4.291	-4.276	-4.261	-4.246	-4.231	-190
-180	-4.231	-4.215	-4.200	-4.185	-4.169	-4.154	-4.138	-4.123	-4.107	-4.091	-4.076	-180
-170	-4.076	-4.060	-4.044	-4.029	-4.013	-3.997	-3.981	-3.965	-3.949	-3.933	-3.917	-170
-160	-3.917	-3.901	-3.885	-3.869	-3.852	-3.836	-3.820	-3.803	-3.787	-3.771	-3.754	-160
-150	-3.754	-3.738	-3.721	-3.705	-3.688	-3.671	-3.655	-3.638	-3.621	-3.604	-3.587	-150
-140	-3.587	-3.571	-3.554	-3.537	-3.520	-3.503	-3.486	-3.469	-3.451	-3.434	-3.417	-140
-130	-3.417	-3.400	-3.382	-3.365	-3.348	-3.330	-3.313	-3.295	-3.278	-3.260	-3.243	-130
-120	-3.243	-3.225	-3.207	-3.190	-3.172	-3.154	-3.136	-3.119	-3.101	-3.083	-3.065	-120
-110	-3.065	-3.047	-3.029	-3.011	-2.993	-2.975	-2.957	-2.938	-2.920	-2.902	-2.884	-110
-100	-2.884	-2.865	-2.847	-2.829	-2.810	-2.792	-2.773	-2.755	-2.736	-2.718	-2.699	-100
-90	-2.699	-2.680	-2.662	-2.643	-2.624	-2.605	-2.587	-2.568	-2.549	-2.530	-2.511	-90
-80	-2.511	-2.492	-2.473	-2.454	-2.435	-2.416	-2.397	-2.378	-2.359	-2.339	-2.320	-80
-70	-2.320	-2.301	-2.282	-2.262	-2.243	-2.223	-2.204	-2.185	-2.165	-2.146	-2.126	-70
-60	-2.126	-2.106	-2.087	-2.067	-2.048	-2.028	-2.008	-1.988	-1.968	-1.948	-1.928	-60
-50	-1.928	-1.908	-1.889	-1.869	-1.850	-1.830	-1.810	-1.790	-1.770	-1.749	-1.729	-50
-40	-1.729	-1.709	-1.689	-1.669	-1.649	-1.628	-1.608	-1.588	-1.568	-1.547	-1.527	-40
-30	-1.527	-1.507	-1.486	-1.466	-1.445	-1.425	-1.404	-1.384	-1.363	-1.343	-1.322	-30
-20	-1.322	-1.301	-1.281	-1.260	-1.239	-1.218	-1.198	-1.177	-1.156	-1.135	-1.114	-20
-10	-1.114	-1.094	-1.073	-1.052	-1.031	-1.010	-0.989	-0.968	-0.947	-0.926	-0.905	-10
0	-0.905	-0.883	-0.862	-0.841	-0.820	-0.799	-0.778	-0.756	-0.735	-0.714	-0.692	0
0	-0.692	-0.671	-0.650	-0.628	-0.607	-0.586	-0.564	-0.543	-0.521	-0.500	-0.478	0
10	-0.478	-0.457	-0.435	-0.413	-0.392	-0.370	-0.349	-0.327	-0.305	-0.284	-0.262	10
20	-0.262	-0.240	-0.218	-0.197	-0.175	-0.153	-0.131	-0.109	-0.088	-0.066	-0.044	20
30	-0.044	-0.022	0.000	0.022	0.044	0.066	0.088	0.110	0.132	0.154	0.176	30
40	0.176	0.198	0.220	0.242	0.264	0.286	0.308	0.330	0.353	0.375	0.397	40
50	0.397	0.419	0.441	0.463	0.486	0.508	0.530	0.552	0.575	0.597	0.619	50
60	0.619	0.642	0.664	0.686	0.709	0.731	0.753	0.776	0.798	0.821	0.843	60
70	0.843	0.865	0.888	0.910	0.933	0.955	0.978	1.000	1.023	1.045	1.068	70
80	1.068	1.090	1.112	1.135	1.158	1.181	1.203	1.226	1.249	1.271	1.294	80
90	1.294	1.316	1.339	1.362	1.384	1.407	1.430	1.453	1.475	1.498	1.521	90

°F	0	1	2	3	4	5	6	7	8	9	10	°F
100	1.521	1.543	1.566	1.589	1.612	1.635	1.657	1.680	1.703	1.726	1.749	100
110	1.749	1.771	1.794	1.817	1.840	1.863	1.886	1.908	1.931	1.954	1.977	110
120	1.977	2.000	2.023	2.046	2.069	2.092	2.115	2.138	2.161	2.184	2.207	120
130	2.207	2.230	2.253	2.276	2.299	2.321	2.344	2.367	2.390	2.413	2.436	130
140	2.436	2.459	2.483	2.506	2.529	2.552	2.575	2.598	2.621	2.644	2.667	140
150	2.667	2.690	2.713	2.736	2.759	2.782	2.805	2.828	2.851	2.874	2.897	150
160	2.897	2.920	2.944	2.967	2.990	3.013	3.036	3.059	3.082	3.105	3.128	160
170	3.128	3.151	3.174	3.197	3.220	3.244	3.267	3.290	3.313	3.336	3.359	170
180	3.359	3.382	3.405	3.428	3.451	3.474	3.497	3.520	3.544	3.567	3.590	180
190	3.590	3.613	3.636	3.659	3.682	3.705	3.728	3.751	3.774	3.797	3.820	190
200	3.820	3.843	3.866	3.889	3.912	3.935	3.958	3.981	4.004	4.027	4.050	200
210	4.050	4.073	4.096	4.119	4.142	4.165	4.188	4.211	4.234	4.257	4.280	210
220	4.280	4.303	4.326	4.349	4.372	4.395	4.417	4.440	4.463	4.486	4.509	220
230	4.509	4.532	4.555	4.578	4.601	4.623	4.646	4.669	4.692	4.715	4.738	230
240	4.738	4.760	4.783	4.806	4.829	4.852	4.874	4.897	4.920	4.943	4.965	240
250	4.965	4.988	5.011	5.034	5.056	5.079	5.102	5.124	5.147	5.170	5.192	250
260	5.192	5.215	5.238	5.260	5.283	5.306	5.328	5.351	5.374	5.396	5.419	260
270	5.419	5.441	5.464	5.487	5.509	5.532	5.554	5.577	5.599	5.622	5.644	270
280	5.644	5.667	5.690	5.712	5.735	5.757	5.779	5.802	5.824	5.847	5.869	280
290	5.869	5.892	5.914	5.937	5.959	5.982	6.004	6.026	6.049	6.071	6.094	290
300	6.094	6.116	6.138	6.161	6.183	6.205	6.228	6.250	6.272	6.295	6.317	300
310	6.317	6.339	6.362	6.384	6.406	6.429	6.451	6.473	6.496	6.518	6.540	310
320	6.540	6.562	6.585	6.607	6.629	6.652	6.674	6.696	6.718	6.741	6.763	320
330	6.763	6.785	6.807	6.829	6.852	6.874	6.896	6.918	6.941	6.963	6.985	330
340	6.985	7.007	7.029	7.052	7.074	7.096	7.118	7.140	7.163	7.185	7.207	340
350	7.207	7.229	7.251	7.273	7.295	7.318	7.340	7.362	7.384	7.407	7.429	350
360	7.429	7.451	7.473	7.495	7.517	7.540	7.562	7.584	7.606	7.628	7.650	360
370	7.650	7.673	7.695	7.717	7.739	7.761	7.783	7.805	7.828	7.850	7.872	370
380	7.872	7.894	7.917	7.939	7.961	7.983	8.005	8.027	8.050	8.072	8.094	380
390	8.094	8.116	8.138	8.161	8.183	8.205	8.227	8.250	8.272	8.294	8.316	390
400	8.316	8.338	8.361	8.383	8.405	8.427	8.450	8.472	8.494	8.516	8.539	400
410	8.539	8.561	8.583	8.605	8.628	8.650	8.672	8.694	8.717	8.739	8.761	410
420	8.761	8.784	8.806	8.828	8.851	8.873	8.895	8.918	8.940	8.962	8.985	420
430	8.985	9.007	9.029	9.052	9.074	9.096	9.119	9.141	9.163	9.186	9.208	430
440	9.208	9.231	9.253	9.276	9.298	9.320	9.343	9.365	9.388	9.410	9.432	440
450	9.432	9.455	9.477	9.500	9.522	9.545	9.567	9.590	9.612	9.635	9.657	450
460	9.657	9.680	9.702	9.725	9.747	9.770	9.792	9.815	9.837	9.860	9.882	460
470	9.882	9.905	9.927	9.950	9.973	9.995	10.018	10.040	10.063	10.086	10.108	470
480	10.108	10.131	10.153	10.176	10.199	10.221	10.244	10.267	10.289	10.312	10.334	480
490	10.334	10.357	10.380	10.402	10.425	10.448	10.471	10.493	10.516	10.539	10.561	490
500	10.561	10.584	10.607	10.629	10.652	10.675	10.698	10.720	10.743	10.766	10.789	500
510	10.789	10.811	10.834	10.857	10.880	10.902	10.925	10.948	10.971	10.994	11.017	510
520	11.017	11.039	11.062	11.085	11.108	11.131	11.154	11.176	11.199	11.222	11.245	520
530	11.245	11.268	11.291	11.313	11.336	11.359	11.382	11.405	11.428	11.451	11.474	530
540	11.474	11.497	11.519	11.542	11.565	11.588	11.611	11.634	11.657	11.680	11.703	540
550	11.703	11.726	11.749	11.772	11.795	11.818	11.841	11.864	11.887	11.910	11.933	550
560	11.933	11.956	11.978	12.001	12.024	12.047	12.070	12.093	12.116	12.140	12.163	560
570	12.163	12.186	12.209	12.232	12.255	12.278	12.301	12.324	12.347	12.370	12.393	570
580	12.393	12.416	12.439	12.462	12.485	12.508	12.531	12.554	12.577	12.600	12.624	580
590	12.624	12.647	12.670	12.693	12.716	12.739	12.762	12.785	12.808	12.831	12.855	590
600	12.855	12.878	12.901	12.924	12.947	12.970	12.993	13.016	13.040	13.063	13.086	600
610	13.086	13.109	13.132	13.155	13.179	13.202	13.225	13.248	13.271	13.294	13.318	610
620	13.318	13.341	13.364	13.387	13.410	13.433	13.457	13.480	13.503	13.526	13.549	620
630	13.549	13.573	13.596	13.619	13.642	13.665	13.689	13.712	13.735	13.758	13.782	630
640	13.782	13.805	13.828	13.851	13.874	13.898	13.921	13.944	13.967	13.991	14.014	640
650	14.014	14.037	14.060	14.084	14.107	14.130	14.154	14.177	14.200	14.223	14.247	650
660	14.247	14.270	14.293	14.316	14.340	14.363	14.386	14.410	14.433	14.456	14.479	660
670	14.479	14.503	14.526	14.549	14.573	14.596	14.619	14.643	14.666	14.689	14.713	670
680	14.713	14.736	14.759	14.783	14.806	14.829	14.853	14.876	14.899	14.923	14.946	680
690	14.946	14.969	14.993	15.016	15.039	15.063	15.086	15.109	15.133	15.156	15.179	690

°F	0	1	2	3	4	5	6	7	8	9	10	°F

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Revised Thermocouple Reference Tables

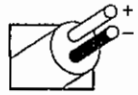
TYPE K

Reference Tables
N.I.S.T. Monograph 175
Revised to ITS-90



Thermocouple Grade

Nickel-Chromium vs. Nickel-Aluminum



Extension Grade

MAXIMUM TEMPERATURE RANGE
Thermocouple Grade
– 328 to 2282°F
– 200 to 1250°C
Extension Grade
32 to 392°F
0 to 200°C

LIMITS OF ERROR
(whichever is greater)
Standard: 2.2°C or 0.75% Above 0°C
2.2°C or 2.0% Below 0°C
Special: 1.1°C or 0.4%

COMMENTS, BARE WIRE ENVIRONMENT:
Clean Oxidizing and Inert; Limited Use in Vacuum or Reducing; Wide Temperature Range; Most Popular Calibration

TEMPERATURE IN DEGREES °F
REFERENCE JUNCTION AT 32°F

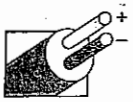
Thermoelectric Voltage in Millivolts

°F	0	1	2	3	4	5	6	7	8	9	10	°F
700	15.172	15.203	15.228	15.258	15.273	15.296	15.329	15.343	15.366	15.390	15.413	700
710	15.413	15.437	15.460	15.483	15.507	15.530	15.554	15.577	15.600	15.624	15.647	710
720	15.647	15.671	15.694	15.717	15.741	15.764	15.788	15.811	15.834	15.858	15.881	720
730	15.881	15.905	15.928	15.952	15.975	15.998	16.022	16.045	16.069	16.092	16.116	730
740	16.116	16.139	16.163	16.186	16.209	16.233	16.256	16.280	16.303	16.327	16.350	740
750	16.350	16.374	16.397	16.421	16.444	16.468	16.491	16.514	16.538	16.561	16.585	750
760	16.585	16.608	16.632	16.655	16.679	16.702	16.726	16.749	16.773	16.796	16.820	760
770	16.820	16.843	16.867	16.890	16.914	16.937	16.961	16.984	17.008	17.031	17.055	770
780	17.055	17.078	17.102	17.125	17.149	17.173	17.196	17.220	17.243	17.267	17.290	780
790	17.290	17.314	17.337	17.361	17.384	17.408	17.431	17.455	17.478	17.502	17.526	790
800	17.526	17.549	17.573	17.596	17.620	17.643	17.667	17.690	17.714	17.738	17.761	800
810	17.761	17.785	17.808	17.832	17.855	17.879	17.902	17.926	17.950	17.973	17.997	810
820	17.997	18.020	18.044	18.068	18.091	18.115	18.138	18.162	18.185	18.209	18.233	820
830	18.233	18.256	18.280	18.303	18.327	18.351	18.374	18.398	18.421	18.445	18.469	830
840	18.469	18.492	18.516	18.539	18.563	18.587	18.610	18.634	18.657	18.681	18.705	840
850	18.705	18.728	18.752	18.776	18.799	18.823	18.846	18.870	18.894	18.917	18.941	850
860	18.941	18.965	18.988	19.012	19.035	19.059	19.083	19.106	19.130	19.154	19.177	860
870	19.177	19.201	19.224	19.248	19.272	19.295	19.319	19.343	19.366	19.390	19.414	870
880	19.414	19.437	19.461	19.485	19.508	19.532	19.556	19.579	19.603	19.626	19.650	880
890	19.650	19.674	19.697	19.721	19.745	19.768	19.792	19.816	19.839	19.863	19.887	890
900	19.887	19.910	19.934	19.958	19.981	20.005	20.029	20.052	20.076	20.100	20.123	900
910	20.123	20.147	20.171	20.194	20.218	20.242	20.265	20.289	20.313	20.336	20.360	910
920	20.360	20.384	20.407	20.431	20.455	20.478	20.502	20.526	20.550	20.573	20.597	920
930	20.597	20.621	20.644	20.668	20.692	20.715	20.739	20.763	20.786	20.810	20.834	930
940	20.834	20.857	20.881	20.905	20.929	20.952	20.976	21.000	21.023	21.047	21.071	940
950	21.071	21.094	21.118	21.142	21.165	21.189	21.213	21.236	21.260	21.284	21.308	950
960	21.308	21.331	21.355	21.379	21.402	21.426	21.450	21.473	21.497	21.521	21.544	960
970	21.544	21.568	21.592	21.616	21.639	21.663	21.687	21.710	21.734	21.758	21.781	970
980	21.781	21.805	21.829	21.852	21.876	21.900	21.924	21.947	21.971	21.995	22.018	980
990	22.018	22.042	22.066	22.089	22.113	22.137	22.160	22.184	22.208	22.232	22.255	990
1000	22.255	22.279	22.303	22.326	22.350	22.374	22.397	22.421	22.445	22.468	22.492	1000
1010	22.492	22.516	22.540	22.563	22.587	22.611	22.634	22.658	22.682	22.705	22.729	1010
1020	22.729	22.753	22.776	22.800	22.824	22.847	22.871	22.895	22.918	22.942	22.966	1020
1030	22.966	22.990	23.013	23.037	23.061	23.084	23.108	23.132	23.155	23.179	23.203	1030
1040	23.203	23.226	23.250	23.274	23.297	23.321	23.345	23.368	23.392	23.416	23.439	1040
1050	23.439	23.463	23.487	23.510	23.534	23.558	23.581	23.605	23.629	23.652	23.676	1050
1060	23.676	23.700	23.723	23.747	23.771	23.794	23.818	23.842	23.865	23.889	23.913	1060
1070	23.913	23.936	23.960	23.984	24.007	24.031	24.055	24.078	24.102	24.126	24.149	1070
1080	24.149	24.173	24.197	24.220	24.244	24.267	24.291	24.315	24.338	24.362	24.386	1080
1090	24.386	24.409	24.433	24.457	24.480	24.504	24.527	24.551	24.575	24.598	24.622	1090
1100	24.622	24.646	24.669	24.693	24.717	24.740	24.764	24.787	24.811	24.835	24.858	1100
1110	24.858	24.882	24.905	24.929	24.953	24.976	25.000	25.024	25.047	25.071	25.094	1110
1120	25.094	25.118	25.142	25.165	25.189	25.212	25.236	25.260	25.283	25.307	25.330	1120
1130	25.330	25.354	25.377	25.401	25.425	25.448	25.472	25.495	25.519	25.543	25.566	1130
1140	25.566	25.590	25.613	25.637	25.660	25.684	25.708	25.731	25.755	25.778	25.802	1140
1150	25.802	25.825	25.849	25.873	25.896	25.920	25.943	25.967	25.990	26.014	26.037	1150
1160	26.037	26.061	26.084	26.108	26.132	26.155	26.179	26.202	26.226	26.249	26.273	1160
1170	26.273	26.296	26.320	26.343	26.367	26.390	26.414	26.437	26.461	26.484	26.508	1170
1180	26.508	26.532	26.555	26.579	26.602	26.626	26.649	26.673	26.696	26.720	26.743	1180
1190	26.743	26.767	26.790	26.814	26.837	26.861	26.884	26.907	26.931	26.954	26.978	1190
1200	26.978	27.001	27.025	27.048	27.072	27.095	27.119	27.142	27.166	27.189	27.213	1200
1210	27.213	27.236	27.259	27.283	27.306	27.330	27.353	27.377	27.400	27.424	27.447	1210
1220	27.447	27.471	27.494	27.517	27.541	27.564	27.588	27.611	27.635	27.658	27.682	1220
1230	27.682	27.705	27.729	27.752	27.775	27.799	27.822	27.845	27.869	27.892	27.916	1230
1240	27.916	27.939	27.962	27.986	28.009	28.032	28.056	28.079	28.103	28.126	28.149	1240
1250	28.149	28.173	28.196	28.219	28.243	28.266	28.289	28.313	28.336	28.359	28.383	1250
1260	28.383	28.406	28.430	28.453	28.476	28.500	28.523	28.546	28.570	28.593	28.616	1260
1270	28.616	28.640	28.663	28.686	28.710	28.733	28.756	28.780	28.803	28.826	28.849	1270
1280	28.849	28.873	28.896	28.919	28.943	28.966	28.989	29.013	29.036	29.059	29.082	1280
1290	29.082	29.106	29.129	29.152	29.176	29.199	29.222	29.245	29.269	29.292	29.315	1290
°F	0	1	2	3	4	5	6	7	8	9	10	°F

°F	0	1	2	3	4	5	6	7	8	9	10	°F
1300	29.315	29.338	29.362	29.385	29.408	29.431	29.455	29.478	29.501	29.524	29.546	1300
1310	29.546	29.571	29.594	29.617	29.640	29.664	29.687	29.710	29.733	29.757	29.780	1310
1320	29.780	29.803	29.826	29.849	29.873	29.896	29.919	29.942	29.965	29.989	30.012	1320
1330	30.012	30.035	30.058	30.081	30.104	30.128	30.151	30.174	30.197	30.220	30.243	1330
1340	30.243	30.267	30.290	30.313	30.336	30.359	30.382	30.405	30.429	30.452	30.475	1340
1350	30.475	30.498	30.521	30.544	30.567	30.590	30.613	30.637	30.660	30.683	30.706	1350
1360	30.706	30.729	30.752	30.775	30.798	30.821	30.844	30.868	30.891	30.914	30.937	1360
1370	30.937	30.960	30.983	31.006	31.029	31.052	31.075	31.098	31.121	31.144	31.167	1370
1380	31.167	31.190	31.213	31.236	31.259	31.283	31.306	31.329	31.352	31.375	31.398	1380
1390	31.398	31.421	31.444	31.467	31.490	31.513	31.536	31.559	31.582	31.605	31.628	1390
1400	31.628	31.651	31.674	31.697	31.720	31.743	31.766	31.789	31.812	31.834	31.857	1400
1410	31.857	31.880	31.903	31.926	31.949	31.972	31.995	32.018	32.041	32.064	32.087	1410
1420	32.087	32.110	32.133	32.156	32.179	32.202	32.224	32.247	32.270	32.293	32.316	1420
1430	32.316	32.339	32.362	32.385	32.408	32.431	32.453	32.476	32.499	32.522	32.545	1430
1440	32.545	32.568	32.591	32.614	32.636	32.659	32.682	32.705	32.728	32.751	32.774	1440
1450	32.774	32.796	32.819	32.842	32.865	32.888	32.911	32.933	32.956	32.979	33.002	1450
1460	33.002	33.025	33.047	33.070	33.093	33.115	33.138	33.161	33.184	33.207	33.230	1460
1470	33.230	33.253	33.275	33.298	33.321	33.344	33.366	33.389	33.412	33.435	33.458	1470
1480	33.458	33.480	33.503	33.526	33.548	33.571	33.594	33.617	33.639	33.662	33.685	1480
1490	33.685	33.708	33.730	33.753	33.776	33.798	33.821	33.844	33.867	33.889	33.912	1490
1500	33.912	33.935	33.957	33.980	34.003	34.025	34.048	34.071	34.093	34.116	34.139	1500
1510	34.139	34.161	34.184	34.207	34.229	34.252	34.275	34.297	34.320	34.343	34.365	1510
1520	34.365	34.388	34.410	34.433	34.456	34.478	34.501	34.524	34.546	34.569	34.591	1520
1530	34.591	34.614	34.637	34.659	34.682	34.704	34.727	34.750	34.772	34.795	34.817	1530
1540	34.817	34.840	34.862	34.885	34.908	34.930	34.953	34.975	34.998	35.020	35.043	1540
1550	35.043	35.065	35.088	35.110	35.133	35.155	35.178	35.201	35.223	35.246	35.268	1550
1560	35.268	35.291	35.313	35.336	35.358	35.381	35.403	35.426	35.448	35.471	35.493	1560
1570	35.493	35.516	35.538	35.560	35.583	35.605	35.628	35.650	35.673	35.695	35.718	1570
1580	35.718	35.740	35.763	35.785	35.807	35.830	35.852	35.875	35.897	35.920	35.942	1580
1590	35.942	35.964	35.987	36.009	36.032	36.054	36.076	36.099	36.121	36.144	36.166	1590
1600	36.166	36.188	36.211	36.233	36.256	36.278	36.300	36.323	36.345	36.367	36.390	1600
1610	36.390	36.412	36.434	36.457	36.479	36.501	36.524	36.546	36.568	36.591	36.613	1610
1620	36.613	36.635	36.658	36.680	36.702	36.725	36.747	36.769	36.792	36.814	36.836	1620
1630	36.836	36.859	36.881	36.903	36.925	36.948	36.970	36.992	37.014	37.037	37.059	1630
1640	37.059	37.081	37.104	37.126	37.148	37.170	37.193	37.215	37.237	37.259	37.281	1640
1650	37.281	37.304	37.326	37.348	37.370	37.393	37.415	37.437	37.459	37.481	37.504	1650
1660	37.504	37.526	37.548	37.570	37.592	37.615	37.637	37.659	37.681	37.703	37.725	1660
1670	37.725	37.748	37.770	37.792	37.814	37.836	37.858	37.881	37.903	37.925	37.947	1670
1680	37.947	37.969	37.991	38.013	38.036	38.058	38.080	38.102	38.124	38.146	38.168	1680
1690	38.168	38.190	38.212	38.235	38.257	38.279	38.301	38.323	38.345	38.367	38.389	1690
1700	38.389	38.411	38.433	38.455	38.477	38.499	38.522	38.544	38.566	38.588	38.610	1700
1710	38.610	38.632	38.654	38.676	38.698	38.720	38.742	38.764	38.786	38.808	38.830	1710
1720	38.830	38.852	38.874	38.896	38.918	38.940	38.962	38.984	39.006	39.028	39.050	1720
1730	39.050	39.072	39.094	39.116	39.138	39.160	39.182	39.204	39.226	39.248	39.270	1730
1740	39.270	39.292	39.314	39.335	39.357	39.379	39.401	39.423	39.445	39.467	39.489	1740
1750	39.489	39.511	39.533	39.555	39.577	39.599	39.620	39.642	39.664	39.686	39.708	1750
1760	39.708	39.730	39.752	39.774	39.796	39.817	39.839	39.861	39.883	39.905	39.927	1760
1770	39.927	39.949	39.970	39.992	40.014	40.036	40.058	40.080	40.101	40.123	40.145	1770
1780	40.145	40.167	40.189	40.211	40.232	40.254	40.276	40.298	40.320	40.341	40.363	1780
1790	40.363	40.385	40.407	40.429	40.450	40.472	40.494	40.516	40.537	40.559	40.581	1790
1800	40.581	40.603	40.624	40.646	40.668	40.690	40.711	40.733	40.755	40.777	40.798	1800
1810	40.798	40.820	40.842	40.864	40.885	40.907	40.929	40.950	40.972	40.994	41.015	1810
1820	41.015	41.037	41.059	41.081	41.102	41.124	41.146	41.167	41.189	41.211	41.232	1820
1830	41.232	41.254	41.276	41.297	41.319	41.341	41.362	41.384	41.405	41.427	41.449	1830
1840	41.449	41.470	41.492	41.514	41.535	41.557	41.578	41.600	41.622	41.643	41.665	1840
1850	41.665	41.686	41.708	41.730	41.751	41.773	41.794	41.816	41.838	41.859	41.881	1850
1860	41.881	41.902	41.924	41.945	41.967	41.988	42.010	42.032	42.053	42.075	42.096	1860
1870	42.096	42.118	42.139	42.161	42.182	42.204	42.225	42.247	42.268	42.289	42.311	1870
1880	42.311	42.333	42.354	42.376	42.397	42.419	42.440	42.462	42.483	42.505	42.526	1880
1890	42.526	42.548	42.569	42.591	42.612	42.633	42.655	42.676	42.698	42.719	42.741	1890
°F	0	1	2	3	4	5	6	7	8	9	10	°F

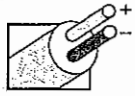
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MAXIMUM TEMPERATURE RANGE
Thermocouple Grade
– 328 to 2282°F
– 200 to 1250°C
Extension Grade
32 to 392°F
0 to 200°C
LIMITS OF ERROR
(whichever is greater)
Standard: 2.2°C or 0.75% Above 0°C
2.2°C or 2.0% Below 0°C
Special: 1.1°C or 0.4%
COMMENTS, BARE WIRE ENVIRONMENT:
Clean Oxidizing and Inert; Limited Use in Vacuum or Reducing; Wide Temperature Range; Most Popular Calibration
TEMPERATURE IN DEGREES °F
REFERENCE JUNCTION AT 32°F



Thermocouple Grade

Nickel-Chromium vs. Nickel-Aluminum



Extension Grade

Revised Thermocouple Reference Tables

TYPE K
Reference Tables
N.I.S.T. Monograph 175
Revised to ITS-90

Thermoelectric Voltage in Millivolts

°F	0	1	2	3	4	5	6	7	8	9	10	°F
1900	42.741	42.782	42.783	42.825	42.826	42.846	42.868	42.891	42.912	42.933	42.955	1900
1910	42.955	42.976	42.998	43.019	43.040	43.062	43.083	43.104	43.126	43.147	43.169	1910
1920	43.169	43.190	43.211	43.233	43.254	43.275	43.297	43.318	43.339	43.361	43.382	1920
1930	43.382	43.403	43.425	43.446	43.467	43.489	43.510	43.531	43.552	43.574	43.595	1930
1940	43.595	43.616	43.638	43.659	43.680	43.701	43.723	43.744	43.765	43.787	43.808	1940
1950	43.808	43.829	43.850	43.872	43.893	43.914	43.935	43.957	43.978	43.999	44.020	1950
1960	44.020	44.041	44.063	44.084	44.105	44.126	44.147	44.169	44.190	44.211	44.232	1960
1970	44.232	44.253	44.275	44.296	44.317	44.338	44.359	44.380	44.402	44.423	44.444	1970
1980	44.444	44.465	44.486	44.507	44.528	44.550	44.571	44.592	44.613	44.634	44.655	1980
1990	44.655	44.676	44.697	44.719	44.740	44.761	44.782	44.803	44.824	44.845	44.866	1990
2000	44.866	44.887	44.908	44.929	44.950	44.971	44.992	45.014	45.035	45.056	45.077	2000
2010	45.077	45.098	45.119	45.140	45.161	45.182	45.203	45.224	45.245	45.266	45.287	2010
2020	45.287	45.308	45.329	45.350	45.371	45.392	45.413	45.434	45.455	45.476	45.497	2020
2030	45.497	45.518	45.539	45.560	45.580	45.601	45.622	45.643	45.664	45.685	45.706	2030
2040	45.706	45.727	45.748	45.769	45.790	45.811	45.832	45.852	45.873	45.894	45.915	2040
2050	45.915	45.936	45.957	45.978	45.999	46.019	46.040	46.061	46.082	46.103	46.124	2050
2060	46.124	46.145	46.165	46.186	46.207	46.228	46.249	46.269	46.290	46.311	46.332	2060
2070	46.332	46.353	46.373	46.394	46.415	46.436	46.457	46.477	46.498	46.519	46.540	2070
2080	46.540	46.560	46.581	46.602	46.623	46.643	46.664	46.685	46.706	46.726	46.747	2080
2090	46.747	46.768	46.789	46.809	46.830	46.851	46.871	46.892	46.913	46.933	46.954	2090
2100	46.954	46.975	46.995	47.016	47.037	47.057	47.078	47.099	47.119	47.140	47.161	2100
2110	47.161	47.181	47.202	47.223	47.243	47.264	47.284	47.305	47.326	47.346	47.367	2110
2120	47.367	47.387	47.408	47.429	47.449	47.470	47.490	47.511	47.531	47.552	47.573	2120
2130	47.573	47.593	47.614	47.634	47.655	47.675	47.696	47.716	47.737	47.757	47.778	2130
2140	47.778	47.798	47.819	47.839	47.860	47.880	47.901	47.921	47.942	47.962	47.983	2140
2150	47.983	48.003	48.024	48.044	48.065	48.085	48.105	48.126	48.146	48.167	48.187	2150
2160	48.187	48.208	48.228	48.248	48.269	48.289	48.310	48.330	48.350	48.371	48.391	2160
2170	48.391	48.411	48.432	48.452	48.473	48.493	48.513	48.534	48.554	48.574	48.595	2170
2180	48.595	48.615	48.635	48.656	48.676	48.696	48.717	48.737	48.757	48.777	48.798	2180
2190	48.798	48.818	48.838	48.859	48.879	48.899	48.919	48.940	48.960	48.980	49.000	2190
2200	49.000	49.021	49.041	49.061	49.084	49.101	49.122	49.142	49.162	49.182	49.202	2200
2210	49.202	49.223	49.243	49.263	49.283	49.303	49.323	49.344	49.364	49.384	49.404	2210
2220	49.404	49.424	49.444	49.465	49.485	49.505	49.525	49.545	49.565	49.585	49.605	2220
2230	49.605	49.625	49.645	49.666	49.686	49.706	49.726	49.746	49.766	49.786	49.806	2230
2240	49.806	49.826	49.846	49.866	49.886	49.906	49.926	49.946	49.966	49.986	50.006	2240
°F	0	1	2	3	4	5	6	7	8	9	10	°F

°F	0	1	2	3	4	5	6	7	8	9	10	°F
2250	50.006	50.026	50.046	50.066	50.086	50.106	50.126	50.146	50.166	50.186	50.206	2250
2260	50.206	50.226	50.246	50.266	50.286	50.306	50.326	50.346	50.366	50.385	50.405	2260
2270	50.405	50.425	50.445	50.465	50.485	50.505	50.525	50.545	50.564	50.584	50.604	2270
2280	50.604	50.624	50.644	50.664	50.684	50.703	50.723	50.743	50.763	50.783	50.802	2280
2290	50.802	50.822	50.842	50.862	50.882	50.901	50.921	50.941	50.961	50.981	51.000	2290
2300	51.000	51.020	51.040	51.060	51.079	51.099	51.119	51.139	51.159	51.178	51.198	2300
2310	51.198	51.217	51.237	51.257	51.276	51.296	51.316	51.336	51.355	51.375	51.395	2310
2320	51.395	51.414	51.434	51.453	51.473	51.493	51.512	51.532	51.552	51.571	51.591	2320
2330	51.591	51.611	51.630	51.650	51.669	51.689	51.708	51.728	51.748	51.767	51.787	2330
2340	51.787	51.806	51.826	51.845	51.865	51.885	51.904	51.924	51.943	51.963	51.982	2340
2350	51.982	52.002	52.021	52.041	52.060	52.080	52.099	52.119	52.138	52.158	52.177	2350
2360	52.177	52.197	52.216	52.235	52.255	52.274	52.294	52.313	52.333	52.352	52.371	2360
2370	52.371	52.391	52.410	52.430	52.449	52.468	52.488	52.507	52.527	52.546	52.565	2370
2380	52.565	52.585	52.604	52.623	52.643	52.662	52.681	52.701	52.720	52.739	52.759	2380
2390	52.759	52.778	52.797	52.817	52.836	52.855	52.875	52.894	52.913	52.932	52.952	2390
2400	52.952	52.971	52.990	53.010	53.029	53.048	53.067	53.087	53.106	53.125	53.144	2400
2410	53.144	53.163	53.183	53.202	53.221	53.240	53.259	53.279	53.298	53.317	53.336	2410
2420	53.336	53.355	53.375	53.394	53.413	53.432	53.451	53.470	53.490	53.509	53.528	2420
2430	53.528	53.547	53.566	53.585	53.604	53.623	53.643	53.662	53.681	53.700	53.719	2430
2440	53.719	53.738	53.757	53.776	53.795	53.814	53.833	53.852	53.871	53.890	53.910	2440
2450	53.910	53.929	53.948	53.967	53.986	54.005	54.024	54.043	54.062	54.081	54.100	2450
2460	54.100	54.119	54.138	54.157	54.176	54.195	54.214	54.233	54.252	54.271	54.289	2460
2470	54.289	54.308	54.327	54.346	54.365	54.384	54.403	54.422	54.441	54.460	54.479	2470
2480	54.479	54.498	54.517	54.535	54.554	54.573	54.592	54.611	54.630	54.649	54.668	2480
2490	54.668	54.687	54.705	54.724	54.743	54.762	54.781	54.800	54.819	54.837	54.856	2490
2500	54.856	54.875	54.894									2500
°F	0	1	2	3	4	5	6	7	8	9	10	°F

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Appendix A-7

CONVERSION TABLES

Temperature Conversion Chart (Courtesy Omega)

Pressure Conversion Factors (selected)

PRESSURE CONVERSION FACTORS

Refer to the *ISA Handbook of Measurement Equations and Tables* for more conversion factors than you will ever need.

psi = in. of H_2O x (0.0361273)

psi = in. of Hg x (0.491154)

psi = mm of H_2O x (0.00142233)

psi = mm of Hg x (0.0193368)

psi = kg/cm^2 x (14.223)

psi = bar x (14.50377)

psi = mbar x (0.014503×10^{-2})

psi = Pa x (1.4503×10^{-4})

psi = kPa x (0.1450377)

psi = millitorr x (1.93368×10^{-5})

Temperature Conversion Chart

°C = ⅝ (°F - 32)
Kelvin = °C + 273.15

°F = 1.8°C + 32
Rankine = °F + 459.67

TABLE EXAMPLE:
To Convert 1000°C to °F, look up 1000 and read left
To Convert 1000°F to °C, look up 1000 and read right

to °F	From	to °C	to °F	From	to °C	to °F	From	to °C	to °F	From	to °C	to °F	From	to °C
	-460	-272.22		-300	-188.89	-252.4	-168	-105.56	+17.6	-6	-21.22	287.6	142	61.11
	-458	-271.11		-308	-187.78	-240.2	-166	-104.44	+21.2	-5	-21.11	291.2	144	62.22
	-454	-270.00		-304	-186.67	-245.2	-164	-103.33	+24.8	-4	-20.00	294.8	146	63.33
	-452	-268.89		-302	-185.56	-241.6	-162	-102.22	+28.4	-2	-18.89	298.4	148	64.44
	-450	-267.78		-300	-184.44	-230.0	-160	-101.11	+32.0	0	-17.78	302.0	150	65.56
	-448	-266.67		-298	-183.33	-234.4	-148	-100.00	+35.6	2	-16.67	305.6	152	66.67
	-446	-265.56		-296	-182.22	-230.2	-146	-98.89	+30.2	4	-15.56	309.2	154	67.78
	-444	-264.44		-294	-181.11	-227.2	-144	-97.78	+42.8	6	-14.44	312.8	156	68.89
	-442	-263.33		-292	-180.00	-223.6	-142	-96.67	+46.4	8	-13.33	316.4	158	70.00
	-440	-262.22		-290	-178.89	-220.0	-140	-95.56	+50.0	10	-12.22	320.0	160	71.11
	-438	-261.11		-288	-177.78	-216.4	-138	-94.44	+53.6	12	-11.11	323.6	162	72.22
	-436	-260.00		-286	-176.67	-212.8	-136	-93.33	+57.2	14	-10.00	327.2	164	73.33
	-434	-258.89		-284	-175.56	-209.2	-134	-92.22	+60.8	16	-8.89	330.8	166	74.44
	-432	-257.78		-282	-174.44	-205.6	-132	-91.11	+64.4	18	-7.78	334.4	168	75.56
	-430	-256.67		-280	-173.33	-202.0	-130	-90.00	+68.0	20	-6.67	338.0	170	76.67
	-428	-255.56		-278	-172.22	-198.4	-128	-88.89	+71.6	22	-5.56	341.6	172	77.78
	-426	-254.44		-276	-171.11	-194.2	-126	-87.78	+75.2	24	-4.44	345.2	174	78.89
	-424	-253.33		-274	-170.00	-191.2	-124	-86.67	+78.8	26	-3.33	348.8	176	80.00
	-422	-252.22	-457.6	-272	-168.89	-187.6	-122	-85.56	+82.4	28	-2.22	352.4	178	81.11
	-420	-251.11	-454.0	-270	-167.78	-184.0	-120	-84.44	+86.0	30	-1.11	356.0	180	82.22
	-418	-250.00	-450.4	-268	-166.67	-180.4	-118	-83.33	+80.6	32	0.00	359.6	182	83.33
	-416	-248.89	-446.8	-266	-165.56	-176.8	-116	-82.22	+93.2	34	1.11	363.2	184	84.44
	-414	-247.78	-443.2	-264	-164.44	-173.2	-114	-81.11	+96.8	36	2.22	366.8	186	85.56
	-412	-246.67	-439.6	-262	-163.33	-169.6	-112	-80.00	+100.4	38	3.33	370.4	188	86.67
	-410	-245.56	-436.0	-260	-162.22	-166.0	-110	-78.89	+104.0	40	4.44	374.0	190	87.78
	-408	-244.44	-432.4	-258	-161.11	-162.4	-108	-77.78	107.6	42	5.56	377.6	192	88.89
	-406	-243.33	-428.8	-256	-160.00	-158.8	-106	-76.67	111.2	44	6.67	381.2	194	90.00
	-404	-242.22	-425.2	-254	-158.89	-155.2	-104	-75.56	114.2	46	7.78	384.8	196	91.11
	-402	-241.11	-421.6	-252	-157.78	-151.6	-102	-74.44	118.4	48	8.89	388.4	198	92.22
	-400	-240.00	-418.0	-250	-156.67	-148.0	-100	-73.33	122.0	50	10.00	392.0	200	93.33
	-398	-238.89	-414.4	-248	-155.56	-144.4	-98	-72.22	125.6	52	11.11	395.6	202	94.44
	-396	-237.78	-410.8	-246	-154.44	-140.8	-96	-71.11	129.2	54	12.22	399.2	204	95.56
	-394	-236.67	-407.2	-244	-153.33	-137.2	-94	-70.00	132.8	56	13.33	402.8	206	96.67
	-392	-235.56	-403.6	-242	-152.22	-133.6	-92	-68.89	136.4	58	14.44	406.4	208	97.78
	-390	-234.44	-400.0	-240	-151.11	-130.0	-90	-67.78	140.0	60	15.56	410.0	210	98.89
	-388	-233.33	-396.4	-238	-150.00	-126.4	-88	-66.67	143.6	62	16.67	413.6	212	100.00
	-386	-232.22	-392.8	-236	-148.89	-122.2	-86	-65.56	147.2	64	17.78	417.2	214	101.11
	-384	-231.11	-389.2	-234	-147.78	-119.2	-84	-64.44	150.8	66	18.89	420.8	216	102.22
	-382	-230.00	-385.6	-232	-146.67	-115.6	-82	-63.33	154.4	68	20.00	424.4	218	103.33
	-380	-228.89	-382.0	-230	-145.56	-112.0	-80	-62.22	158.0	70	21.11	428.0	220	104.44
	-378	-227.78	-378.4	-228	-144.44	-108.4	-78	-61.11	161.6	72	22.22	431.6	222	105.56
	-376	-226.67	-374.8	-226	-143.33	-104.8	-76	-60.00	165.2	74	23.33	435.2	224	106.67
	-374	-225.56	-371.2	-224	-142.22	-101.2	-74	-58.89	168.8	76	24.44	438.8	226	107.78
	-372	-224.44	-367.6	-222	-141.11	-97.6	-72	-57.78	172.4	78	25.56	442.4	228	108.89
	-370	-223.33	-364.0	-220	-140.00	-94.0	-70	-56.67	176.0	80	26.67	446.0	230	110.00
	-368	-222.22	-360.4	-218	-138.89	-90.4	-68	-55.56	179.6	82	27.78	449.6	232	111.11
	-366	-221.11	-356.8	-216	-137.78	-86.8	-66	-54.44	183.2	84	28.89	453.2	234	112.22
	-364	-220.00	-353.2	-214	-136.67	-83.2	-64	-53.33	186.8	86	30.00	456.8	236	113.33
	-362	-218.89	-349.6	-212	-135.56	-79.6	-62	-52.22	190.4	88	31.11	460.4	238	114.44
	-360	-217.78	-346.0	-210	-134.44	-76.0	-60	-51.11	194.0	90	32.22	464.0	240	115.56
	-358	-216.67	-342.4	-208	-133.33	-72.4	-58	-50.00	197.6	92	33.33	467.6	242	116.67
	-356	-215.56	-338.8	-206	-132.22	-68.8	-56	-48.89	201.2	94	34.44	471.2	244	117.78
	-354	-214.44	-335.2	-204	-131.11	-65.2	-54	-47.78	204.8	96	35.56	474.8	246	118.89
	-352	-213.33	-331.6	-202	-130.00	-61.6	-52	-46.67	208.4	98	36.67	478.4	248	120.00
	-350	-212.22	-328.0	-200	-128.89	-58.0	-50	-45.56	212.0	100	37.78	482.0	250	121.11
	-348	-211.11	-324.4	-198	-127.78	-54.4	-48	-44.44	215.6	102	38.89	485.6	252	122.22
	-346	-210.00	-320.8	-196	-126.67	-50.8	-46	-43.33	219.2	104	40.00	489.2	254	123.33
	-344	-208.89	-317.2	-194	-125.56	-47.2	-44	-42.22	222.8	106	41.11	492.8	256	124.44
	-342	-207.78	-313.6	-192	-124.44	-43.6	-42	-41.11	226.4	108	42.22	496.4	258	125.56
	-340	-206.67	-310.0	-190	-123.33	-40.0	-40	-40.00	230.0	110	43.33	500.0	260	126.67
	-338	-205.56	-306.4	-188	-122.22	-36.4	-38	-36.89	233.6	112	44.44	503.6	262	127.78
	-336	-204.44	-302.8	-186	-121.11	-32.8	-36	-37.78	237.2	114	45.56	507.2	264	128.89
	-334	-203.33	-299.2	-184	-120.00	-29.2	-34	-36.67	240.2	116	46.67	510.8	265	130.00
	-332	-202.22	-295.6	-182	-118.89	-25.6	-32	-35.56	244.4	118	47.78	514.4	268	131.11
	-330	-201.11	-292.0	-180	-117.78	-22.0	-30	-34.44	248.0	120	48.89	518.0	270	132.22
	-328	-200.00	-288.4	-178	-116.67	-18.4	-28	-33.33	251.6	122	50.00	521.6	272	133.33
	-326	-198.89	-284.8	-176	-115.56	-14.8	-26	-32.22	255.2	124	51.11	525.2	274	134.44
	-324	-197.78	-281.2	-174	-114.44	-11.2	-24	-31.11	258.8	126	52.22	528.8	276	135.56
	-322	-196.67	-277.6	-172	-113.33	-7.6	-22	-30.00	262.4	128	53.33	532.4	278	136.67
	-320	-195.56	-274.0	-170	-112.22	-4.0	-20	-28.89	266.0	130	54.44	536.0	280	137.78
	-318	-194.44	-270.4	-168	-111.11	-0.4	-18	-27.78	269.6	132	55.56	539.6	282	138.89
	-316	-193.33	-266.8	-166	-110.00	+3.2	-16	-26.67	273.2	134	56.67	543.2	284	140.00
	-314	-192.22	-263.2	-164	-108.89	+6.8	-14	-25.56	276.8	136	57.78	546.8	286	141.11
	-312	-191.11	-259.6	-162	-107.78	+10.4	-12	-24.44	280.4	138	58.89	550.4	288	142.22
	-310	190.00	-256.0	-160	106.67	+14.0	-10	-23.33	284.0	140	60.00	594.0	290	143.33

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to °F	From	to °C	to °F	From	to °C	to °F	From	to °C	to °F	From	to °C	to °F	From	to °C
557.6	292	144.44	870.8	456	241.11	1832.0	100	537.78	3398.0	1870	1021.1	4964.0	2740	1504.4
561.2	294	145.56	874.4	458	242.22	1850.0	101	543.33	3416.0	1880	1026.7	4982.0	2750	1510.0
564.8	296	146.67	878.0	470	243.33	1868.0	102	548.89	3434.0	1890	1032.2	5000.0	2760	1515.6
568.4	298	147.78	881.6	472	244.44	1886.0	103	554.44	3452.0	1900	1037.8	5018.0	2770	1521.1
572.0	300	148.89	885.2	474	245.56	1904.0	104	560.00	3470.0	1910	1043.3	5036.0	2780	1526.7
575.6	302	150.00	886.8	476	246.67	1922.0	105	565.56	3489.0	1920	1048.9	5054.0	2790	1532.2
579.2	304	151.11	892.4	478	247.78	1940.0	106	571.11	3506.0	1930	1054.4	5072.0	2800	1537.8
582.8	306	152.22	896.0	480	248.89	1958.0	107	576.67	3524.0	1940	1060.0	5090.0	2810	1543.3
586.4	308	153.33	899.6	482	250.00	1976.0	108	582.22	3542.0	1950	1065.6	5108.0	2820	1548.9
590.0	310	154.44	903.2	484	251.11	1994.0	109	587.78	3560.0	1960	1071.1	5126.0	2830	1554.4
593.6	312	155.56	906.8	486	252.22	2012.0	110	593.33	3578.0	1970	1076.7	5144.0	2840	1560.0
597.2	314	156.67	910.4	488	253.33	2030.0	111	598.89	3596.0	1980	1082.2	5162.0	2850	1565.6
600.8	316	157.78	914.0	490	254.44	2048.0	112	604.44	3614.0	1990	1087.8	5180.0	2860	1571.1
604.4	318	158.89	917.6	492	255.56	2066.0	113	610.00	3632.0	2000	1093.3	5198.0	2870	1576.7
608.0	320	160.00	921.2	494	256.67	2084.0	114	615.56	3650.0	2010	1098.9	5216.0	2880	1582.2
611.6	322	161.11	924.8	496	257.78	2102.0	115	621.11	3668.0	2020	1104.4	5234.0	2890	1587.8
615.2	324	162.22	928.4	498	258.89	2120.0	116	626.67	3686.0	2030	1110.0	5252.0	2900	1593.3
618.8	326	163.33	932.0	500	260.00	2138.0	117	632.22	3704.0	2040	1115.6	5270.0	2910	1598.9
622.4	328	164.44	935.6	502	261.11	2156.0	118	637.78	3722.0	2050	1121.1	5288.0	2920	1604.4
626.0	330	165.56	939.2	504	262.22	2174.0	119	643.33	3740.0	2060	1126.7	5306.0	2930	1610.0
629.6	332	166.67	942.8	506	263.33	2192.0	120	648.89	3758.0	2070	1132.2	5324.0	2940	1615.6
633.2	334	167.78	946.4	508	264.44	2210.0	121	654.44	3776.0	2080	1137.8	5342.0	2950	1621.1
636.8	336	168.89	950.0	510	265.56	2228.0	122	660.00	3794.0	2090	1143.3	5360.0	2960	1626.7
640.4	338	170.00	953.6	512	266.67	2246.0	123	665.56	3812.0	2100	1148.9	5378.0	2970	1632.2
644.0	340	171.11	957.2	514	267.78	2264.0	124	671.11	3830.0	2110	1154.4	5396.0	2980	1637.2
647.6	342	172.22	960.8	516	268.89	2282.0	125	676.67	3848.0	2120	1160.0	5414.0	2990	1643.3
651.2	344	173.33	964.4	518	270.00	2300.0	126	682.22	3866.0	2130	1165.6	5432.0	3000	1648.9
654.8	346	174.44	968.0	520	271.11	2318.0	127	687.78	3884.0	2140	1171.1	5450.0	3010	1654.4
658.4	348	175.56	971.6	522	272.22	2336.0	128	693.33	3902.0	2150	1176.7	5468.0	3020	1660.0
662.0	350	176.67	975.2	524	273.33	2354.0	129	698.89	3920.0	2160	1182.2	5486.0	3030	1665.6
665.6	352	177.78	978.8	526	274.44	2372.0	130	704.44	3938.0	2170	1187.8	5504.0	3040	1671.1
669.2	354	178.89	982.4	528	275.56	2390.0	131	710.00	3956.0	2180	1193.3	5522.0	3050	1676.7
672.8	356	180.00	986.0	530	276.67	2408.0	132	715.56	3974.0	2190	1198.9	5540.0	3060	1682.2
676.4	358	181.11	989.6	532	277.78	2426.0	133	721.11	3992.0	2200	1204.4	5558.0	3070	1687.8
680.0	360	182.22	993.2	534	278.89	2444.0	134	726.67	4010.0	2210	1210.0	5576.0	3080	1693.3
683.6	362	183.33	996.8	536	280.00	2462.0	135	732.22	4028.0	2220	1215.6	5594.0	3090	1698.9
687.2	364	184.44	1000.4	538	281.11	2480.0	136	737.78	4046.0	2230	1221.1	5612.0	3100	1704.4
690.8	366	185.56	1004.0	540	282.22	2498.0	137	743.33	4064.0	2240	1226.7	5702.0	3150	1732.2
694.4	368	186.67	1007.6	542	283.33	2516.0	138	748.89	4082.0	2250	1232.2	5792.0	3200	1760.0
698.0	370	187.78	1011.2	544	284.44	2534.0	139	754.44	4100.0	2260	1237.8	5882.0	3250	1787.8
701.6	372	188.89	1014.8	546	285.56	2552.0	140	760.00	4118.0	2270	1243.3	5972.0	3300	1815.6
705.2	374	190.00	1018.4	548	286.67	2570.0	141	765.56	4136.0	2280	1248.9	6062.0	3350	1843.3
708.8	376	191.11	1022.0	550	287.78	2588.0	142	771.11	4154.0	2290	1254.4	6152.0	3400	1871.1
712.4	378	192.22	1040.0	560	293.33	2606.0	143	776.67	4172.0	2300	1260.0	6242.0	3450	1898.9
716.0	380	193.33	1058.0	570	298.89	2624.0	144	782.22	4190.0	2310	1265.6	6332.0	3500	1926.7
719.6	382	194.44	1076.0	580	304.44	2642.0	145	787.78	4208.0	2320	1271.1	6422.0	3550	1954.4
723.2	384	195.56	1094.0	590	310.00	2660.0	146	793.33	4226.0	2330	1276.7	6512.0	3600	1982.2
726.8	386	196.67	1112.0	600	315.56	2678.0	147	798.89	4244.0	2340	1282.2	6602.0	3650	2010.0
730.4	388	197.78	1130.0	610	321.11	2696.0	148	804.44	4262.0	2350	1287.8	6692.0	3700	2037.8
734.0	390	198.89	1148.0	620	326.67	2714.0	149	810.00	4280.0	2360	1293.3	6782.0	3750	2065.6
737.6	392	200.00	1166.0	630	332.22	2732.0	150	815.56	4298.0	2370	1298.9	6872.0	3800	2093.3
741.2	394	201.11	1184.0	640	337.78	2750.0	151	821.11	4316.0	2380	1304.4	6962.0	3850	2121.1
744.8	396	202.22	1202.0	650	343.33	2768.0	152	826.67	4334.0	2390	1310.0	7052.0	3900	2148.9
748.4	398	203.33	1220.0	660	348.89	2786.0	153	832.22	4352.0	2400	1315.6	7142.0	3950	2176.7
752.0	400	204.44	1238.0	670	354.44	2804.0	154	837.78	4370.0	2410	1321.1	7232.0	4000	2204.4
755.6	402	205.56	1256.0	680	360.00	2822.0	155	843.33	4388.0	2420	1326.7	7322.0	4050	2232.2
759.2	404	206.67	1274.0	690	365.56	2840.0	156	848.89	4406.0	2430	1332.2	7412.0	4100	2260.0
762.8	406	207.78	1292.0	700	371.11	2858.0	157	854.44	4424.0	2440	1337.8	7502.0	4150	2287.8
766.4	408	208.89	1310.0	710	376.67	2876.0	158	860.00	4442.0	2450	1343.3	7592.0	4200	2315.6
770.0	410	210.00	1328.0	720	382.22	2894.0	159	865.56	4460.0	2460	1348.9	7682.0	4250	2343.3
773.6	412	211.11	1346.0	730	387.78	2912.0	160	871.11	4478.0	2470	1354.4	7772.0	4300	2371.1
777.2	414	212.22	1364.0	740	393.33	2930.0	161	876.67	4496.0	2480	1360.0	7862.0	4350	2398.9
780.8	416	213.33	1382.0	750	398.89	2948.0	162	882.22	4514.0	2490	1365.6	7952.0	4400	2426.7
784.4	418	214.44	1400.0	760	404.44	2966.0	163	887.78	4532.0	2500	1371.1	8042.0	4450	2454.4
788.0	420	215.56	1418.0	770	410.00	2984.0	164	893.33	4550.0	2510	1376.7	8132.0	4450	2482.2
791.6	422	216.67	1436.0	780	415.56	3002.0	165	898.89	4568.0	2520	1382.2	8222.0	4550	2510.0
795.2	424	217.78	1454.0	790	421.11	3020.0	166	904.44	4586.0	2530	1387.2	8312.0	4600	2537.8
798.8	426	218.89	1472.0	800	426.67	3038.0	167	910.00	4604.0	2540	1393.3	8402.0	4650	2565.6
802.4	428	220.00	1490.0	810	432.22	3056.0	168	915.56	4622.0	2550	1398.9	8492.0	4700	2593.3
806.0	430	221.11	1508.0	820	437.78	3074.0	169	921.11	4640.0	2560	1404.4	8582.0	4750	2621.1
809.6	432	222.22	1526.0	830	443.33	3092.0	170	926.67	4658.0	2570	1410.0	8672.0	4800	2648.9
813.2	434	223.33	1544.0	840	448.89	3110.0	171	932.22	4676.0	2580	1415.6	8762.0	4850	2676.7
816.8	436	224.44	1562.0	850	454.44	3128.0	172	937.78	4694.0	2590	1421.1	8852.0	4900	2704.4
820.4	438	225.56	1580.0	860	460.00	3146.0	173	943.33	4712.0	2600	1426.7	8942.0	4950	2732.2
824.0	440	226.67	1598.0	870	465.56	3164.0	174	948.89	4730.0	2610	1432.2	9032.0	5000	2760.0
827.6	442	227.78	1616.0	880	471.11	3182.0	175	954.44	4748.0	2620	1437.8	9122.0	5050	2787.8
831.2	444	228.89	1634.0	890	476.67	3200.0	176	960.00	4766.0	2630	1443.3	9212.0	5100	2815.6
834.8	446	230.00	1652.0	900	482.22	3218.0	177	965.56	4784.0	2640	1448.9	9302.0	5150	2843.3
838.4	448	231.11	1670.0	910	487.78	3236.0	178	971.11	4802.0	2650	1454.4	9392.0	5200	2871.1
842.0	450	232.22	1688.0	920	493.33	3254.0	179	976.67	4820.0	2660	1460.0	9482.0	5250	2898.9
845.6	452	233.33	1706.0	930	498.89	3272.0	180	982.22	4838.0	2670	1465.6	9572.0	5300	2926.7
849.2	454	234.44	1724.0	940	504.44	3290.0	181	987.78	4856.0	2680	1471.1	9662.0	5350	2954.4
852.2	456	235.56	1742.0	950	510.00	3308.0	182	993.33	4974.0	2690	1457.7	9752.0	5400	2982.2
856.4	458	236.67	1760.0	960	515.56	3326.0	183	998.89	4892.0	2700	1482.2	9842.0	5450	3010.0
860.0	460	237.78	1778.0	970	521.11	3344.0	184	1004.40	4910.0	2710	1487.8	9932.0	5500	3037.8
863.6	462	238.89	1796.0	980	526.67	3362.0	185	1010.0	4928.0	2720	1493.3	10,002.0	5550	3065.6
867.2	464	240.00	1814.0	990	532.22	3380.0	186	1015.6	4946.0	2730	1498.9	10,112.0	5600	3093.3

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Appendix B

ANSWERS TO CHAPTER REVIEW QUESTIONS

CHAPTER 1 REVIEW QUESTIONS

1. Match the term on the left with the definition on the right.

E	Calibration	A. permissible deviation from specified value
H	Instrument Range	B. upper and lower values specified for facility
B	Calibration Range	C. algebraic difference between the upper and lower range value
F	Accuracy	D. adjustment used to produce a parallel shift of the input-output curve
A	Tolerance	E. comparison of instrument to a known value
I	Traceability	F. percent error
D	Zero	G. characterizes the dispersion of the values that could reasonably be attributed to the measurand
C	Span	H. upper and lower values specified by manufacturer
G	Uncertainty	I. measurement related to standards through an unbroken chain of comparisons

2. Which of the following errors is typically not correctable?

C. Linearity

3. Why should a calibration technician have:

   A. Honesty and Integrity?

   Calibrations must be performed in accordance with procedures and must be properly documented even when the calibration department may be understaffed and production schedules demand unrealistic completion requirements.

   B. Attention to detail?

   The minute a technician is not paying attention to detail, safety and proper performance are jeopardized.

   C. Excellent documentation practices?

   If it isn't documented, it wasn't done. The impression of quality is determined by the content and appearance of documentation.

   D. Understanding of processes?

   Understanding processes, particularly how the instrument monitors and controls the process, will enable the technician to more efficiently calibrate process instrumentation without disrupting the process. Also, the technician is more capable of identifying and troubleshooting the cause of process problems.

4. What are the advantages of performing a field calibration? Disadvantages?

ADVANTAGES OF FIELD CALIBRATION	DISADVANTAGES OF FIELD CALIBRATION
1. May save time 2. May identify and allow trouble-shooting of installation problems 3. Performed in actual ambient environment	1. Availability of instrument for cleaning and inspection is limited 2. Difficult (and sometimes impossible) work environment 3. Utilities (electrical, air, vacuum) may not be available

5. What are the advantages of performing a bench calibration? Disadvantages?

ADVANTAGES OF BENCH CALIBRATION	DISADVANTAGES OF BENCH CALIBRATION
1. Removed, cleaned, inspected 2. Better work environment 3. Fixed calibration setup and utilities (electrical, air, vacuum) available	1. May take more time (removal, reinstallation) 2. Minimizes identification of installation problems 3. Not performed in actual ambient environment where installed

6. What are the advantages of performing a loop calibration? Disadvantages?

ADVANTAGES OF LOOP CALIBRATION	DISADVANTAGES OF LOOP CALIBRATION
1. Entire loop, including sensor, is verified within tolerance 2. Mistakes on re-connect minimized 3. More efficient use of time to do one calibration for loop as opposed to one calibration for each loop instrument	1. Wrong instrument may be adjusted to bring the loop within calibration 2. Not as compatible with multifunction calibrators used for "paperless" data collection

7. What are the advantages of performing an individual instrument calibration? Disadvantages?

ADVANTAGES OF INDIVIDUAL CALIBRATION	DISADVANTAGES OF INDIVIDUAL CALIBRATION
1. Correct instrument will be adjusted 2. More compatible with multifunction calibrators	1. Entire loop is not verified within tolerance 2. Mistakes on re-connect 3. Less efficient use of time to do one calibration for each loop instrument as opposed to one calibration for the loop

8. What are the advantages of classifying instruments by their "importance/criticality" to a process?

   - *may be helpful in assigning calibration frequencies*
   - *investigate out-of-tolerance calibrations more efficiently*

9. Arrange the traceability hierarchy below, beginning with lowest level and ending with the highest. C, B, E, A, D

   A. Primary Standards
   B. Working Standards
   C. Process Instrument
   D. NIST (or recognized national standard)
   E. Secondary Standards

CHAPTER 2 REVIEW QUESTIONS

1. List the advantages of each procedure development method below.

   A. straight from technical manual

      - *Very little time/resources required to develop procedures*
      - *Technically accurate and detailed instructions for the specific instrument*

   B. generic procedure for an instrument type

      - *Limits number of procedures to a manageable level*
      - *Can be a good starting point for new facility startup until more effort can be devoted to procedure development*

   C. procedure developed for a specific manufacturer/model or specific instrument in the plant

      - *Calibrations are performed the same by all technicians*
      - *May take into account the affect on process*
      - *Technically accurate and detailed instructions for the specific instrument*

2. If test equipment with the specified accuracy is not available to perform the calibration, what should you do?

   Notify supervisor.

3. If a step of a calibration procedure cannot be followed as specified or a procedure does not exist, what actions would you take?

   Place instrument in a safe condition and notify supervisor.

4. What elements of a calibration data sheet reflect that the calibration is NIST traceable?

   Record the unique identification of any test standards used to perform the calibration and, if required by procedure, record the calibration due date of the standard.

5. For an **established** facility, what is the most likely resource for determining the calibration frequency of a new instrument?

   Calibration history of the specific instrument and identical mfg/model.

6. For a **new** facility, what is the most likely resource for determining initial calibration frequency?

   Past experience.

7. What justification can be used to increase the calibration interval (perform less often) of all instruments of the same type?

   Once the calibration history of several calibrations have been performed on similar instruments and the as-found data has been well within tolerance, the calibration interval can be increased (with proper documentation of the analysis and proper approvals per your company procedures).

8. What event(s) can lead to decreasing the calibration interval (perform more often)?

   Failed "as-found" data on successive calibrations (or on even one calibration in some cases).

9. Match the resources on the left with content on the right.

C P&ID	A. includes general instrument specifications for the facility
D Instrument Specification Sheet	B. includes instrument range capability and instrument accuracy
E Loop diagram	C. detailed overview of a process system
A Project specifications	D. detailed device requirements
F Manufacturer's Specifications	E. includes all associated electrical and piping connections

10. What is the purpose of a calibration seal?

Cover adjustments so that any unauthorized adjustments are detected.

11. What is the minimum information required on a calibration status label?

- *Instrument Identification (such as Tag Number, Instrument ID#, or serial number)*
- *Date of calibration*
- *Next calibration due date*
- *Technician who performed the calibration (initials, employee ID, etc.)*

12. When should a Limited Calibration status label be used?

For instruments that are not calibrated throughout the range of the indication.

13. What criteria must be met for test equipment to be used for calibration?

- *Specified in the calibration procedure (appropriate range and accuracy for calibration performed)*
- *Within its calibration periodicity (do not use beyond calibration due date with approved extension)*

- *Recorded on Calibration Data Sheet (to ensure process instrument calibration is traceable to NIST)*

CHAPTER 3 REVIEW QUESTIONS

Situation for Questions 1 – 10: TT-300 is installed in a process tank, which is currently in production. This is the first calibration after initial startup. Use the references in Appendix A to answer the following questions.

1. What is the correct calibration range and manufacturer's specified accuracy for TT-300? (Appendix A-3)

   Calibration Range specified is 0.0°C to 150.0°C – 4.00 mA to 20.00 mA (from row 112 of the Instrument Specification for TT-300) and the manufacturer's specified accuracy is 0.2% span (from row 53 of same specification).

2. Select the correct procedure(s) that could be used for this calibration of TT-300? (Appendix A-4)

   SOP-CAL-08, Calibration/Calibration Check of RTD Input Electronic Temperature Transmitters.

3. What local/remote indications should be recorded during calibration of TT-300? (Appendix A-1)

   Indication on TIC-300 (the components in the loop are the RTD (TE300), transmitter (TT300), Indication controller (TIC-300), I/P (TY-300) and the control valve (TV300). The only indication in the loop is the TIC.

4. What must be done prior to removing RTD for calibration of TT-300? (initial conditions of SOP)

   - *Determine if temperature control loop performs control function which could produce undesirable operating results while simulating test signals (from the P&ID we can see that this instrument loop controls the cooling water supply to the jacket of Reactor R-300. Therefore, if reactor R-300 is in operation you would need to coordinate with production to determine appropriate conditions to perform the calibration. In this case you would likely wait until system is not in operation).*

- *Determine if RTD is in a well or not (P&ID indicates the RTD is installed in a well)*

5. What are the correct resistance values to input for 10%, 50%, and 90%? (refer to the correct RTD Table in App A-6)

   From the Instrument Specification for TE-300, row 25 specifies 0.00385 ohm/ohm/C. Therefore:

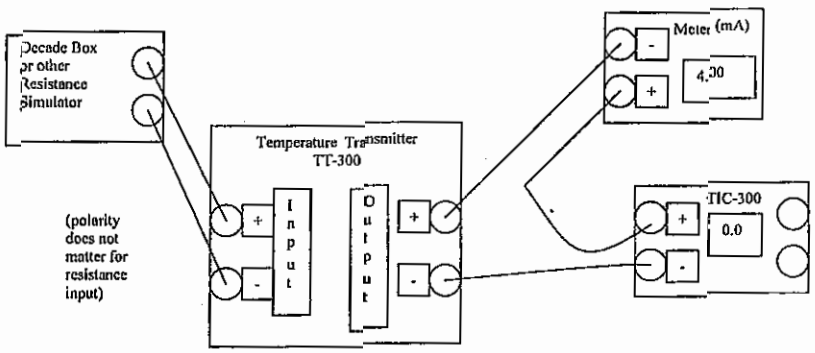
   10% = 15.0C = 105.85 ohms
   50% = 75.0C = 128.99 ohms
   90% = 135.0C = 151.71 ohms

6. What are the expected transmitter output values for the resistance inputs from step 5?

   10% output = 5.60 mA
   50% output = 12.00 mA
   90% output = 18.40 mA

   The transmitter output is 4.00 to 20.00 mA (span of 16 mA). 10% of 16mA is 1.60 mA. The 1.60 mA must be added to the zero value of 4.00 mA. The result is an expected output of 5.60 mA for the 10% value. Fifty percent of 16mA is 8.00 mA, and 8.00mA added to 4.00 mA results in 12.00 mA at the 50% value. Ninety percent of 16mA is 14.40mA, added to 4 mA results in 18.40 mA at the 90% value.

7. Indicate the correct test equipment hookup for calibration of TT-300?



8. Assume a calibration tolerance of ± 0.5°C/0.05 mA. With the results indicated what instrument(s) in the loop require(s) adjustment?

transmitter TT-300

% INPUT	TT-300 mA OUTPUT	TIC-300 INDICATION
10%	5.70	15.9
50%	12.10	75.9
90%	18.50	135.9

9. What type of error is indicated by the results of question 8?

A. Zero error.

10. Following adjustment, all as-left data is within tolerance. What must be performed to place the instrument loop back in service?

- *Disconnect test equipment*
- *Reconnect RTD and mA output*
- *Install RTD in well*
- *Place controller in automatic*
- *Notify appropriate personnel of work performed*

Situation for questions 11 – 15: You will be performing an initial calibration of TT-200 on the bench, prior to installation. Use the references in Appendix A to answer the following questions.

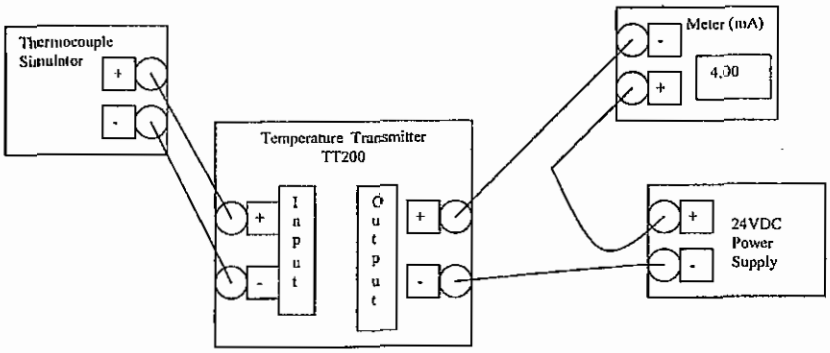
11. What are the correct reference temperatures and corresponding millivolt values to input for a calibration check at the following test points? *(The Instrument specification for TT-200 indicates range is 32.0°F to 212.0°F and the input sensor is Type T thermocouple. Note: it is typically not necessary to look up millivolt values since calibrators will read out in temperature).*

% INPUT	TEMPERATURE	MILLIVOLTS
0%	32.0°C	1.279 mv
25%	77.0°C	3.222 mv
50%	122.0°C	5.325 mv
75%	167.0°C	7.566 mv
100%	212.0°C	9.930 mv

12. What are the expected transmitter outputs for the following simulated inputs?

% INPUT	OUTPUT (mA)
0%	4.00
25%	8.00
50%	12.00
75%	16.00
100%	20.00

13. Indicate the correct test equipment hookup for calibration of TT-200.



14. Assume a calibration tolerance of ± 1.0°F / 0.08 mA. With the results indicated, what must be done to bring the instrument to within tolerance? *adjust span*

% INPUT	OUTPUT (mA)
0%	4.00
25%	7.98
50%	11.96
75%	15.94
100%	19.92

15. Following adjustment, all as-left data is within tolerance. What steps remain to complete the bench calibration?

Disconnect test equipment and restore transmitter to original condition.

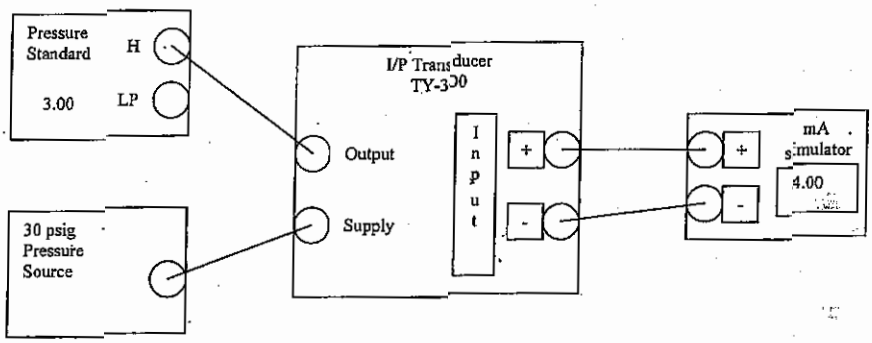
4. From your answer to question 1 and the procedure selected in question 3, what are the correct input values to simulate TY-300?

   4.0 mA, 12.0 mA, and 20.0 mA

5. What are the expected output values for the inputs applied from question 4?

   3.0 psig, 9.0 psig, and 15.0 psig, respectively (assume direct acting I/P).

6. Indicate the correct test equipment hookup for calibration of TY-300.



7. With the results indicated what instrument in the loop requires adjustment?

   I/P

% INPUT	OUTPUT (PSIG)	VALVE POSITION
0	3.2	2%
50	9.5	54%
100	15.8	100%

8. How would the calibration required in the previous step be performed?

   With 4 mA applied to the input adjust the zero for an output of 3.0 psig. Apply 20.0 mA to the input and adjust the span for 15.0 psig output. Repeat zero and span until no further adjustment is required.

9. If a control valve is being checked during an I/P calibration and the I/P is properly calibrated, what is the most likely cause of an improper valve position (no positioner is installed)?

   Bench set of the control valve spring is not matched with the output of the I/P.

10. What is the purpose of a valve positioner?

    Increase or decrease the air pressure from the I/P to assure proper positioning and finer control of the control valve.

11. Is the action (direct or reverse) verified during a positioner calibration?

    Yes.

12. What is the basic procedure for calibration of a positioner?

    Apply pressure for the desired test points to the input of the positioner and record corresponding valve position. This can be combined with the I/P calibration by teeing in the pressure standard and leaving the positioner connected to the output loop.

13. What is the basic procedure for calibration of a control valve?

    Apply pressure for the desired test points to the input of the control valve and record corresponding valve position. This can be combined with the I/P calibration by teeing in the pressure standard and leaving the control valve connected to the output loop.

CHAPTER 8 REVIEW QUESTIONS

1. What are the ideal ΔmV input values for a pH meter at 4, 7, and 10 pH at 25°C?

   177.48 mV, 0 mV, and –177.48 mV, respectively.

2. What buffer solution is used to standardize a pH meter?

   7.0 buffer solution.

3. What buffer is preferred for calibration of the slope? Why?

   4.0 buffer solution because pH buffer solutions above 7.0 pH are less stable and have a limited shelf life.

4. What is the purpose of a pH diagnostic test?

   To evaluate electrode performance.

5. What are the basic steps of a pH diagnostic test?

   - *Short the input and adjust for reading of 0.0 mv*
   - *Insert pH electrode in 7.00 pH buffer solution and record this mv reading as the asymmetry potential (AP)*
   - *Insert pH electrode in 4.01 buffer solution and record mv reading*
   - *Calculate slope*

6. What is the diagnostic test acceptance criterion that determines whether the electrode should be replaced?

   If AP is > ± 40 mV and/or the slope is < 91%.

7. What test equipment is used to simulate a conductivity signal to a conductivity meter?

   Resistance simulator such as a decade box.

8. How is the resistance value input for a conductivity instrument calibration determined?

   Resistance is the inverse of conductivity (Resistivity = 1 ÷ conductivity).

9. How is the 25°C temperature simulated during a conductivity or resistivity instrument calibration?

   A resistor of the specific resistance to simulate 25°C for the specific conductivity instrument is connected in place of the temperature sensor leads (or a separate decade box could be connected set to the required resistance).

10. Why is it important to know the cell constant for a conductivity calibration?

The cell constant is used as a multiplication factor for the indicated conductivity. If the cell constant is not known or accounted for, the calibration will likely result in significant error.

INDEX

acceptance criteria 21
accuracy 2
    ratio 3
actuator 94
air pressure 49
as-found data 23, 54, 68, 69
as-left data 23, 69
atmospheric pressure 49

bench calibration 11
Bourdon pressure gauge 53

calibration 1
    bench 11
    control valve 97
    d/p transmitter 83
    data sheets 21
    equipment combined accuracy 6
    field 11
    flowmeter 82
    I/P transducer 95
    individual instrument 10
    intervals 28
    loop 11
    pH 105
    procedure number 22
    procedures, specific 19
    range 1, 22
    seal 38
    status labels 30
    tolerance 2, 22
    valve positioner 96
capacitance 62
    level instrument 71
    probe 63
capacitor 62

cell constant 107
classification of instruments 12
conductivity 108
    measuring 107
control system technician (CST) 8
    characteristics 9
control valve 94
    calibration 97
coriolis mass flowmeter 81
correct fittings 51
critical 12

d/p transmitter calibration 83
dial thermometer 36
differential pressure 61, 78
distributed control system (DCS) 42
documentation 17
dry legs 66

elevated zero 64
EPA 13
error
    combined zero and span 7
    linearization 8
    span 6
    zero 7

field calibration 11
filled-bulb sensors 35
final control element 91
flow instrumentation 77
flowmeter
    calibration 82
    coriolis mass 81
    differential pressure 78
    magnetic 79

turbine 81
vortex-shedding 80

generic procedure 18
gravimetric method 87
gravity, specific 64

hysteresis 51

I/P transducer 91, 92
calibration 95
individual instrument calibration 10
instrument
identification number 21
range 2
specification forms 26
tag number 21
instrumentation
accuracy and reliability 17
interface 56

leakage 51
level instruments 61
linearization error 8
liquid head measurement 65
loop
calibration 11
diagrams 25

magnetic flowmeter 79
calibration 84
manufacturer's specifications 27
master meter 86
measurement by weight 87
motion-balance positioner 93

nameplate data 22
National Institute of Standards and Technology (NIST) 4
non-critical 12
normally closed (NC) 44
normally open (NO) 44

OSHA 13
out-of-tolerance standard 4

P&IDs 25
pH 102, 108
calibration 105
electrode output 104
pneumatic signal 94
preliminary operating point check 41
pressure 49
air 49
atmospheric 49
calibrating 50
gauge, Bourdon 53
gauges 52
switch, calibrating 55
transmitter, calibrating 55
process instrument combined accuracy 6
programmable logic controller (PLC) 42
project specifications 27
prover 86

radio frequency (RF) 62
reference only 12
resistance temperature detectors (RTDs) 34
resistivity 107
RTD calibration check 38

safety 20, 29
considerations 52
signal conversion 35
span 2
error 6
standardize 103
suppressed zero 64

technical manual procedure 18
temperature 33, 103, 108
controller loop 42
sensors 34
switch 44
transmitter 40
test
equipment 36
procedure 21
standards 23
thermistors 35
thermocouples 34
tolerance 2
traceability 4, 5
trip 44

turbine flowmeter 81

ultrasonic 63, 64
  level transmitter 72
uncertainty 5
unit of measure 50

valve positioner
  calibration 96
vapor 66
vortex-shedding flowmeter 80

wet legs 65

zero error 7
zero point 103